



— BUREAU OF —
RECLAMATION

Pole Hill Powerplant Upgrade

Controls, Protection, and Station Service Upgrade

Colorado-Big Thompson Project, Colorado

Missouri Basin Region

Solicitation No. **Property of the**
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**POLE HILL POWERPLANT
CONTROLS, PROTECTION, AND STATION SERVICE UPGRADE
COLORADO-BIG THOMPSON PROJECT, COLORADO**

FOREWORD

- A. Work is located at Pole Hill Powerplant, approximately 17 miles west of Loveland, Colorado in Larimer County. Work consists primarily of replacement of protection, controls, instrumentation, and station service switchgear and transformer. The powerplant site's latitude and longitude coordinates are approximately 40 degrees 21 minutes 54 seconds north by 105 degrees 19 minutes 30 seconds west (40.365083, -105.325194).

1.02 PRINCIPAL COMPONENTS OF WORK

- A. Powerplant:
1. Replace protective relays, plant controls, control panels, auto-synchronizer, annunciator, meters, instrumentation, and station service.
 2. Relocate station service switchboard to the turbine level.
 3. Replace low voltage electrical panels and transformer.
 4. Relocate one air compressor.
 5. Replace two existing station service transformers with a single station service transformer.
- B. Gatehouse:
1. Replace low voltage electrical panels and transformer, instrumentation, penstock gate controls, and battery systems.
 2. Replace approximately 2,000 feet of communication conduit and fiber optic cable between the gate house and the powerplant.

PRE-BID SITE VISIT: A PREBID SITE VISIT WILL BE HELD ON {insert date}, {insert location}, BEGINNING AT {insert time}. PROSPECTIVE BIDDERS SHOULD MAKE ARRANGEMENTS BY CONTACTING [] AT THE TELEPHONE NUMBER LISTED ABOVE. ATTENDEES ARE TO PROVIDE AND WEAR SAFETY BOOTS, HARD HATS, SAFETY VESTS.

FOR DATE AND PLACE OF BID OPENING, SEE "SOLICITATION, OFFER, AND AWARD," STANDARD FORM 1442.

**FOR INFORMATION REGARDING BUREAU OF RECLAMATION'S PUBLICATION
"RECLAMATION SAFETY AND HEALTH STANDARDS" INCLUDING REVISIONS
AT <https://www.usbr.gov/safety/rshs/> WHICH IS APPLICABLE TO WORK UNDER
THIS CONTRACT, SEE CLAUSE AT WBR 1452.223-81, SAFETY AND HEALTH.**

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**CONTINUATION OF
SECTION B – SUPPLIES OR SERVICES AND PRICES/COSTS**

**POLE HILL POWERPLANT
CONTROLS, PROTECTION, AND STATION SERVICE UPGRADE
COLORADO-BIG THOMPSON PROJECT, COLORADO TURBINE**

B.1 PRICE SCHEDULE

- (a) Offers will be considered for award on the following Price Schedule, but no offer will be considered for award on only a part of the Price Schedule.
 - (1) Offerors shall complete Price Schedule. Offers for less than the total of the Price Schedule will be considered nonresponsive and will not be evaluated for award.
 - (2) Offers will be evaluated based on the total of the Price Schedule.
- (b) Offers are subject to the terms and conditions of this solicitation.
- (c) Quantities in the Price Schedule are estimated quantities for comparison of offers only. Except as provided in the contract clause at FAR 52.211-18, Variation in Estimated Quantity, no claim shall be made against the Government for overruns or underruns. The contract clause at FAR 52.211-18, Variation in Estimated Quantity does not apply to lump sum items.
- (d) See contract clause at WBR 1452.232-81, Payment for Mobilization and Preparatory Work.
- (e) Definitions:
 - (1) CLIN: Contract Line Item Number.

PRICE SCHEDULE

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
1	WBR 1452. 232-81	Mobilization and Preparatory Work	For the lump sum of		\$
2	01 31 30	Contract Document Management System (CDMS)	For the lump sum of		\$
3	01 78 30	Spare Parts	For the lump sum of		\$
4	01 79 20	Training	For the lump sum of		\$
5	22 15 00	Governor Compressed-Air Systems	For the lump sum of		\$
6	25 00 01	Automation Control and Monitoring Systems	For the lump sum of		\$
7	26 04 10	Demolition and Removal of Electrical Equipment	For the lump sum of		\$
8	26 04 10	Removal of Medium Voltage Power Cable Systems	300 lin ft	\$	\$
9	26 05 10	Conductors and Cables	lin ft	\$	\$
10	26 05 13	Medium-Voltage Power Cable	300 lin ft	\$	\$
11	26 05 33	Electrical Conduit	lin ft	\$	\$
12	26 05 33	Core Drilling Holes in Concrete	4 lin ft	\$	\$
13	26 05 33	Cable Vaults	ea	\$	\$
14	26 05 36	Cable Trays	27 lin ft	\$	\$
15	26 05 39	Engine Generator Terminal Box	For the lump sum of		\$
16	26 09 18	Control Boards CSA for Unit 1	For the lump sum of		\$
17	26 12 19	Pad Mounted Transformer	For the lump sum of		\$
18	26 22 12	Dry-Type Transformer	For the lump sum of		\$
19	26 24 13	Switchboards	For the lump sum of		\$
20	26 24 24	Distribution Panelboards	For the lump sum of		\$
21	26 24 23	600 Volt Motor Control Center	For the lump sum of		\$
22	26 27 40	Wiring Devices	For the lump sum of		\$

PRICE SCHEDULE

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
23	26 29 13	Enclosed Contactors	For the lump sum of		\$
24	26 33 46	125v DC Station Battery, Battery Rack and Charger	For the lump sum of		\$
25	26 36 23	Automatic Transfer Switch	For the lump sum of		\$
26	27 05 43	Outdoor Conduit and Accessories	For the lump sum of		\$
27	27 20 01	Fiber Optic Cable and Accessories	For the lump sum of		\$
28	28 05 11	Place holder			
29	33 09 10	Instrumentation and Control Equipment	For the lump sum of		\$

TOTAL FOR PRICE SCHEDULE: \$ _____

END OF SECTION

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SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

1.01 LOCATION

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- C. Replace approximately 2,000 feet of communication conduit and fiber optic cable between the gate house and the powerplant.

1.03 SPECIFICATIONS REQUIREMENTS

- A. Requirements in Division 1, General Requirements, apply to Divisions 2 through 53.
- B. Imperative statements in these specifications are Contractor requirements unless otherwise stated.
- C. Where specifications are written in streamlined form, the words "shall be" are included by inference where a colon (:) is used within a sentence or phrase.

1.04 DEFINITIONS

- A. When specifications use a word or term defined in Federal Acquisition Regulations (FAR), definition of the word or term shall be in accordance with FAR sections in effect at the time solicitation was issued.

1.05 ACRONYMS

- A. The following acronyms apply to specifications Divisions 1 through 53:
1. CO: Contracting Officer.
 2. COR: Contracting Officer's Representative.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 14 10

USE OF SITE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1. Reclamation Cleaning Manual Inspection and Cleaning Manual for Equipment and Vehicles to Prevent the Spread of Invasive Species, available at: <https://www.usbr.gov/mussels/prevention>

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 14 10-1, Land Use and Landscape Rehabilitation Plan:

1. For each Contractor use site on Government land.
 - a. Show use location and extent of impact. Uses include, but are not limited to, the following:
 - 1) Buildings and service areas including offices, shops, warehouses, storage areas, fuel and oil storage areas, and fabrication yards.
 - 2) Parking areas, temporary roads, and haul routes.
 - 3) Utilities including air, power, and water lines; fire hydrants; and compressor station.
 - 4) First-aid and medical facilities.
 - 5) Areas for processing, storing, and disposing of waste materials from construction operations.
 - 6) Temporary fences.
 - b. Describe methods to preserve, protect, and repair, vegetation (e.g., trees, shrubs, and grass) and other landscape features on or adjacent to jobsite, which are not to be removed and which do not interfere with work required under this contract. Include methods to mark work area limits, protect disturbed areas, and prevent erosion.

- c. Describe methods to protect, and repair if damaged, existing improvements and utilities at or near jobsite.
 - d. Describe methods for removing temporary structures and facilities, cleanup, and rehabilitating site after completion of construction activities.
- 2. Submit revised drawings of changes in use of Government land made during design and erection stages or after use of Government land is in operation.

1.04 PROJECT CONDITIONS

- A. See Figure 01 14 10A –Pole Hill Powerplant Area Overview for site overview and loading and unloading areas.
- B. Government land areas indicated on Figure 01 14 10B –Pole Hill Powerplant Contractor Use Area Closeup may be used for required construction facilities.
 - 1. Protect and maintain stockpiled material in accordance with State, County, and local requirements, and requirements of this contract.
 - 2. Coordinate with COR for locations of dedicated Government parking spaces within shared Contractor / Government parking areas.
- C. When private land is used for construction facilities, or other construction purposes, make necessary arrangements associated with use of private land.
- D. Location, construction, operation, maintenance, and removal of construction facilities on Government land are subject to approval.
- E. Do not interfere with work of other contractors or the Government in vicinity, or with reservations made by the Government for use of such land.
- F. Housing for construction personnel is not permitted on Government land.

Pole Hill Powerplant

Controls, Protection, and Station Service Upgrade

Colorado-Big Thompson Project, Colorado

Solicitation No.

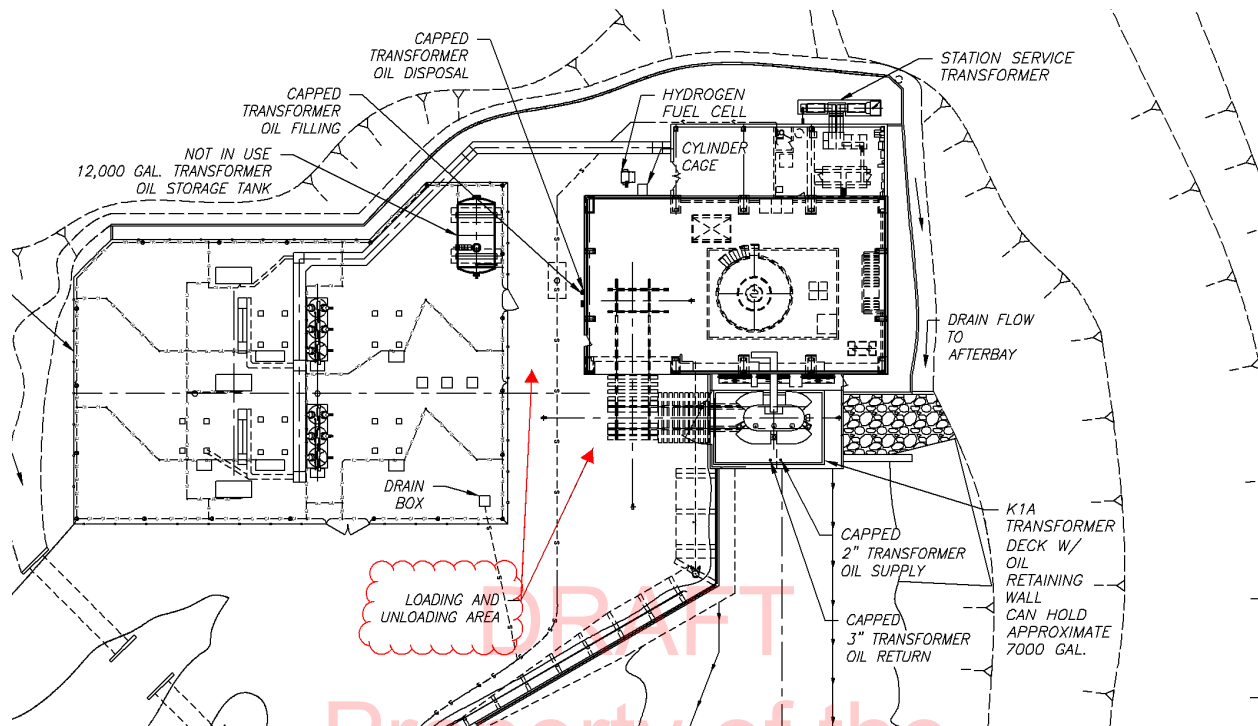


Figure 01 14 10A - Pole Hill Powerplant Area Overview



Figure 01 14 10B – Pole Hill Powerplant Contractor Use Area Closeup

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CLEANING

A. Construction Equipment:

1. Before bringing on site, clean construction equipment, to remove dirt, vegetation, and other organic material to prevent introduction of noxious weeds, and invasive plant and animal species.
2. Contractor cleaning procedures shall result in equipment being cleaned as well or better than procedures described in Reclamation Cleaning Manual.
3. COR will inspect construction equipment following procedures described in Reclamation Cleaning Manual before allowing equipment onsite.

3.02 RESTORATION

- A. Restore disturbed areas to pre-construction conditions.
- B. After completion of work, regrade and scarify Government land used for construction purposes and not required for completed installation so that surfaces blend with natural terrain and are in a condition that will facilitate revegetation, provide unimpeded surface drainage, and prevent erosion.
- C. Seed disturbed areas of Government land used for construction purposes and not required for completed installation in accordance with Section 32 92 20 – Seeding.

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SECTION 01 14 12

WORK RESTRICTIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 PROJECT CONDITIONS

A. Onsite work hours restricted to Monday through Saturday, 6:30 a.m. to 5:00 p.m., unless otherwise authorized by COR.

1. Request additional work hours, for COR approval, in writing at least 7 days in advance of need.
2. Request additional shifts, for COR approval, in writing at least 21 days in advance of need.

B. No onsite work shall be performed on Federal Holidays.

C. Access to powerplant will be controlled.

1. Keys will not be issued to Contractor.
2. Coordinate entry to facility with COR.
3. Contractor personnel shall be onsite to accept and store deliveries.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

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SECTION 01 14 30

INTERRUPTION OF SERVICE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work which require interruption of services.

1.02 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 14 30-1, Interruption of Service Request:
1. Equipment that will be out of service.
 2. Commencement time of work.
 3. Duration of work.
 4. Area where work will be done (by number or other description).
 5. Outline procedures for accomplishing work including:
 - a. Specific safety precautions.
 - b. Type and location of barricades.
 - c. Warning signs.
 - d. Protective grounds and devices.
 - e. Number of personnel and their classification.
 - f. Description of equipment.
 6. Information indicating that required equipment and materials are on site or data indicating shipping dates of equipment and materials not on site.

1.03 PROJECT CONDITIONS

- A. Coordinate and schedule interruption of services with COR:
1. Coordinate work with Government operations.
 2. No specific interruption of service request will be considered unless:
 - a. COR has reasonable assurance that materials and equipment required for work will be onsite.

- b. Contractor is prepared to perform work on date and during period requested.
 - c. Submittals required to complete the work are approved.
 - 3. Allowable time and available periods will govern interruption of service requests, work scheduling, and onsite delivery of materials.
- B. When work cannot be performed during an approved interruption of service period, notify COR.
- C. Show construction interruption of services greater than 4 hours in construction schedule.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 31 19
PROJECT MANAGEMENT AND COORDINATION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work.

1.02 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 31 19-1, Weekly Coordination Meeting Documentation:
1. Provide agenda 3 Federal working days before meeting time.
 2. 3-week look-ahead schedule covering the following as applicable:
 - a. Shop fabrication review items and tasks.
 - b. Construction review items and tasks.
 3. Work completed to date.
 4. Meeting minutes from the previous weekly coordination meeting to include:
 - a. Date, time, and list of attendees.
 - b. Summary of topics discussed.
 - c. List of action items, including:
 - 1) Description of action item.
 - 2) Responsible party.
 - 3) Completion due date.

1.03 PROJECT COORDINATION

- A. The following meetings are considered significant, but do not relieve Contractor from responsibility of attending other meetings required by the specifications.
1. Weekly Coordination Meetings:
 - a. Attendees as required:
 - 1) Contractor's project manager, project engineer, superintendent, and Quality Control Supervisor, and pertinent subcontractors.
 - 2) COR, project manager, project engineer, safety personnel, and other required Reclamation staff.

b. Purpose:

- 1) Review drawings, data, and technical requirements of specifications. Schedule, Progress, and Coordination:
 - a) Review safety, progress made, schedule, project issues, submittals, Request for Information (RFI), and resolution necessary to meet specifications requirements.
 - b) Review dates that activities were started and completed, remaining duration, and percent complete.
 - c) Review submittals register, submittals currently in review and approval, and meeting submittals due dates or delivery time frames.
 - d) Provide 3-week look-ahead schedule at least 3 Federal working days before start of meeting.
 - e) Review drawings.
 - f) Provide meeting minutes from the previous weekly meeting.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 31 30
CONTRACT DOCUMENT MANAGEMENT SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Contract Document Management System:
1. Payment: Lump-sum price offered in the Price Schedule.

1.02 DEFINITIONS

- A. CDMS: Contract Document Management System.
1. Contract Documents include, but are not limited to:
 - a. Specifications.
 - b. Contract Drawings.
 - c. Submittals.
 - d. RFCs: Requests for Change (Contractor generated document).
 - e. RFIs: Requests for Information (Contractor generated document).
 - f. RFPs: Requests for Proposal (Government generated document).
 - g. CMs: Contract Modifications.
 - h. Contractor Proposals.
 - i. Value Engineering Proposals.
 - j. Government Inspection Reports.
 - k. Invoices and Progress Payments.
 - l. Contract Schedules.
 - m. Meeting Agendas and Meeting Minutes.
 - n. Letters and Memos.

1.03 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 31 30-1, Approval Data:
1. System provider experience, number of systems operating, references.
 2. Documentation of system capabilities.
 3. Instructions for system use.

C. RSN 01 31 30-2, Final Data:

1. Reclamation supplied USB Drive with uploads to CDMS.

1.04 QUALIFICATIONS

- A. System provider: At least 5 years successful performance in providing required services.

1.05 SYSTEM DESCRIPTION

- A. CDMS shall be capable of generating, storing, tracking, categorizing, and managing Contract Documents.
- B. Access:
1. Web-based:
 - a. Contract Documents are stored and accessed by authorized individuals via an internet site.
 - b. Compatible with web browsers Microsoft Edge, Apple Safari, Mozilla Firefox, and Google Chrome.
 2. Ability to upload Contract Documents and make available for user download in supported file formats, minimum:
 - a. Adobe Acrobat.
 - b. MS Word.
 - c. MS Excel.
 - d. MS Project.
 - e. MS PowerPoint.
 - f. MP4 for Videos.
 - g. Autodesk AutoCAD Civil 3D Version 2021.
 3. Allow Contract Documents to be prepared by Contractor or by Government.
 - a. Preparing organization will control access of documents. See Document Security and Backup paragraph below.
 4. Contractor and Government shall each have a project manager for CDMS.
 - a. Each project manager shall have capabilities to assign users within their organization and to assign user rights which control access to documents based on user class and document type.
 - b. Each organization shall be capable of defining its access hierarchy.
 - c. Inclusion of users will be prerogative of organizational project manager.

- d. Government project manager shall be able to create and modify project properties (i.e., Contractor, project location, description, bid amount, and project directory).
 - 5. Email notifications shall be automatically sent to selectable users when new documents are submitted and available for viewing, or alerts are generated. Examples include:
 - a. Updates to project information (e.g., entry of a new submittal);
 - b. Changes to project information (e.g., approval of a submittal);
 - c. Alerts (e.g., submittal under review for greater than 15 days without action).
 - 6. Ability to link files. Examples include:
 - a. RFP linked to a CM.
 - b. Submittal linked to a Specification Section or Drawing.
- C. Reports:
 - 1. Generate reports that list and sort documents by status. Examples include:
 - a. Submittals that have been approved.
 - b. Invoices paid to date.
 - c. RFIs under review.
 - 2. Generate alerts when documents requiring action approach or exceed allowable time.
 - 3. Generate customizable summary reports. Examples include:
 - a. Submittal Turn-Around.
 - b. RFI Turn-Around.
 - c. CM Processing Turn-Around.
 - d. Total CM Costs.
 - e. Total CM Request Exposure.
 - 4. Printable, exportable as searchable pdf or printer friendly HTML, and exportable to Excel, XML, or CSV.
 - 5. Ability to link from an item in a report directly to item by clicking on it.
 - a. For example, in a submittal report, clicking on a submittal that is 15 days old takes you directly to information on that submittal.
 - 6. Capable of producing a complete and logically organized set of documents within CDMS in both .pdf and .xml format.
- D. Document Security and Backup:

1. Secure Server Language (SSL) encryption for secured data exchange between browser and server (i.e., secure access to documents and information).
 2. Allow document access security until authoring party chooses to share it with others.
 - a. Show parties that have access to a document.
 - b. Extent of document sharing shall be determined by authoring party.
 - c. Once documents are shared, track documents so that changes cannot occur without a record of changes.
 3. Backup/Archiving to FTP site of documents to which an organization has access on a regular (minimum monthly) basis.
 - a. Backups shall be in pdf format.
 4. Documents shall be downloadable to FTP sites where Government and other organizations can each access their own data confidentially.
- E. General Tracking and Control:
1. Documents shall be grouped into categories (Submittals, RFC, RFI, RFP, CM, Payments, Daily Inspection Reports, Meeting Minutes).
 2. CDMS shall track:
 - a. When documents were received or returned, as well as status of documents (e.g., Under Review; Returned, Approved or Rejected).
 - b. Document changes including who entered or changed document and date change was made.
 - c. When a reviewer has opened an item.
- F. Submittal Tracking and Control:
1. Ability to create master list of submittals (“Schedule of Submittals”).
 2. Customizable standard submittal form for entering information:
 - a. Required submittal number (RSN), title, description, due date, and specification section.
 - b. Upload submittal data in electronic format.
 3. Track date received, date response due, date returned, and approval status.
 - a. Status designations may be modified to fit Government standard designations – A/AWC/NA (Approved/Approved with Comments/Not Approved).
 4. Supply Submittal Tracking form for Government to enter review comments and action taken. Allow customizing of submittal tracking form.

- a. Allow for Government to route submittal to third parties (e.g., consultant, internal design, or operations and maintenance groups) by generating design-review transmittals with each submittal package item, to track status of individual sub-items within submittal package, to track status of individual sub-items within the submittal package, and to route each sub-item to appropriate reviewers.
 5. Ability to generate automatic notification after specified days without response.
- G. RFI Tracking and Control:
1. Supply standard RFI form to enter information.
 2. Ability to create or attach documents to the RFI form or links to other documents within CDMS.
 3. Ability for Government to route RFI to appropriate reviewers and for reviewers to enter comments into standard response form.
- H. Inspection Reports Tracking and Control:
1. Ability to attach Government standard inspection and report forms into CDMS and ability to create customizable inspection forms in CDMS. Typical forms:
 - a. Daily Inspection Report (DIR).
 - b. Survey requests.
 - c. Plant inspection reports.
 - d. Testing reports.
 2. Ability to link or attach photos to forms and link forms to other documents within CDMS.
- I. Contract Modifications:
1. Supply standard Contractor Proposal form with ability to attach documents and link to other documents within CDMS.
 2. Supply ability to upload Government's standard CM form and RFP forms, or create an acceptable alternative, to enter information.
 3. Ability to attach documents to these forms, and link to other documents within CDMS.
 4. Ability for Contractor to respond to RFP and CM form with Cost and Time impacts.
 5. Ability to track date CM issued and date responded, with automatic notification after specified days without response.
- J. Cost Tracking and Payment Request Management:
1. Ability to import payment bid items cost from a MS Excel spreadsheet.

2. Ability to edit percent complete and payment amount each month. Format to be spreadsheet (e.g., items available for editing at once, rather than needing to edit one item at a time, save, open next item and edit that).

K. Correspondence:

1. Ability to generate letters and memos and to:
 - a. Attach documents to letters and memos.
 - b. Reference other documents within CDMS.
 - c. Automatically assign sequential numbers to letters and memos.
2. Ability to generate responses to letters and memos.
 - a. System automatically generates memo suffixes (e.g., first response to Memo 32 to be 32.1, second response to be 32.2).
3. Official correspondence from Government to Contractor will contain Government's official letterhead and corresponding logos.
4. Ability to generate meeting agendas and meeting minutes, and to:
 - a. Attach documents to minutes.
 - b. Automatically assign sequential meeting numbers.

L. Miscellaneous Tracking and Controls. Ability to:

1. Generate reports tracking RFCs, RFIs, RFPs, CMs, Submittals, Payments, Memos, DIR's, survey requests, plant inspection reports and testing reports.
2. Filter logs based on criteria:
 - a. Submittals: Specification Section, Days Out, Status (Approved – Complete, Approved with Comments – Resubmit, Received, Not Approved).
 - b. RFI's: Days out, Status (Open/Closed), Specification Section/Dwg No., Responsibility/Ball-In-Court.
3. Sort Logs based on criteria (e.g., for Submittal, Specification Section, Days Out, Status, or other criteria determined by COR).
4. Supply forms to match Government's CM Form and Progress Payment Form or other forms when needed.
5. Track Cost Exposure with links to RFI and CM requests.
6. Measure productivity benchmarks on a project such as computing average review days for RFIs and submittals, categorize CMs, and track total CM percentages by category.
7. Keyword search documents in database.
8. Allow online Payment Requests (electronic invoicing) by Contractor.

M. Budget and Funding Source Tracking:

1. Ability to enter funding sources and assign contract line items to multiple funding sources.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 IMPLEMENTATION

A. CDMS shall be in place and operating within 45 days of Award:

1. CDMS operation includes COR, or other authorized Government representative, having program control to assign users and user rights to Government personnel to access appropriate areas of system.

B. Input Schedule of Submittals before making submittals. Include:

1. RSNs listed in Table 01 33 00A – List of Submittals.
2. Scheduled submittal date.

C. Provide a summary table report of submittal status monthly or as requested by COR. Include:

1. Required Submittal Number (RSN).
2. Section Title.
3. Submittal Title.
4. Date Submitted.
5. Date of Response.
6. Approval Status (Approved – Complete, Approved with Comments – Resubmit, Received, Not Approved).
7. Days in Review.
8. Due Date.

D. CDMS shall be operational until final project closeout.

END OF SECTION

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SECTION 01 32 20

CONSTRUCTION PROGRAM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Developing Baseline Schedule: Include in price offered in the Price Schedule for mobilization and preparatory work.
 2. Updating and Using Construction Program: Include in prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

- A. Cost Loading: Contractor earnings based on payments from Government to Contractor.
- B. Project Calendar(s): Cross reference of numerical work days with calendar days. Project calendars serve as the basis for day/date conversion and assign work days and non-workdays.
- C. Resources: Equipment, labor or crews, materials, subcontractors, fabricators, manufacturers, and consultants.
- D. Schedule: Critical Path Method (CPM) of planning and scheduling a construction project where activities are arranged based on activity relationships and network calculations determine when activities can be performed and the critical path of the project.

1.03 REFERENCE STANDARDS

- A. Associated General Contractors of America (AGC)
1. AGC Manual-04 Construction Planning and Scheduling Manual
- B. Bureau of Reclamation (Reclamation)
1. PM Framework Reclamation's Project Management Framework Report, available at:
<https://www.usbr.gov/pm/docs/Project-Management-Framework-Report.pdf>

1.04 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.

1. Database files in format compatible with Microsoft Project (release 2010 or newer).
2. Schedule and analysis data.
3. Upon request, provide information and data used to develop and maintain the Construction Program to the CO.

B. RSN 01 32 20-1, Baseline Schedule:

1. Include:
 - a. Construction Program/Schedule database.
 - b. Narrative explanation of project schedule development:
 - 1) Describe intended work sequence addressing principal features of work, contract, and intermediate milestones.
 - 2) Include impacts of resource limitations on project schedule development. Include planned number of work hours per day and hours per day for subcontractors.
 - 3) Description of project calendar(s), to include workdays per week, non-workdays due to holidays.
 - c. Gantt chart (bar chart) for project.
 - d. Activity report including logic constraints consisting of predecessors, successors, and constraint dates.
 - e. Tabular listings of resources, resource limitations, and production rates used to produce baseline schedule:
 - 1) Correlate resources with schedule assumptions and offered price.
 - 2) Tabular listing of resources does not require resource loading in a schedule database.

C. RSN 01 32 20-2, Updated Schedule Reports:

1. Government disapproval will be considered failure of Contractor to comply with contract requirements. Payment due date may be delayed until 14 days after submittal contents are approved in accordance with subparagraph (a)(i)(A) of clause at FAR 52.232-27, Prompt Payment for Construction Contracts.
2. Include:
 - a. Construction Program database with updated activity and milestone data.
 - b. Narrative report:
 - 1) State status of project including data date represented by project.
 - 2) Discuss contract milestones and significant activities started, continued, or completed during update period.

- 3) Discuss contract milestones and significant activities scheduled to start, continue, or be completed during next update period.
 - 4) Definition of project calendars if revised from baseline calendars.
 - 5) If negative float exists, cite specific actions and conditions which caused the "behind schedule" condition and provide proposed course of action to complete the project within the specified delivery time.
 - 6) List of Contractor-initiated changes to current schedule. State reason for action taken and unresolved issues relating to the Construction Program.
- c. Gantt chart for project.
 - d. Activity value report: As detailed below.
 - e. Earned value report: As detailed below.

1.05 SCHEDULE

- A. Develop, maintain, and use approved Construction Program to plan, monitor, evaluate, and report accomplishment of work.
- B. Prepare construction schedules using Critical Path Method outlined in AGC Manual.
- C. Failure to include an element of the work will not release Contractor from completing required work under the contract.
- D. Construction Program:
 1. Prepare detailed activity network for accomplishing required work organized by work breakdown structure.
 2. Prepare schedule based on required sequence and interdependence of activities.
 3. Include work of subcontractors, Government interfaces, and contract milestones.
 4. Include contract title, contract number, and Contractor's name on each sheet.
 5. Adjust Construction Program/schedule for seasonal weather conditions.
 6. Define activities to a level of detail resulting in their durations being no greater than 20 workdays unless otherwise accepted by CO.
 - a. Durations for administrative activities (e.g., submittals and reviews, fabrication, manufacturing) and typical level of effort activities identified in the contract will not be subject to the workday duration limitation.
 7. Use durations in units of whole workdays.
 8. Include activities for Government reviews and approvals of submittals that are critical to the project completion data.

9. Update schedule with changes directed by contract modifications or approved changes to sequencing.
- E. Baseline Schedule:
1. Represents Contractor's as-planned approach to accomplishing the work.
 2. Include planned start and completion dates for features of work.
- F. Updated Schedule:
1. Enter actual progress.
 2. Assign a unique project file name for each schedule update.
- G. Cost Loading:
1. Allocate contract value to schedule activities per CLIN amounts in accordance with contract requirements and Measurement and Payment statements in specifications.
 2. Provide a Schedule of Values for Lump Sum CLIN's. Coordinate content for Schedule of Values with COR prior to submitting Cost-Loaded Schedule.
 3. Assign values to appropriate activities.
 - a. Total of activity values equal total contract value.
 - b. Every activity does not require a value. For example, submittals, submittal reviews, and manufacturing typically do not have values.
 4. Include separate schedule activities to depict amount and timing of payment, if necessary, such as Delivery activities.
 5. Activity Value Report:
 - a. Show expense allocation for each activity. Include for each activity:
 - 1) Activity ID.
 - 2) Activity Name.
 - 3) CLIN.
 - 4) Planned Cost.
 - 5) Actual Cost.
 - 6) Budget at Completion.
 - 7) Remaining Cost.
 6. Earned Value Report: Generate estimated monthly earned value of activities following guidance in PM Framework.
 - a. Show project cash flow (i.e., summation of activities) for project duration.
 - b. Show cumulative monthly cash flow for project duration, including:

- 1) Budgeted cost of work scheduled or the planned value (PV) of work at monthly intervals.
 - 2) Budget at Completion (BAC): Contract cost.
 - 3) Estimated monthly placement (i.e., S-curve with estimated costs).
 - 4) Actual cost (AC) of work performed for each month and the current cumulative amount.
 - 5) Estimated physical percent complete of activity.
 - 6) Earned Value (EV) of work performed for each month and the current cumulative amount.
 - 7) Cost variance (CV).
 - 8) Schedule variance (SV).
 - 9) Cost performance index (CPI).
 - 10) Schedule performance index (SPI).
 - 11) Estimate to complete (ETC).
 - 12) Estimate at completion (EAC).
 - 13) Variance at completion (VAC).
7. Assign contract values for contract modification activities.
- H. Contract Change or Delay:
1. Submit a time impact evaluation within 14 days of contract change or delay, showing effects of change order or delay on the progress schedule.
 2. Time extensions for performance will be considered only to the extent that the Contractor's current scheduled milestone dates exceed the contract milestone dates.
- I. Float is not for the exclusive use by or benefit of either the Government or the Contractor.

1.06 FAILURE TO COMPLY

- A. Failure to comply with requirements of this section shall be grounds for determination by CO that Contractor is not prosecuting the work with sufficient diligence to ensure completion within the specified time.
- B. CO may terminate Contractor's right to proceed with the work, or separable part of it, in accordance with default terms of this contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 33 00

SUBMITTALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
 - 1. Include in prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

- A. Days: Calendar days.
- B. Required Submittal Number (RSN): Identifies items to be submitted together as a complete submittal.
- C. Submittals: Considered “shop drawings” within the terms of clause at FAR 52.236-21, Specifications and Drawings for Construction.

1.03 REFERENCE STANDARDS

- A. National Institute of Building Sciences (NIBS)
 - 1. NIBS NCS-14 United States National CAD Standards, Version 6

1.04 SUBMITTAL REQUIREMENTS

- A. In case of conflict between requirements of this section and requirements included elsewhere in these specifications, requirements included elsewhere take precedence.
- B. General:
 - 1. Prepare in English.
 - 2. Label with contract number and title, and RSN.
 - 3. Measurement Units: US Customary Units.
- C. Drawings:
 - 1. Minimum Identification in Title Block:
 - a. Contract number and title.
 - b. Contractor’s or supplier’s title and drawing number.
 - c. Date.
 - 2. Reserve 3- by 3-inch space next to title block for review stamps.

3. Size: ANSI D size (22 inches by 34 inches).
4. Draw to scale using computer drafting.
 - a. In accordance with NIBS NCS.
 - b. Electronic file format: Compatible with AUTOCAD, Version 2021 or later.
 - c. Compile using “eTransmit” utility in AUTOCAD.
 - d. Government will provide electronic AutoCAD format template.
 - e. Title block and sheet format:
 - 1) As shown on Standard Drawing 40-D-7102.
 - 2) Government will supply template.
 - 3) Government will supply specific title block information.
 - f. Show as-built changes, including revision dates, made during installation. Indicate changes by clouding.

D. Product Data:

1. Mark manufacturer's data for commercial products or equipment, such as catalog cut sheets.
 - a. Identify manufacturer's name, type, model, size, and characteristics.
 - b. Illustrate the product or equipment meets requirements of specifications.
 - c. Mark items to be furnished in manner that will photocopy (no highlighter).
 - d. Strike through items that do not apply.

E. Certifications:

1. Certifications by a registered professional: Signed and sealed by registered professional.
2. Manufacturer's certifications: Signed by authorized representative of manufacturer.

F. Photographs and Isometric Images:

1. Professional quality:
 - a. Include contract number, name of equipment, and view title.
 - b. Digital files in .jpeg.

G. Videos:

1. Include digital file in MP4 format.
2. Minimum resolution: 1080i or 1080p.

H. Samples and Color Selection Submittals:

1. Submit samples to COR:

Bureau of Reclamation
Eastern Colorado Area Office
11056 W. County Rd 18E
Loveland, CO 80537-9711
2. Provide copy of transmittal letter with samples and on CDMS.
3. Label with complete manufacturer's product and color identification.
4. Include type and quantity of materials specified in the referenced section in each "set" of samples.
5. Samples: Representative of product to be installed. Xerographic sheets for color selection are not acceptable.
6. Color Chips: Sample paint chips. Ink color reproductions are not acceptable.
7. Label each sample, sample kit, set of color chips, of color chart with contract number and title.
8. Government will select architectural color and pattern after product approval.

1.05 SUBMITTAL PROCEDURES

- A. Submit only checked submittals. Submittals without evidence of Contractor's approval will be returned for resubmission.
- B. Submit complete sets of required materials for each RSN as specified in "Submittals Required" column in Table 01 33 00A – List of Submittals. Complete set includes all listed items for RSNs with multiple parts.
- C. Submit in accordance with "Sets to be sent:" columns in Table 01 33 00A – List of Submittals.
 1. Submittals identified with "CDMS" in the "Sets to be sent" column shall be submitted electronically in accordance with Section 01 31 30 – Contract Document Management System.
 2. E-mail address for submittals identified with "electronic" in the "Sets to be sent" column shall be sent to:
 - a. Government will provide with Notice of Award.
- D. Include the following information in transmittal letters:
 1. Contract number and title.
 2. RSN for submittal.
 3. Responsible code.

4. Identify submittal as initial or resubmittal.
- E. Resubmittal of submittals not approved:
1. Mark changes such that they are readily identifiable and show revision date.
 - a. Resubmitted drawings shall include a revision block with date.
 2. Describe reasons for significant changes in transmittal cover letter.
 3. Resubmit returned submittals within 7 calendar days after receiving comments.
 4. Requirements for initial submittals apply to resubmittals.
 5. Resubmittals shall be complete submittals without reference to the original submittal.

1.06 REVIEW OF SUBMITTALS

- A. Time Required:
1. Submittal review will require 28 days for review of each submittal or resubmittal, unless otherwise specified.
 2. Time required for review of each submittal or resubmittal begins when complete sets of materials required for a particular RSN are received and extends through digital signature date stamp and upload date in CDMS.
- B. Time in Excess of Specified:
1. CO may extend contract completion date to allow additional time for completing work affected by excess review time.
 - a. Time extension will be to extent that excess review time caused delay to contract completion date.
 - b. Time extension will not exceed time used in excess of specified number of days for review of submittals or resubmittals.
 - c. Concurrent days of excess review time resulting from review of 2 or more separate submittals or resubmittals will be counted only once in extending contract completion date.
 2. No time extension will be allowed if Contractor fails to make complete action submittals in sequence and within time periods specified.
 3. Adjustment for delay will be made only to extent that:
 - a. Approval was required under contract, and
 - b. Requests for approval were properly and timely submitted and were approved.
 4. Adjustment will be subject to terms of paragraphs (b) and (c) of clause at FAR 52.242-14, Suspension of Work; however, no such delay shall be deemed to be a "suspension order" as the term is used in that clause.

C. Return of Submittals:

1. Return of submittals will be by CDMS response or hard copy, as applicable.
2. Response letter will indicate whether submittal is approved, approved with comments, or not approved.
 - a. Submittals that are not approved shall be revised and resubmitted.
 - b. Do not change designs without approval of CO after drawings, documentation, and technical data have been approved.
 - c. Exception: Schedules will be either accepted, partially accepted or not accepted.
3. Request For Information (RFI): Response will be provided within 28 calendar days.
4. General Correspondence to the COR: Response will be provided within 14 calendar days.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
1	Performance and Payment Bonds – Construction 52.228-15	Performance and Payment Bonds	Within 10 calendar days after award	CO	1 electronic	0	0
2	Additional Bond Security 52.228-2	Additional Bond Security	Within 10 calendar days after award	CO	1 electronic	0	0
3	Insurance-Work on a Government Installation 1452.228-70	1) Written certification that the required insurance has been obtained 2) Current certification of insurance for each subcontractor	Before commencing onsite work	CO	1 electronic 1 electronic	0	0
4	Liability Insurance 1452.228-70	Acceptable Certificate of Liability Insurance	Prior to commencement of work under the contract	CO	1 electronic	0	0
5	Release of Claims 1452.204-70	Release of Claims (DI-137) against the Government	After completion of the work and prior to final payment	CO	1 electronic	0	0
6	Personal Identity Verification 52.204-09	SF-85 and OF-306 for each employee and subcontractor employee working on site	Within 10 days after receipt of Notice to Proceed or 10 days before commencing work (for new hire)	CO	1 electronic	0	0
7	Labor Standards Data 52.222-11	1) List of subcontractors	Within 14 days after award of contract	COR	1 electronic	1 electronic	0
8	Labor Standards Data 52.222-11	2) Statement and Acknowledgement Form (SF-1413) for each subcontract	Within 14 days after award of subcontract	COR	1 electronic	1 electronic	0
9	Payrolls and Basic Records 52.222-08	Copies of all payrolls	Weekly for each week of contract performance	COR	1 electronic encrypted	1 electronic encrypted	0
10	Site Security WBR 1452.237-80	List of contractor and subcontractor employees	Submitted and approved prior to arriving at worksite.	COR	CDMS		
11	Site Security WBR 1452.237-80	List of vehicles and equipment to be brought on site	Submitted and approved prior to arriving at worksite.	COR	CDMS		

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
12	Superintendence by the Contractor 52.236-6	Site Superintendent	Submitted and approved at least 28 days before mobilization	COR	CDMS		
01 14 10-1	Use of Site	Land Use and Landscape Rehabilitation Plan	Submitted and approved at least 28 days before mobilization	COR	CDMS		
01 14 30-1	Interruption of Service	Interruption of Service Request	Submitted and approved at least 28 days before mobilization	COR	CDMS		
01 31 19-1	Project Management and Coordination	Weekly Coordination Meeting Documentation	Weekly, 3 Federal working days prior to meeting	COR	CDMS		
01 31 30-1	Contract Document Management System	Approval Data	Within 14 days of Award and approved before implementation	COR	1	1	1
					CDMS		
01 31 30-2	Contract Document Management System	Final Data	Within 14 days of completion of work	COR	1	1	1
					CDMS		
01 32 20-1	Construction Program	Baseline Schedule	Submitted within 28 days after preconstruction meeting	COR	CDMS		
01 32 20-2	Construction Program	Updated Schedule Reports	Submitted and approved for monthly progress payments	COR	CDMS		
01 35 10-1	Safety Data Sheets	Complete LHM and SDS	At least 28 days before jobsite delivery of hazardous material	COR	CDMS		
01 35 10-2	Safety Data Sheets	Updated LHM and SDS	At least 28 days before jobsite delivery of hazardous material not previously listed	COR	CDMS		
01 35 20-1	Safety and Health	Safety Program	Submitted and accepted before mobilization. See Section 3 of RSHS	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
01 35 20-2	Safety and Health	Monthly Accident Summary Reports	Monthly	COR	CDMS		
01 35 20-3	Safety and Health	Respirator User Documentation	Submitted and accepted at least 28 days before beginning onsite work	COR	CDMS		
01 35 20-4	Safety and Health	Crane Operation Certificates	Submitted and accepted at least 28 days before beginning onsite work	COR	CDMS		
01 35 20-5	Safety and Health	Exposure Assessment Form	Submitted and accepted at least 28 days before beginning onsite work	COR	CDMS		
01 35 20-6	Safety and Health	Other Training Certificates	Submitted and accepted at least 28 days before beginning onsite work	COR	CDMS		
01 35 30-1	Contractor's Onsite Safety Personnel	Qualifications	Submitted and accepted at least 21 days before beginning onsite work	COR	CDMS		
01 35 30-2	Contractor's Onsite Safety Personnel	Safety Inspection Reports	Weekly	COR	CDMS		
01 35 30-3	Contractor's Onsite Safety Personnel	CIH Reports	Monthly	COR	CDMS		
01 46 00-1	Quality Procedures	Contractor Quality Control (CQC) Program	Submitted and approved at least 28 days prior to mobilization	COR	CDMS		
01 46 00-2	Quality Procedures	Contractor's Daily Reports	Weekly	COR	CDMS		
01 46 00-3	Quality Procedures	Summary Monthly Test Results	Monthly	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
01 56 10-1	Protection of Existing Installations	Plan for Protecting Existing Installations	Submitted and approved at least 28 days before beginning mobilization	COR	CDMS		
01 56 10-2	Protection of Existing Installations	Initial Construction Site Digital Recording	At least 21 days before beginning onsite work	COR	CDMS		
01 56 10-3	Protection of Existing Installations	Post-Construction Site Digital Recording	Within 14 days of completion of work	COR	CDMS		
01 57 20-1	Environmental Controls	Copy of applicable Air Quality Permit	Submitted and approved at least 28 days before beginning onsite work	COR	CDMS		
01 57 30-1	Water Pollution Control	Pollution Prevention Plan	Submitted and approved at least 28 days before beginning onsite work	COR	CDMS		
01 57 30-2	Water Pollution Control	Spill Prevention, Control, and Countermeasure (SPCC) Plan	Submitted and approved at least 28 days before beginning onsite work	COR	CDMS		
01 61 10-1	Buy American	Buy American Certificates	Submitted and approved at least 28 days before beginning onsite work	COR	CDMS		
01 74 00-1	Cleaning and Waste Management	Waste Production and Disposal Plan	Submitted and approved at least 28 days before beginning onsite work	COR	CDMS		
01 74 00-2	Cleaning and Waste Management	Waste Production and Disposal Records	Within 7 days of waste disposal	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
01 74 00-3	Cleaning and Waste Management	Environmental Consultant Resume	Submitted and approved at least 28 days before beginning onsite work	COR	CDMS		
01 74 00-4	Cleaning and Waste Management	Environmental Site Assessment	Within 14 days of completion of work	COR	CDMS		
01 78 30-1	Project Record Documents	Final Drawings	Within 14 days of substantial completion	COR	CDMS		
02 80 10-1	General Requirements for Hazardous Materials Abatement	Resumes and Qualifications	Submitted and approved at least 28 days before beginning asbestos abatement	COR	CDMS		
02 80 10-2	General Requirements for Hazardous Materials Abatement	Worker Background Exposure Test Results	Submitted and approved at least 14 days prior to beginning abatement.	COR	CDMS		
02 80 10-3	General Requirements for Hazardous Materials Abatement	Working Level Test Results	Weekly	COR	CDMS		
02 80 10-4	General Requirements for Hazardous Materials Abatement	Final Clearance	Submitted and approved within two days of sampling	COR	CDMS		
02 82 20-1	Removal and Disposal of Asbestos Containing Materials	Qualifications and Resumes	Submitted and approved at least 28 days before beginning asbestos abatement	COR	CDMS		
02 82 20-2	Removal and Disposal of Asbestos Containing Materials	General Abatement Contractors History	Submitted and approved at least 28 days before beginning asbestos abatement	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
02 82 20-3	Removal and Disposal of Asbestos Containing Materials	Asbestos Survey	Submitted and approved 28 days prior to asbestos abatement mobilization.	COR	CDMS		
02 82 20-4	Removal and Disposal of Asbestos Containing Materials	Asbestos Abatement Plan	Submitted and approved 28 days prior to asbestos abatement mobilization.	COR	CDMS		
02 82 20-5	Removal and Disposal of Asbestos Containing Materials	Notice of Asbestos Abatement	Submitted and approved within 7 days of filing the notice.	COR	CDMS		
02 82 20-6	Removal and Disposal of Asbestos Containing Materials	Bills of Lading and Certificate of Disposal	Submitted and approved within 14 days of disposal.	COR	CDMS		
02 83 33-1	Removal and Disposal of Coatings Containing Regulated Metals	Qualifications	Submitted and approved at least 28 days before beginning disposal of equipment with coatings containing hazardous materials	COR	CDMS		
02 83 33-2	Removal and Disposal of Coatings Containing Regulated Metals	Recycled Materials Handling and Disposal Plan	Submitted and approved at least 28 days before beginning disposal of equipment with coatings containing hazardous materials	COR	CDMS		
02 83 33-3	Removal and Disposal of Coatings Containing Regulated Metals	Bills of Lading and Recycling Certificates	Submitted and approved within 14 days of shipping materials.	COR	CDMS		
02 83 37-1	Recycling of Universal Waste	Qualifications	Submitted and approved at least 28 days before beginning work	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
02 83 37-2	Recycling of Universal Waste	UW Handling and Disposal Plan	Submitted and approved at least 28 days before beginning work	COR	CDMS		
02 83 37-3	Recycling of Universal Waste	Shipping Papers	Within 14 days after disposal	COR	CDMS		
02 83 37-4	Recycling of Universal Waste	Disposal Certificates	Within 14 days after disposal	COR	CDMS		
02 87 20-1	Removal and Disposal of Used Oil	Qualifications	Submitted and approved at least 28 days before beginning work	COR	CDMS		
02 87 20-2	Removal and Disposal of Used Oil	Oil Sampling, Handling and Disposal Plan	Submitted and approved at least 28 days before beginning work	COR	CDMS		
02 87 20-3	Removal and Disposal of Used Oil	Oil Characterization Report	Submitted and approved at least 28 days before beginning work	COR	CDMS		
02 87 20-4	Removal and Disposal of Used Oil	Manifest for Reclamation Signature	Submitted and signed at least 14 days before shipping.	COR	CDMS		
02 87 20-5	Removal and Disposal of Used Oil	Manifests, Recycling Report, Bills of Lading, Disposal Certificates	Within 28 days of shipping materials.	COR	CDMS		
03 62 20-1	Nonshrink Grout for Equipment and Metalwork	Packaged Nonshrink Grout Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
03 81 10-1	Concrete Removal	Concrete Removal Plan	Submitted and approved at least 28 days before beginning work	COR	CDMS		
07 84 00-1	Firestopping	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
07 84 00-2	Firestopping	Instructions	Submitted and approved at least 28 days before beginning work	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
22 15 00-1	Governor Compressed-Air Systems	Approval Product Drawings and Data:	Submitted and approved at least 28 days before beginning work	COR	CDMS		
22 15 00-2	Governor Compressed-Air Systems	Test Plan	Submitted and approved at least 28 days before beginning work	COR	CDMS		
22 15 00-3	Governor Compressed-Air Systems	Final Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
25 00 01-1	Automation Control and Monitoring Systems	Resumes and Qualifications - Automation Control and Monitoring System	Submitted and approved at least 28 days before beginning work	COR	CDMS		
25 00 01-2	Automation Control and Monitoring Systems	Software Development Documentation for Generator Unit	Submitted and approved at least 56 days before beginning work	COR	CDMS		
25 00 01-3	Automation Control and Monitoring Systems	Factory Acceptance Test Plans and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
25 00 01-4	Automation Control and Monitoring Systems	Factory Acceptance Test Reports - Automation Control and Monitoring System for Generator Unit	Submitted and approved at least 28 days before beginning work	COR	CDMS		
25 00 01-5	Automation Control and Monitoring Systems	Field and Commissioning Test Plans and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
25 00 01-6	Automation Control and Monitoring Systems	Operation and Maintenance Manual; Generator Automation Control and Monitoring System	Submitted and approved at least 14 days before Commissioning	COR	CDMS		

Table 01 33 00A – List of Submittals

** CO indicates Contracting Officer, COR indicates Contracting Officer's Representative, and TSC indicates Technical Service Center.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
25 00 01-7	Automation Control and Monitoring Systems	Commissioning Test Procedures for Generator	Submitted and approved at least 28 days before beginning work	COR	CDMS		
25 00 01-8	Automation Control and Monitoring Systems	Field and Commissioning Test Reports for Generator	Within 5 days of testing	COR	CDMS		
25 12 10-1	Programmable Logic Controllers	System Description and Operational Documentation	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 04 10-1	Electrical Demolition and Removal	Electrical Demolition and Removal Plan	Submitted and approved at least 56 days before beginning work	COR	CDMS		
26 05 02-1	Basic Electrical Materials and Methods	As-Found Existing Condition Drawings	Submitted and approved at least 42 days before beginning work	COR	CDMS		
26 05 10-1	Conductors and Cables	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 05 10-2	Conductors and Cables	Field Test Reports and Cable Termination Sheets	Within 7 days of testing	COR	CDMS		
26 05 10-3	Conductors and Cables	As-Built Drawings	At least 14 days before Commissioning				
26 05 20-1	Grounding and Bonding	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 05 20-2	Grounding and Bonding	As-Built Drawings	At least 14 days before Commissioning	COR	CDMS		
26 05 20-3	Grounding and Bonding	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 05 13-1	Medium-Voltage Power Cable System	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
26 05 13-2	Medium-Voltage Power Cable System	Factory Production Test Certifications	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 05 13-3	Medium-Voltage Power Cable System	Field Test Notification	At least 7 days before testing	COR	CDMS		
26 05 13-4	Medium-Voltage Power Cable System	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 05 33-1	Raceways and Boxes	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 05 33-2	Raceways and Boxes	As-Built Drawings	At least 14 days before Commissioning	COR	CDMS		
26 05 36-1	Cable Trays	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 05 36-2	Cable Trays	As-Built Drawings	At least 14 days before Commissioning	COR	CDMS		
26 05 39-1	Engine Generator Terminal Box	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 05 39-2	Engine Generator Terminal Box	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 05 90-1	Wiring Checkout and Tests	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 09 16-1	Protection Systems	Approval Drawings and Data – Government Format	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 16-2	Protection Systems	Software Development and Cyber Security Documentation	Submitted and approved at least 56 days before beginning work	COR	CDMS		

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
26 09 16-3	Protection Systems	Final Software Files	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-1	Control Boards	Resumes and Qualifications	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-2	Control Boards	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-3	Control Boards	Proposed Location for Turbine Level Distributed I/O Cabinet and Network Communication Cabinet	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-4	Control Boards	Factory Acceptance Test (FAT) Plans and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-5	Control Boards	Factory Acceptance Test Reports	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-6	Control Boards	Installation Plans and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-7	Control Boards	Field and Commissioning Test Plans and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 09 18-8	Control Boards	Field and Commissioning Test Reports	Within 7 days of testing	COR	CDMS		
26 09 18-9	Control Boards	Final Documents	At Commissioning	COR	CDMS		
26 12 19-1	Pad Mounted Transformer	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
26 12 19-2	Pad Mounted Transformer	Factory Test Reports	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 12 19-3	Pad Mounted Transformer	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 12 19-4	Pad Mounted Transformer	As-Built Drawings	At least 14 days before Commissioning	COR	CDMS		
26 12 19-5	Pad Mounted Transformer	Operation and Maintenance Instruction Book	At Commissioning	COR	CDMS		
26 22 12-1	Dry-Type Transformer	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 22 12-2	Dry-Type Transformer	Field Test Reports.	Within 7 days of testing	COR	CDMS		
26 22 12-1	Dry-Type Transformer	Operation and Maintenance Instruction Book	At Commissioning	COR	CDMS		
26 24 13-1	Switchboards	Approval Drawings and Data in Manufacturer's Format	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 13-2	Switchboards	Approval Drawings in Government Format	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 13-3	Switchboards	Factory Test Report	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 13-4	Switchboards	Check Prints	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 13-5	Switchboards	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 24 13-6	Switchboards	As-Built Drawings	At least 14 days before Commissioning	COR	CDMS		

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
26 24 13-7	Switchboards	Operation and Maintenance Instruction Book	At Commissioning	COR	CDMS		
26 24 24-1	Distribution Panelboards	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 24-2	Distribution Panelboards	Factory Test Reports	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 24-3	Distribution Panelboards	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 24 24-4	Distribution Panelboards	Operation and Maintenance Instruction Books	At Commissioning	COR	CDMS		
26 24 43-1	600 Volt Motor Control Center	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 43-2	600 Volt Motor Control Center	Approval Drawings	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 43-3	600 Volt Motor Control Center	Schematic Diagrams	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 43-4	600 Volt Motor Control Center	Time-Current Characteristic Curves	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 43-5	600 Volt Motor Control Center	Schematic and Wiring Diagram Check Prints.	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 43-6	600 Volt Motor Control Center	Factory Test Reports	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 24 43-7	600 Volt Motor Control Center	Field Test Reports	Within 7 days of testing	COR	CDMS		

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
26 24 43-8	600 Volt Motor Control Center	Operation and Maintenance Instruction Book	At Commissioning	COR	CDMS		
26 27 40-1	Wiring Devices	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 29 13-1	Enclosed Contactors	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 29 13-2	Enclosed Contactors	Factory Test Reports	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 29 13-3	Enclosed Contactors	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 29 13-4	Enclosed Contactors	Operation and Maintenance Instruction Books	At Commissioning	COR	CDMS		
26 33 46-1	125v DC Battery and Charger	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 33 46-2	125v DC Battery and Charger	Proposed location of New Battery Chargers	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 33 46-3	125v DC Battery and Charger	Factory Test Report	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 33 46-4	125v DC Battery and Charger	Operation and Maintenance Instruction Books	At Commissioning	COR	CDMS		
26 36 23-1	Automatic Transfer Switch	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
26 36 23-2	Automatic Transfer Switch	As-Built Drawings	At least 14 days before Commissioning	COR	CDMS		

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
26 36 23-3	Automatic Transfer Switch	Field Test Reports	Within 7 days of testing	COR	CDMS		
26 36 23-4	Automatic Transfer Switch	Operation and Maintenance Instruction Books	At Commissioning	COR	CDMS		
27 05 43-1	Outside Plant Raceways for Communications Systems	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
27 05 43-2	Outside Plant Raceways for Communications Systems	Installation Plan and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
27 20 01-1	Fiber Optic Cable and Accessories	Approval Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
27 20 01-2	Fiber Optic Cable and Accessories	Installation Plan and Procedures	Submitted and approved at least 28 days before beginning work	COR	CDMS		
27 20 01-3	Fiber Optic Cable and Accessories	Field Test Reports	Within 7 days of testing	COR	CDMS		
27 20 01-4	Fiber Optic Cable and Accessories	Final Drawings	At Commissioning	COR	CDMS		
27 21 29-1	Network Switches	Approval Drawings and Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
28 05 11-1	Cybersecurity Requirements for Electronic Safety and Security	Preliminary Cybersecurity Documentation	Submitted and approved at least 28 days before beginning work	COR	CDMS		
28 05 11-2	Cybersecurity Requirements for Electronic Safety and Security	Preliminary Configuration Documentation	Submitted and approved at least 28 days before beginning work	COR	CDMS		
28 05 11-3	Cybersecurity Requirements for Electronic Safety and Security	Factory Acceptance Test Procedure	Submitted and approved at least 28 days before beginning work	COR	CDMS		

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	No. of Sets to be sent: **		
					CO	COR	TSC
28 05 11-4	Cybersecurity Requirements for Electronic Safety and Security	Factory Acceptance Test Report	Submitted and approved at least 28 days before beginning work	COR	CDMS		
28 05 11-5	Cybersecurity Requirements for Electronic Safety and Security	Site Acceptance Test Procedure	Submitted and approved at least 28 days before beginning work	COR	CDMS		
28 05 11-6	Cybersecurity Requirements for Electronic Safety and Security	Site Acceptance Test Report	Submitted and approved at least 28 days before beginning work	COR	CDMS		
28 05 11-7	Cybersecurity Requirements for Electronic Safety and Security	As-Installed Cybersecurity Documentation	At Commissioning	COR	CDMS		
28 05 11-8	Cybersecurity Requirements for Electronic Safety and Security	Final Documentation	At Commissioning	COR	CDMS		
33 09 10-1	Instrumentation and Control for Water Utilities	Product Data	Submitted and approved at least 28 days before beginning work	COR	CDMS		
33 09 10-2	Instrumentation and Control for Water Utilities	Installation Instructions	Submitted and approved at least 28 days before beginning work	COR	CDMS		
33 09 10-3	Instrumentation and Control for Water Utilities	Operation and Maintenance Manuals	At Commissioning	COR	CDMS		

END OF SECTION

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SECTION 01 33 26
ELECTRICAL DRAWINGS AND DATA

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for items requiring electrical drawings and data.

1.02 REFERENCE STANDARDS

- A. American Society of Mechanical Engineers (ASME)
1. ASME Y14.1-2012 Decimal Inch Drawing Sheet Size And Format
- B. Institute of Electrical and Electronic Engineers (IEEE)
1. IEEE 315-1975(1993) Graphic Symbols For Electrical And Electronics Diagrams (Including Reference Designation Letters)
 2. IEEE C37.2-2008 Electrical Power System Device Function Numbers, Acronyms, and Contact Designations
 3. IEEE Y32.9-1972(1989) Graphic Symbols for Electrical Wiring and Layout Diagrams Used in Architecture and Building Construction
- C. National Electrical Manufacturers Association (NEMA)
1. NEMA ICS 1-2000(2008) Industrial Control and Systems: General Requirements

1.03 SUBMITTAL PROCEDURES

- A. In addition to the requirements in Section 01 33 00 - Submittals, prepare electrical drawings and data submittals in accordance with the following.
- B. Drawings, general:
1. Device designations and symbols:
 - a. Conform to IEEE 315, IEEE C37.2, IEEE Y32.9, and NEMA ICS-1.
 - b. As indicated on Standard Drawing 104-D-1150.

2. Size: D-size prints as defined in ASME Y14.1 made from original drawings.
3. Indicate changes on revised drawings to distinguish them from previous submittals. Describe reasons for significant changes in submittal letters.

C. Approval Data:

1. Bill of Material List:
 - a. Provide information on manufacturer, style, type, rating, quantity, and other identifying information for each device or item of equipment.
 - b. Provide unique reference number for each device or item of equipment listed on bill of material.
2. Nameplate Lists: Provide information on type of material, size, and engraved lettering.
3. Manufacturer's Technical Catalog Data:
 - a. Provide technical data for each device or item of equipment.
 - b. Include manufacturer's name and address; catalog number, type, style, or model number; electrical ratings; and dimensions.
 - c. Where several items are listed on same sheet, indicate items being submitted for approval.
 - d. Demonstrate proposed device or item of equipment meets specifications requirements.
 - e. Mark catalog data sheet with appropriate bill of material (BOM) item number.
 - f. Assemble catalog data sheet into one enclosing cover with bill of material (BOM) index.
4. Time-Current Characteristic and Equipment Coordination Curves:
 - a. Provide time-current characteristic curves for adjustable and non-adjustable protective devices.
 - b. Provide equipment coordination curves for:
 - 1) Transformer inrush current.
 - 2) Transformer full load current.
 - 3) Transformer thermal and mechanical damage.
 - 4) Motor inrush current.
 - 5) Motor full load current.
 - c. Original plot on log-log graph paper.
 - d. Illustrate degree of selectivity being provided.
5. Job Hazard Analysis (JHA):

- a. Provide when the Government will be performing factory inspection.
- b. Prepare JHA in accordance with 29 CFR 1910, Subpart I.
- c. As a minimum include the following in JHA:
 - 1) Factory address and name of site visit contact person.
 - 2) Date of JHA.
 - 3) Purpose of JHA, for example “Visitor Safety Procedures”.
 - 4) Hazards.
 - a) Types of hazards at factory.
 - b) Safety procedures in or around high-voltage test labs.
 - c) Safety procedures on factory floor.
 - 5) Personal Protective Equipment.
 - a) List of equipment required.
 - b) List of equipment provided by factory.
 - 6) Emergency response.
 - a) Name and phone number of Factory Safety Officer.
 - b) Employees trained in emergency conditions (CPR and First Aid).
 - c) Factory fire protection/evacuation procedures.
 - d) Nearest hospital or emergency room.
 - 7) Name, title and signature of factory official approving JHA.

D. Approval Drawings in Government Format:

- 1. General:
 - a. Furnish with Government title block and in form as indicated on Standard Drawing 40-D-7102.
 - b. Government will provide specific title block information when drawings are submitted.
 - c. Government may revise conductor or device designations on Contractor’s drawings when drawings are approved at no change in contract price or completion time.
- 2. Equipment Layout Drawings:
 - a. Indicate dimensions of equipment.
 - b. Indicate location of devices and items of equipment including nameplates, terminal blocks, wiring ducts, bus, conduit entries, and other features in their relative physical location.

- c. Indicate method of securing equipment to deck, floor, or wall.
 - d. Identify each device and item of equipment with a bill of material reference number.
- 3. Schematic Diagrams:
 - a. In accordance with Standard Drawing 104-D-1150.
 - b. Format:
 - 1) Vertical ladder diagram form.
 - 2) Label rungs with sequential numbers starting with number 1.
 - 3) Locate rung numbers adjacent and to left of associated rung.
 - 4) Provide unique rung numbers for each circuit.
 - c. Indicate device terminal and terminal block designations.
 - d. Indicate functional operation.
 - e. Describe unusual or nonstandard operation.
 - f. Indicate ratings and/or values of devices.
 - g. Indicate contacts, including spare contacts, for motor contactors, auxiliary relays, timers, and protective relays.
 - h. Indicate cross-referencing between other schematic and wiring diagrams.
 - i. Do not include current and potential transformer circuits on drawing with direct-current control and protective relaying circuits.
 - j. Conductor designations: As indicated on drawings or as developed by Contractor. Contractor-developed conductor designations shall consist of:
 - 1) Circuit prefix.
 - 2) Rung number.
 - 3) Unique rung wire letter such as 5P15C (“5”-Unit Number, “P”-Protection Circuit, “15”-Rung Number, “C”-Unique Rung Wire Letter).
 - k. Base instrument current and potential transformer circuit conductor designations upon device designations indicated on single-line diagram drawings.
 - l. Show switch developments for control, selector, and limit switches.
- 4. Wiring Diagram Drawings:
 - a. Correlate with schematic diagram drawings.
 - b. Indicate point-to-point wiring in accordance with Standard Drawing 104-D-1165.

- c. Indicate equipment as mounted on back, side, swing, and door panels. Show panel as viewed from wiring side of panel.
- d. Identify each item of equipment with designation indicated on schematic diagram.
- e. Identify control and protective devices with IEEE device number as indicated on specifications drawings or on Standard Drawing 104-D-1150.
- f. Identify components such as fuses and resistors by value.
- g. Indicate wiring of devices and items of equipment including terminal number.
- h. Indicate cables, cable and individual wire designations, and connections to external circuits.
- i. Cable designations: Similar to designations on Standard Drawing 104-D-1165 and match schematic diagrams.
- j. Show no more than one panel per drawing.

E. Approval Drawings in Manufacturer's Format:

- 1. General: Furnish with manufacturer's standard format, except as noted below.
- 2. Equipment Layout Drawings:
 - a. Indicate dimensions of equipment.
 - b. Indicate location of devices and items of equipment including nameplates, terminal blocks, wiring ducts, bus, conduit entries, and other features in their relative physical location.
 - c. Indicate method of securing equipment to deck, floor, or wall.
 - d. Identify each device and item of equipment with a bill of material reference number.
- 3. Schematic Diagrams:
 - a. Format:
 - 1) Vertical ladder diagram form.
 - 2) Label rungs with sequential numbers starting with number 1.
 - 3) Locate rung numbers adjacent and to left of associated rung.
 - 4) Provide unique rung numbers for each circuit.
 - b. Indicate functional operation.
 - c. Describe unusual or nonstandard operation.
 - d. Indicate ratings and/or values of devices.
 - e. Indicate contacts, including spare contacts, for each motor contactor coil, auxiliary relay, and timer.

- f. Indicate a complete cross-reference between other schematic and wiring diagrams.
 - g. Indicate switch developments for each selector switch and limit switch.
 - h. Identify devices and items of equipment by reference to bill of materials item number and with appropriate device number in IEEE and NEMA standards.
 - 4. Wiring Diagrams:
 - a. Indicate equipment as mounted on each back panel, side panel, and door. Each panel shall be indicated as viewed from wiring side of panel.
 - b. Identify each item of equipment with the device designation indicated on schematic diagrams.
 - c. Indicate point-to-point wiring in accordance with Standard Drawing 104-D-1165.
 - d. Indicate cables, cable and individual wire designations, and connections to all external circuits.
 - e. Indicate no more than one panel per drawing.
- F. Check Prints:
 - 1. Schematic and wiring diagrams which indicate "As-built" condition of equipment at time of shipment.
 - 2. Mark as changes and revisions are made during installation and checkout of equipment.
 - 3. Keep drawings at the jobsite and mark as work progresses
 - 4. Maintain separate, identical set of prints for use in preparing final drawings.
- G. As-built Drawings:
 - 1. Mark prints of electrical specifications drawings to indicate as-built changes made to equipment and systems during construction.
 - 2. Provide for grounding and conduit systems.
 - a. Indicate as-built changes made during construction.
 - b. Indicate dimensions of:
 - 1) Location of grounding electrodes.
 - 2) Direct buried grounding cables.
 - 3) Direct buried conduit.
 - 3. Include revision dates.
 - 4. Mark the drawings in the following colors:

- a. Red - Additions to original drawings.
- b. Green - Deletions to original drawings.
- c. Blue - Notations necessary for explanation of as-built markings.

H. Test Reports:

- 1. Typed, 8 1/2- by 11-inch sheets.
- 2. Certified.
- 3. Submit multi-page reports in bound folder or three-ring binder.

I. Operation and Maintenance Instructions Books:

- 1. Print copies:
 - a. Assemble each set of material into one or more books with enclosing covers.
- 2. Electronic copies:
 - a. Adobe Acrobat (.pdf) format on DVD discs.
 - b. Conform to print copies.
 - c. Bookmark file(s) to aid in navigating.
- 3. Provide approved copies of the following items:
 - a. Index sheet at front of each book which provides page or index tab number information for each device or item of equipment in book.
 - b. Manufacturer's operation and maintenance procedures.
 - c. Installation details, as necessary.
 - d. Catalog data sheets for each device or item of equipment.
 - e. List of recommended spare parts and components.
 - f. Parts lists for replacement parts.
 - g. Manufacturer's full-size circuit breaker and fuse time-current curves.
 - h. Manufacturer's full-size transformer and motor equipment coordination curves.
 - i. Copies of factory test reports.
 - j. Copies of bills of materials
 - k. Copies of drawings.
 - 1) Revised to reflect approval comments.
 - 2) Maximum size: 11- by 17-inch.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 10

SAFETY DATA SHEETS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

A. LHM: List of Hazardous Materials.

B. SDS: Safety Data Sheet.

1. Referred to as Material Safety Data Sheets in the clause at 52.223-3, Hazardous Material Identifications and Material Safety Data – Alternate 1.

1.03 APPLICATION

A. For the purposes of this contract, “delivered under this contract” in paragraph (b) of clause at FAR 52.223-3, Hazardous Material Identification and Material Safety Data – Alternate 1, includes:

1. Materials delivered to Government.
2. Materials incorporated into work.
3. Materials used by Contractor during contract performance at jobsite.

1.04 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 35 10-1, Complete LHM and SDS.

C. RSN 01 35 10-2, Updated LHM and SDS:

1. Comply with paragraph (e) of clause at FAR 52.223-3, Hazardous Material Identification and Material Safety Data – Alternate 1.

1.05 DELIVERY

A. Do not deliver hazardous materials to jobsite which are not included on original or previously updated LHM and SDS before receipt of updated LHM and SDS by COR.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 20 SAFETY AND HEALTH

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. American Conference of Governmental Industrial Hygienists (ACGIH)

1. Threshold Limit Values (TLVs) Available at www.acgih.org

B. Bureau of Reclamation (Reclamation)

1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>
2. FIST 1-1 (2019) Hazardous Energy Control Program available at https://www.usbr.gov/power/data/fist_pub.html

C. Occupational Safety and Health Administration (OSHA)

1. 29 CFR Part 1910 Occupational Safety and Health Standards.
2. 29 CFR Part 1926 Safety and Health Regulations for Construction. Available at www.osha.gov

D. Society for Protective Coatings (SSPC)

1. SSPC-C3 Lead/Hazardous Paint Removal Course
2. SSPC-C5 Lead/Hazardous Paint Removal Refresher Course

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 35 20-1, Safety Program:

1. Submit a comprehensive, job-specific safety and health program in accordance with Section 01 33 00 – Submittals and in accordance with the requirements of

Section 3 of the Reclamation Safety and Health Standards (RSHS) and these specifications.

2. Cover aspects of onsite and applicable off-site operations and activities associated with this contract.
3. Follow outline in Appendix 3C of RSHS Section 3.
4. The Safety Program will not be accepted for review by COR unless it addresses, in order, lettered and numbered per Section 3 of the RSHS, a comprehensive narrative for each applicable item in outline and program elements contained in these specifications.
5. Generic company safety programs are not acceptable. The Safety Program shall be site specific for this contract.
6. The Contractor's Safety Program must be submitted and accepted by Reclamation prior to commencing onsite work, including mobilization.
7. The Safety Program Statement of Responsibility must state that "the contractor is responsible for ensuring that all work will be in compliance with Reclamation Safety and Health Standards and these specifications."
8. The Safety and Health Program shall include, but not be limited to, the following program elements:
 - a. Safety and Health Program Guidance – General Requirements in accordance with paragraph 1.03.B.4. of this section and RSHS Section 3, Appendix 3C.
 - b. Comply with OSHA, RSHS, local safety programs, and other referenced safety and health standards.
 - 1) The most stringent shall apply.
 - c. Perform work in accordance with the specific safety program elements and referenced standards.
 - d. In the site-specific safety and health program, address sections that do not apply to this project as 'not applicable.'
 - e. Statement of Policy:
 - 1) Contractor's Safety and Health Policy.
 - f. Statement of Safety and Health Responsibilities.
 - g. Statement of Compliance with Regulations, Standards, and Codes:
 - 1) Specifically state the intention to comply with the RSHS, OSHA, and any other regulatory standard. The most stringent shall apply.
 - h. Statement of Subcontractor Compliance:
 - 1) Contractor shall state that they will ensure compliance of their employees and subcontractors.

- i. Safety Inspection Procedures:
 - 1) Specify in accordance with this specification and Section 3 of the RSHS. List procedures for reporting to COR and Reclamation and the use of required forms.
- j. Accident Investigation and Reporting Procedures:
 - 1) Specify in accordance with this specification and Section 3 of the RSHS. List procedures for reporting to COR and Reclamation and the use of required forms.
- k. Applicable Emergency Plans:
 - 1) Detail coordination plan for outside emergency services including response time and a site visit where a rescue requires special capabilities or equipment.
 - 2) Address detailed emergency contacts and procedures to be used during an onsite emergency.
 - 3) Additional emergency plans and procedures as applicable to the location and potential hazards such as but not limited to the remoteness of worksite, potential hazards of worksite, and limitations of outside assistance.
 - 4) Include procedures for weather related emergencies such as high wind, lightning, tornado, and wildfire.
- l. Confined Space Procedures:
 - 1) Include a detailed confined-space entry plan for operations that require employees to enter confined spaces.
 - 2) The detailed confined-space entry program shall be site-specific and in accordance with OSHA 29 CFR 1926.1200 thru 1926.1213 and these specifications.
 - 3) Utilize OSHA's definition of IDLH in the 1926.1202 standard on this project. Immediately dangerous to life or health (IDLH) means any condition that would interfere with an individual's ability to escape unaided from a permit space and that poses a threat to life or that would cause irreversible adverse health effects. A ventilation system alone does not alleviate the need for a rescue team to be prepared to enter an IDLH atmosphere due to the potential for failure of a ventilation system.
 - 4) Include engineering controls, detailed ventilation plans, personal protective equipment and air monitoring procedures for each confined space.
 - 5) Include non-entry and entry rescue plan.
 - 6) Include the Contractor's Confined Space Program.

- 7) If permit-required spaces will be entered, a trained and evaluated rescue team will be required onsite. If outside emergency services are to be relied upon for rescue and/or transporting of injured personnel, employer must first secure an agreement with outside personnel to include what role they will have, response time, and GPS coordinates if helicopter will be used. The rescue team must be immediately available (onsite) during all entries into a permit-required confined space. Non-entry rescue is required unless the retrieval equipment would increase the overall risk of entry or would not contribute to the rescue of the entrant.
- 8) For non-entry rescue, each authorized entrant must use a chest or full body harness, with a retrieval line attached at the center of the entrant's back near shoulder level, above the entrant's head, or at another point which the employer can establish presents a profile small enough for the successful removal of the entrant. The other end of the retrieval line must be attached to a mechanical device or fixed point outside the permit space in such a manner that rescue can begin as soon as the rescuer becomes aware that rescue is necessary. A mechanical device must be available to retrieve personnel from vertical type permit spaces more than 5 feet (1.52 meters) deep.
- 9) The rescue team will consist of no less than 2 rescue personnel and one attendant (3 total) located outside of the confined space during all permit-required confined space entries. Additional rescue personnel will be required when the employer is unable to demonstrate a successful rescue of an unconscious victim from the space utilizing 2 rescue persons.
- 10) If the Contractor intends to utilize a rescue service, the Contractor is responsible for ensuring that the rescue service meets the requirements in this specification and the requirements of 29 CFR 1926.1211(a) thru (d). Supporting documentation of the rescue services' training and abilities must be made available and be acceptable to the COR.
- 11) The rescue team must be trained in all potential types of rescue that might be necessary, including an oxygen deficient atmosphere, on the specific permit-required confined space and in accordance with 29 CFR 1926 Subpart AA (1926.1200 thru 1926.1213).
- 12) Confined space training documentation is required for authorized entrant and additional documentation of training for rescue personnel. General confined space training for authorized entrants will not suffice for rescue personnel. The training for rescue personnel shall be specific to rescue.

- 13) The rescue personnel training shall include the hazards of the space, use of SCBA or supplied air with SCBA backup while performing rescue, current first aid and CPR, use of rescue and monitoring equipment, use of PPE, and all other aspects of 29 CFR 1926.1211.
- 14) The confined space rescue team will be required to perform a practice rescue in each permit-required confined space prior to the start of work in the permit-required confined space. The practice rescue(s) must be documented and include the space, names of rescuers, and methods used. The practice rescue shall be for a worst-case scenario rescue with an unconscious victim.
- 15) The Contractor must ensure that rescue team members practice making permit space rescues before attempting an actual rescue, and at least once every 12 months, by means of simulated rescue operations in which they remove dummies, manikins, or actual persons from the actual permit spaces or from representative permit spaces.
- 16) If the Contractor is to provide rescue services through trained employees, the Contractor must maintain training records for the rescue team that include the employee's names, the name of the trainers and the dates of the training. These records must be available to the COR for review.
- 17) Whenever the confined space has the potential for gases and vapors, an oxygen deficient atmosphere, or any other hazard that could cause the space to become an IDLH atmosphere due to an upset or failure of ventilation equipment, the rescue team will be trained, medically evaluated, fit tested and will utilize a full facepiece pressure demand SCBA certified by NIOSH for a minimum service life of thirty minutes, or a combination full facepiece pressure demand supplied-air respirator (SAR) with auxiliary self-contained air supply. All coating operations within a confined space will require the rescue team to meet this respiratory criteria.
- 18) No employee shall enter the permit-required confined space until a mock rescue has been performed. The mock rescue shall be for the most serious incident that could potentially occur (such as unconscious victim or IDLH atmosphere).
- 19) When there is a potential for employee exposure to a substance for which a Safety Data Sheet (SDS) or other similar written information is required to be kept at the worksite, that SDS or written information must be attached to the PRCS entry permit.
- 20) If an injured entrant is exposed to a substance for which a Safety Data Sheet (SDS) or other similar written information is required

to be kept at the worksite, that SDS or written information must be made available to the medical facility treating the exposed entrant.

- 21) Coatings and other flammables in confined spaces:
 - a) Include controls for hazards such as static electricity and control of ignition sources and sparks.
 - b) Include the use of intrinsically safe equipment or classified hazardous electrical for hazardous atmospheres.
 - c) Flammables that are not being immediately used shall not be kept in the permit-required confined space.

m. Lockout/Tagout Procedures:

- 1) Include the contractor's written Hazardous Energy Control Program (HECP). Comply with FIST 1-1.
- 2) Comply with OSHA's 29 CFR 1910.147 for the control of hazardous energy (lockout/tagout).
- 3) The Contractor will not be allowed to place locks on Reclamation equipment. The Contractor will be required to utilize a group lockout procedure when working under the protection of Reclamation equipment. Include the Contractor's group lockout procedures for coordination with the facility and its equipment.
- 4) The group lockout shall utilize a procedure which affords the employees a level of protection equivalent to that provided by the implementation of a personal lockout device. Tags and master lists will not be used in place of locks.
- 5) Employee locks on group lockout device will be required to have the employee's name for immediate identification. Employee locks are to be removed by the employee when the employee is not working in the protected area.
- 6) Comply with facility Hazardous Energy Control Program (HECP) procedures and coordinate activities with Reclamation staff.

n. Electrical Safety:

- 1) Include safe work procedures at the site, electrically qualified persons, grounding techniques, use of mobile generators, construction lighting, power tool safety, and use of personal protective equipment.
- 2) Include the use of intrinsically safe equipment or classified hazardous electrical for potential hazardous atmospheres as needed.
- 3) Include controls for hazards such as static electricity as needed.

- 4) Protect all flexible cords and cables from damage. Flexible extension cords must be 3-wire grounded type, designated for hard service or extra hard service, free of splices and must be listed by the Underwriters Laboratories.
 - 5) Provide ground fault protection for all temporary wiring installations, tools, and equipment.
- o. Fire Protection Plans – Hot Work and Fire Prevention:
- 1) Include a plan to address hazards associated with coating application, hot work, and other activities.
 - 2) Include list of workplace fire hazards, potential ignition sources, safe work procedures, engineering controls and equipment to be used to prevent fires during any coating application, hot work or applicable process.
 - 3) Include the type and location of suppression equipment or systems.
 - 4) Include an offsite assistance agreement in accordance with RSHS Section 3, Contractor Safety Program.
 - 5) When working in areas with the potential for a hazardous atmosphere from gases or vapors, include a plan to control potential ignition sources such as but not limited to lighting, motors, heating devices, fans, static electricity (e.g., clothing, abrasive blasting, and coating spray application), and small tools. This may be addressed in the site-specific confined space program if applicable.
- p. Medical:
- 1) Facilities:
 - a) List medical facilities and services, emergency contacts, phone numbers, GPS coordinates, and other pertinent information as pertains to worksite and location.
 - 2) Training:
 - a) Onsite Contractor/Subcontractor personnel shall have current First Aid/CPR certifications. Training shall be in accordance with the American Red Cross or the American Heart Association. Online First Aid and CPR training is not acceptable.
 - 3) Certifications:
 - a) List required medical certifications for specific tasks such as crane operator (license requirement), truck driver commercial driver's license (CDL).

- b) List specific tasks or positions requiring a medical evaluation or medical monitoring.
- 4) Physician:
 - a) List physician or medical facility and contact information.
- 5) Ambulance:
 - a) Name, location, and telephone number. Post this information in a conspicuous area accessible by all employees. Include specific directions and coordinates to be given to emergency personnel.
- 6) Physical Qualification of Employees:
 - a) List specific positions and tasks that may/do have physical requirements and what those requirements are. Examples: Crane Operators, Confined Space Rescue Team members, and Truck Drivers.
- 7) Records:
 - a) Documentation of successful medical evaluations.
 - b) Do not include protected medical information (HIPPA) or personal employee information (PII) in this section or in required submittals.
 - c) Medical evaluations for respirator users should include a document, signed by the evaluating personnel, stating that the employee can or cannot wear a respirator and any limitations of respirator use.
- q. Communications:
 - 1) Job Hazard Analysis:
 - a) Provide a list of proposed JHA's in the Safety Program for each anticipated work task.
 - b) A detailed JHA shall be completed on site prior to starting each work task. Work will not begin on the phase of the work until a JHA is acceptable to the Onsite Government Representative.
 - c) Activities involving hazardous materials shall have the appropriate Safety Data Sheet(s) attached to the JHA.
 - 2) Employee Training:
 - a) List training that is required for employees on site and position specific training in addition to orientation training.
 - b) Review and include Reclamation required training in the RSHS.

- c) Include training for Confined Space Rescue Team, Confined Space Entrants, Competent Person training as required by specific standards (e.g., scaffolding, Permit Required Confined Spaces, and asbestos abatement).
- 3) Safety Meetings:
 - a) See RSHS Section 3 and this specification for required meetings such as Contractor Safety Program Review meeting, Joint Safety Policy Meetings, Supervisory Safety Meetings and Toolbox Safety Meetings.
- 4) Onsite Training:
 - a) List training that will be provided on site to employees.
- 5) Supervisor Training:
 - a) See RSHS Section 3 and this specification for required Supervisor training.
- r. Occupational Health:
 - 1) Procedures and Equipment to Minimize Hazards:
 - a) Include procedures used to reduce employee exposures such as but not limited to noise, respiratory, dermal, and environmental hazards.
 - b) Identify respirator user positions, tasks, and operations with recognized respiratory hazards.
 - c) The plan shall address exposure to respirable silica associated with concrete work as follows:
 - i. Use of tight-fitting half-face air purifying respirators with P100 cartridges in the absence of an onsite exposure assessment.
 - d) Hearing Conservation:
 - i. Include a task- and site-specific hearing conservation plan.
 - ii. Identify the type of hearing protection and verify that workers are in a Hearing Conservation Program if required.
 - 2) Testing Program for Employees and Work Environments:
 - a) Include items such as air monitoring, noise monitoring, wipe sampling, and other relevant testing, as required.
 - 3) Qualified Personnel:

- a) Identify tasks or positions requiring qualified personnel and what those qualifications are (certifications, medical qualifications, licensing).
- b) Do not list personnel by name in this section (PII).
- 4) Personal Protective Equipment (PPE) and Clothing:
 - a) Include a detailed plan for each task that specifies which personal protective equipment shall be used for all construction activities. At a minimum, hard hats, safety glasses, hard toe shoes, and Class II high visibility vests are required on Reclamation construction projects.
 - b) Include a detailed plan for each task that specifies which personal protective equipment shall be used for all construction activities (PPE Hazard Assessment).
 - c) Onsite personnel shall wear Class II High Visibility Vests or the equivalent such as Class II Jackets.
 - d) Class III High Visibility clothing shall be worn when working between sunset and sunrise (night). Class III clothing is required for head, arms, legs, torso, and feet for employees working within highway right-of-way or near traffic. Night requirements begin one half hour before sunset and end one half hour after sunrise.
 - e) Safety glasses are required, except when other types of eye protection are being utilized for a specific task which does not allow for the use of regular safety glasses (e.g., welding and cutting).
 - f) The employer must ensure that the PPE is verified by the manufacturer for the intended use and protection.
 - g) Provide appropriate PPE at no cost to the employees.
- 5) Ventilation Plans:
 - a) Include detailed plans with design calculations for ventilation in areas such as but not limited to confined spaces. This section can be contained in the confined space section if it is addressed only in confined spaces.
 - b) Approved by CIH in accordance with Section 01 35 30 – Contractor's Onsite Safety Personnel.
- s. Machinery and Mechanical Equipment:
 - 1) Procedures and Equipment to Minimize Hazards:
 - a) Include testing program for employees and work environments.

- b) Include mobile and stationary equipment. Equipment that is leaking fluids or fails the initial inspection shall be repaired immediately or removed from the site immediately.
 - c) Include in the program the requirement in Sections 18, 19, and 20 of the RSHS.
 - d) Reference the appropriate Appendices in the RSHS regarding inspection and testing of equipment during mobilization and prior to start of work.
 - e) Reference the types of equipment that the contractor intends to use on site and any pertinent requirement that the contractor must meet for inspection, testing, and maintenance of the equipment.
 - f) List the training and certification requirements that the contractor requires for specific types of equipment (such as cranes and forklifts).
- 2) Inspection Procedures:
- a) See RSHS Section 20 for requirements.
- 3) Maintenance Procedures:
- a) See RSHS Section 20 for requirements.
- 4) Operating Personnel.
- a) See RSHS Section 20. Include applicable licensing and certification requirements, DOT requirements, forklift operator certifications, and other pertinent qualifications.
- 5) Protective Safety Devices and Certifications.
- a) Protective safety devices and current certifications are required for each type of equipment as listed by the RSHS, Operators Manual and OSHA. Inoperable safety devices will require that equipment is taken out of service until the devices are repaired.
 - i. Includes alarms, pressure relief valves, anti-two block devices.
- 6) Elevators and Aerial Lifts:
- a) Include types of equipment to be used. Identify if the equipment is powered by electric, diesel, propane or gas if to be used indoors or in closed in areas.
- 7) Crane, Hoisting and Rigging Safety:
- a) Include operator and rigging safety, materials handling, signaling and communication procedures.

- b) All Operators, Riggers and Signalpersons shall be qualified in accordance with OSHA 1926 Subpart CC.
 - i. Crane Operators shall be certified by an accredited crane operator testing organization such as NCCCO or equivalent.
 - c) Reclamation requires an onsite verified load test in accordance with RSHS Section 19 prior to use of cranes.
- 8) Excavations and Demolition:
- a) Tunnels and Shafts: Not Applicable.
 - b) Blasting: Not Applicable.
 - c) Other Excavations and Demolition: Provide a detailed excavation safety plan addressing all aspects of excavation operations. At a minimum, include:
 - i. Specific duties of the competent person.
 - ii. Slide protections.
 - iii. Type of Support systems to be used.
 - iv. Inspections.
 - v. Access.
 - vi. Control of ground water.
 - vii. Refer to and follow all applicable regulations including OSHA and RSHS Section 22.
- 9) High Scaling: Not Applicable
- 10) Haulage:
- a) Haul roads:
 - i. Intended haul roads and routes.
 - b) Equipment and procedures:
 - i. Intended equipment and procedures.
- t. Working Surfaces:
- 1) Access:
- a) Fall Protection and Prevention:
 - i. Detail safe work procedures, training, and engineering controls for work in elevated areas where employees are exposed to a fall.
 - ii. Use engineered fall protection systems, fall restraint, and/or fall arrest systems for work in

elevated areas. Warning line and safety monitoring systems or any combination thereof are not permitted.

- b) Rope Supported Work: Not Applicable.
- c) Ladders:
 - i. See RSHS for specific requirements.
 - ii. Extension ladders shall rise 42 inches above landing surface.
 - iii. A-frame type step ladders shall not be used to access other levels.
- d) Platforms, stairways, and ramps.
- e) Uneven or Non-Uniform Surfaces.
- 2) Scaffolding Safety and Design:
 - a) Include a scaffolding plan indicating what type of scaffolding will be used at the site and provide drawings, design specifications, manufacturer information, fall protection procedures and falling object protection.
 - b) Prepared or reviewed and signed by a qualified Professional Engineer (P.E.) except when manufactured systems and standard designs are used.
 - c) Reclamation requires fall protection at 6 feet on unguarded scaffolding.
- 3) Safety Nets: Not Applicable.
- u. Protection of the Public:
 - 1) Signs and Barricades:
 - a) Specific to project needs.
 - b) Include road closure signs, barriers, and detours as needed.
 - c) Include roadway signage for trucks entering/leaving roadway and other signage warranted for protection of public along haul routes, recreation sites, and publicly accessible areas where equipment is operating.
 - 2) Flagging Procedures:
 - a) Specific to project needs. May be included in Traffic Control Procedures if required.
 - b) Where trucks and equipment are crossing a public roadway or entering an area where visibility of oncoming vehicles or

the trucks and equipment is limited, certified flaggers shall be required and the junction.

3) Traffic Control:

- a) For each operation requiring the use of heavy equipment, in addition to the use of backup alarms and high visibility vests, include in the Safety and Health Program and the JHA, specific steps to ensure the protection of onsite personnel (internal traffic control) from the hazards of being struck by or caught between moving equipment while working around such equipment. Possible methods of controlling said hazards include radio communications, backup cameras, proximity devices, and designated observers.

v. Electrical Facilities. Not Applicable.

w. Additional Requirements:

1) Hazard Communication:

- a) Include a written Hazard Communication plan.
- b) Include an inventory (list) of all hazardous chemicals that the contractor plans to bring on site.
- c) The safety and health practices, procedures, and protective equipment used for new and existing products shall be based upon the health hazards identified in the SDS.

2) Protection Against Drowning:

- a) The safety program shall include a plan to minimize the hazards associated with working adjacent to water.
- b) Include in the plan the protections such as, but not limited to, rescue boats, life jackets and ring buoys that the contractor intends to use if working over or near water.

C. RSN 01 35 20-2, Monthly Accident Summary Reports:

- 1. Form 7-2218 or other acceptable form in accordance with Section 3 of RSHS.

D. RSN 01 35 20-3, Respirator User Documentation:

- 1. Submit documentation of medical qualifications less than 12 months old for all respirator users at the site.
- 2. Submit results of fit tests less than 12 months old for all users of tight-fitting negative-pressure respirators at the site.
- 3. Submit documentation of respiratory protection training within the last 12 months for all respirator users at the site.

- E. RSN 01 35 20-4, Crane Operation Certificates:
1. Qualifications for all operators, riggers, and signalpersons.
- F. RSN 01 35 20-5, Exposure Assessment Form:
1. Completed by a Certified Industrial Hygienist (CIH).
- G. RSN 01 35 20-6, Other Training Certificates:
1. First Aid/CPR certifications for personnel on site.
 - a. Training shall be in accordance with the American Red Cross or the American Heart Association.
 - b. Online first aid and CPR training is not acceptable.
 2. Documentation of hazard communication training for all Contractor employees specific to the chemicals that will be used on site.
 3. Documentation of fall protection and prevention training for employees who work in elevated areas.
 4. Documentation of lockout/tagout training for all employees who will work in any area protected by lockout/tagout and hazardous energy control procedures.
 5. Training certifications and experience for scaffolding competent person if applicable.
 - a. Include site-specific training documentation.
 6. Training certification for Competent Person acquired within the last 3 years for excavation competent person if applicable.
 - a. Include site-specific training documentation.
 7. Documentation of training within the last 12 months for all Contractor employees involved with the following hazardous materials work:
 - a. Removal of coatings containing lead and other regulated metals in accordance with SSPC-C3 and SSPC-C5.
 - b. Training in the hazards of coal tar and in the means and methods to reduce hazards exposure during coating removal.
 - c. Asbestos training in accordance with the Environmental Protection Agency requirements.
 8. Training certifications and experience for scaffolding competent person if applicable.

1.04 SAFETY AND HEALTH

- A. Comply with RSHS, OSHA, local safety programs, and other referenced safety and health standards or industry standards (the most stringent will apply). Perform work in accordance with the specific safety program elements and referenced standards.

- B. Provide and maintain a work environment and procedures that will safeguard the public, Government personnel, and Contractor employees from Contractor operations and activities that are unsanitary, hazardous, or dangerous to health and safety.
- C. Comply specifically with all requirements contained in RSHS Section 3 – Contractor Requirements.
- D. Do not begin onsite work, including mobilization, until the COR accepts the Safety Program.
- E. Investigate and report all accidents, incidents and near misses in accordance with RSHS Section 3.5.
- F. When the Contractor fails or refuses to promptly correct a compliance directive, the Contracting Officer may issue an order to stop all or part of the work.
1. When satisfactory corrective action is taken, an order to resume work will be issued.
 2. Contractor shall not be entitled to extension of time or to claim for damage or to additional compensation by reason of either the directive or the stop order.
 3. Failure of the CO to order discontinuance of the Contractor's operations shall not relieve the Contractor of the responsibility for the safety of personnel and property.
- G. Disputes regarding safety and health issues will be settled in accordance with RSHS Section 3.
- H. Provide medical and emergency assistance in a timely manner that is consistent with recognized emergency protocols.
1. In instances where local emergency responders are not readily available, the responsibility to provide emergency services shall default back to the Contractor.
- I. Ensure that Subcontractors comply with the requirements set forth in this specification, written programs, the regulatory standards, and the industry standards incorporated by reference in the RSHS and OSHA standards.
- J. Where there is conflict between the standards, the more stringent or restrictive shall be utilized.
- K. Participate fully in the Contractor Safety Program Review meeting, according to RSHS Section 3.2.3.2.
1. Contractor Safety Program Review meeting will take place before the start of construction.
 2. Include subcontractor management representatives.

3. Safety Personnel shall be present, in order to discuss in detail, the Safety Program and Plans.
 4. The COR, Contractor's principal onsite representative, and designated members of respective staffs must participate in the scheduled monthly safety meeting (JPSM) according to RSHS Section 3.
- L. Minimum work crew on the construction site shall consist of no less than 2 people.
- M. ~~Pandemic Preventive Measures:~~
1. ~~Implement regulations and guidance from Federal, State, and local health agencies into workplace specific plans.~~
 2. ~~Hierarchy of Controls.~~
 - a. ~~Develop controls that are equivalent to the level of exposure risk occupation:~~
 - 1) ~~Engineering Controls (e.g., barriers, ventilation).~~
 - 2) ~~Administrative Controls (e.g., social distancing, scheduling).~~
 - 3) ~~Personal Protective Equipment (e.g., respirators, gloves).~~
 3. ~~Promote frequent and thorough hand washing, including by providing workers, customers, and worksite visitors with a place to wash their hands.~~
 4. ~~Schedule and maintain regular housekeeping practices, including routine cleaning and disinfecting of surfaces, equipment, and other elements of the work environment.~~
 5. ~~Implement "social distancing" in the workplace.~~
 6. ~~Develop flexible pandemic flu attendance and sick leave policies. Require sick workers to stay home.~~
 7. ~~Establish flexible working hours or possible staggered shift schedule.~~
 8. ~~Prompt Identification and Isolation of Sick People:~~
 - a. ~~Train and encourage employees to self monitor for signs and symptoms if possibly exposed.~~
 - b. ~~Develop policies and procedures for employees to report when sick or experiencing symptoms.~~
 - c. ~~Develop policies and procedures for immediately isolating people with signs or symptoms. Train workers on implementation.~~
 9. ~~Infected Employee Return to Work:~~
 - a. ~~Provide medical clearance or show clearance of symptoms plus waiting period as recommended by local, State, and Federal guidance.~~
 - b. ~~Follow local, State, and Federal guidance for returning to work.~~

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 30
CONTRACTOR'S ONSITE SAFETY PERSONNEL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

- A. ABIH: American Board of Industrial Hygiene.

1.03 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>

1.04 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 35 30-1, Qualifications:

1. Contractor's Onsite Safety Representative.
2. Part-time Onsite Safety and Health Professional.
3. Certified Industrial Hygienist (CIH).
 - a. Copy of current ABIH certificate.
4. Independent testing laboratory:
 - a. NLLAP laboratory accreditation number.
 - b. NVLAP laboratory accreditation number.
 - c. Name, address, and telephone number.

C. RSN 01 35 30-2, Safety Inspection Reports:

1. Prepare detailed weekly safety inspection reports listing noted deficiencies, digital photos, abatement dates, and follow-up action for all jobsite activities.
2. Include findings of jobsite walk-through with Government representative.

D. RSN 01 35 30-3, CIH Reports:

1. Detailed monthly summary report for any hazardous agents related to work activities.
 - a. Summary of exposure monitoring results and sampling activities and interpretation of the data to confirm the work is being performed in accordance with the submitted approval plans.
2. Include noted deficiencies, digital photos, abatement dates and follow-up action.
3. Submit written report to COR within 7 days of inspection.

1.05 QUALIFICATIONS**A. Contractor's Onsite Safety Representative:**

1. Competent supervisory employee with current safety and health related training, experience, and duties on at least 3 projects of similar nature, size, complexity, and work to be performed as the current project.
2. OSHA Construction 30-Hour Training Course including applicable elective subjects. For example: Concrete and Masonry Construction; Confined Space Entry; Cranes, Derricks, Hoists, Elevators, and Conveyors; Ergonomics; Excavations; Fire Protection and Prevention; Materials Handling, Storage, Use and Disposal; Motor Vehicles, Mechanized Equipment and Marine Operations; Rollover Protective Structures and Overhead Protection; and Signs; Signals and Barricades; Powered Industrial Vehicles; Safety and Health Programs; Scaffolds; Steel Erection; Tools - Hand and Power; Welding and Cutting.
3. Current first aid and CPR certification.
4. Include resume with current telephone numbers of references, description of safety representative responsibilities, and copies of training certifications.

B. Contractor's Part-time Onsite Safety and Health Professional:

1. Qualified by virtue of education, training, certification, and experience in which professional status in the construction safety field has been established.
 - a. Possesses at least a 4-year bachelor's degree from an accredited college or university in one of the following:
 - 1) Safety Engineering.
 - 2) Safety and Occupational Health.
 - 3) Industrial Hygiene (IH).
 - b. Accredited as one of the following:
 - 1) Registered Professional Safety Engineer.
 - 2) Certified Safety Professional (CSP).
 - 3) Associate Safety Professional (ASP).

- 4) Construction Health & Safety Technician (CHST).
 - 5) Occupational Health and Safety Technologist (OHST).
 - 6) CIH certified by the Board of Certified Safety Professionals /American Board of Industrial Hygiene (ABIH).
2. Knowledgeable of current safety trends and industry practices; can exercise technical expertise.
3. Include resume with current telephone numbers of references, description of safety professional responsibilities on similar construction projects, and copies of training certifications, degree, and registration.
4. Safety Professional requirements may be met by retaining appropriate level of services of an acceptable safety consultant.

C. Certified Industrial Hygienist (CIH):

1. Identify an individual meeting the following requirements:
 - a. Certified by the American Board of Industrial Hygiene.
 - b. Trained in recognizing complex safety, occupational health, and environmental issues.
 - c. Knowledgeable of current safety trends and industry practices; can exercise technical expertise.
 - d. Demonstrates professional level knowledge of exposure hazards associated with the specific construction activities and can provide best practices on mitigation and controls.
2. Include resume, description of CIH responsibilities on similar construction projects, and copies of training certifications.
3. For hazardous materials work, CIH shall be independent from the Contractor.

1.06 AUTHORITIES, DUTIES, AND RESPONSIBILITIES

- A. The authorities, duties, and responsibilities of onsite safety personnel include, but are not limited to, the following:
- B. Contractor's Onsite Safety Representative:
 1. Effectively implement Contractor's Safety Program.
 2. Coordinate all emergency response activities.
 3. Recommend controls for observed hazards or predicted hazards.
 4. Has full authorization to correct unsafe acts on the spot, including the authority to stop work to correct safety and health problems.
 5. Lead weekly safety meetings at the jobsite.

6. Onsite during any and all construction activities.
 7. Assembles, reviews, and signs each Job Hazard Analysis (JHA).
 - a. Review JHA with employees.
 8. Verifies the correct use of respiratory protection and personal protective equipment.
 9. Conduct and prepares weekly safety inspections in conjunction with Contractor's part-time Onsite Safety and Health Professional and onsite Government representative listing noted deficiencies, their abatement dates, and follow up actions. The report shall be available for review onsite.
 10. Provide direct oversight and assemble a detailed report of all incident reviews, near-misses, accidents, and emergency response or rescue actions.
 11. Onsite Safety Representative may have other non-safety duties.
- C. Contractor's Part-time Onsite Safety and Health Professional:
1. Onsite Safety and Health Professional's job shall be hazard recognition, control, and accident prevention.
 2. Shall NOT be supervised by the onsite superintendent, but shall report directly to the corporate manager/owner.
 3. Has full authorization to correct unsafe acts on the spot, including the authority to stop work to correct safety and health problems.
 4. Visit the site monthly and inspects the site with a Government representative listing noted deficiencies, their abatement dates, and follow up actions. The report shall be available for review onsite.
 5. Provide guidance and oversight to Contractor's Onsite Safety Representative.
 6. Participate in the planning and preparation of all activities including but not limited to:
 - a. Development and approval of JHA.
 - b. Review of equipment and materials.
 - c. Employee training.
 - d. Observing and reporting on activities.
 7. Participate in the planning, preparation, and execution of high exposure activities.
 8. Duties include, but are not limited to the following:
 - a. Before start of construction, identify existing and predictable unsanitary, hazardous, or dangerous conditions.
 - b. Approve site-specific Contractor safety and health plans and program prior to submission.

- c. Coordinate all elements contained in the Contractor safety and health program and these specifications, including safety training.
- d. May act as the qualified person for supervising erection and accepting final configuration and perform daily checks of fall protection, excavation and scaffolding systems in accordance with the regulatory standards.
- e. Review and finalize all incident reviews, near-misses and accident reports in accordance with RSHS Section 3.
- f. Attend the Safety Program Review Meeting and Conduct the monthly Joint Safety Policy Meetings in accordance with the RSHS Section 3.

D. Certified Industrial Hygienist (CIH):

- 1. Visit site during start-up of coating removal and application activities and a minimum of one time per week during all coating removal and application activities to confirm that the work is being performed in accordance with submitted plans.
- 2. Review and sign the Written Compliance Program for Worker Protection, Containment System Plan, Emissions Monitoring Plan, Waste Characterization, Handling, and Disposal Plan, Heavy Metal Exposure Certifications and Test Results, Working Level Test Results, Post-Construction Test Results, and Certificate of Disposal.
- 3. Conduct or coordinate, review, and sign all coatings background sampling, surface sampling, waste stream sampling, personal exposure monitoring, area air monitoring, peripheral air monitoring, and clean up monitoring.
- 4. Conduct functional testing of engineering controls, critical barriers, containments, and secondary containments.
- 5. Has full authorization to correct unsafe acts, stop work, prevent exposures, control contamination, engineering controls and containment failures.
- 6. Perform exposure assessments to select or verify personal protective equipment or respiratory protection inside and outside containments.
- 7. Prepare detailed monthly CIH inspection reports in accordance with RSN 01 35 30-3.
- 8. Attend the Safety Program Review Meeting in accordance with RSHS Section 3.
 - a. Assist Onsite Supervisor in implementing Hazardous Materials safety and health requirements during the performance of the required work.
 - b. Available for emergencies.
 - c. Direct air monitoring.

E. Competent Person:

- 1. Duties include, but are not limited to:

- a. Control entry to and from the Regulated Area.
- b. Ensure engineering controls in use are in proper operating condition and are functioning properly.
- c. Ensure employees wear appropriate PPE, are trained in the use of appropriate methods for exposure control, use proper hygiene methods and decontamination procedures.

1.07 QUALITY ASSURANCE

- A. The effectiveness of the Contractor's onsite safety personnel performing their respective duties and responsibilities will be subject to continued review and approval by the CO.
 1. Should these efforts be deemed insufficient the Contractor may be required to provide the services of qualified, full-time personnel at no additional cost to the Government.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 42 10

REFERENCE STANDARDS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

- A. Referenced editions of standard specifications, codes, and manuals form a part of this specification to the extent referenced.
- B. These specifications take precedence when conflicting requirements occur between specifications and referenced standard.

1.03 JOBSITE REFERENCE STANDARDS

- A. Maintain at fabrication site, access to referenced standard specifications, codes, and manuals required for work in progress at fabrication site. Make available for use by the Government.
- B. Maintain onsite, access to referenced standard specifications, codes, and manuals required for onsite work in progress. Make available for use by the Government.

1.04 AVAILABILITY

- A. Code of Federal Regulations (CFR):
1. Available online, authorized by the National Archives and Records Administration (NARA) and the Government Printing Office (GPO), at <https://www.govinfo.gov/help/cfr>.
- B. Federal Specifications, Standards, and Commercial Item Descriptions; and Military Specifications:
1. Copies of Federal Specifications, Standards, and Commercial Item Descriptions may be obtained from GSA Federal Supply Service, see the provision at FAR 52.211-1, Availability of Specifications Listed in the GSA Index of Federal Specifications, Standards and Commercial Item Descriptions, FPMR Part 101-29.
 2. Copies of Military Specifications may be obtained from Department of Defense, see the provision at FAR 52.211-2, Availability of Specifications, Standards, and Data Item Descriptions Listed in the Acquisition Streamlining and Standardization Information System (ASSIST).

C. Bureau of Reclamation (Reclamation) Documents:

1. Reclamation Safety and Health Standards (RSHS) may be downloaded at <https://www.usbr.gov/safety/rshs/index.html>.
2. Reclamation manuals and other publications including significant scientific, technical, and engineering works are available from the National Technical Information Service (NTIS). Information regarding availability and pricing may be obtained by contacting NTIS at the following address:

United States Department of Commerce
National Technical Information Service
5285 Port Royal Road
Springfield, VA 22161
703-487-4650 or 1-800-553-6847

D. Industrial and Governmental Documents:

1. When a reference has a joint designation (e.g., ANSI/IEEE) these specifications generally cite the proponent organization (e.g., IEEE).
2. Addresses for obtaining industrial and governmental (other than Federal and Reclamation specifications and standards) specifications, standards, and codes are listed in Table 01 42 10A – Addresses for Specifications, Standards, and Codes.

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
AA	Aluminum Association 1400 Crystal Drive, Suite 430 Arlington VA, 22202 www.aluminum.org	703-358-2960
ACGIH	American Conference of Governmental Industrial Hygienists 1330 Kemper Meadow Dr. Suite 600 Cincinnati, Ohio 45240 www.acgih.org	(513) 742-2020
ACI	American Concrete Institute 38800 Country Club Dr. Farmington Hills, MI 48331-3439 www.aci-int.org	(248) 848-3700
AEIC	Association of Edison Illuminating Companies P.O. Box 58130 Washington, DC 20037 https://aeic.org	(205)-257-3839

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
AGC	Associated General Contractors of America 333 John Carlyle Street, Suite 200 Alexandria, VA 22314 www.agc.org	703-548-3118
AISC	American Institute of Steel Construction 130 East Randolph, Suite 2000 Chicago, IL 60601-2001 www.aisc.org	(312) 670- 2401
ANSI	American National Standards Institute 1899 L Street, NW, 11 th Floor Washington, DC 20036 www.ansi.org	(202) 293-8020
ASME	American Society of Mechanical Engineers Two Park Avenue New York, NY 10016-5990 www.asme.org	800-843-2763
ASSE	American Society of Sanitary Engineering 18927 Hickory Creek Drive, Suite 220 Mokena, IL 60448 www.asse-plumbing.org	(708) 995 3019
ASTM	ASTM International P.O. Box C700 100 Barr Harbor Drive West Conshohocken, PA 19428-2959 www.astm.org	610-832-9585
AWS	American Welding Society 8669 NW 36 Street, Suite #130 Miami, FL 33166-6672 www.aws.org	800-443-9353
BICSI	Building Industry Consulting Service International 8610 Hidden River Parkway Tampa, FL 33637-1000 www.bicsi.org	1.813.979.1991 or 800.242.7405

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
CCR	Colorado Code of Regulations Colorado Secretary of State 1700 Broadway, Suite 550 Denver, CO 80290 www.sos.state.co.us/CCR/	303-894-2200
ICEA	Insulated Cable Engineers Association, Inc. P.O. Box 1568 Carrollton, GA 30112 http://www.icea.net/	(800) 447-3352
ICRI	International Concrete Repair Institute 1000 Westgate Drive, Suite 252 St. Paul, Minnesota 55114 www.icri.org	651-366-6095 Fax: 651-290-2266
IEC	International Electrotechnical Commission 3, rue de Varembe, 1st floor P.O. Box 131 CH - 1211 Geneva 20 – Switzerland www.iec.ch	+41 22 919 02 11 Fax +41 22 919 03 00
IEEE	Institute of Electrical and Electronics Engineers 3 Park Avenue, 17th Floor New York, NY 10016-5997 www.ieee.org	212-419-7900
ISA	The International Society of Automation 67 T.W. Alexander Drive PO Box 12277 Research Triangle Park, NC 27709 www.isa.org	919 549-8411 Fax 919 549-8288
ISO	International Organization for Standardization ISO Central Secretariat Chemin de Blandonnet 8 CP 401 1214 Venier, Geneva, Switzerland www.iso.org	+41 22 749 01 11
MSS	Manufacturers Standardization Society 127 Park St., NE Vienna, VA 22180-4602 www.mss-hq.com	(703) 281-6613

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
NACE	AMPP: Association for Material Protection and Performance (formerly known as NACE International) 15835 Park Ten Place Houston, Texas 77084 www.ampp.org	800-797-6223 281-228-6200
NEIS	National Electrical Installation Standards (NEIS) 1201 Pennsylvania Ave. NW, Suite 1200, Washington, DC 20004 www.necanet.org	Phone: (202) 991-6300 Fax: (202) 217-4171
NEMA	National Electrical Manufacturers Association 1300 N 17th St., Suite 900 Arlington, VA 22209 www.nema.org	(703) 841-3200
NEMI	National Environment Methods Index (NEMI) www.nemi.gov/home/	nemi@usgs.gov
NETA	International Electrical Testing Association 3050 Old Centre Ave., Suite 101 Portage, MI 49024 www.netaworld.org	(269) 488 6382
NFPA	National Fire Protection Association One Batterymarch Park P.O. Box 9101 Quincy, MA 02269-9101 www.nfpa.org	(800) 344-3555 (617) 770-3000
NIBS	National Institute of Building Sciences 1090 Vermont Avenue, NW., Suite 700 Washington, DC 20005-4905 www.nibs.org	202-289-7800
OSHA	U. S. Department of Labor Occupational Safety & Health Administration 200 Constitution Avenue NW Washington, DC 20210 www.osha.gov	800-321-OSHA 800-321-6742

Table 01 42 10A – Addresses for Specifications, Standards, and Codes

Acronym	Name and Address	Telephone
SSPC	AMPP: Association for Material Protection and Performance (formerly known as SSPC: The Society for Protective Coatings) 15835 Park Ten Place Houston, Texas 77084 www.ampp.org	800-797-6223 281-228-6200
TIA/EIA	Telecommunications Industry Association/Electronic Industries Alliance 1320 North Courthouse Road, Suite 200 Arlington, VA 22201 www.tiaonline.org	(703) 907-7700
UL	Underwriters Laboratories Inc. 750 Anthony Trail Northbrook, IL 60062-2096 www.ul.com	(847) 272-8800

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 46 00

QUALITY PROCEDURES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
 - 1. Include in prices offered in the Price Schedule for other items of work.
- B. Progress Payments: If test reports are not submitted in a timely manner, Contractor will be considered non-compliant and delaying that phase of work to which testing applies. CO may retain appropriate amounts of applicable progress payments.

1.02 DEFINITIONS

- A. Quality Assurance: Inspection and tests performed by Government to ensure compliance with terms of the contract.
- B. Quality Control: Activities performed by Contractor to ensure work conforms to contract requirements.
 - 1. The clause at FAR 52.246-12, Inspection of Construction requires Contractor to establish an inspection system to ensure quality.
 - 2. Quality Control includes activities in addition to specified Contractor Quality Testing to ensure work conforms to contract requirements.
- C. Contractor Quality Testing: Specified tests shall be performed by Contractor.
 - 1. Government may use the test results for Quality Assurance.
 - 2. Contractor may use test results as part of Contractor quality control.
 - a. Government anticipates that these tests will be part of Contractor's Quality Control Program; however, the tests do not relieve the Contractor of providing adequate quality control measures in accordance with the clause at FAR 52.246-12, Inspection of Construction.

1.03 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 46 00-1, Contractor Quality Control (CQC) Program:
 - 1. Designation and resume of authorized individual as the Quality Control Supervisor to develop and manage quality control processes.

2. Identification of independent testing firm hired to provide Quality Control testing, including:
 - a. Agency accreditation to perform specified testing or agency qualifications to perform specified testing.
 - b. Resumes of personnel performing tests.
 - c. Samples of report forms.
 3. Address methods, procedures, frequencies, and scheduling for performing quality control testing in the following sections:
 - a. Section 09 96 20 – Coatings.
 - b. Section 33 11 20 – Pipe Rehabilitation.
 - c. Section 35 21 92 – Pipe Removal and Installation.
- C. RSN 01 46 00-2, Contractor's Daily Reports:
1. Submit daily reports on a weekly basis.
- D. RSN 01 46 00-3, Summary Monthly Test Results.

1.04 QUALIFICATIONS

- A. Designate a Quality Control Supervisor; this employee may have additional duties.
- B. Quality Control Supervisor: Individual with minimum 5 years of experience in charge of quality control for construction similar in nature to that required by this contract, with authority to stop work if necessary.

1.05 TESTING ACCEPTANCE

- A. Calibrate measuring devices, laboratory equipment, and instruments at established intervals.
- B. Material acceptance based on the passing of both quality control testing and quality assurance testing.
- C. Government quality assurance testing results will prevail in the event of a discrepancy between Contractor's test results and Government test results.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CONTRACTOR QUALITY CONTROL PLAN

- A. Acceptance of the CQC Plan is required prior to the start of construction. Acceptance is conditional and will be predicated on continued satisfactory performance during construction. Government reserves the right to require Contractor to make changes in the CQC Plan and operations including removal of personnel and completed work, as necessary, to obtain the quality required by these specifications.
- B. Contractor Quality Control Testing
 - 1. Employ an accredited independent agency to perform sampling, testing, and reporting as required in the following sections:
 - a. Section 09 96 20 – Coatings
 - 2. Provide testing results immediately when available after testing but no later than 24 hours after conducting tests.

3.02 CONTRACTOR QUALITY PROCEDURES

- A. At a minimum, utilize the following definable features of work for development and application of quality procedures and planning and include preparatory, initial, and follow-up phases for each:
 - 1. Removal and control of water including cofferdam.
 - 2. Removal of steel pipe, removal and disposal of coatings, surface preparation.
 - 3. Pipe rehabilitation.
- B. Preparatory Phase: Performed prior to beginning work on each definable feature or work, after all required plans/documents/materials are approved/accepted.
 - 1. Required Preparatory Phase actions:
 - a. Notify the COR, in writing, 7 days in advance for beginning any of the required preparatory phase actions.
 - b. Review all approved submittals and identify any missing information or required submittals.
 - c. Review of each paragraph of applicable specifications and references.
 - d. Review of contract plans.
 - e. Verification that materials and equipment have been tested, submitted, and approved.
 - f. Verification that provisions have been made for required quality control inspection and testing.

- g. Examination of the work area to ensure that all required preliminary work has been completed.
 - h. Physical examination of required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings or submitted data.
 - i. Review of appropriate activity hazard analysis.
 - j. Discussion of procedures for constructing the work including the review of repetitive deficiencies.
 - k. Notify COR in advance for beginning any of the required action of the preparatory phase as required in the quality control specifications.
 - 2. Includes a "Preparatory Meeting" conducted by the Quality Control Manager and attended by the superintendent, other quality control personnel, the foreman responsible for the definable feature, and the COR if available.
 - a. Results of the preparatory phase actions shall be discussed in the Preparatory Meeting and shall be documented by separate minutes prepared by the Quality Control Manager and attached to the daily Contractor's Quality Control Report.
- C. Initial Phase: Accomplished at the beginning of a definable feature of work.
 - 1. Verify that control for work developed in the Preparatory Meeting is implemented and the work is performed to the level of workmanship required.
 - 2. Required Initial Phase actions:
 - a. Notify the COR, in writing, 7 calendar days in advance of beginning any of the required action of the initial phase.
 - b. Review minutes of "Preparatory Meeting."
 - c. Check preliminary work.
 - d. Verify adequacy of controls to ensure full contract compliance.
 - e. Establish levels of workmanship.
 - f. Resolve all differences.
 - g. Review of safety information shall include compliance with the safety program and job hazard analysis. Review the job hazard analysis with workers.
 - 3. Include an "Initial Phase Meeting" conducted by the Quality Control Manager and attended by the superintendent, other quality control personnel, the foreman responsible for the definable feature, and COR if available.
 - a. Results of the initial phase actions shall be discussed in the Initial Phase Meeting and shall be documented by separate minutes prepared by the Quality Control Manager and attached to the Contractor's Daily Report.

4. Repeat initial phase for each new crew on the work site and at any time established level of workmanship is not being met.

D. Follow-up Phase:

1. Perform daily checks to ensure continuing compliance with contract requirements, including safety and control testing, until completion of each definable feature of work.
2. Make checks a matter of record in the Contractor's Quality Control documentation.
3. Conduct final follow-up checks and correct deficiencies prior to the start of additional features of work.
4. Quality control personnel shall continually refer to standards set in the Preparatory and Initial Phases.

3.03 CONTRACTORS DAILY REPORT

A. The Contractor's Daily Report shall include the following information:

1. Contractor's representative on site during shift.
2. Record of all the Contractor's and subcontractor's activities throughout the shift regardless of how minor or complex.
3. Safety violations noted during the shift, including actions taken, names of people involved in the infraction, and individual responsible for the correction.
4. Significant discussions held between Government and Contractor personnel or other parties involved with the project.
5. Daily temperature (maximum/minimum), wind conditions, estimated humidity, sunshine, and precipitation (type and intensity).
6. Record of quantities installed, specific stations or locations of work, and description by CPM program activity (if applicable). Show labor classification and equipment actually used during the shift, work the labor force and equipment is performing, and the period of time labor and equipment is performing. Note equipment that is idle or being repaired.
7. List of materials and equipment delivered to the job and any materials that were refused or wasted during the shift.
8. Daily Labor Summary Table displaying employees name, classification, activity, and hours worked.
9. Daily Equipment Summary Table displaying equipment, make and model, activity, hours worked, hours idle, and hours in repair status.

3.04 QUALITY ASSURANCE

- A. During the course of the work, Government may perform quality assurance tests. Tests performed by the Government will be used to assure compliance with contract requirements and not as a replacement for specified Contractor quality testing.
1. Upon request, make measuring and testing devices available for use by the Government for verification tests.

3.05 COMPLIANCE

- A. Contractor's Quality Control Supervisor:
1. Shall be at project site during work activities and shall have a complete understanding of work being performed.
 2. The effectiveness of the Contractor's Quality Control Supervisor in prosecuting the Quality Control Program will be subject to continued review and approval by the CO.
 3. Should the Contractor's Quality Control Supervisor's efforts be deemed insufficient the Contractor may be required to replace the Quality Control Supervisor at no additional cost to Government.
- B. Contractor's Quality Control Program:
1. The effectiveness of the Contractor's Quality Control Program will be subject to continued review and approval by the CO.
- C. Contractor's Testing Agency Personnel:
1. The effectiveness of the Testing Agency Personnel in prosecuting the required testing will be subject to continued review and approval by the CO.

END OF SECTION

SECTION 01 51 00

TEMPORARY UTILITIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C2-2023 National Electrical Safety Code (NESC)

1.03 TEMPORARY ELECTRICITY

A. Electric power will be available for work performed under these specifications.

1. Power is available at maintenance receptacles. Provide necessary adapters.
 - a. Single-phase, 120-Volt. Standard convenience outlet.
 - b. Three-phase, 480-Volt, 50 amps. One outlet located within plant near pipe gallery.
 - c. Three-phase, 480-Volt, 50 amps. One outlet located outside on top of deck near downstream side of plant.
2. No charge will be made for this power.

B. Provide other electric power required for construction.

C. Provide generators, transmission circuits, distribution circuits, transformers, and other electrical equipment and facilities required for obtaining power and distributing power to points of use.

D. Comply with IEEE C2 clearances and spacing for temporary communications and supply lines.

1.04 TEMPORARY WATER

A. Provide water required for construction purposes.

B. Use water which meets specified requirements for water used in permanent work.

C. Provide means of conveying water to points of use.

1.05 TELEPHONE

- A. No landline is available.
- B. Mobile service is intermittent.

1.06 TEMPORARY COMPRESSED AIR

- A. Provide compressed air required for construction purposes.
- B. Provide means of conveying compressed air to points of use.

1.07 SANITARY FACILITIES

- A. Restroom facilities will not be available to Contractor personnel.
- B. Provide temporary sanitary facilities for Contractor personnel.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 REMOVAL

- A. Remove temporary equipment and facilities upon completion of work under this contract.

END OF SECTION

SECTION 01 55 00

VEHICULAR ACCESS AND PARKING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REGULATORY REQUIREMENTS

- A. Meet requirements established by jurisdictional authority for use of existing roadways and haul routes; including seasonal or other limitations or restrictions, payment of excess size and weight fees, and posting of bonds conditioned upon repair of damage.
- B. Comply with applicable regulations for haul routes over public highways, roads, or bridges.

1.03 PROJECT CONDITIONS

- A. Rights-of-way for access to work from existing roads will be established by Government.
1. In accordance with clause at FAR 52.236-10, Operations and Storage Areas, use only established roadways, parking areas, and haul routes; or temporary roadways, parking areas, or haul routes constructed by Contractor when and as authorized by CO.
 2. Subject to clause at FAR 52.249-10, Default (Fixed-Price Construction), unavailability of transportation facilities or limitations thereon shall not become a basis for claims for damages or extension of time for completion of work.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Materials to maintain and repair existing roadways, parking areas, and haul routes: In accordance with requirements of jurisdictional authority.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Investigate condition of available public or private roads for clearances, restrictions, bridge-load limits, bond requirements, and other limitations that affect or may affect access and transportation operations to and from jobsite.

3.02 ESTABLISHED ROADWAYS AND PARKING AREAS

- A. Established roadways and parking areas are available for Contractor's use subject to existing restrictions and approval of the COR.
- B. Designated area of existing parking facilities may be used by construction personnel.
- C. Do not allow heavy vehicles or construction equipment in established parking areas.
- D. Parking Areas:
 - 1. Arrange for temporary parking areas for construction personnel use.
 - 2. Provide additional offsite parking when site space is not adequate.
 - 3. Locate in accordance with Section 01 14 10 – Use of Site.

3.03 HAUL ROUTES

- A. Use existing available public highways, roads, or bridges as haul routes subject to applicable local regulations.
- B. Minimize interference with or congestion of local traffic.
- C. Provide barricades, flaggers, and other necessary precautions for safety of public where haul routes cross public highways or roads.

3.04 MAINTENANCE

- A. Maintain roadways, parking areas, and haul routes in a sound, smooth condition until completion and acceptance of all work under this contract.
- B. As approved by COR, defer until latest practicable date within specified completion period, placement of surfacing on roads or parking areas subject to heavy and deteriorating use by Contractor's construction operations or equipment.
- C. Maintain surfacing of gravel-surfaced roads and parking areas in a smooth condition until completion and acceptance of work under this contract.
- D. Snow removal for convenience of Contractor or to facilitate work operations of Contractor is considered normal required maintenance.

3.05 REPAIR

- A. Promptly repair ruts, broken pavement, potholes, low areas with standing water, and other deficiencies to maintain road surfacing and drainage in original or specified condition.

3.06 REMOVAL

- A. Remove materials used to construct temporary roadways, parking areas, and haul routes prior to contract completion. Recycle salvageable materials as approved by COR.

END OF SECTION

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SECTION 01 56 10
PROTECTION OF EXISTING INSTALLATIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work, except as specified.
 2. Costs for repair of installations damaged by the Contractor's operations are the Contractor's responsibility.

1.02 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 56 10-1, Plan for Protecting Existing Installations.
- C. RSN 01 56 10-2, Initial Construction Site Digital Recording.
- D. RSN 01 56 10-3, Post-Construction Site Digital Recording.

1.03 PROJECT CONDITIONS

- A. Drawings included in these specifications show items of existing materials and equipment but may not show all equipment and materials existing at the jobsite.
- B. Obtain the location of embedded conduit, pipe, cable, ground mat, and other buried items before performing drilling or cutting of concrete.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 EXAMINATION

- A. Photograph and video record existing project conditions prior to mobilization and after completion of the work. Include:
1. Interior and exterior views of each area in vicinity of the work to show complete system.

2. Initial construction site conditions establishing existing conditions of site access, site features, roads, structures, utilities, vegetation, stockpiles, stored materials, and similar conditions that may be vulnerable to damage or deterioration during construction.
 3. Post construction conditions. Use locations identified in initial recording for site access, site features, roads, structures, utilities, vegetation, stockpiles, stored materials, and similar conditions.
- B. Notify COR minimum of 3 business days in advance. COR will be present during documentation.
- C. Provide close up of damage or defects to existing facilities. Documentation of existing conditions shall be used as the standard for restoration. Include:
1. Maximize detail while providing context for surroundings.
 2. Refer to Section 01 33 00 – Submittals for minimum resolution and file format requirements.
 3. Audio track with video describing conditions, including date, time, people present, location, view compass directions, and observations throughout.

3.02 PROTECTION

- A. Protect existing facilities from harm due to Contractor's operations. Protection shall be subject to approval of the Government.
- B. Arrange protective installations to permit operation of existing equipment and facilities by the Government while work is in progress.
- C. Do not discharge anything but clear water into building drainage system.
- D. Prevent dust from entering ventilating systems.

3.03 ENCLOSURES

- A. Enclose work area to prevent dust, spalls, chips, grit, and other foreign material from endangering personnel and contaminating or damaging equipment.
- B. Provide enclosures to confine the Contractor's operations to immediate work area.
- C. Enclosures shall be subject to approval of the Government.

3.04 REMOVAL OF PROTECTIVE INSTALLATIONS

- A. Remove protective installations after purpose has been served. Materials furnished by the Contractor to provide protection remain property of the Contractor.

3.05 REPAIR

- A. Repair, at Contractor's expense, damage to existing installations due to Contractor's operations or Contractor's failure to provide proper protection. At the Government's option, damage may be repaired by the Government, and the Contractor will be backcharged repair costs.

END OF SECTION

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SECTION 01 56 15

PROTECTION OF EXISTING UTILITIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1. RSHS-2024 Reclamation Safety and Health Standards,
including revisions posted at
<http://www.usbr.gov/safety/RSHS/rshs.html>

B. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C2-2023 National Electrical Safety Code (NESC)

1.03 PROJECT CONDITIONS

A. Drawings included in these specifications show existing utilities, but may not show all utilities existing at the jobsite.

B. Obtain location of buried conduit, pipe, cable, ground mat, and other buried items before excavating.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 INTERFERENCE WITH OPERATION OR MAINTENANCE

A. Do not interfere with operation or maintenance service on utilities, existing on date offers are received.

1. Provide for access to utilities in a manner satisfactory to owners and operators and the Government.

B. Provide required temporary structures; make necessary repairs, replacements, or similar operations; and furnish indemnity or other bonds.

3.02 CLEARANCES

- A. Contractor shall provide in accordance with RSHS.

END OF SECTION

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SECTION 01 57 20 ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work, except as specified.
 2. Costs for damages and work stoppage are Contractor's responsibility.

1.02 REFERENCE STANDARDS

- A. Bureau of Reclamation (Reclamation)
1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>

1.03 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 57 20-1, Copy of applicable Air Quality Permit:
1. Air Quality Permits are required for certain construction-related activities including, but not limited to, earthmoving, sandblasting, aggregate processing, welding, spray-coating operations, or other processes which discharge pollutants into the open air.
 2. Air Quality Permits, and information concerning the requirements, are available from: Colorado Department of Public Health and Environment, Air Pollution Control Division, <https://cdphe.colorado.gov/apens-and-air-permits>

1.04 REGULATORY REQUIREMENTS

- A. Comply with Federal, State, and local laws and regulations.
- B. Comply with RSHS.
- C. Conform to most stringent requirement in cases of conflict between specifications, regulatory requirements, and RSHS.
- D. Contractor shall be responsible for damages resulting from dust originating from Contractor operations in accordance with clause at FAR 52.236-7, Permits and Responsibilities.

- E. The CO may stop construction activity in violation of Federal, State, or local laws and additional expenses resulting from work stoppage will be responsibility of Contractor.

1.05 DUST CONTROL

- A. Provide dust control and abatement during performance of work.
- B. Prevent, control, and abate dust pollution on rights-of-way provided by Government or elsewhere during performance of work.
- C. Provide labor, equipment, and materials, and use efficient methods to prevent dust nuisance or damage to persons, property, or activities.

1.06 AIR POLLUTION CONTROL

- A. Use reasonably available methods and devices to prevent, control, and otherwise minimize atmospheric emissions or discharges of air contaminants.
- B. Do not operate equipment and vehicles that show excessive exhaust gas emissions until corrective repairs or adjustments reduce such emissions to acceptable levels.

1.07 NOISE CONTROL

- A. Prevent, control, and abate harmful noise levels.
- B. Comply with applicable Federal, State, and local laws and regulations regarding harmful noise levels.
- C. Noise barriers and equipment mufflers required.

1.08 LIGHT CONTROL

- A. Direct stationary floodlights shall shine downward at an angle less than horizontal.
- B. Shield floodlights so that floodlights will not be a nuisance to surrounding areas.
- C. Correct lighting control problems when they occur.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 57 30 WATER POLLUTION CONTROL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>

B. Code of Federal Regulations (CFR)

1. 40 CFR, Part 112 Oil Pollution Prevention

C. Public Law

1. Sections 311, 402, and 404 Clean Water Act (Public Law 92-500, as amended)

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 57 30-1, Pollution Prevention Plan:

1. As required for discharge of pipe rehabilitation curing water, wastewater, or other pollutants.
 - a. Include copy of wastewater discharge permit.
2. As required by the stormwater permit for discharges from construction sites.
 - a. Include copy of stormwater discharge permit, as applicable.

C. RSN 01 57 30-2, Spill Prevention, Control, and Countermeasure (SPCC) Plan:

1. Submit when SPCC Plan is required in accordance with 40 CFR, Part 112.
 - a. SPCC Plan is required where release of oil and oil products could reasonably be expected to enter into or upon navigable waters of the United States or adjoining shorelines in quantities that may be harmful (40 CFR, Part 110), and aggregate on site oil storage capacity is over 1,320

gallons. Only containers with capacity of 55 gallons and greater are included in determining on site aggregate storage capacity.

2. Reviewed and certified by a registered professional engineer in accordance with 40 CFR, Part 112, as required by Section 311 of the Clean Water Act (Public Law 92-500 as amended).

1.04 REGULATORY REQUIREMENTS

A. Construction Safety Standards:

1. Comply with sanitation and potable water requirements of Section 7 of RSHS.

B. Laws, Regulations, and Permits:

1. Perform construction operations to comply, and ensure subcontractors comply, with:
 - a. Applicable Federal, State, and local laws, orders, regulations, and Water Quality Standards concerning control and abatement of water pollution; and terms and conditions of applicable permits issued by permit issuing authority.
 - b. If conflict occurs between Federal, State, and local laws, regulations, and requirements, the most stringent shall apply.

C. Contractor Violations:

1. If noncompliance should occur, verbally report noncompliance within 2 hours to the CO. Submit specific written information within 2 days including corrective actions.
2. Violation of applicable Federal, State, or local laws, orders, regulations, or Water Quality Standards may result in the CO stopping site activity until compliance is ensured.
3. Contractor shall not be entitled to extension of time, claim for damage, or additional compensation by reason of such a work stoppage.
4. Corrective measures required to bring activities into compliance shall be at the Contractor's expense.

1.05 REQUIRED PERMIT

A. Wastewater Discharge Permit:

1. Prior to discharging wastewater or other pollutants, secure a permit to discharge pollutants as required under section 402 of the Clean Water Act (Public Law 92-500 as amended).
2. Terms and Conditions: Comply with terms and conditions as stated in the permit.
3. Monitoring and Treatment:

- a. Provide monitoring and water treatment, if necessary, to achieve compliance with permit conditions.
 - b. Provide recordkeeping required of the section 402 permittee, as stated in the section 402 permit.
 4. Sampling: Include sampling and monitoring required of the Contractor to meet section 402 requirements, as well as required laboratory tests to determine effluent characteristics.
 5. Monitoring Results:
 - a. Provide monitoring results to COR at least 2 weeks prior to submittal deadline to appropriate State or Environmental Protection Agency (EPA) Regional Administrator.
 - b. Send copies of all information transmitted to appropriate agency to COR.
- B. Stormwater Discharge Permit Associated with a Construction Site, as applicable:
 1. Required when total area of disturbance is equal to or greater than 1 acre.
 2. Notice of Intent (NOI):
 - a. Both Reclamation and Contractor shall sign the NOI to obtain coverage under a stormwater general permit to control stormwater discharges from the construction site as required under section 402 of the Clean Water Act (Public Law 92-500, as amended).
 3. Pollution Prevention Plan:
 - a. Prepare a Pollution Prevention Plan as required to obtain and maintain the permit.
 - b. Comply with terms and conditions to obtain and maintain this stormwater discharge permit.
 4. Monitoring and Water Treatment:
 - a. Provide monitoring and water treatment, if necessary, to achieve compliance with applicable Water Quality Standards.
 - b. Provide the recordkeeping by the stormwater discharge permit associated with construction activity.

1.06 CONTRACTOR RESPONSIBILITIES

- A. Permits:
 1. Any permits obtained by Reclamation are exceptions to the clause at FAR 52.236-7, Permits and Responsibilities, which requires Contractor to obtain necessary licenses and permits.
- B. Monitoring:

1. Conduct monitoring to meet the requirements of the permits which may include:
 - a. Sampling.
 - b. Site inspections.
 - c. Required laboratory tests to determine effluent characteristics.
- C. Reporting Results:
 1. Government will report required monitoring results to appropriate agencies. The section 402 wastewater discharge permit has specific reporting requirements for the permittee for noncompliance when effluent limitations are exceeded.
- D. Recordkeeping:
 1. Retain records and data required by permits.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 POLLUTION CONTROLS

- A. Control pollutants by use of sediment and erosion controls, wastewater and stormwater management controls, construction site management practices, and other controls including Federal, State, and local control requirements.
- B. Sediment and Erosion Controls:
 1. Establish methods for controlling sediment and erosion which address vegetative practices, structural control, silt fences, straw dikes, sediment controls, and operator controls as appropriate.
 2. Institute stormwater management measures as required, including velocity dissipators, and solid waste controls which address controls for building materials and offsite tracking of sediment.
- C. Wastewater and Stormwater Management Controls:
 1. Pollution prevention measures:
 - a. Use methods of unwatering, excavating, or stockpiling earth and rock materials which include prevention measures to control silting and erosion, and which will intercept and settle runoff of sediment-laden waters.
 - b. Prevent wastewater from general construction activities such as drainwater collection, drilling, grouting, or other construction operations, from

entering flowing or dry watercourses without the use of approved turbidity control methods.

- c. Divert stormwater runoff from upslope areas away from disturbed areas.

2. Turbidity prevention measures:

- a. Use methods for prevention of excess turbidity which include, but are not restricted to, intercepting ditches, settling ponds, gravel filter entrapment dikes, flocculating processes, recirculation, combinations thereof, or other approved methods that are not harmful to aquatic life.
- b. Wastewaters discharged into surface waters shall meet conditions of Section 402, the National Pollutant Discharge Elimination System (NPDES) permit.

D. Construction Site Management:

1. Contractor construction operations:

- a. Perform construction activities by methods that will prevent entrance, or accidental spillage, of solid matter, contaminants, debris, or other pollutants or wastes into streams, flowing or dry watercourses, lakes, wetlands, reservoirs, or underground water sources.
 - 1) Pollutants and wastes include, but are not restricted to: refuse, garbage, cement, sanitary waste, industrial waste, hazardous materials, radioactive substances, oil and other petroleum products, aggregate processing tailings, mineral salts, and thermal pollution.

2. Stockpiled or deposited materials:

- a. Do not stockpile or deposit excavated materials or other construction materials, near or on, stream banks, lake shorelines, or other watercourse perimeters where they can be washed away by high water or storm runoff, or can encroach upon the watercourse.

3. Petroleum product storage tanks management:

- a. Place oil or other petroleum product storage tanks at least 20 feet from streams, flowing or dry watercourses, lakes, wetlands, reservoirs, and other water sources.
- b. Do not use underground storage tanks.
- c. Construct storage area dikes at least 12-inches high or graded and sloped to permit safe containment of leaks and spills equal to storage tank capacity located in the area plus sufficient freeboard to contain the 25-year rainfall event.
 - 1) Line diked areas with an impermeable barrier at least 50 mils thick.

4. Areas for refueling operations: Lined with impermeable barrier at least 10 mils thick covered with 2 to 4 inches of soil.

END OF SECTION

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SECTION 01 57 35

OIL SPILL PREVENTION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for items of work that require oil spill prevention measures.
 2. Contractor is responsible for damages resulting from oil spills resulting from work performed under this contract.

1.02 REFERENCE STANDARDS

- A. ASTM International (ASTM)
1. ASTM D2103-10 Polyethylene Film and Sheeting

1.03 REGULATORY REQUIREMENTS

- A. Comply with Federal, State, and local regulations for oil-spill and control regulations.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Polyethylene Sheeting: ASTM D2103.
1. Thickness: 6 mils, minimum.

PART 3 EXECUTION

3.01 ONSITE SPILL PREVENTION

- A. Contractor is responsible for spills or leaks during performance of this contract which occur as a result of, or are contributed to by the actions of Contractor's agents, employees, or subcontractors.
- B. Perform following before draining and transferring oil or removing bushing operations:
1. Place polyethylene sheeting below barrels, hoses, and connections used for draining oils.

2. Overlap, triple-fold, and tape adjoining sheets to trap spilled oil and to not contaminate surfaces.
 3. Inspect hoses and gaskets used for pumping for holes, cracks or deterioration. Inspect fittings for proper connections. Repair or replace damaged hoses, gaskets, and fittings to Government's satisfaction.
 4. Place drip pans under pumps and connections.
 5. Place pumps and hoses so they are not a tripping hazard.
 6. Have available onsite sorbent materials and cleanup equipment. Includes but not limited to the following:
 - a. Shovels.
 - b. Brooms.
 - c. Department of Transportation recovery drums.
 - d. Rags.
 - e. Vermiculite, minimum: 50 pounds.
 - f. Sorbent booms and sorbent pads.
 - g. Solvents.
- C. Monitor pumping operations at all times.
- 3.02 SPILLS OR LEAKS**
- A. Stop source of spill or leak.
 - B. Stop migration of spill or leak.
 - C. Place berm of sorbent material around perimeter of spill.
 - D. Solidify free standing oil.

3.03 REPORTING

- A. Report spills or leaks, regardless of quantity, to Government within 4 hours of their discovery.
- B. Make follow-up written report to Government not later than 7 days after initial report. Include in a narrative form, as a minimum, the following:
 1. Description of item spilled. Include identity, quantity, manifest number.
 2. Whether or not amount spilled is EPA or State reportable. If so, when it was reported.
 3. Exact time and location of spill including a description of the area involved.

4. Containment procedures initiated.
 5. Summary of communications with media or Government officials other than Government.
 6. Description of clean-up procedures employed or to be employed at the site.
- C. Disposal location and methods.

3.04 CLEANUP

- A. Clean up spills or leaks to satisfaction of Government and in a manner that complies with applicable Federal, State, and local laws and regulations.
- B. Pump spills not practical for solidifying into Department of Transportation drums. Label and date drums.
- C. Place contaminated porous solids; such as, but not limited to; soil, asphalt, wood, and paper into Department of Transportation removable head drums. Label and date drums.
- D. Decontaminate non-porous materials using a solvent, such as xylene, toluene, hexane, or kerosene.
1. Place rags and solvents used in decontamination in Department of Transportation removable head drums.
 - a. Label and date drums.
- E. If spill exceeds onsite control and cleanup capabilities, hire outside subcontractor specializing in environmental cleanup to perform cleanup.
- F. For cleanup operations, document extent of spill, effectiveness of cleanup operations, and point in time when the environment has been decontaminated as approved by Government.
- G. Perform cleanup at no additional cost to Government.

3.05 DISPOSAL

- A. Handle and dispose of spilled or leaked materials contaminated with 50 ppm or greater polychlorinated biphenyls in accordance with Section 02 84 40 - Removal and Disposal of Polychlorinated Biphenyls (PCBs).
- B. Handle and dispose of spilled or leaked materials not contaminated or contaminated with less than 50 ppm polychlorinated biphenyls in accordance with applicable Federal, State, and local regulations.

END OF SECTION

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SECTION 01 60 00

PRODUCT REQUIREMENTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. When the Price Schedule includes a separate item for furnishing a material, include cost of furnishing, hauling, storing, and handling in the price offered in the Price Schedule for the item.
 2. When the Price Schedule does not include a separate item for furnishing a material, include cost of furnishing, hauling, storing, and handling in the price offered in the Price Schedule for work for which the material is required.

1.02 DEFINITIONS

- A. Essential Characteristics: As used in these specifications, the term "essential characteristics" is synonymous with the term "salient characteristics."
- B. Salient Characteristics: Those qualities of an item that are essential to ensure that the intended use of the item can be satisfactorily realized.

1.03 REFERENCE STANDARDS

- A. American Society of Mechanical Engineers (ASME)
1. ASME B1.1-2019 Unified Inch Screw Threads, (UN and UNR Thread Form)
 2. ASME B1.20.1-2013(R2018) Pipe Threads, General Purpose, Inch
- B. Bureau of Reclamation (Reclamation)
1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Transport and handle manufactured products in accordance with manufacturer's instructions.
- B. Store and protect manufactured products in accordance with manufacturer's instructions and RSHS. Obtain instructions from the manufacturer before delivery of materials to jobsite. Maintain a copy of instructions at jobsite.

- C. Protect materials from adverse effects of moisture, sunlight, ultraviolet light, or weather during storage at jobsite.
- D. Remove and replace damaged items with new items.
- E. Store curing compounds, sealants, adhesives, paints, coatings, sealers, joint compounds, grouts, and similar products at the temperature and environmental conditions recommended by manufacturer.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Provide materials required for completion of work.
- B. Provide type and quality described in these specifications. Make diligent effort to procure specified materials from all available sources.
- C. Furnish new materials conforming to referenced standards unless otherwise specified.
- D. For materials not covered by these or referenced specifications, furnish materials of standard commercial quality.
- E. If materials to be used deviate from or are not covered by recognized specifications and standards, submit, for approval, justification for and exact nature of the deviation, and complete specifications for materials proposed for use.
- F. Make parts accurately to standard gauge where possible.
 - 1. Use unified screw threads conforming to ASME B1.1 or B1.20.1 for threads, including but not limited to those of bolts, nuts, screws, taps, pipes, and pipefittings.
 - 2. For internal connections only, the Contractor may deviate from ASME standards, provided a complete set of taps and dies are furnished as required to facilitate repair or replacement.
- G. Permanently mark fasteners with a symbol identifying the manufacturer and with symbol(s) indicating grade, class, type, and other identifying marks in accordance with reference or applicable standard.

2.02 SUBSTITUTIONS

- A. If materials required by these specifications become unavailable, because of Government priorities or other causes, substitute materials may be used.
- B. Obtain written approval to use substitute materials from the CO. State in the request for approval the amount of the adjustment, if any, to be made in favor of the Government.

- C. The Government's determination as to whether substitution will be permitted and as to what substitute materials may be used, shall be final and conclusive.
- D. If approved substitute materials are of less value to the Government or involve less cost to the Contractor than specified material, a contract adjustment will be made in favor of the Government. Where the amount involved or the importance of substitution warrants, a deductive modification to the contract will be issued.
- E. No payments in excess of prices offered in the Price Schedule will be made because of substitution of one material for another or because of use of one alternate material in place of another.

2.03 WORKMANSHIP

- A. Accurately manufacture and fabricate materials in accordance with best modern practice and requirements of these specifications, notwithstanding minor errors or omissions therein.
- B. Use liberal factors of safety and adequate shock-absorbing features in designs, especially for parts subjected to variable stress or shock, including alternating or vibrating stress or shock.
- C. Include provisions which prevent components from loosening for shock-absorbing features and parts subject to vibration.

2.04 GOVERNMENT CONTRACT SOURCE QUALITY ASSURANCE

- A. Materials will be subject to inspection in accordance with clause at FAR 52.246-12, Inspection of Construction at following locations, as determined by the CO:
 - 1. At place of production or manufacture.
 - 2. At shipping point.
 - 3. At jobsite.
- B. To allow sufficient time to provide for inspection, submit at time of issuance, copies of purchase orders, including drawings and other pertinent information, covering material on which inspection will be made as advised by the CO, or submit other evidence if such purchase orders are issued verbally or by letter.
- C. Inspection of materials at locations specified above or waiving of inspection shall not be construed as being conclusive as to whether materials and equipment conform to contract requirements under the clause at FAR 52.246-12, Inspection of Construction, nor shall the Contractor be relieved thereby of the responsibility for furnishing materials meeting the requirements of these specifications.
- D. Acceptance of materials will be made only at the jobsite.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install products in accordance with manufacturer's recommendations unless otherwise specified.

3.02 GOVERNMENT CONTRACT FIELD QUALITY ASSURANCE

- A. Final inspection and acceptance of materials will be made only at the jobsite after installation and testing.

END OF SECTION

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SECTION 01 61 10
BUY AMERICAN

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

- A. Buy American Material: In accordance with FAR 52.225-11, Buy American-Construction Materials under Trade Agreements.

1.03 SUBMITTAL

- A. Include applicable certification for material in each RSN listed in Table 01 33 00A – List of Submittals in accordance with FAR 52.225-11, Buy American-Construction Materials under Trade Agreements.
- B. Submit the following in accordance with Section 01 33 00 - Submittals.
- C. RSN 01 61 10-1, Buy American Certificates:
1. Prior to project closeout provide applicable Buy American Certificate for materials used on the project.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 CERTIFICATION

- A. Deliver applicate certificate in accordance with 52.225-11 Buy American-Construction Materials under Trade Agreements to COR with material delivered to the site.
- B. Reclamation will not accept material without proper certification.

END OF SECTION

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SECTION 01 74 00 CLEANING AND WASTE MANAGEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work, except as specified.
 2. Cost of environmental site assessments are Contractor's responsibility.

1.02 DEFINITIONS

- A. Hazardous Waste: Defined as hazardous by 40 CFR 261.3; or by other Federal, State, or local laws or regulations.

1.03 REFERENCE STANDARDS

- A. Bureau of Reclamation (Reclamation)
1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>
- B. Code of Federal Regulations (CFR)
1. 40 CFR 261.3 Definition of Hazardous Waste
 2. 49 CFR 171-179 Transportation – Hazardous Waste Regulations

1.04 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 74 00-1, Waste Production and Disposal Plan:
1. For each type of waste, list estimated quantity and planned disposal location.
- C. RSN 01 74 00-2, Waste Production and Disposal Records:
1. For each type of waste, list quantity and disposal location.
 2. Include certifications that waste was properly disposed.
 3. Hazardous wastes manifests.

4. Bills of Lading and recycling records for recycled waste containing hazardous materials.
- D. RSN 01 74 00-3, Environmental Consultant Resume:
1. Describe experience on similar project.
- E. RSN 01 74 00-4, Environmental Site Assessment.

1.05 QUALIFICATIONS

- A. Environmental Consultant: Minimum 2 years' experience in conducting environmental site assessments for similar construction.

1.06 REGULATORY REQUIREMENTS

- A. Comply with Federal, State, and local laws and regulations.
- B. Comply with RSHS.
- C. Conform to most stringent requirement in cases of conflict between specifications, regulatory requirements, and RSHS.

1.07 PROJECT CONDITIONS

- A. Report waste materials discovered at jobsite to COR.
1. Cease work in areas where waste may be hazardous until waste materials are investigated by the Government.
 2. If waste is hazardous, the CO may order delays in time of performance or changes in work, or both.
 3. If such delays or changes are ordered, an equitable adjustment will be made in the contract in accordance with applicable clauses of the contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROGRESS CLEANING

- A. Keep work and storage areas free from accumulations of waste materials and rubbish.

3.02 FINAL CLEANUP

- A. Remove temporary plant facilities, buildings, concrete footings and slabs, rubbish, unused materials, and other similar materials which are not part of permanent work.
- B. Leave premises "broom clean."

3.03 DISPOSAL

- A. Nonhazardous Waste Materials:
 - 1. Disposal by Removal:
 - a. Reuse or recycle waste materials whenever possible.
 - b. Dispose of nonhazardous waste materials that are not reused or recycled at appropriately permitted disposal facilities.
 - 2. Do not burn waste materials.
 - 3. Do not bury waste materials.
- B. Hazardous Waste Disposal:
 - 1. Dispose by removal from jobsite.
 - 2. Recycle hazardous waste whenever possible.
 - 3. Dispose of hazardous waste materials that are not recycled at appropriately permitted treatment or disposal facilities.
 - 4. Transport hazardous waste in accordance with 49 CFR 171-179.
- C. Certification: Certify that wastes are disposed of in accordance with Federal, State, and local regulations.

3.04 SITE ASSESSMENT

- A. Upon completion of work, perform site assessment at following areas for work done under these specifications:
 - 1. Hazardous waste accumulation areas.
 - 2. Petroleum dispensing and storage areas where aggregate storage of petroleum at jobsite was over 110 gallons.
 - 3. Hazardous material storage areas.
- B. Employ qualified environmental consultant to perform assessments.
- C. Demonstrate and document by appropriate analytical sampling that site contamination is less than State action cleanup levels.

3.05 RECORDS

- A. Keep records of types and amounts of waste materials produced.
- B. Keep records of waste material disposal.

END OF SECTION

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SECTION 01 78 30
PROJECT RECORD DOCUMENTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Spare Parts:
 - 1. Payment: Lump sum price offered in the Price Schedule.
- B. Cost:
 - 1. Include cost of final drawings in prices offered in the Price Schedule for other items of work.

1.02 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals.
- B. RSN 01 78 30-1, Final Drawings:
 - 1. Final Drawings:
 - a. Revised to reflect approval comments and as-built condition of installed equipment at time of contract completion.
 - b. Provide original plot of all submitted drawings.
 - c. Provided in AutoCAD drawing (.dwg) format and Adobe Acrobat (.pdf).
 - 1) Drawings or output data files produced on a Computer Aided Design and Drafting (CADD) system and furnished to Government shall be compatible with 2020 Autodesk products, .dwg file format, manufactured by Autodesk, Inc., Sausalito, CA.
 - d. ASCII point files shall be included, with a point descriptor glossary.
 - 2. For Information Only (FIO) Drawings: Revision date and details.

1.03 RECORD DRAWINGS

- A. Maintain 2 sets of full-size prints of contract drawings marked to show accurate and complete records of as-built conditions. Keep drawings at the jobsite and mark as work progresses.
 - 1. Mark and dimension to show variations between actual construction and that indicated or specified in contract documents.
 - a. Include buried or concealed construction and utilities.

- b. Include existing items and utility lines revealed during construction which differ from those shown on contract drawings.
- 2. Mark to define construction actually provided where choice of materials or methods is permitted in specifications, or where variations in scope or character of work from that of the original contract are authorized.
- 3. If FIO drawings require changes, mark with the appropriate revisions.
- B. Use standard drafting practice to represent changes and include supplementary notes, legends, and details necessary to clearly portray as-built construction.
- C. Mark drawings in the following colors:
 - 1. Red: Additions.
 - 2. Green: Deletions.
 - 3. Blue: Notations necessary for explanation of markings.
- D. Allow Government to review drawings at all times.
- E. Upon completion of work, sign drawings.
 - 1. Only certify FIO drawings which require changes.

1.04 SPARE PARTS

- A. Reference individual Specification Section articles and paragraphs for required spare parts.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 79 20

TRAINING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Training:
1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

- A. Bureau of Reclamation (Reclamation)
1. RSHS Reclamation Safety and Health Standards, including revisions, available at <https://www.usbr.gov/safety/rshs/index.html>

1.03 QUALIFICATIONS

- A. Training Staff:
1. Competent with and knowledge of the specific equipment that training is provided for, and the material presented:
 - a. Minimum 5 years' experience on equipment of similar type and function as the equipment.
 - b. Minimum 2 years' experience on the specific type and models of equipment.
 2. Fluent in English.
 3. Instructor qualifications.

1.04 COORDINATION

- A. Coordinate instruction schedule and location with Government.
- B. Training Schedule:
1. Tuesdays through Thursdays, 9:00 am to 2:00 pm.
 2. Exclude Mondays, Fridays, Saturdays, Sundays, and Federal holidays from potential training days.

1.05 TRAINING REQUIREMENTS

- A. General:

1. Training provided on a minimum of the following topics, content shall be facility and equipment specific:
 - a. Overview of how components interact and function in overall operation of the facility.
 - b. Safety and protection.
 - c. How to use and operate components, including control and monitoring fundamentals and theory.
 - d. Understanding equipment feedback.
 - e. Navigation of equipment user interface.
 - f. Operation and maintenance.
 - g. Purpose regarding the Automation Control System.
 - 1) Instruct operators in the basic knowledge of the Automation Control and Monitoring System.
 - 2) Demonstrate operational sequences.
 - 3) Discuss and demonstrate the required calibrations, adjustments, operations, possible program changes and maintenance required for the PLCs, HMIs, Distributed I/O assemblies, and the Data Historian Server.
 - 4) Discuss and demonstrate troubleshooting/repairs on the Automation Control and Monitoring System.
 - 5) Discuss and demonstrate the maintenance and troubleshooting of fault codes, theory of operation and adjustment of control system parameters.
 - a) Alarms and events.
 - b) How alarms are generated.
 - c) Emergency response.
 - h. Testing and troubleshooting.
 - i. Hardware.
 - j. Software.
 - 1) PLC
 - 2) Instrumentation Devices
 - 3) Control Devices
 - k. Network.
 - l. Programming and altering system.
 - m. Final Drawings.

- n. Operation and maintenance manual ready for training course.
 2. Training Courses Quantity: 1
 3. Training Course Duration: 8 hours minimum per course unless specified otherwise.
 4. Date: Government will provide at least 14 days advance notice of preferred training dates.
 5. Total attendees at each training course: 10 maximum unless otherwise noted.
- B. Goal:
1. Each attendee can perform intended function on completed system.
 2. Operators can interpret feedback from system to operate and run facility.
 3. Engineers, electricians, and technicians can modify and maintain system.
 4. Training shall cover items contained in operating and maintenance instructions.
- C. Training materials in pdf and original file format. Government will retain Contractor-provided training materials. Government reserves the right to reproduce and modify Contractor-provided training materials.
- D. Recording of Training Session:
1. Provide audio and color video recording of training (.mp4 format).
 2. Provide necessary equipment and staffing to record classroom portions of training.
 3. Ensure screen presentations, white board diagrams, and flip chart paper diagrams developed during class are visible and legible on video recording.
 4. Ensure audio for both instructor and class participants are understandable.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 TRAINING

- A. Approved Training Materials:
1. Class Presentation:
 - a. Agenda
 - b. Presentation and associated notes.

- c. Generic and customized handouts.
 - d. Personal experience examples.
 - e. In progress as-built drawings.
 - f. Operation and maintenance manual for specific equipment.
 - g. Photos with descriptions of onsite installed equipment.
- B. Where manufacturer instructions are insufficient, develop and provide procedures for system operation.
- C. Provide one copy of bound, hard-copy training materials for each attendee for each session.
- D. Government will furnish facilities for training and will retain training materials for future reference. Coordinate the final site with the COR.
- E. Onsite for hands-on training: Use actual installed equipment for training demonstration.

END OF SECTION

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SECTION 02 80 10

GENERAL REQUIREMENTS FOR HAZARDOUS MATERIALS ABATEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in price in the Price Schedule for Demolition and Removal of Electrical Equipment.

1.02 DEFINITIONS

- A. Asbestos Air Monitoring Specialist: See 5 CCR 1001-10, Part B.
- B. Asbestos Competent Person: See 29 CFR Part 1926.1101.
- C. CDPHE: Colorado Department of Public Health and Environment.
- D. NIOSH: National Institute for Occupational Safety and Health.
- E. NMAM: NIOSH Manual of Analytical Methods.
- F. OSHA: Occupational Safety and Health Administration.
- G. PCM: Phase Contrast Microscopy.

1.03 REFERENCE STANDARDS

- A. Code of Colorado Regulations (CCR)
1. 5 CCR 1001-10, Part B Regulation Number 8, Control of Hazardous Air Pollutants, Asbestos Control
- B. Code of Federal Regulations (CFR)
1. 29 CFR Part 1926.103 Respiratory Protection
 2. 29 CFR Part 1926.1101 Asbestos
- C. National Institute of Occupational Safety and Health (NIOSH)
1. NIOSH (NMAM) 7400-19 Asbestos and Other Fibers by PCM

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 02 80 10-1, Resumes and Qualifications:**1. Independent Testing Laboratory:**

- a. See 1.05A.2.
- b. Include name, contact person, address, and telephone number.
- c. Note that qualifications for the Air Monitoring Specialist are in RSN 02 82 20-1.

C. RSN 02 80 10-2, Worker Background Exposure Test Results.

1. See 1.07A.2.
2. Do not submit RSNs 2 through 4 prior to RSN 1 being accepted.

D. RSN 02 80 10-3, Working Level Test Results.

1. See 1.07A.3.

E. RSN 02 80 10-4, Final Clearance.

1. See 1.07A.4.

1.05 QUALITY**A. Qualifications:**

1. Asbestos Monitoring Specialist: See RSN 02 82 20-1.
2. Independent Testing Laboratory:
 - a. Certified in analyzing air samples using NIOSH (NMAM) Method 7400. Certified in accordance with 5 CCR 1001-10, Part B:
 - 1) Registered in the state of Colorado as an accredited laboratory for asbestos air sample analysis (list available here: <https://cdphe.colorado.gov/indoor-air-quality/asbestos/certified>).

1.06 SITE CONDITIONS**A. Existing Conditions:**

1. Materials contain asbestos: See Section 02 82 20 - Removal and Disposal of Asbestos Containing Materials.

1.07 ADMINISTRATIVE REQUIREMENTS**A. Asbestos Monitoring Specialist:**

1. General:
 - a. Performs as the Asbestos Competent Person.

- b. Independent from Asbestos Abatement Contractor according to 5 CCR 1001-10, Part B.
 - c. Be onsite for duration of asbestos abatement activities.
 - d. Ensure workers use proper hygiene methods and decontamination procedures.
 - e. Perform or direct air monitoring, visual inspection, and final air clearance testing, review and provide results.
 - f. Available for emergencies.
- 2. Background Test Results:
 - a. Air monitoring sample analysis by NIOSH 7400.
 - b. Map indicating location of each sample.
- 3. Working Level Test Results:
 - a. Air monitoring by NIOSH 7400. Provide analytical results.
 - b. Air Monitoring Specialist verification that Colorado asbestos air monitoring levels were not exceeded.
 - c. Map indicating location of each sample.
- 4. Perform or oversee final clearance.
 - a. Visual inspection.
 - b. Air sample analysis by NIOSH 7400.
 - c. Compare with pre-construction test results for increased contaminant levels. Include:
 - 1) Air Monitoring Specialist verification that Colorado asbestos air monitoring levels were not exceeded.
 - 2) Acceptance for wipe sample analytical comparison between pre- and post-construction.

1.08 SAFETY AND HEALTH

A. For items not covered in this section, refer to:

- 1. Section 01 35 20 - Safety and Health.
- 2. Section 01 35 30 - Contractor's Onsite Safety and Health Personnel.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 ASBESTOS

- A. Follow accepted Worker Protection Plan.

END OF SECTION

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SECTION 02 82 20
REMOVAL AND DISPOSAL OF ASBESTOS CONTAINING MATERIALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in the price offered in the Price Schedule for Demolition and Removal of Electrical Equipment.

1.02 DEFINITIONS

- A. ACM: Asbestos Containing Material. See 5 CCR 1001-10, Part B.
- B. CDPHE: Colorado Department of Public Health and Environment.
- C. Handling: Includes containment, collection, and storage.
- D. PLM: Polarized Light Microscopy.

1.03 REFERENCE STANDARDS

- A. Code of Colorado Regulations (CCR)
1. 5 CCR 1001-10, Part B Regulation Number 8, Control of Hazardous Air Pollutants, Asbestos Control
 2. 6 CCR 1007-2, Part 1, Section 5 Solid Waste Disposal Site and Facilities, Asbestos Waste Management
- B. Code of Federal Regulations (CFR)
1. 49 CFR Sub B, Ch I, Sub C Hazardous Materials Regulations

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 02 82 20-1, Qualifications and Resumes:
1. Certificates, see 1.05.
 - a. Air Monitoring Specialist.
 - b. Asbestos Abatement Project Designer.
 - c. Asbestos Abatement Supervisor.
 - d. Asbestos Abatement Worker.

- e. Asbestos Building Inspector.
 - f. General Abatement Contractor.
 - g. Independent Testing Laboratory.
 - 2. Qualifications, see 1.05:
 - a. Independent Testing Laboratory including name, address, and telephone number.
 - b. Solid Waste Disposal Facility including name, address, and telephone number.
 - 3. Resumes:
 - a. Asbestos Abatement Project Designer.
 - b. Asbestos Abatement Supervisor.
 - c. Asbestos Building Inspector.
- C. RSN 02 82 20-2, General Abatement Contractors History:
 - 1. Do not submit this or any submittals beyond this until RSN 02 82 20-1 has been accepted.
 - 2. Signed statement containing:
 - a. Record of citations issued by regulatory agencies relating to asbestos activities. Include project names, dates, and resolutions.
 - b. List of penalties incurred through non-compliance with asbestos regulations.
 - c. Termination of asbestos-related contract. Include project names, dates, and reasons for terminations.
 - 3. If there are none, provide negative declaration.
- D. RSN 02 82 20-3, Asbestos Survey:
 - 1. See 1.08C.1.
- E. RSN 02 82 20-4, Asbestos Abatement Plan:
 - 1. Do not submit RSN 02 82 20-4 until RSN 02 82 20-3 is accepted.
 - 2. See 1.08D.1.
- F. RSN 02 82 20-5, Notice of Asbestos Abatement:
 - 1. See 1.08B.1.
- G. RSN 02 82 20-6, Bills of Lading and Certificate of Disposal:
 - 1. Documents that ACMs are shipped, received, and properly disposed of.

2. In accordance with 1.08D.1.e.

1.05 QUALIFICATIONS

A. Asbestos Abatement Project Designer:

1. Minimum 1 year experience as an Asbestos Abatement Project Designer in Colorado.
2. Current Colorado Asbestos Abatement Project Designer certification.

B. Asbestos Abatement Supervisor:

1. Minimum 1 year experience as Asbestos Supervisor in State where abatement is occurring.
2. Current State of Colorado Asbestos Abatement Supervisor certification.

C. Asbestos Abatement Worker:

1. Minimum 1 year experience as Asbestos Abatement Worker.
2. Current State of Colorado Asbestos Abatement Worker certification.

D. Asbestos Building Inspector:

1. Minimum 1 year experience as an Asbestos Building Inspector in Colorado.
2. Current State of Colorado Asbestos Building Inspector certification.

E. General Abatement Contractor:

1. Minimum 1 year experience in similar type of abatement work in Colorado.
2. Current State of Colorado General Abatement Contractor certificate.

F. Asbestos Air Monitoring Specialist: Current State of Colorado Asbestos Air Monitoring Specialist certificate.

G. Solid Waste Disposal Facility: Certified to dispose of ACM in accordance with 6 CCR 1007-2, Part 1, Section 5.

H. Independent Testing Laboratory: Current CDPHE registration for bulk asbestos analysis by PLM.

I. Transporter: Licensed and placarded in accordance with 49 CFR Subtitle B, Chapter I, Subchapter C.

1.06 SAFETY AND HEALTH REGULATIONS

A. Refer to Section 02 80 10 - General Requirements for Hazardous Materials Abatement.

1.07 SITE CONDITIONS

- A. Electrical equipment may contain asbestos. These materials could not be accessed during the Hazardous Materials Survey. See Section 51 02 82 - Hazardous Materials Survey and Analytical Results for tables, figures, and analytical results where materials were tested and were not.

1.08 ADMINISTRATIVE REQUIREMENTS

- A. Asbestos Abatement Supervisor:
1. On jobsite during asbestos abatement activity.
 2. Ensure engineering controls in use are in proper operating condition and are functioning properly.
- B. General Abatement Contractor:
1. Notice of Asbestos Abatement: CDPHE Asbestos Abatement Notification Permit Application Form.
 2. Contain and store of ACM in accordance with 5 CCR 1001-10, Part B.
 3. Transportation and Disposal Tasks, obtain required transportation permits.
- C. Asbestos Building Inspector:
1. Asbestos Survey: Conduct inspection of area not surveyed but noted in Section 51 02 82 - Hazardous Materials Survey and Analytical Results, in accordance with 5 CCR 1001-10, Part B.
- D. Asbestos Project Designer:
1. Develop and sign Asbestos Abatement Plan.
 - a. Containment System, Enclosed Containment:
 - 1) Descriptions, detail project drawings, and site layout.
 - 2) Include containment techniques, local exhaust ventilation system locations, decontamination units, temporary waste storage facility, location of temporary utilities, and boundaries of regulated areas.
 - b. Emissions: Reporting release procedure.
 - c. Clearance testing: equipment, analytical method, and locations of equipment (stationary and personal)
 - d. Map indicating the proposed location of staging equipment, containment, air monitoring, and decontamination.
 - e. Waste Handling and Disposal:
 - 1) Handling, storage, transporting, and disposal of project-generated waste and cleaning of reusable items.

- 2) Packaging and storage in accordance with 5 CCR 1001-10, Part B.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 ASBESTOS SURVEY

- A. In accordance with 1.08C.1.

3.02 ASBESTOS ABATEMENT

- A. In accordance with Approved Asbestos Hazard Abatement Plan and 1.04E.

END OF SECTION

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SECTION 02 83 33
REMOVAL AND DISPOSAL OF EQUIPMENT WITH COATINGS CONTAINING
REGULATED METALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in the price offered in the Price Schedule for Demolition and Removal of Electrical Equipment.

1.02 DEFINITIONS

- A. CDPHE: Colorado Department of Public Health and Environment.
- B. Handling: Includes containment, collection, storage, and transportation.
- C. Regulated Metals: Resource and Recovery Act (RCRA) 8 metals which includes arsenic, barium, cadmium, chromium, lead, mercury, selenium, and silver.

1.03 REFERENCE STANDARDS

- A. Code of Colorado Regulations (CCR)
1. 6 CCR 1007-2, Part 1, Section 8 Recycling and Beneficial Use
- B. Code of Federal Regulations (CFR)
1. 49 CFR Chapter III Federal Motor Carrier Safety Administration, Department of Transportation

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals. Do not submit RSN 02 83 33-2 or RSN 02 83 33-3 until RSN 02 83 33-1 has been accepted.
- B. RSN 02 83 33-1, Qualifications.
1. See 1.05.
- C. RSN 02 83 33-2, Recycled Materials Handling and Disposal Plan.
1. See 1.08A.
- D. RSN 02 83 33-3, Bills of Lading and Recycling Certificates:

1. See 1.08B.

1.05 QUALITY

A. Qualifications.

1. Industrial Recycling Facility:
 - a. Current Colorado business license.
 - b. Registered in Colorado as an Industrial Recycling Facility with CDPHE in accordance with 6 CCR 1007-2, Part 1, Section 8.
2. Transporter: Certified in accordance with 49 CFR Chapter III.

1.06 SAFETY AND HEALTH REGULATIONS

A. In accordance with:

1. Section 01 35 20 - Safety and Health.
2. Section 01 35 30 - Contractor's Onsite Safety Personnel.

1.07 SITE CONDITIONS

- ### **A. Existing coatings contain Hazardous materials. See Section 51 02 82 - Hazardous Materials Survey and Analytical Results.**

1.08 ADMINISTRATIVE REQUIREMENTS

- ### **A. Recycled Materials Handling and Disposal Plan:**
1. Handling and recycling of project-generated recyclable materials in accordance with 6 CCR 1007-2, Part 1, Section 8.
 - a. Quantities and types of waste to be disposed.
 - b. Industrial Recycling Facility name, contact person, telephone number, and address.
- ### **B. Bills of Lading and Recycling Certificates: From Transporter and Industrial Recycling Facility.**

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION**3.01 WASTE HANDLING AND DISPOSAL**

- A. Comply with approved Recycled Materials Handling and Disposal Plan:

END OF SECTION

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SECTION 02 83 37
REMOVAL AND DISPOSAL OF UNIVERSAL WASTE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in the price offered in the Price Schedule for Demolition and Removal of Electrical Equipment.

1.02 DEFINITIONS

- A. Destination Facility: per 6 CCR Part 273.9.
- B. UW: Universal Waste. See 6 CCR Part 273.9. Includes batteries, mercury-containing devices, lamps, and electronic devices and components.

1.03 REFERENCE STANDARDS

- A. Colorado Code of Regulations (CCR)
1. 6 CCR Part 273.9 Standards for Universal Waste Management, Definitions
 2. 6 CCR 1007-3, Part 273 Subp B Standards for Small Quantity Handlers of Universal Waste
 3. 6 CCR 1007-3, Part 273, Subp C Standards for Large Quantity Handlers of Universal Waste
 4. 6 CCR 1007-3, Part 273, Subp D Standards for Universal Waste Transporters
 5. 6 CCR 1007-3, Part 273, Subp E Standards for Destination Facilities
- B. Code of Federal Regulations (CFR)
1. 49 CFR Chap I, Subchapter C Hazardous Materials Regulations

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals. Do not submit RSN 02 83 37-2 through RSN 02 83 37-5 until RSN 02 83 37-1 has been accepted.
- B. RSN 02 83 37-1, Qualifications:
1. For all: Address, contact name, and telephone number.
 2. Destination Facility: See 1.05A.

3. UW Handler: See 1.05B.
 4. UW Transporter: See 1.05C.
- C. RSN 02 83 37-2, UW Handling and Disposal Plan:
1. See 1.08A.
 2. Do not ship materials until this RSN is accepted.
- D. RSN 02 83 37-3, Shipping Papers:
1. See 1.08C.
- E. RSN 02 83 37-4, Disposal Certificates:
1. Copy of shipping papers signed by Destination Facility indicating disposal of UW mercury and recycling of UW electronics.
 2. See 1.08C.

1.05 QUALIFICATIONS

- A. Destination Facility:
1. Meet requirements of 6 CCR Part 273 Subpart E.
 2. EPA Site Identification number.
 3. If more than one facility is needed to recycle or dispose of UW, provide contact information and qualifications for all facilities.
- B. UW Handler:
1. Meet requirements of one or the other:
 - a. 6 CCR 1007-3, Part 273 Subpart B for Small Quantity Handlers.
 - b. 6 CCR 1007-3, Part 273 Subpart C for Large Quantity Handlers.
 - c. Documentation that employees who handle or have responsibility for UW have been trained in accordance with the regulations listed above.
 2. Current EPA Identification number.
- C. UW Transporter:
1. Meet requirements of 6 CCR 1007-3, Part 273.18 Subpart D.
 2. Current EPA Site Identification number.
 3. Registration per 49 CFR Chapter I, Subchapter C.

1.06 SAFETY AND HEALTH REGULATIONS

- A. In accordance with Section 01 35 20 - Safety and Health.

1.07 PROJECT CONDITIONS

- A. UW exists at the site in the form of:
 - 1. Mercury containing equipment.
 - 2. Batteries containing lead.
 - 3. Electronics. Colorado regulates electronics recycling as UW.
 - 4. See Section 51 02 82 - Hazardous Materials Survey and Analytical Results for tables and figures noting the locations of the above materials.
- B. Contractor will be given Facility's EPA Site Identification number after contract award.
- C. Contractor is responsible for handling UW and will be considered a generator in accordance with "Large Quantity Handler of Universal Waste" or "Small Quantity Handler of Universal Waste", 6 CCR Part 273.9.

1.08 ADMINISTRATIVE REQUIREMENTS

- A. UW Handling and Disposal Plan:
 - 1. Include information from RSN 02 83 37-1.
 - 2. Determine whether Contractor is Large or Small Quantity Handler in accordance with 6 CCR Part 273 Subpart B or Subpart C. Assume facility is a Small Quantity Handler.
 - 3. Determine packaging and marking requirements plus temporary storage requirements and location. Storage location shall be approved via this plan.
 - 4. Obtain notice from Destination Facility owner or operator, that Destination Facility:
 - a. Has required disposal permits.
 - b. Will accept UW to be shipped under this contract.
 - 5. Mercury spill/release:
 - a. Spill response plan and notification procedures.
 - b. Note that once a spill occurs, it shall be managed as hazardous waste until proven to be otherwise in accordance with 6 CCR 1007-3, Part 273.17, subpart B (Part 273.17) or 6 CCR 1007-3, Part 273, subpart C (Part 273.37), as applicable.
- B. Prepare shipment papers, in accordance with:
 - 1. 49 CFR Chapter I, Subchapter C.
 - 2. 6 CCR 1007-3, Part 273, Subpart B or Subpart C.
- C. Shipping papers and permits, signed during transportation and disposal:

1. Prior to offsite transportation, shipping papers signed by UW Transporter.
2. Upon UW arrival at Destination Facility, shipping papers signed by Destination Facility.
3. Shipping papers signed by Destination Facility, indicating disposal or recycling.
4. Transportation permits.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 UW HANDLING AND DISPOSAL

- A. Comply with approved UW Handling and Disposal plan.

END OF SECTION

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SECTION 02 87 20
REMOVAL AND DISPOSAL OF USED OIL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in the price offered in the Price Schedule for Demolition and Removal of Electrical Equipment.

1.02 DEFINITIONS

- A. DOT: U.S. Department of Transportation.
- B. Handling: Includes containment, collection, and storage.
- C. Hazardous Waste: As defined in 6 CCR 1007-3 Part 260.
- D. NELAP: National Environmental Laboratory Accreditation Program.
- E. NVLAP: National Voluntary Laboratory Accreditation Program.
- F. RCRA: Resource Conservation and Recovery Act.
- G. TSCA: Toxic Substances Control Act.
- H. TSDF: Treatment, Storage, and Disposal Facility.
- I. Used Oil: As defined in 6 CCR 1007-3 Part 279.

1.03 REFERENCE STANDARDS

- A. American Society for Testing and Materials (ASTM)
1. ASTM D92-24 Flash and Fire Points by Cleveland Open Cup Tester
 2. ASTM D93-25 Flash Point by Pensky-Martens Closed Cup Tester
 3. ASTM D923-15(2023) Sampling Electrical Insulating Fluids
- B. Colorado Code of Regulations (CCR)
1. 6 CCR 1007-3 Part 260 Hazardous Waste Management System: General

2. 6 CCR 1007-3 Part 261 Hazardous Waste - Identification and Listing of Hazardous Waste
 3. 6 CCR 1007-3 Part 262 Hazardous Waste - Standards Applicable to Generators of Hazardous Waste
 4. 6 CCR 1007-3 Part 263 Hazardous Waste - Standards Applicable to Transporters of Hazardous Waste
 5. 6 CCR 1007-3 Part 264 Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
 6. 6 CCR 1007-3 Part 279 Standards for the Management of Used Oil
- C. Code of Federal Regulations (CFR)
1. 40 CFR Part 761 Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions, Toxic Substances Control Act (TSCA)
 2. 49 CFR Chapter I, Subchapter C Hazardous Materials Regulations
- D. Environmental Protection Agency (EPA) Hazardous Waste Test Methods (SW-846)
1. Method 1010 Test Methods for Flash Point by Pensky-Martens Closed Cup Tester
 2. Method 6010B, C, or D Inductively Coupled Plasma-Atomic Emission Spectrometry
 3. Method 8021 Aromatic and Halogenated Volatiles by Gas Chromatography Using Photoionization and/or Electrolytic Conductivity Detectors
 4. Method 8260B, C, or D Volatile Organic Compounds by Gas Chromatography/Mass Spectrometry (GC/MS)
 5. Method 9076 Test Method for Total Chlorine in New and Used Petroleum Products by Oxidative Combustion and Microcoulometry
- E. National Environment Methods Index (NEMI)
1. NEMI 4500-Cl Amperometric Titration Method

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 02 87 20-1, Qualifications:

1. Qualifications: See 1.05.
 2. Provide name, contact person, address, and telephone number.
 3. Do not submit RSN 02 87 20-2 or others until this RSN has been accepted.
- C. RSN 02 87 20-2, Oil Sampling, Handling and Disposal Plan:
1. See 1.08B and include information from the accepted RSN 02 87 20-1.
 2. Do not collect oil for analysis until this RSN has been accepted.
- D. RSN 02 87 20-3, Oil Characterization Report:
1. Copy of laboratory analytical results including the chain of custody form.
- E. RSN 02 87 20-4, Manifest for Reclamation Signature:
1. See 1.08E.
 2. Only required if approved RSN 02 87 20-3 indicates that the oil is a hazardous waste.
- F. RSN 02 87 20-5, Manifests, Recycling Report, Bills of Lading, Disposal Certificates:
1. Final manifest and Certificate of Disposal: See 1.08E.2 and 1.08E.3.
 2. Bills of Lading and Recycling: see 1.08F and 1.08G.

1.05 QUALITY

- A. Independent Testing Laboratory:
1. Other methods may be substituted with prior approval. Submit under RSN 02 87 20-1.
 2. Accredited under:
 - a. NELAP or NVLAP for EPA and ASTM analytical methods.
 - b. NEMI for NEMI methods.
 3. Flammability:
 - a. ASTM D92 or
 - b. ASTM D93 (EPA Method 1010A or B).
 4. Metals in used oil (arsenic, cadmium, chromium, lead): EPA Method 6010 (B, C, or D).
 5. Halogens:
 - a. Volatile Organics:
 - 1) EPA Method 8021A or B or
 - 2) EPA Method 8260 (B, C, or D).

- b. Chlorine:
 - 1) EPA Method 9076 or
 - 2) NEMI 4500-Cl.

B. Hazardous Waste Transporter:

- 1. Approved in accordance with 6 CCR 1007-3 Part 263.
- 2. Registration per 49 CFR Chapter I, Subchapter C.
- 3. Current EPA Identification Number.
- 4. Permit from Colorado Public Utilities Commission to transport Hazardous Materials.

C. TSDF for Hazardous Waste:

- 1. Meets requirements of 6 CCR 1007-3 Part 264.
- 2. Current EPA Identification Number.

D. Used Oil Processing Facility: Current EPA Identification Number.

E. Used Oil Transporter:

- 1. Meet requirements of 49 CFR Chapter I, Subchapter C.
- 2. Current EPA Identification Number.

1.06 SITE CONDITIONS

- A. Station service transformer oil does not contain PCBs above the regulatory level per 40 CFR Part 761 as the transformers have the federal "No PCBs" sticker on them. See Section 51 02 82 - Hazardous Materials Survey and Analytical Results.
- B. Facility owner EPA Identification number will be provided to Contractor after award.
- C. If oil is found to be a hazardous waste, the Contractor is a co-generator of Hazardous Waste in accordance with 6 CCR 1007-3 Part 262.

1.07 SAFETY AND HEALTH REGULATIONS

- A. In accordance with Section 01 35 20 - Safety and Health.

1.08 ADMINISTRATIVE REQUIREMENTS

- A. Contractor is responsible for characterizing, transporting, and disposing of Hazardous Waste and / or Used Oil.
 - 1. Obtain EPA Identification Number.
- B. Provide Oil Sampling, Handling, and Disposal Plan.

1. Sampling plan for Oil Characterization Report: See 1.08C.1.
 2. Handling and recycling or disposing of Used Oil or Hazardous Waste.
 - a. Include with items below:
 - 1) Name, contact person, telephone number, and address.
 - 2) EPA Identification number and permit.
 - b. Disposing of Oil as Hazardous Waste:
 - 1) TSDF: Written agreement to accept waste generated under this contract.
 - 2) Hazardous Waste Transporter.
 - c. Recycling Used Oil:
 - 1) Used Oil Processing Facility.
 - 2) Used Oil Transporter.
- C. Provide Oil Characterization Report.
1. Collect transformer oil samples according to:
 - a. Sampling: ASTM D923. Collect a separate sample from each transformer.
 - b. Collection containers, sample stabilization, shipping requirements, and analytical methods: See 1.05A.3 through 1.05A.5.
 2. Determine whether Oil is a:
 - a. Hazardous Waste in accordance with 6 CCR 1007-3 Part 262 or
 - b. Used Oil in accordance with 6 CCR 1007-3 Part 279.
- D. Prior to Transporting:
1. Hazardous Waste:
 - a. Before transporting, obtain notice from TSDF owner or operator, that facility:
 - 1) Has required permits.
 - 2) Will accept waste to be shipped under this contract.
 2. Oil, Non-hazardous:
 - a. Before transporting Oil, obtain notice from Used Oil Processing Facility owner or operator, that facility:
 - 1) Has required permits.
 - 2) Will accept Used Oil shipped under this contract.
- E. Manifests:

1. Prepare manifests before transporting Hazardous Waste for offsite treatment, storage, or disposal in accordance with:
 - a. Hazardous Waste: 6 CCR 1007-3 Part 262.
 - b. 49 CFR Chapter I, Subchapter C.
 2. Obtain signed manifests and disposal documentation:
 - a. At time of collection, signed by Generators and Transporter.
 - b. At time of delivery to TSDF, signed by TSDF.
 - c. At time of disposal, signed by TSDF.
 - d. Certificate of Disposal or Notice of Acceptance for Hazardous Waste from TSDF.
 3. Acceptance of Hazardous Waste at the approved TSDF does not meet the definition of final treatment nor final disposal under the contract.
 - a. Obtain documentation to prove final treatment or final disposal of has been accomplished.
 - b. Attach documentation to the Certificate of Disposal or Notice of Acceptance.
- F. Bills of Lading: Prepare Bill of Lading, in accordance with 6 CCR 1007-3 Part 279, before transporting Used Oil for recovery.
- G. Recycling Reports:
1. Provide information on the total amount of recycled oil during the preceding 3 month period.
 2. Break down date and quantity.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 WASTE CHARACTERIZATION, HANDLING, AND DISPOSAL

- A. Do not combine different oils or untested oils of same type.
- B. Manage Oil as Hazardous Waste until acceptance of Oil Characterization Report indicates that the oil is a Used Oil and not a hazardous waste.
- C. Comply with approved Oil Handling and Disposal Plan.

- D. Storing Hazardous Waste:
1. Do not mix different types of Hazardous Waste.
 2. Seal, label, and store in accordance with:
 - a. 6 CCR 1007-3 Part 262.
 - b. 49 CFR Chapter I, Subchapter C.
 - c. Use DOT approved containers for liquids.
- E. Storing Used Oil: Store in DOT approved containers for liquids, separate from Hazardous Waste areas.
- F. Transport Hazardous Waste and Used Oil in accordance with approved Oil Handling and Disposal Plan.
- G. Other materials, non-hazardous: Comply with Section 01 74 00 - Cleaning and Waste Management.
- H. Oil spill prevention, refer to Section 01 57 35 – Oil Spill Prevention.

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SECTION 03 62 20
NONSHRINK GROUT FOR EQUIPMENT AND METALWORK

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for items of work requiring nonshrink grout.

1.02 REFERENCE STANDARDS

- A. ASTM International (ASTM)
1. ASTM C150/C150M-19a Portland Cement
 2. ASTM C1107/C1107M-17 Packaged Dry, Hydraulic-Cement Grout (Nonshrink)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 03 62 20-1, Packaged Nonshrink Grout Data:
1. Manufacturer's product data.
 2. Manufacturer's certification that product meets specifications.
 3. Manufacturer's instructions:
 - a. Surface preparation, mixing, installation, finishing, curing, and protection recommendations.
 - 1) Include specific recommendations based on anticipated jobsite temperatures and environmental conditions.
 - 2) Include manufacturer's recommended methods for float finishing grout.
 - 3) Include manufacturer's recommended minimum temperature and protection period for protecting grout from freezing and damage after placement.

PART 2 PRODUCTS

2.01 PACKAGED NONSHRINK GROUT

- A. Pre-mixed commercial grout mixture:
 - 1. ASTM C1107.
- B. Water:
 - 1. Clean.
 - 2. Free of detrimental quantities of silt, organic matters, salts, or other impurities.
 - 3. As recommended by grout manufacturer.
- C. Mix: In accordance with manufacturer's instructions to consistency recommended by manufacturer for application.

PART 3 EXECUTION

3.01 PREPARATION

- A. Clean and prepare base concrete surface to which nonshrink grout will be bonded:
 - 1. At locations where existing concrete is removed in accordance with Section 03 81 10 – Concrete Removal
 - 2. All other locations in accordance with manufacturer's instructions or as directed by COR:
 - a. Roughen surface.
 - b. Remove dirt, laitance, loose or defective concrete, curing compound, coatings, and other foreign material.
 - c. Wash with water.
- B. Lubricate concrete surface by washing with water immediately before placing nonshrink grout when delay occurs between washing and placing nonshrink grout.

3.02 PLACING

- A. Place nonshrink grout in accordance with manufacturer's instructions to completely fill space to be grouted.
- B. Use forms, where required, to confine nonshrink grout.

3.03 CURING

- A. Cure exposed surface of packaged nonshrink grout by method recommended by manufacturer.

- B. Do not apply loads sooner than 72 hours after placement and only after nonshrink grout has attained compressive strength of at least 3,000 lb/in².

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SECTION 03 81 10

CONCRETE REMOVAL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost of concrete removal in the prices offered in the Price Schedule for items of work requiring concrete removal and disposal.

1.02 REFERENCE STANDARDS

- A. American Concrete Institute
1. ACI 318-19 Code Requirements for Structural Concrete and Commentary
- B. Bureau of Reclamation (USBR)
1. USBR M-47 Guide to Concrete Repair, 2015
- C. International Concrete Repair Institute (ICRI)
1. ICRI 310.2R-2013 Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, Polymer Overlays and Concrete Repair

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 03 81 10-1, Concrete Removal Plan:
1. Methods and equipment used to locate embedded metalwork and reinforcement within concrete to be removed.
 2. Methods, equipment, and sequence used for cutting and removing concrete.
 3. Describe cleanup operations, equipment, and equipment locations.
 4. Plans to prevent damage to remaining concrete.
 5. Temporary barriers and enclosures to protect building occupants and contain dust during work.

1.04 EXISTING CONDITIONS

- A. Existing concrete consists of cement, sand, and broken rock or gravel.

- B. Information drawings provide general locations of embedded reinforcement, pipe, conduit, and metalwork. Actual locations may vary.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

- A. Cease operations and notify COR if concrete removal operations:
1. Contact electrical conduit containing energized circuits.
 2. Crack concrete to remain in place.
- B. Do not resume operations until directed by COR.
- C. Coordinate performance of noisy, malodorous, or dusty work with the COR.
- D. Control waste water in accordance with:
1. Approved plan.
 2. Section 01 57 30 - Water Pollution Control.

3.02 EXAMINATION

- A. Locate and mark embedded reinforcement, pipe, conduit and metalwork within concrete removal areas before beginning concrete removal.

3.03 PREPARATION

- A. Provide temporary barriers and enclosures to protect building occupants and equipment.
- B. Location of concrete removal on drawings is approximate. Verify and adjust concrete removal areas with COR in the field.
- C. Prior to starting removal work, obtain written permission from COR.

3.04 REMOVAL

- A. Initially remove damaged, deteriorated, loosened, or unbonded portions of existing concrete by approved method. Follow with approved mechanical means if necessary.
- B. Produce surfaces and edges suitable for required construction.
- C. Use methods which will not damage concrete or reinforcement to remain in place.

- D. Do not use jackhammers in excess of 30 pounds or, dry sandblasting, without COR approval.
1. Points on jackhammer bits shall cleanly break the concrete.
 2. Replace or sharpen jackhammer bits without points.
- E. Do not pry on concrete to be removed or existing reinforcement.
- F. Blasting not permitted.
- G. Do not use bush hammers or scabblers.
- H. Saw cutting:
1. Fully encompass removal area.
 2. Produce clean, sharp edges.
 3. Saw cut depth: Shown on drawings or as necessary to remove curb to level of existing deck elevation.
 - a. Do not damage reinforcing steel.
 - b. If embedment or other conditions prevent a minimum 1-inch saw cut depth, use a grinder to create a minimum 1/4-inch perpendicular face.
 4. Limit number of corners.
 5. Use diamond blade or diamond wire saws.
 6. Make cuts normal to exposed concrete surface except as indicated on drawings.
 7. Prevent kerfs from crossing.
 8. Do not extend saw kerfs beyond specified limits of removal.
- I. Reinforcement:
1. Remove concrete shadows around circumferences of reinforcing bars.
 2. When corroded reinforcement steel bars are exposed, remove additional concrete along corroded bars until a continuous length of 2-in of bar free from corrosion is exposed.
 - a. Assess limit of active corrosion on a visual basis.
 - b. Edges of additional areas removed shall be cut square as specified below.
 - c. Remove additional concrete along bar to accommodate couplers or lap splices for replacement reinforcement.
 - d. Obtain approval from COR prior to additional concrete removal.
 3. When removing concrete in and around reinforcing steel to remain in place, use 15-pound or smaller chipping hammers, high or ultra-high pressure water jetting.

- a. For removal away from saw cut edges, cleanly break concrete with pointed jackhammer bits. Replace or sharpen jackhammer bits that are not pointed.
- b. Complete removal near repair boundaries with hammers fitted with spade bits.

J. Cleaning and Surface Preparation:

1. Primary Cleaning - Remove dust and debris resulting from concrete removal.
 - a. Use contained shotblasting, wet sandblasting, or water blasting to remove weakened or damaged concrete microfractured surfaces.
 - b. If water blasting, use pressure sufficiently high to prepare surface. Pressures up to 15,000 psi may be necessary.
 - c. Remove materials that weaken the bond between remaining concrete and repair material.
 - d. Include saw cut or ground faces.
 - e. After primary cleaning, protect surface from contaminants.
2. Concrete substrate surface roughness:
 - a. Surfaces to receive nonshrink grout: Equivalent to CSP 5 or higher in accordance with ICRI 310.2R, or as stated by nonshrink grout manufacturer's instructions, whichever is higher.
 - b. Surfaces to remain exposed concrete: Equivalent to CSP 3 or lower in accordance with ICRI 310.2R.
3. Clean and allow surfaces to dry thoroughly, unless specific repair technique requires application of materials to a saturated surface.
4. Do not use acids for cleaning or preparing concrete surfaces for repair.

3.05 REPAIR

- A. Repair concrete and reinforcement outside of prescribed removal lines damaged or loosened during cutting and removal operations. Repair or replace:
1. In accordance with M-47 as approved by COR.

3.06 DISPOSAL

- A. Dispose of removed materials in accordance with Section 01 74 00 - Cleaning and Waste Management.

END OF SECTION

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SECTION 07 84 00

FIRESTOPPING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include cost of firestopping in prices offered in the Price Schedule for items of work requiring firestopping.

1.02 REFERENCE STANDARDS

A. ASTM International (ASTM)

1. ASTM E84-08a Surface Burning Characteristics of Building Materials
2. ASTM E814-08b Fire Tests of Through-Penetration Fire Stops

B. Factory Mutual Global (FM)

1. FM Approval Guide www.approvalguide.com

C. Underwriters Laboratories, Inc. (UL)

1. UL FRD-2008 Fire Resistance Directory
2. UL 723-2003 Surface Burning Characteristics of Building Materials
3. UL 1479-2003 Fire Tests of Through-Penetration Firestops

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 07 84 00-1, Approval Data:

1. Firestopping schedule which lists fire-rated assemblies, penetrations, and firestopping system or design to be used for each type of penetration.
2. Manufacturer's drawings for each firestopping system or design.
 - a. Show construction compliance with UL FRD or FM P7825.
 - b. Reference UL system or FM design number.
3. Manufacturer's product data for each material.
 - a. Include evidence of UL classification or FM approval for systems or designs to be used.
 - b. Include evidence of compliance with specified fire hazard criteria.

C. RSN 07 84 00-2, Instructions:

1. Manufacturer's storage, surface preparation, and installation instructions for each firestopping material.

1.04 PERFORMANCE CRITERIA

A. Fire Resistance Criteria:

1. Provide firestopping systems or designs listed in UL FRD or FM Approval Guide for use with required fire-rated assembly design and type of penetration.
2. Provide systems or designs with "F" and "T" ratings equal to or better than fire resistance rating of penetrated assembly when tested in accordance with UL 1479 or ASTM E814.

B. Fire Hazard Criteria:

1. Provide materials with flame spread rating of 25 or less, smoke development rating of 50 or less, and fuel contribution of 50 or less when tested in accordance with UL 723 or ASTM E84.

1.05 QUALIFICATIONS

A. Manufacturer's Representative:

1. Arrange for representative from firestopping manufacturer to perform specified field inspection.

1.06 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials to jobsite in manufacturer's original unopened packaging with labels and seals intact.
- B. Store materials in protected area in accordance with manufacturer's instructions.

1.07 ENVIRONMENTAL REQUIREMENTS

- A. Comply with manufacturer's environmental restrictions.

PART 2 PRODUCTS

2.01 FIRESTOPPING MATERIALS

A. Provide materials having following characteristics:

1. Capable of maintaining an effective barrier against flame, smoke, gases, and temperature rise during required fire resistance period.
2. In compliance with UL FRD or FM P7825.

3. Do not contain asbestos.
4. Do not emit toxic gases during installation or fire conditions.
5. For cable trays: Removable and replaceable to accommodate penetration changes.

B. Cable Wall Penetrations:

1. Fire Wall Penetration Sleeve:
 - a. Appropriate for a 9-inch concrete wall.
 - b. Fire rating: 4 hours per ASTM E 814.
2. Fire Barrier Composite Sheet:
 - a. CS 195+, as manufactured by 3M, Website: www.3m.com/firestop; or equal, with the following characteristics:
 - 1) Appropriate for use with fire wall penetration sleeve.
 - 2) Fire rating: 4 hours per ASTM E 814.
3. Fire Barrier Wall Sealant:
 - a. CP 25WB+, as manufactured by 3M, Website: www.3m.com/firestop; or equal, with the following essential characteristics:
 - 1) Designed for use with fire wall penetration sleeve and fire barrier sheet.
 - 2) Fire rating: 4 hours per ASTM E 814.

2.02 ACCESSORIES

- A. Materials furnished or recommended by firestopping manufacturer.

PART 3 EXECUTION

3.01 LOCATIONS

- A. Provide firestopping at following locations:
1. Penetrations through fire-rated wall assemblies by pipes, ducts, cables, conduit, and other systems.
 2. Construction joints at intersection of fire-rated wall assemblies and roof deck structures.
 3. Cable Tray shown on the drawings.

3.02 PREPARATION

- A. Prepare clean, dry contact surfaces in accordance with manufacturer's instructions.

- B. Remove dirt, petroleum products, rust, corrosion, and loose materials.
- C. Remove other substances which could affect proper installation or adhesion of materials or could affect required fire resistance.

3.03 INSTALLATION

- A. Install in accordance with approved system or design and manufacturer's instructions.

3.04 CONTRACTOR FIELD QUALITY TESTING

- A. After completion of firestopping, perform mutual inspection of firestopping installations with manufacturer's representative and OGR.
- B. Correct defects identified by inspection.

3.05 CLEANING

- A. Remove excess materials protruding from penetrations.
- B. Remove excess materials from adjacent surfaces by methods recommended by manufacturer.

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SECTION 22 15 00 GOVERNOR COMPRESSED-AIR SYSTEMS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Governor Compressed-Air Systems:

1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1. ASME A13.1-2023 Scheme for the Identification of Piping Systems
2. ASME B16.11-2021 Forged Fittings, Socket-Welding and Threaded
3. ASME B16.18-21 Cast Copper Alloy Solder Joint Pressure Fittings
4. ASME B16.22-21 Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
5. ASME B31.1-2024 Power Piping

B. American Society of Sanitary Engineering (ASSE)

1. ASSE 1079-2024 Performance Requirements for Dielectric Pipe Unions

C. ASTM International (ASTM)

1. ASTM A53/A53M-2024 Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
2. ASTM B306-20 Copper Drainage Tube (DWV)

D. Manufacturers Standardization Society (MSS)

1. MSS SP-58-2025 Pipe Hangers and Supports

1.03 SUBMITTALS

A. RSN 22 15 00-1, Approval Product Drawings and Data:

1. Commercial product data for pipe, fittings and pipe hangers.
 2. Shop drawings, detailing pipe layout drawings
- B. RSN 22 15 00-2, Test Plan:
1. Test plan:
 - a. Include step by step instructions, time estimates, and list of required equipment.
- C. RSN 22 15 00-3, Final Drawings and Data:
1. Previously approved data, submitted under RSN 22 15 00-1, in booklet form.
 2. Shop Drawings:
 - a. Provide full size drawings on BOR title block.
 - b. Provide in AutoCAD (.dwg) and Portable Document Format (PDF).
 3. Performance Test Report

1.04 COORDINATION

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- A. Contractor shall field verify locations of existing systems in work area. If any modifications to existing systems are required to complete work, contractor shall receive COR approval in advance.
 - B. Contractor shall coordinate pipe routing with COR. Contractor shall receive COR approval of pipe routing prior to installation of pipe.
 - C. Contractor shall coordinate the outage of the governor air system with COR and Client. Outage shall conform to specification 01 14 30 – Interruption of Service.

1.05 SYSTEM DESCRIPTION

- A. Governor air compressor is 8 CFM at 300 psi under normal operating conditions.
- B. Remove existing compressed air and drain pipe and fittings as necessary to facilitate the relocation of governor actuator compressed air system.
- C. Provide and install pipe and fittings for the piping associated with governor actuator compressed air system on elevation 6581.00:
 1. Compressor relocation: shown on drawings 245-D-60155 through 245-D-60156.
 2. Compressed air piping: from compressor to governor cabinet actuator with isolation valves as shown on drawings 245-D-60155 through 245-D-60156.
 3. Condensate drain piping: drain line for compressor to floor drain on elevation 6564.75, as shown on drawings 245-D-60155 through 245-D-60156.

PART 2 PRODUCTS

2.01 PIPING COMPONENTS

A. Compressed Air:

1. Piping, Fittings and Valves: Provide and install in accordance with ASME B31.1. Plastic materials not permitted.
2. Galvanized Steel:
 - a. Pipe: ASTM A53 schedule 40,
 - b. Fittings:
 - 1) ASME B16.11 forged, threaded fittings, Class 3000.
3. Dielectric Fittings: Provide dielectric protection at connection of dissimilar metals.
 - a. Description:
 - 1) Standard: ASSE 1079.
 - 2) Pressure Rating: 300 pounds per square inch gauge minimum.

B. Condensate Drain:

1. Piping, tubing, Fittings and Valves: Provide and install in accordance with ASME B31.1. Plastic materials not permitted.
 - a. Material Schedule:
 - 1) Tube: ASTM B306 copper drainage tube (DWV).
 - 2) Fittings:
 - a) Cast copper, solder joints; ASME B16.18, pressure fittings.
 - b) Wrought copper, solder joint fittings; ASME B16.22, pressure fittings.
2. Dielectric Fittings: Provide dielectric protection at connection of dissimilar metals.
 - a. Description:
 - 1) Standard: ASSE 1079.

2.02 PIPE HANGERS AND SUPPORTS

- A. Pipe Hangers: Provide and install in accordance with ASME B31.1.
- B. MSS-SP-58, Types 1 through 12, factory-fabricated components

- C. Vertical Support: Steel riser clamp.

2.03 HANGER RODS

- A. Mild steel threaded both ends, threaded on one end, or continuous threaded.
- B. Rods for trapeze hangers shall be a minimum 3/8-inch. The use of pipe hooks, chains, perforated iron strapping, or wire for pipe supports are NOT permitted.

2.04 VALVES

- A. Ball Valves, Series BV2M, by DWYER INSTRUMENTS, LLC, 2855 Metropolitan Pl Pomona, CA 91767; or equal, having the following essential characteristics:
1. Working Pressure Rating: 500 psig minimum.
 2. Connection: FNPT.
 3. Rated for water or gas applications.

2.05 CONCRETE ANCHORS

- A. Adhesive Anchors: Drilled type, installed with a chemical adhesive system:
1. Seismic qualified per IBC and AC308 for seismic design category A through F and cracked concrete conditions.
 2. Adhesive: HIT-HY 200 epoxy adhesive manufactured by HILTI, P.O. Box 21148, Tulsa, OK 74121; or equal, having the following essential characteristics:
 - a. Injectable two-component epoxy adhesive.
 - b. Dual packaging system designed for automatic mixing during injection.
 - c. Weather resistant.
 - d. Resistant against high temperatures.
 - e. Suitable for use in diamond cored or pneumatic drilled holes.
 - f. Extended temperature range from -40 to 104-degrees F.

2.06 PLASTIC TAPE PIPE MARKERS

- A. Flexible, colored vinyl film tape with pressure sensitive adhesive backing.
- B. Printed markings in accordance with ASME A13.1.
- C. Identify service type and flow direction.
- D. Match existing color scheme.

2.07 VALVE TAGS

- A. Valve Tags:
 - 1. Round brass.
 - 2. Attachment: Brass beaded chain.
- B. Compressor isolation valve label: "GS-01".

2.08 COATINGS

- A. Coat exposed galvanized steel pipe per specification section 09 96 20 Coatings.

PART 3 EXECUTION

3.01 GENERAL

- A. Protection: Properly cap open ends of pipe at end of each day's work or other stopping point throughout construction. Tightly cover equipment and protect against dirt, water, chemical, and mechanical damage.

3.02 PIPING INSTALLATION

- A. Piping:
 - 1. Provide installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing in accordance with ASME B31.1.
 - 2. Install nipples, unions, special fittings, and valves with pressure ratings same as or higher than system pressure rating used in applications below unless otherwise indicated.
 - 3. Install drain piping with ¼" per foot slope downward in direction of flow.
 - 4. Install piping straight and true to bear evenly on hangers and supports.
 - 5. Install piping at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
 - 6. Keep interior and ends of new piping thoroughly cleaned of dirt and foreign matter during installation.
 - 7. Blow out entire piping assembly with compressed air after assembly to ensure piping is free of all foreign materials before connection to equipment.
 - 8. Inspect piping before placing installation.
 - 9. Where piping penetrates a fire rated wall or floor, provide fire-stopping insulation so that the assembly when complete retains the fire rating of construction penetrated.

10. Provide insulating type union at connections between ferrous and non-ferrous water piping, valves and piping accessories and where required for connections to electrical equipment.
11. Install an adequate number of unions in piping systems to allow for future maintenance activities requiring disassembly and reassembly of piping systems.
12. Provide fittings for changes in direction of piping and for connections. Make changes in piping sizes through tapered reducing pipe fittings. Bushings are not permitted.
13. Identification:
 - a. Install appropriate plastic tape pipe markers and plastic nameplates for all exposed piping components, valves, and equipment.
 - b. Install after completion of coatings.
 - c. Plastic tape pipe markers:
 - 1) Align with axis of piping. Install completely around pipe with markings in clear view.
 - 2) Distance between identification is not to exceed 20 feet on centers on straight runs, including risers and drops.
 - 3) Provide adjacent to each valve and tee, at each side of penetration of structure or enclosure, and at each obstruction.
 - d. Plastic nameplates:
 - 1) Fasten securely to equipment or other appropriate location with stainless steel mechanical fasteners or adhesive.
 - 2) Install with markings in clear view.

B. Pipe Hanger and Supports:

1. Install pipe hangers and supports in accordance with ASME B31.1. Arrange piping supports so no excessive bending stresses are induced into the piping from concentrated loads between supports. Install insulating material between dissimilar alloys.
2. Adequately support system piping in order to maintain its integrity during fire conditions.
3. Support the system without sagging, including supports at each offset or change in pipe direction and at the ends of pipe branches over five feet in length.
4. Support vertical piping with riser clamps secured to the piping and resting on the building structure at each floor.
5. Use hangers with 1-1/2 inches minimum vertical adjustment.

6. Support riser piping independently of connected horizontal piping.
7. Provide supports that allow free axial movement and only support the weight of the piping or tubing. Provide additional hangers or brackets to support valves, flanges, specialties, etc., to prevent excessive deflection.
8. Prime coat exposed steel hangers and supports. Hangers and supports located in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed.
9. Install supports to provide a minimum 1/2-inch clear space between finished covering and adjacent work.
10. Maximum spacing between pipe hangers and supports per MSS SP-58.

3.03 CONTRACTOR FIELD QUALITY TESTING

- A. Test conducted by Contractor and witnessed by COR.
- B. Notify COR in writing at least 14 working days prior to blow-out and testing. Include the following:
 1. Copy of approved Test Plan.
 2. Date and time the system shall be ready for testing.
 3. Period of time required to complete testing.
- C. Piping:
 1. Inspection, examination and pneumatic testing in accordance with ASME B31.1
 2. Prior to testing ensure piping system is free of debris by performing an air blow-out.
- D. Pressure Test:
 1. Test Pressure: 1.5 times the working pressure of the system.
 2. Test time: 30-minute minimum pressure hold time.
 3. Test only the newly installed piping and valves. Do not test the existing compressed air pipe.
 4. Isolate Governor from pipe with isolation valve prior to starting test.
- E. Contractor to supply all equipment, valves, gauges, etc. necessary to perform pipe pressure test.

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SECTION 25 00 01

AUTOMATION CONTROL AND MONITORING SYSTEMS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Automation Control and Monitoring Systems:
1. Payment: Lump sum price offered in the Price Schedule for Automation Control and Monitoring Systems.
 - a. Includes cost of the following:
 - 1) Programmable Logic Controllers (PLCs) and Distributed I/O Modules.
 - 2) Human Machine Interface and Graphic Displays.
 - 3) Accessories, cables, conductors, associated with the PLCs.
 - 4) PLC power supplies, PLC I/O cards, PLC network cards, PLC time synchronization cards.
- B. Spare Parts:
1. Payment: Lump sum price offered in the Price Schedule for Spare Parts.
- C. Cost:
1. Include cost of training in the lump sum price offered in the Price Schedule for Training.

1.02 REFERENCE STANDARDS

- A. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE 1010 – 2022 Guide for Control of Hydroelectric Power Plants
 2. IEEE 1046 – 1996 Distributed Digital Control and Monitoring for Power Plants
 3. IEEE 1249 – 2013 Guide for Computer-Based Control for Hydroelectric Power Plant Automation
 4. IEEE C37.1 – 2007 Standard for SCADA and Automation Systems
- B. International Society of Automation (ISA)
1. ISA 101.01 – 2015 Human Machine Interfaces for Process Automation Systems

2. ISA TR101.02 – 2019 HMI Usability and Performance
 3. ISA 18.1 – 2024 Annunciator Sequences and Specification
 4. ISA 18.2 – 2016 Management of Alarm Systems for the Process Industries
 5. ISA 18.1 – 2024 Annunciator Sequences and Specification
- C. National Fire Protection Association, Inc. (NFPA)
1. NFPA 70 – 2023 National Electrical Code (NEC)

1.03 DEFINITIONS AND ACRONYMS

- A. CPU: Central Processing Unit.
- B. FAT: Factory Acceptance Testing.
- C. HMI: Human Machine Interface.
- D. I/O: Input/Output.
- E. IRIG: Inter-Range Instrumentation Group Timecodes.
- F. NTP: Network Time Protocol.
- G. PLC: Programmable Logic Controller.
- H. RTU: Remote Terminal Unit.
- I. SFP: Small Form-Factor Pluggable.
- J. VDC: Volts Direct Current.

1.04 AUTOMATION CONTROL AND MONITORING SYSTEM DESCRIPTION

- A. The Automation Control and Monitoring System shall be comprised of one programmable logic controller (PLC), multiple human machine interfaces (HMIs) and distributed I/O. The unit control board shall include a PLC, HMI, and display screens for Annunciation. The Turbine level and Gatehouse shall include distributed I/O and HMI's which connect to the main PLC via ethernet network switches.
- B. The system shall include but not limited to, distributed Input/Outputs (I/Os) assemblies, power supplies, input modules, output modules, communications devices, and any other required equipment to provide a fully functional system. The system shall receive inputs with an internal control logic program as well as control output relay operations to perform local, automatic, and remote-control functions via the HMIs. Refer to specification drawings 245-D-60124 (sheets 1- 4).

- C. Design, furnish (equipment, labor, and material), install, program, calibrate, adjust, document, test, and commission a complete automation system for plant wide systems integration, including remote RTU testing.
1. Changes to the RTU programming will be performed by the Government.
 2. Excitation System circuit outputs from C1B cabinets shall be wired to inputs at the main PLC.
- D. Automation Control, Monitoring, and Annunciation:
1. The Automation Control and Monitoring System shall control all electrically operated equipment, monitor all processes, and annunciate troubles and alarms. Electrical equipment control, intercommunication, and protection systems shall be integrated into the Automation Control and Monitoring System. As part of a graphic user interface system, provide historical data of all operations for the generator.
 2. The PLC equipment provides the logic control for the Automation Control and Monitoring Systems. Unless noted otherwise, all user-initiated interfaces to the PLCs are performed through the HMIs, control switches, or the RTU.
 3. Control functions are to be executed by the PLC logic programming, when 143C is set to Local-Manual, Local-Auto, or Supervisory.
 4. The existing Pole Hill PLC annunciator system, comprising two 48-window panels for generator and station annunciation, shall be decommissioned. All visual and audible alarm functions shall be replaced by and fully integrated into the new Automation Control and Monitoring System.

1.05 RELATED WORK

- A. Coordinate features of the Automation Control and Monitoring System to be compatible with the following Sections:
1. Section 25 12 10 – Programmable Logic Controllers.
 2. Section 26 09 16 – Protection Systems.
 3. Section 26 09 18 – Control Boards.
 4. Section 27 05 43 – Outside Plant Raceways for Communications Systems.
 5. Section 27 20 01 – Fiber Optic Cable and Accessories.
 6. Section 27 21 29 – Network Switches.
 7. Section 28 05 11 – Cybersecurity for Requirements for Electronic Safety and Security

1.06 AUTOMATION CONTROL AND MONITORING SYSTEM DESIGN REQUIREMENTS AND PERFORMANCE REQUIREMENTS

- A. Automation Control network architecture:

1. Refer to specification sections stated above.
 2. Network Switch communication connecting main PLC, Turbine level Distributed I/O, Gatehouse Distributed I/O, and existing operator RTU.
 3. Communication for the generator unit shall be a star network between generator PLC, HMIs, meters, RTU, and distributed I/O modules.
- B. Design a hydroelectric generator Automation Control and Monitoring System.
1. Control and monitor synchronous generator, exciter, power circuit breakers, brakes, valves, gates, cooling water pumps, lubrication oil pumps, thrust and jacking motors, fire suppression system, meters, protection devices, levels, temperatures, pressures, and other pertinent systems.
 2. The Automation Control and Monitoring System shall collect trouble and alarm data throughout the facility and replicate the existing generator and station service annunciation systems.
 - a. The system shall have sequence of events recording with precise time stamps for all alarms and events.
 - b. The annunciation interface tabs for the generator and station service shall provide drill-down capability within alarm windows to identify the exact source of each contact closure. It is preferred to have similar number of windows as existing Annunciators with drill down capability using a touchscreen at the control board HMI screen.
 3. Send data, generator operational status, indication, alarms, trips, and notifications to existing RTU PLC in cabinet outside of control room on the generator floor.
 4. Automation Control System shall accept remote commands from existing RTU and hardwired control switches on CSA3F control board in Control room.
- C. Hardwired Controls:
1. In the event of automation system failure (PLC and/or HMI), the generator can still be operated in Local-Manual mode for shutdown purposes.
 - a. Emergency stop/trip switches located on the control panel.
 - b. Emergency Trip.
 - 1) Direct hardwired trip signals to Generator Lockout Relay (186G).
 - 2) Independent of automation logic to guarantee operation during system failure.
 - c. Governor Control.
 - 1) Manual operation of turbine gates to bring the unit to a stop.
 - 2) Emergency trip overrides governor commands for rapid shutdown.
 - d. Excitation System.

- 1) Maintains safe excitation shutdown during manual trip.
 - 2) Excitation outputs wired to PLC for monitoring, but trip is hardwired.
 - e. Protection Systems.
 - 1) Digital multi-function relays (e.g., SEL 700G) remain active for overspeed, overcurrent, and other abnormal conditions.
 - 2) Trips act directly on lockout circuits without relying on automation.
 - f. Breaker Control.
 - 1) Manual open/close breakers via hardwires switches.
 - 2) Emergency trip overrides all modes.
 - g. Manual operation of the generator brake at the control panel.
- D. Data Historian Server for the annunciation system, field devices, and protection system devices.
- E. New Network Distribution Cabinet (CP1N) assembly with communications devices and associated components. Refer to Section 26 09 18 – Control Boards and 27 21 29 – Network Switches.
- F. Provide password protection for all PLCs, network switches, and HMI panels, in accordance with Section 28 05 11 – Cybersecurity for Requirements for Electronic Safety and Security.

1.07 SUBMITTALS

- A. In accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.
- B. RSN 25 00 01-1, Resumes and Qualifications – Automation Control and Monitoring System.
 1. Include specialized personnel resumes.
- C. Approval Drawings and Data in Government Format – Generator Unit: Most of the automation control devices will be installed on the new control boards, refer to required submittals in Section 26 09 18 - Control Boards.
 1. The control board's approval drawings and data submittal shall include but not be limited to the following regarding the new Automation Control and Monitoring System devices:
 - a. Network Communications Diagrams.

- D. Installation Plan and Procedures: Refer to required submittal in Section 26 09 18 – Control Boards.
- E. RSN 25 00 01-2, Software Development Documentation for Generator Unit:
1. I/O List
 2. Master Alarm List, including the minimum following information for each alarm:
 - a. Alarm identifier.
 - b. Alarm priority.
 - c. Alarm group.
 - d. Alarm setpoints.
 3. Data Historian Server: Historian database of all points, system generated points as well as I/O points, to be recorded by the historian.
 - a. Alarm Sequence description, alarm list for the PLC-based HMI annunciators.
 4. Network switch configuration data.
 - a. IP addresses for all devices.
 - b. Subnets.
 - c. MAC addresses of all devices.
 - d. Port numbers.
 5. Firewall configuration.
 6. System Overview.
 - a. Provide high-level, narrative description of PLC code.
 - b. Include ladder logic diagram of PLC code.
 - c. Narrative shall reference specific variables, subroutines, I/O from PLC code.
 7. Licenses.
 8. Provide preliminary software applications for new control board testing.
 9. Network asset configuration management listing associated with appropriate Tag Names.
 - a. Reference Section 25 00 01 – Automation Control and Monitoring Systems, Part 2.
 10. High-resolution, representative screenshots in color:
 - a. Unit control HMI displays and pop-ups.
 - b. Window generator and station service annunciation displays and pop-ups.

- F. RSN 25 00 01-3, Factory Acceptance Test Plans and Procedures: Refer to Section 26 09 18 – Control Boards and Section 28 05 11 – Cybersecurity Requirements for Electronic Safety and Security.
- G. RSN 25 00 01-4, Factory Acceptance Test Reports - Automation Control and Monitoring System for Generator Unit: Refer to Section 26 09 18 – Control Boards and Section 28 05 11 – Cybersecurity Requirements for Electronic Safety and Security.
- H. RSN 25 00 01-5, Field and Commissioning Test Plans and Procedures – Refer to Section 26 05 90 – Wiring Checkout and Tests.
1. Describe the methods, software, and testing metrics which demonstrate network connectivity, network packet captures, throughput bandwidth measurements, and network locations where these measurements will take place.
 2. Describe the methods for I/O verification and PLCs logic verification.
 3. Describe the methods used for the Data Historian Server.
- I. RSN 25 00 01-6, Operation and Maintenance Manual; Generator Automation Control and Monitoring System: Refer to Section 01 78 30 – Project Record and Documents.
- J. RSN 25 00 01-7, Commissioning Test Procedures for Generator: Refer to Section 26 05 90 – Wiring Checkout and Tests.
- K. RSN 25 00 01-8, Field and Commissioning Test Reports for Generator: Refer to Section 26 05 90 – Wiring Checkout and Tests.
1. Include results for all network communication paths.
 2. Completed, signed, and approved electrical commissioning forms.

1.08 QUALIFICATIONS

- A. Manufacturer:
1. Regularly engaged in manufacturing similar items.
- B. On-site Representative:
1. Minimum of 5 years of experience testing and commissioning electrical control systems.
 2. In depth knowledge of this application and fully capable of troubleshooting and modifying control system hardware and software, including making modifications during on-site testing as necessary.
 3. Technically qualified to supervise installation and testing work.
 4. Present or represented at work site during installation period.

5. Report, in writing to Project Manager, work not in accordance with manufacturer's recommendations or other conditions possibly resulting in unsatisfactory job.
 6. Maintain daily log of accomplished work.
- C. Project Designer and Project Programmer:
1. Minimum of 3 years of experience designing similar systems utilizing similar components. At least two prior projects including design of controls and alarm systems for power plants. The Project Programmer shall have experience with Rockwell Studio 5000.

1.09 DESIGN COORDINATION MEETINGS

- A. Conduct design coordination meetings after Award through completion of project in accordance with Section 01 31 19 – Project Management and Coordination.

1.10 EXTRA MATERIALS

- A. Spare Parts:
1. Deliver spare parts to locations as determined by Contract Officer Representative.
 2. Deliver spare parts with shipment of the unit control boards.
 3. Furnish spare parts listed in Table 25 00 01A – Spare Parts for Automation Control and Monitoring System. Quantities listed are total spare parts required, not spare parts for each panel.

Table 25 00 01A – Spare Parts for Automation Control Systems

Item	Description	Quantity
1	Solid-state hard drive	1 of each type

PART 2 PRODUCTS

2.01 CONTROL BOARD PLCs

- A. Refer to specification Section 25 12 10 – Programmable Logic Controller and Section 26 09 18 – Control Boards.

2.02 PLC APPLICATION PROGRAM

- A. HMI at entrance of powerplant shall not have any control capability and shall be used for monitoring only.

- B. Control selector switch (143C) shall be located on the new control board panel (CSA3F). The control modes for these switches are as follows:
1. Local Manual – no automation control function. PLC should continue to monitor and send alarms.
 2. Local Auto - sequences controlled by PLC automation controls based on plant control board switches, HMI input, and inputs from plant equipment and devices.
 3. Supervisory – sequences controlled by PLC automation controls based on commands from the Casper Control Center (CCC) Remote Terminal Unit or hardwired emergency controls on CSA3F cabinets.
 4. Backup Supervisory – This shall be included in the logic for future use, see drawing 245-D-60126.
- C. Adjustable Parameters:
1. Provide a password protected configuration screen(s) on the HMI for authorized personnel to adjust automation control parameters. These adjustments are intended for configuration purpose only and not for routine operator inputs. (Admin level access only).
 2. Include the following items at a minimum:
 - a. PLC control timer setpoints.
 - b. Alarm and trip thresholds.
 - c. Analog input scaling and calibration setpoints.
 - d. Password management in accordance with Section 28 05 11 – Cybersecurity for Requirements for Electronic Safety and Security.
 - e. Time and date settings.
 - f. Equipment operating setpoints.
 - g. Deadband ranges.
 - h. Delays.
- D. Inputs and Outputs
1. Analog and discrete I/O points as described in Section 51 25 00 – Input Output List.
 - a. Includes Annunciation input points to be gathered at nearest I/O input point of network.
- E. Generator Start and Stop Sequences in Local Auto:
1. PLC will automatically sequence the start and stop process in accordance with Manual Start and Stop process as documented in the Pole Hill Powerplant Standing Operating Procedure.
 2. PLC ignores RTU commands and RTU setpoint adjustments in Local Auto.

3. PLC ignores Start, Stop, and VAR adjustments from the control switches located in the CSA3F cabinets in Local Auto.
4. Excitation System shall maintain 0 MVARs.
5. Control switches on CSA3F will not respond in Local Auto, except for emergency trip.
6. HMIs continue displaying any running information.
7. HMI on CSA3F board shall allow operational changes or setpoint adjustments.
 - a. Other HMIs shall only allow for monitoring, except for emergency stop.

F. Generator Start and Stop Sequences in Supervisory:

1. PLC digitally receives commands from existing Casper Control Center RTU system and processes them similarly as Local Auto commands.
2. PLC shall receive VAR adjustments signals from RTU cabinet and passes them to the Excitation System for processing.
3. Control switches on CSA3F will not respond in Remote, except for emergency stop.
4. HMIs on CSA3F continues displaying any running information.
5. HMIs shall not permit operational changes or setpoint adjustments.
 - a. Functions that can be performed while in Supervisory:
 - 1) Generator Start
 - 2) Generator Stop
 - 3) Generator Voltage Raise/Lower.
 - 4) Generator Gate Limit Raise/Lower.
 - 5) Generator Emergency Stop.
 - 6) 13.8KV Generator PCB PH-122 Trip.
 - 7) 115KV PCB PH-164 Open/Close.
 - 8) 115KV PCB PH-264 Open/Close.

G. Generator Start and Stop Sequences in Local Manual:

1. Control switches on CSA3F shall respond in Local Manual.
2. PLC ignores Start, Stop, and other adjustments from RTU cabinets in Local Manual.
3. HMIs continue displaying any running information.
4. HMI on CSA3F board shall allow operational changes or setpoint adjustments.

- H. Emergency Stop/Trip issued by 105CS, HMIs on CSA3F, Unit PLC, or existing RTU system shall be executed regardless of control mode state of 143C.
1. Use of PLC interlocks within hard-wired emergency control circuits is not permitted.
- I. Automatic Load Control (ALC):
1. ALC loads/unloads generator after unit is started and synchronized.
 2. In Local-Manual: Automatic Load Control is turned ON/OFF by using Auto Load Control Soft Switch (143ALCS) in HMI.
 3. In Local-Auto and Supervisory: Automatic Load Control is turned ON/OFF when auxiliaries start/stop, respectively.
 4. Float Load Control:
 - a. During startup/shutdown, ramp rate shall be between 2 - 3MW/min
 - b. Float load control to maintain Forebay water elevation of approximately 7418.00 ft. at intake:
 - 1) At elevation 7417.8 ft:
 - a) Pulse Speed Adjust, decreases load to increase water in Forebay.
 - b) Pulse to last 20 cycles, then wait 8 seconds before allowing another pulse.
 - 2) At elevation 7418.2 ft:
 - a) Pulse Speed Adjust, increases load to reduce water in Forebay.
 - b) Pulse to last 20 cycles, then wait 8 seconds before allowing another pulse.
- J. Generator Synchronization in Auto:
1. With Generator Synchronizer Switch (125SS) in the AUTO position, the new SEL 700G shall auto synchronize the generator to the bus.
- K. Generator Synchronization in Manual:
1. With Generator Synchronizer Switch (125SS) in the MANUAL position, synchronization shall occur through operator adjustments to the AUTO-VOLT-ADJ Control Switch (190ECS) using the voltmeters and synchroscope as indication for proper synchronization.
- L. Coordinate new capabilities of equipment provided by Contractor with COR.
- M. Incorporate the existing Excitation System into new start, stop and running sequencing of generator as documented in the Pole Hill Powerplant Standing Operating Procedure.

N. Control Mode Transitions:

1. Control mode transitions of 143C shall be permissible only at one of the following states: complete dead-stop, auxiliaries fully started, or unit fully running.
2. When a control mode transition is attempted outside of the permissible operational states, alarm on the unit annunciator and ignore the transition attempt (maintain last valid 143C control mode).

O. PLC Watchdog

1. To monitor that the PLC is responsive, an internal watchdog timer relay will trip the Generator lockout relay when its timer is expired.
2. The PLC shall continuously refresh the timer on the watchdog relay.
3. Watchdog signal to 186G is blocked when 143C is in Local Manual.

P. Maintenance Test Mode:

1. Provide a password protected test mode from the HMI requiring technician level access for debugging purposes.
2. Allow the technician to select individual PLC inputs and outputs and force their state to a set value.

2.03 AUTOMATION CONTROL APPLICATION SOFTWARE**A. Application Programming:**

1. PLC Programming Language.
 - a. PLC code written in Ladder Logic.
 - b. Grouping of programming sections into subroutines.
 - c. Supports variable names using common text of 32 characters or more.
- d. Software required to view and make program changes shall be provided as a part of the contract including any licenses, dongles, activation keys, etc. as needed.
 - 1) Compatible with Microsoft Windows version 11 operating systems.
- e. PLC programming documentation:
 - 1) Sufficiently documented to provide readability for technicians and engineers.
 - 2) Add comments describing program revision level; program sections, modules, and functions; variable names and directions to navigate through the code where an external module is called or a jump to another location takes place.
 - 3) Comments written in clear, concise, English using sentence structure and proper case.

- 4) Comment rate should be 20 percent or more (i.e., one comment line for every 5 program lines).
2. Provide preliminary software applications for new control board testing.
3. Include updated software with check prints that are shipped with the new control boards to site.

2.04 ANNUNCIATION

- A. Annunciation display windows shall be designed to provide operators with clear, intuitive, and effective control and monitoring in accordance with ISA-101.01.
- B. Graphic displays shall adhere to high-performance design standards, including light gray backgrounds with color used sparingly to alert operators to abnormal conditions. Graphics shall be simple 2D representations.
- C. The display annunciation windows shall have windows arranged in a matrix configuration (rows and columns). Each window will be permanently assigned to a specific alarm or a designated group of alarm points, ensuring clarity and accuracy in monitoring system alerts.
- D. Provide a tab section for annunciation in the HMI of the automation control and monitoring system to access window annunciation.
 - 1. The windows shall also be toggleable to show alarm history in the digital annunciation window panels.
- E. Dedicated Annunciator Display:
 - 1. Provide one large industrial display for annunciation, mounted and centered between the top section of control board panel CSA3F and CSA4F.
- F. Design to include but not limited to centralized monitoring and real-time notifications.
- G. Breakout existing grouped alarms for individual monitoring within the automation control and monitoring system.
- H. Acknowledgement of alarms and warnings either individually or all at once.
- I. Alarm States with corresponding pushbuttons:
 - 1. Normal operation: HMI annunciator does not have any trouble or alarm windows illuminated.
 - 2. Acknowledge/Silence:
 - a. The HMI window that is in trouble and/or alarm shall blink, and horns shall sound (both on the HMI and the existing horns) until the operator acknowledges the alarm by pressing the acknowledge pushbutton.

- b. Once the acknowledged/silenced pushbutton is pressed, the HMI window in trouble/alarm shall stop blinking but stay illuminated until the problem is corrected and HMI annunciator reset.
 3. Return to Normal/Reset: After the trouble/alarm is located and corrected, the HMI annunciator can be reset. If all issues are resolved, there should be no window illuminated.
 4. Test: Provide annunciator test feature which allows the operator to confirm all HMI annunciator windows illuminate and sounds horn and operates properly via Test pushbutton.
- J. Pushbuttons:
 1. Acknowledge (alarm light goes solid and the horn is silenced).
 2. Reset.
 3. Test (horn will sound).
 4. Silence (Alarm still flashes and horn stops).
- K. Industrial Display Panel Mount: ID-K43Aw 43" 4K Display Panel Synergy Global Technology Inc., 1103-A Fulton Place, Fremont, CA 94539, USA; having the following essential characteristics:
 1. Size: 43"-inch minimum (diagonal).
 2. Bezel Material: Stainless Steel.
 3. IP-65 minimum.
 4. Anti-reflective glass.
 5. Screen:
 - a. Native Resolution: 3840 x 2160 @ 60Hz.
 - b. Brightness: 350 cd/m2.
 6. Temperature 0-to-55-degree Celsius minimum.
 7. Humidity: 20 - 90 percent non-Condensing.
 8. DC Power: 125 VDC.
- L. Design: New annunciator system network to replace existing annunciator circuits. Refer to Section 25 12 10 – Programmable Logic Controllers and Section 26 09 18 – Control Boards.

2.05 ALARM MANAGEMENT SYSTEM

- A. Provide an alarm management system for the annunciation systems in accordance with the requirements and recommendations in ISA 18.2.
- B. Identify the alarms and develop alarm prioritization and alarm grouping schemes.

1. Identify a hierarchy of alarm priorities and the criteria used to designate each level of the hierarchy.
 2. Group alarms based on area, or system, which the alarm impacts.
 3. Displays of alarm points shall include information on alarm prioritization and alarm grouping.
- C. Document the following for each alarm, in accordance with ISA 18.2:
1. Alarm type.
 2. Alarm priority.
 3. Alarm group.
 4. Alarm setpoint.
- D. Provide master alarm database to access alarm points in the automation control and monitoring system.
- E. Time stamped logs for alarm history, event history, diagnostic and audit history for the alarm management system.
- F. Alarms:
1. Alarms and indications of abnormal situations shall be clear, prominent, and consistently distinguishable.
 2. Provide the following minimum alarm displays:
 - a. Alarm summary, including:
 - 1) Sorting by priority.
 - 2) Sorting by group.
 - 3) Sorting by time stamp.
 - b. Alarm status.
 - c. Alarm log.
 - d. Alarm identifier detail.
 - e. First-out.
 - f. Alarm indication on process.
 - g. Alarm banner showing recent alarms.
 3. Displays shall provide the operator the ability to:
 - a. Silence audible alarm indicators (without acknowledging the alarm).
 - b. Acknowledge alarms.
 - c. Modify alarm attributes.

- G. No animation except for specific alarm related graphics.

2.06 TURBINE LEVEL DISTRIBUTED I/O CABINET (111TDIO)

- A. The assemblies shall be installed, wired, and tested in new control enclosure at the Turbine level.
- B. The cabinets shall be fabricated, populated, wired, and tested by the control board manufacturer.
- C. Refer to Section 26 09 18 – Control Boards.

2.07 GATEHOUSE DISTRIBUTED I/O RACK (111GHDIO)

- A. The assemblies shall be installed, wired, and tested on existing communications rack in Gatehouse Communications Closet.
- B. The panel shall be fabricated, populated, wired, and tested by the control board manufacturer.
- C. Refer to Section 26 09 18 – Control Boards.
- D. Refer to Section 27 05 43 – Outside Plant Raceways for Communications Systems.

2.08 TOUCH SCREEN HMI DISPLAYS

- A. Refer to Part 2 Products of Specifications Section 25 12 10 – Programmable Logic Controller.

2.09 DATA HISTORIAN SERVER

- A. Location: in the Main Control Room.
- B. System Components:
 - 1. Historian SE Server: Central data repository with support for up to 60,000 tags
 - 2. Data Collection Interfaces:
 - a. FactoryTalk Live Data Interface.
 - b. Ethernet/IP Connector.
 - 3. Analysis Tools:
 - a. Asset Framework.
 - b. Event Frames.
 - c. Notification Service.
 - d. Data Archive.

H. Data to be collected:

1. Start and stop sequences.
2. Annunciation HMI Alarm log.
3. Close or Open digital output relays.
4. Analog I/O for trending.

I. Data Historian monitoring of alarms and operational data collection.

1. Expand data acquisition capabilities to log protection relay's annunciation in Data Historian employing industry standard network security.

J. Access Control:

1. Only authorized personnel to have access for modifying relay settings including user accounts and permissions.
2. Limit to authorized users on an as-needed basis with permission verification as defined by system administrator.

K. Authentication:

1. Implement strong authentication for only authorized users.

2.10 TIME SYNCHRONIZATION

- A. Refer to Section 25 12 10 – Programmable Logic Controllers, Section 26 09 18 – Control Boards and Section 26 09 16 – Protection Systems.

- B. The Annunciation, Protection System, Excitation System, HMIs, PLC, and ethernet switches shall be synchronized to the same master clock. A SEL 3400 shall distribute the existing IRIG-B signal to the ethernet network, where time synchronization will be provided via NTP. The PLC shall be capable of accepting the master NTP signal.

1. Minimize signal latency across network to ensure accurate NTP synchronization across all devices.

2.11 MANAGED INDUSTRIAL ETHERNET SWITCH

- A. Refer to Section 27 21 29 – Network Switches.

2.12 SOURCE QUALITY CONTROL

- A. Government may witness each factory acceptance test.

1. Location: Control Board manufacturer's facility.

- B. Coordinate factory acceptance testing of automation systems with factory testing of:

1. Section 25 12 10 – Programmable Logic Controllers

2. Section 26 09 16 – Protection Systems
 3. Section 26 09 18 - Control Boards
 4. Section 28 05 11 – Cyber Security Requirements for Electronic Safety and Security
 5. Section 27 20 01 – Fiber Optic Cable and Accessories
 6. Section 27 21 29 – Network Switches
- C. Prior to Factory Acceptance Test of the new Control Boards, the Contractor shall perform burn-in test of the equipment for a minimum of 72 hours. The Automation Control and Monitoring System would be a part of the burn-in period.
- D. The purpose of the Factory Acceptance Test of the Control Boards with the Automation Control and Monitoring System is to demonstrate the functionality, performance and stability of the hardware and software. All Automation Control and Monitoring System components shall be tested during this time. Other devices and equipment not part of the Control Boards shall be tested on-site at the facility.
- E. The following items are examples of testing that shall be performed during the Control Board FAT:
1. The test shall verify that hardware and software meet the requirements of the specifications and drawings.
 2. Point to point checks of all connections and application simulation tests that demonstrate automation controls will operate as described in these specifications. This includes but is not limited to simulation and control algorithm tests, to demonstrate proper programming sequences.
 - a. Simulation of each discrete data point at the terminal block.
 - b. Simulation of each analog data signal using current or voltage source.
 3. Perform 100 percent I/O checkout, closed loop simulation of binary/modulating logic, and validations of all logic and displays.
 4. Display of all alarms and warnings on the HMI shall be demonstrated for each I/O point.
 5. Operation of communication between PLC, Distributed I/O devices, meters, and network switches.
 6. HMI display screens and data display.
 7. Simulation of unit's start and stop sequences in Local Manual, Local Auto and Supervisory control modes. This includes the ability to introduce out of sequence events to test program stability.
 - a. Includes simulation of HMI status functions in each control mode.
 8. Simulation of autoloader control demonstrating proper raise and lower output commands based on simulated user inputs.

9. Simulation of autoloader control demonstrating proper raise and lower output commands based on float load controller or simulated MW inputs.
 10. Demonstration of Maintenance Test Mode capability.
- F. Current transformers, voltage potential transformers and RTD feeds to digital multi-function protection relays shall be simulated where required for full testing and demonstration purposes.
- G. Testing and demonstration shall include the following communication network information:
1. Passive network packet captures for the entire session for each network path.
 2. Throughput bandwidth measurement for each network path.

2.13 NETWORK ASSET CONFIGURATION MANAGEMENT LIST

- A. Spreadsheet format.
- B. The Network Asset Configuration Management List includes the following minimum information:
1. List of all equipment located at the facility and connected to the communication networks in the system in separate rows and in the first columns and on the same row for each piece of equipment:
 - a. Serial number.
 - b. Model number.
 - c. Firmware revision numbers.
 - d. Communication addresses.
 - e. Default passwords.
 2. In a separate spreadsheet tab for each device, include a list of all active and inactive electronic access ports in the first column and include the following minimum information in subsequent columns, on each row, for each port:
 - a. Enabled or disabled status.
 - b. Communication protocol used on this port.
 - c. If the port is enabled, list the service, node, device, and/or system requiring this communication port to be enabled.
 3. Spreadsheet rows shall be sortable based on any column's contents.

2.14 LICENSING AGREEMENT

- A. Licenses procured as part of this work become the property of the Government upon completion of the work.

1. Studio 5000 full perpetual license. Model: 9324M-RLDT81 or equal.
 - a. Qty: 2.
 - b. Sharable via dongle.
- B. Any changes/updates to these licenses shall be provided to the Government after commissioning and system is in normal operation. Licenses shall have no expiration.
- C. Licenses include but are not limited to the PLC Licenses, HMI Licenses, and the Data Historian Server Licenses.
- D. Include all licenses in the O&M Instruction books. Refer to specification Section 01 78 30 – Project Record Documents.
- E. Upgrade Service: update software to latest version at project completion for all devices. This software upgrade shall be performed after commissioning and system is in normal operation, in accordance with Section 01 33 00 - Submittal Table 01 33 00A.
 1. At substantial completion of the project, turn current licenses for software over to the Government and install the latest version, upgrade, or service pack for software.
 2. Provide the final software programs on portable hard drive with full access rights.
- F. Include the respective software supplier's Comprehensive Support Contract for software covering a full three-year warranty period following substantial completion which shall provide no cost software upgrades, service packs and tech support from the software supplier.

PART 3 EXECUTION

3.01 GENERAL

- A. Install communications system and automation system in accordance with approved submittals.
- B. Format detailed instructions and checklists in accordance with 01 33 26 – Electrical Drawings and Data.
- C. All instructions and checklists submitted and approved under the Installation Plans and Procedures of the sections listed in Part 1.

3.02 INSTALLATION

- A. Perform installation in accordance with requirements in NFPA 70, reference drawings, and in accordance with manufacturer's recommendations.

- B. Cable and Wiring: Provide cable and wiring in accordance with Sections 26 05 02 – Basic Electrical Materials and Methods, 26 05 10 – Conductors and Cables, 27 20 01 – Fiber Optic Cable and Accessories and 27 21 29 – Network Switches.
- C. Make grounding and bonding connections in accordance with Section 26 05 26 – Grounding and Bonding and NFPA 70.

3.03 FIELD AND COMMISSIONING TESTS

- A. Refer to specification Section 26 05 90 – Wiring Checkout and Tests.
- B. Provide development, integration and installation services required to complete and turn over fully functional systems.
- C. Perform bench and throughput testing of all installed systems and equipment.
- D. Verify communication links between devices.
- E. Confirm that the Automation Control and Monitoring System is functional for the overall project.
- F. Provide field service engineer, programmer, and/or technicians to perform field checkout and testing; and to make programming adjustments and calibrations, wiring changes, and equipment or device replacements required for successful commissioning of new equipment. Also includes required adjustments, calibrations and programming of existing interface devices, equipment, and circuits.
 - 1. Update daily-logs, on-site Progress As-Built Drawings, and any other required documentation.
 - 2. Checkout of alarms verifying proper alarm designation on touch screens, and in alarm logs and reports.
 - 3. Demonstrate operation of graphic displays and HMIs.
 - 4. Adjust logic, settings, and HMIs.
 - 5. Check process timing sequences and adjust where necessary to improve system operations.
 - 6. Verify logic and operation of the various systems.
 - 7. This item applies to unit protection and controls as well the overall project.

3.04 DELIVERY OF EQUIPMENT FOR TESTING

- A. Complete set of automation control software shall be installed in the respective devices located in the new control boards. This software shall be tested prior to the Factory Acceptance Tests of the new Control Boards. During the final FAT witnessed by the Government, there should be minimal software, program, and calibration changes.

3.05 TRAINING

- A. Refer to specifications Section 01 79 20 – Training.

3.06 FINAL AS-BUILTS

- A. Final As-built Drawings Government Format: refer to Specification Section 01 78 30 and Section 01 33 26 – Electrical Drawings and Data.

3.07 O&M INSTRUCTION BOOK

- A. Submit Final Operation and Maintenance (O&M) Instruction Books as outlined in Specification Section 01 78 30 – Project Record Documents.

END OF SECTION

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SECTION 25 12 10 PROGRAMMABLE LOGIC CONTROLLERS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include cost of the following in the lump sum price offered in the Price Schedule for Automation Control and Monitoring Systems:
 - a. Includes:
 - 1) Programmable Logic Controllers (PLCs) and Distributed I/O Modules.
 - 2) Human Machine Interface and Graphic Displays.
 - 3) Accessories, cables, conductors, associated with the PLCs.
 - 4) PLC power supplies, PLC I/O cards, PLC network cards
2. Training:
 - a. Include training in the lump sum price offered in the Price Schedule for Training.
3. Spare Parts:
 - a. Include spare parts in the lump sum price offered in the Price Schedule for Spare Parts.

1.02 REFERENCE STANDARDS

A. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C37.1 – 2022 Standard for SCADA and Automation Systems
2. IEEE C37.90.1 – 2024 Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus
3. IEEE C37.90.2 – 2024 Standard for Relays, Relay Systems, and Control Devices used for Protection and Control of Electric Power Apparatus - Radiated Electromagnetic Interference

Withstand Capability Requirements and Tests

4. IEEE 802 – 2024 Standard for local and Metropolitan Area Networks: Overview and Architecture
 5. IEEE 802C – 2017 Standard for Local and Metropolitan Area Networks: Overview and Architecture
Amendment 2: Local Medium Access Control (MAC) Address Usage
 6. IEEE 802 – 2024 Standard for local and Metropolitan Area Networks: Overview and Architecture
 7. IEEE 1249 – 2013 Guide for Computer-Based Control for Hydroelectric Power Plant Automation
- B. International Electrotechnical Commission (IEC)
1. IEC 61000-4-11 – 2022 Electromagnetic compatibility (EMC) - Part 4-11: Testing and measurement techniques - Voltage dips, short interruptions and voltage variations immunity tests for equipment with input current up to 16 A per phase
 2. IEC 61131-2 – 2017 Programmable Controllers – Part 2: Equipment Requirements and Tests
 - 3.
- C. International Society of Automation (ISA)
1. ISA 101.01 – 2015 Human Machine Interfaces for Process Automation Systems
 2. ISA TR101.02 – 2019 HMI Usability and Performance
 - 3.
- D. National Electrical Manufacturers Association
1. NEMA ICS 1 – 2022 Industrial Control and Systems: General Requirements
 2. NEMA ICS 2 – 2020 (R 2020) Controllers, Contactors and Overload Relays Rated 600 V
- E. National Fire Protection Association, Inc. (NFPA)

1. NFPA 70 – 2026

National Electrical Code (NEC)

1.03 DEFINITIONS AND ACRONYMS

- A. A: Amps
- B. AI: Analog Input
- C. AO: Analog Output
- D. DI: Digital Input
- E. DO: Digital Output
- F. FAT: Factory Acceptance Testing
- G. GB: Giga Byte
- H. HMI: Human Machine Interface
- I. I/O: Input/Output
- J. IP: Internet Protocol
- K. LA Power: Local Actuator Power
- L. mA: Milliamps
- M. MOD Power: Modular Power
- N. ms: Milliseconds
- O. MB: Mega Byte
- P. PLC: Programmable Logic Controller
- Q. RTU: Remote Terminal Unit
- R. SA Power: Sensor/Actuator Power
- S. SCADA: Supervisory Control and Data Acquisition
- T. SPD: Surge Protection Device
- U. VAC: Voltage Alternating Current
- V. VDC: Voltage Direct Current
- W. W: Watts

- X. Automation Control: Application of technology to monitor and control processes.
- Y. Communications: A network of communication equipment and associated protocols
- Z. Control system diagram: conceptual diagram of the functional interrelationship of subsystems, usually in block form.
- AA. CPU: Central Processing Unit.
- BB. Dielectric test: tests made to determine ability of insulating materials and spacings to withstand specified over voltages for a specified time without flashover or puncture.
- CC. DRAM: Dynamic Random-Access Memory.
- DD. Drift: as applied to devices such as pressure switches, temperature switches etc., a change in operating characteristics over a specified number of operations or time and specified environmental conditions.

1.04 SUBMITTALS

- A. In accordance with Sections 01 33 00 – Submittals, 01 33 26 – Electrical Drawings and Data, and 01 78 30 - Project Record Documents.
- B. RSN 25 12 10-1, System Description and Operational Documentation
- C. Approval Drawings and Data - Government Format: Refer to specification requirements in Section 26 09 18 – Control Boards.

1.05 LICENSING AGREEMENT

- A. Refer to Specification Section 01 78 30 – Project Record Documents and Section 25 00 01 – Automation Control and Monitoring Systems.

1.06 DESIGN REQUIREMENTS

- A. Provide one main PLC for the generator unit in (CSA), one distributed I/O unit in the turbine level, and one distributed I/O unit for Gatehouse controls. All distributed I/O units to communicate with the main PLC in (CSA). Annunciation circuits shall be input into whichever I/O module is in closest proximity, either the main PLC or distributed I/O. Refer to Section 26 09 18 – Control Boards.
- B. PLCs:
 - 1. Allen Bradley ControlLogix PLCs.
 - 2. Includes data encryption and/or cryptographic hashes into data storage and communications where determined appropriate.

- C. Provide Human Machine Interfaces (HMI).
1. Automation Control and Monitoring System HMI Touch Screen.
- D. Refer to Section 25 00 01 – Automation Control and Monitoring System for Control Logic Description and Operating System.
- E. HMI Philosophy development:
1. Analog indicators.
 2. HMI Trends.
 3. HMI Colors.
- F. HMI Style development:
1. High-performance design in accordance with ISA 101-01 and ISA TR101.02.
 2. Pop-up information-driven display.
 3. Develop at a minimum four display hierarchies:
 - a. Level 1, Overview.
 - b. Level 2, Unit Control.
 - c. Level 3, Equipment, and device detail.
 - d. Level 4, Process level support and configuration.
 4. Process lines.
 5. Equipment representation.
- G. HMI interface development:
1. Process flow and navigation.
 2. Static texts and live data.

1.07 EXTRA MATERIALS

- A. Spare Parts:
1. Deliver spare parts to locations as determined by Contract Officer Representative.
 2. Deliver spare parts within seven days prior to commissioning.
 3. Furnish total spare parts listed in Table 25 12 10A – Spare Parts.

Table 25 12 10A - Spare Parts

Item	Description	Quantity
1	PLC/CPU	1 of each type
2	Power Supply	2 of each type

3	I/O Modules	2 of each type
4	PLC Backplane	1
5	Communication Modules	2 of each type
6	Control HMI Screen	1

PART 2 PRODUCTS

2.01 PROGRAMMABLE LOGIC CONTROLLERS (PLCs)

A. Locations:

1. Control Boards CSA2F.
 - a. Reference drawing 245-D-60137.
2. All modules, except for the PLC, also applies to the Distributed I/O at Gatehouse and Turbine level cabinets.
 - a. Refer to Section 26 09 18 – Control Boards

B. Programmable Logic Controller Type: 1756-L83E ControlLogix as manufactured by Rockwell Automation, 1201 South Second Street, Milwaukee, WI 53204, or approved equal; having the following essential characteristics:

1. Chassis: As large as required for all modules and meets size requirements for new panel on CSA2F.
 - a. The contractor shall add a distributed I/O chassis on CSA2F interior of front panel if more space for modules is required.
2. Din Rail Mountable
3. Provide capability of programming control logic using the following programming languages in accordance with IEC 61131-3:
 - a. Ladder diagram.
4. Environmental:
 - a. Temperature: 32 to 140 degrees Fahrenheit.
 - b. Humidity: 5 to 95 percent non-condensing.
 - c. Vibration: 2g at 10 to 500 Hertz.
5. Memory Storage: 10 megabytes minimum.
6. Secure Digital (SD) card non-volatile memory storage capability.

- C. PLC Digital Input Module: Model 1756-IH16I, manufactured by Rockwell Automation, Inc., 1201 S 2nd St, Milwaukee, Wisconsin, 53204; or equal, having the following essential characteristics:
1. DC Input Module
 2. Inputs: 16 inputs
 3. Voltage Category: 120 VDC.
 4. Din Rail Mountable
 5. Removable terminal block housing.
 6. Certified in accordance with IEC 60068-2-1 and IEC 60068-2-2.
 - a. Operating Temperature: Plus 32 to Plus 140 degrees F.
 - b. Storage Temperature: Minus 40 to Plus 185 degrees F.
 7. Certified in accordance with IEC 60068-2-30 for relative humidity of 5 to 95 percent, non-condensing.
- D. PLC Digital Output Module: Model 1756-OH8I, manufactured by Rockwell Automation, Inc., 1201 S 2nd St, Milwaukee, Wisconsin, 53204; or equal, having the following essential characteristics:
1. Outputs: 8 points, individually isolated
 2. Voltage Category: 125 VDC
 3. Din Rail Mountable
 4. Removable terminal block housing.
 5. Certified in accordance with IEC 60068-2-1 and IEC 60068-2-2.
 - a. Operating Temperature: Plus 32 to Plus 140 degrees F.
 - b. Storage Temperature: Minus 40 to Plus 185 degrees F.
 6. Certified in accordance with IEC 60068-2-30 for relative humidity of 5 to 95 percent, non-condensing.
- E. PLC Analog Input Module: Model 1756-IF16, manufactured by Rockwell Automation, Inc., 1201 S 2nd St, Milwaukee, Wisconsin, 53204; or equal, having the following essential characteristics:
1. Inputs: 16
 2. Accuracy: 0.15 percent full scale or better.
 3. Input Range, Voltage: ± 10 V
 4. Input Range, Current: 4 to 20 mA
 5. Din Rail Mountable
 6. Removable terminal block housing.

7. Certified in accordance with IEC 60068-2-1 and IEC 60068-2-2.
 - a. Operating Temperature: Plus 32 to Plus 140 degrees F.
 - b. Storage Temperature: Minus 40 to Plus 185 degrees F.
 8. Certified in accordance with IEC 60068-2-30 for relative humidity of 5 to 95 percent, non-condensing.
- F. PLC Analog RTD Input Module: Model 1756-IRT8I, manufactured by Rockwell Automation, Inc., 1201 S 2nd St, Milwaukee, Wisconsin, 53204; or equal, having the following essential characteristics:
1. Inputs: 8 channels
 2. RTD Sensors supported:
 - a. 100, 200, 500, 1000 Ω platinum, $\alpha=385$
 100, 200, 500, 1000 Ω platinum, $\alpha=3916$
 120 Ω nickel, $\alpha=672$
 100, 120, 200, 500 Ω nickel, $\alpha=618$
 10 Ω copper 427
 3. Resolution:
 - a. 24 bits
 0 to 510 Ω : 0.06 m Ω /count
 0 to 1020 Ω : 0.12 m Ω /count
 0 to 2040 Ω : 0.25 m Ω /count
 0 to 4080 Ω : 0.50 m Ω /count
- G. PLC Analog Output Module: Model 1756-OF8, manufactured by Rockwell Automation, Inc., 1201 S 2nd St, Milwaukee, Wisconsin, 53204; or equal, having the following essential characteristics:
1. Outputs: 8 voltage or current
 2. Output Range, Voltage: ± 10 V
 3. Output Range, Current: 0 to 20 mA
 4. Din Rail Mountable
 5. Removable terminal block housing.
 6. Certified in accordance with IEC 60068-2-1 and IEC 60068-2-2.
 - a. Operating Temperature: Plus 32 to Plus 140 degrees F.
 - b. Storage Temperature: Minus 40 to Plus 185 degrees F.
 7. Certified in accordance with IEC 60068-2-30 for relative humidity of 5 to 95 percent, non-condensing.
- H. Supports both local I/O on common backplane and distributed I/O via ethernet network.

- I. Communication Capability:
1. In accordance with:
 - a. IEEE 754, 802.3, and 1815.
 - b. TIA/EIA 485 and 598.
 - c. Refer to Network Diagram drawing #####
 2. Ethernet/IP.
 3. Modbus TCP.
 4. DNP.
 5. SNTP.
 6. RJ-45 Ethernet 10baseT connection for technician to monitor, download, and upload application programs.
 7. Modbus RTU mode protocol over RS-485 connection.
- J. Empty slots in the chassis shall include cover plates.
- K. All Modules shall include Removable Terminal Blocks.
- L. General requirements:
1. The PLC equipment shall utilize logic functions, memory, timing functions, software, I/O points, and communication capabilities for the control and/or monitoring functions of the system described and shown in the drawings.
 - a. The Control PLC assemblies shall provide accurate and reliable control of the Automation Control and Monitoring System.
 - b. The PLC programming shall include the necessary logic to enable local automatic operation, local manual and supervisory operation including startup, shutdown, and fault detection.
 - c. Support communication protocols for integration with the existing excitation system, protection system, automation control and monitoring system, and existing SCADA RTU system.
 2. Coordinate PLC assemblies and HMI screen application programming in accordance with specification Section 25 00 01 – Automation Control and Monitoring System.
 3. Failure of PLCs and HMI screens shall be fail-safe and alarmed.
 4. Provide the ability to replace PLC I/O cards, power supplies, CPU, hard drives, and HMI without need to disconnect or disturb plant wiring.
 5. In accordance with IEEE 802, IEEE 802C, and IEEE 1249.
- M. PLC assembly shall be micro-processor based, capable of the following:

1. Receiving binary and analog inputs.
2. Controlling binary and analog output functions.
3. Perform data handling operations and communicate with external network devices.
4. Withstand susceptibility tests in accordance with IEC 61131-2, IEEE C37.90.1 and IEEE C37.90.2.

N. Modular PLC

1. Expandable through additional hardware and/or software.
2. Modules grouped together in a subpanel.
3. Mounting backplane (subpanel) shall provide the communications mechanism integrating individual modules.
 - a. The use of wires connectors between modules will not be accepted.

O. PLCs, I/O modules, communication modules, backplanes, and other components furnished shall be of the same manufacturer.

2.02 AUTOMATION CONTROLS DIGITAL CENTRAL PROCESSING UNIT

A. Automation Controls Digital Central Processing Unit:

1. CPU shall be a microprocessor-based unit that provides the following:
 - a. Self-contained.
 - b. Scanning.
 - c. Application program execution.
 - d. Storage of application programs.
 - e. Storage of numerical values related to the application process and logic.
 - f. I/O bus traffic control.
2. Speed:
 - a. Processor shall be able to simultaneously run all tasks with a scan time equal to or greater than the time taken for the controller to complete a control loop.
 - b. The processor loading shall be less than 60 percent while running all specified tasks.
3. Processor program stored in non-volatile memory, so that the program will be saved indefinitely with no external power applied to the processor board.
4. PLC failure detection performed by watchdog relay in Section 25 00 01 – Automation Control and Monitoring Systems.
5. Communications:

- a. Ability to communicate with various systems and devices as shown on network diagrams.
 - b. Utilize the manufacturer's standard communication architecture and communications protocol.
 - 1) Ethernet IP protocols preferred between Rockwell PLCs and SCADA system.
 - c. Allow programming of the PLC locally or from the Main Control Room.
 - 1) Coordinate network access to PLC with Project Manager.
6. Indications:
- a. Colored LED indications shall display the operating mode of the processor, i.e., running, programming mode, failure condition, communications active, etc.
7. Application Programming:
- a. Refer to specifications Section 25 00 01 – Automation Control and Monitoring System.

2.03 POWER

- A. Provide power distribution system to all supplied components, including overcurrent and surge protection, to support full operation of the system. This distribution system shall include analog loop power supplies and contact interrogation voltages for dry contact inputs where required.
- B. Control power shall be made available through a 125 VDC distribution terminals inside the new control boards.
- C. Power Supply shall have self-diagnostic capability to determine internal faults and annunciate alarms.

2.04 COMMUNICATION MODULES

- A. EtherNet/IP Adapter: Model 1756-EN4TR/A manufactured by Rockwell Automation, Inc., 1201 S 2nd St, Milwaukee, Wisconsin, 53204; having the following essential characteristics:
 - 1. Network port: 2 minimum, ethernet TCP/IP
 - 2. Compatible with ControlLogix 5580 and backplane.

2.05 PLC APPLICATION PROGRAM

- A. Refer to specification requirements in Section 25 00 01 – Automation Control and Monitoring System.

2.06 INPUT/OUTPUT (I/O)

- A. Provide I/O in accordance with the I/O tables located on drawings and specifications Section 51 25 00 – Input Output Lists.

2.07 TOUCH SCREEN CONTROL HMI DISPLAYS

- A. IP-66 minimum.
- B. Dimensions: 19" screen minimum.
- C. Screen:
1. Backlit, 16.7 million colors.
 2. Luminance: 250 cd/m².
 3. Multi-touch capacitive technology.
- D. Rated Voltage:
1. Supply Voltage: 125VDC.
 2. Power Supply: 125VDC to 24VDC power supply as required in the PLC and HMI circuit.
- E. Automatic dimming feature when not in use.
- F. Response time: 14ms, minimum.
- G. Temperature -10 to 50 degree Celsius minimum.
- H. Humidity: At least 90 percent non-condensing.
- I. NEMA 250 type 12.
- J. Provide object library in HMI design with predefined scaling, alarms, and engineering units.
- K. HMI design to give operators access to devices and components comprising of the overall control system.
1. General Requirements:
 - a. Design a menu system that provides an easy mechanism to move between different windows with no more than three movements.
 - b. Screen updates of 500msec or faster.
 - c. In accordance with IEEE C37.1 and ISA 101.01.
- L. Control HMI Touch Screen:

1. Provide a series of screens for the control and monitoring of electrical devices and networking components.
2. One HMI for the unit control and monitoring system.
3. Three HMIs for monitoring only.
4. Design unit control screen that has parameters an operator would normally need and see when starting the generator.
5. Design to provide the following screens:
 - a. Start Permissive.
 - b. Start and Stop Sequences.
 - c. Parameters and Timers.
 - d. Maintenance Testing.
6. Design a network communications single line overlaying relevant network connections. Show the following information:
 - a. Network communication status.
 - b. Number of successful and unsuccessful poll attempts.
 - c. Number of control attempts on the device level.
 - d. LAN statistics and performance.
7. Variable tags shall show a sequence of characters not to be confused with the actual reading when data is not updating.

M. Annunciation Screen Tab:

1. Design Alarm Annunciator screen displaying real-time alarms that can be sorted, filtered, individually disabled/enabled, and silenced based upon multiple criteria such as alarm name, group, and time. Provide the following annunciation display:
 - a. Historical alarm display functioning similarly to an annunciator window.
2. Provide capability to display the following Master Alarm List including:
 - a. Alarm identifier.
 - b. Alarm priority.
 - c. Alarm group
 - d. Alarm setpoints.

2.08 HMI PHILOSOPHY, STYLE, AND INTERFACE DEVELOPMENT

A. HMI Philosophy development:

1. Analog indicators.
 - a. Display the range of the instrument being displayed, and in that range will be an indicator of the normal process limits.

- b. Above and below the indicator shall be white bands indicating abnormally high and low process values respectively.
 - c. Above and below the white bands are dark gray bands which indicate Highest and Lowest process values.
 - 2. HMI Trends.
 - a. Make trends available and accessible in all HMI applications.
 - b. All available information shall be easy to access for operators.
 - 1) The Y-axis and X-axis of trends shall be automatically ranged to a predetermined scale.
 - 2) Provide capability to alter the time-base and ranges without the requirement of keystrokes, making use of dragging the cursor to re-range.
 - 3. HMI Colors
 - a. Develop a high-performance HMI with USBR colors used and applied consistently as follows:
 - 1) Use of Colors:
 - a) Green for open, de-energized or stopped status.
 - b) Red for closed, energized, or running status.
 - c) Yellow for alert or condition out of normal.
 - b. Background color light gray to mitigate glare, contrast, color interference and operator fatigue.
 - c. Foreground colors shall be kept to a minimum with attention to line thickness.
 - d. Redundant indicators such as text and colors shall be used to help indicate abnormal conditions and draw the operator's attention quickly.
- B. HMI Style development:
 - 1. High-performance design in accordance with ISA 101-01 and ISA TR101.02.
 - 2. Pop-up information-driven display:
 - a. Information about each alarm shall be visible to operators.
 - 3. Process lines.
 - a. Process lines for static equipment shall be dark gray or black, with line thickness used to differentiate significance.
 - b. When using different line styles for solid, dotted and dashed.
 - 1) Limited to a maximum of three styles with the same line thickness.
 - c. Process lines shall enter and exit the screen consistently.

4. Equipment representation.
 - a. The exterior shall be outlined in black or dark gray with the interior shaded without gradients and uniform in color.
- C. HMI interface development:
 1. Process flow and navigation:
 - a. The process flow development shall be consistent in layout across all screens.
 - b. Screens shall include equipment designation identification, device function number, and alarm Tags (For Level 4 only).
 2. Static texts and live data:
 - a. Utilize simple font and color that is easily displayed on screen. Font size shall be large as practical.
 - 1) Text and numbers should not move or blink, making it difficult to read.
 - b. Abbreviations must be consistent across the entirety of all systems.
 - c. Use larger texts to identify duplicated equipment such as multiple motors.
 - d. Isolated words, titles, device, and equipment designations shall all be uppercase.
 - e. Live data shall be depicted distinctly from static text.
 - f. If live data is shown in a table, align the numbers on the decimal point and do not display leading zeros.
 - g. When required, the units of measurement are displayed in lower contrast and adjacent to the live data value.

2.09 HMI APPLICATION

- A. Refer to Section 25 12 10 – Programmable Logic Controllers, Section 25 00 01 – Automation Control and Monitoring Systems.
- B. Unit Human Machine Interface (HMI):
 1. The Contractor shall design HMI display screens and shall submit color screen shots of each applicable HMI screen and pop-up window.
 2. Providing one software application for both the HMI development and PLC programming is preferred.
 3. Graphically convey information to operators in accordance with these specifications, Section 25 12 10 – Programmable Logic Controller, ISA 101.01, and ISA TR101.02.
 4. Customizable screens, minimum, showing:

- a. Status of all I/O.
 - 1) Provide operator ability to see the value of any digital or analog I/O value.
- b. Generator Unit operational timers and auxiliary timers.
- c. Overall generator single line diagram
- d. Real time status of all unit breakers, disconnect switches, and auxiliary circuits.
- e. Alarming, Alarm History.
- f. Trouble and Trouble History.
- g. Communication status screen showing communication connections to the PLC and their real time status.
- h. Maintenance Test screen.
- i. Control Screen for starting and stopping generator and auxiliaries.
- j. Start Sequence Screen with status of each start permissive.
- k. Stop Sequence Screen with status of each event.
- l. Analog Gauge screens that display analog quantities with a gauge or meter type of graphic and visually displays the process quantity in reference to the high and low limits for that variable.
- m. Configuration Screens displaying the adjustable parameters for the application.
- n. Trend Screen that provides at a minimum:
 - 1) A Quick Trend screen that displays a list of analog values that when selected will pop-up a single variable trend.
 - 2) Provide a minimum of 5 selectable time periods such as, 30 seconds, 5 minutes, 30 minutes, 5 hours, and 30 hours.
- 5. Record all alarm statuses and user operations to the Data Historian Server.
 - a. Store information when alarms occur and record in chronological sequence.
 - b. Time stamped.

2.10 AUTOMATION CONTROL AND MONITORING HMI APPLICATION

- A. Refer to specification requirements in Section 25 00 01 – Automation Control and Monitoring Systems.
- B. Open Architecture (PC based) Software.
 - 1. The Contractor shall design HMI display screen. The Contractor shall submit color screen shots of each applicable HMI screen and pop-up window.

2. Providing one software application for the HMI development programming.
3. Communicate directly with the new generator unit control PLCs and existing SCADA RTU PLC.
4. Customizable screens, minimum, showing:
 - a. Status of all I/O.
 - b. Provide independent remote, local acknowledge and reset buttons.
 - c. Real time status of all alarms and statuses.
 - d. Alarming logging with time stamps.
 - e. Alarm reporting.
 - f. Communication status screen showing communication connections to PLCs within the same network and their real time status.
 - g. Trend Screen that provides at a minimum a Quick Trend screen that displays a list of analog values.
5. Unit and SS & Switchyard Annunciation Tab Screens (HMI):
 - a. The Contractor shall design HMI display screens and color screen shots of each applicable HMI screen and pop-up window.
 - b. Communicate directly with the distributed I/O assemblies and the unit's PLC. The gatehouse distributed I/O assembly shall be in the new gatehouse distributed I/O cabinet. The turbine level distributed I/O assembly shall be in the new distributed I/O cabinet on the turbine level.
 - c. Graphically convey information to operators in accordance with these specifications, Section 25 12 10 – Programmable Logic Controller, ISA 101.01, ISA TR101.02 and ISA TR18.2.1.
 - d. Alarms presented by worded alarm messages or by equipment and device designation.
 - e. Alarm indications shall flash, colored until acknowledged.
 - f. Alarming, Alarm History.
 - g. Trouble and Trouble History.
 - h. Provide Manual Reset capability through the annunciator HMI.
6. Provide Level 4 alarm Sequencing including the following:
 - a. Alarm designation.
 - b. Sequence diagram.
 - c. Sequence table.
 - d. Number of alarm points: total, active, and spare.
7. Display field contact operation: Normally Open (close to alarm), Normally Closed (open to alarm).

8. Provide network diagrams and single line diagrams comprising the Annunciation Systems.
9. Provide the following software pushbuttons:
 - a. Lock-in sequence feature that retains the alarm state until acknowledged.
 - b. Automatic Reset: Automatic return acknowledged alarms to normal when abnormal conditions return to normal.
 - c. Acknowledge: Manual acknowledges alarms.
 - d. Manual Reset: Operator manually resets individual alarms until the process conditions return to the normal and the manual reset pushbutton is operated.
 - e. Test: pushbutton to initiate sequence for testing of individual alarms.
 - f. Silence: Alarm still flashes and horn stops.
10. Alarms and Alarm History:
 - a. Provide additional alarms for PLC internal variables that have traveled out of range.
 - b. Provide alarms for loss of communications with any connected devices.
 - c. Provide alarms associated with the automation logic start and stop sequences.
 - d. Keep an alarm history log of the past 99 or more alarms.
 - e. Alarm history can only be cleared via a computer connection to the PLC, not by the HMI.
 - f. Provide time stamp on all alarms.
 - g. Time resolution: within 0.1 seconds.
 - h. Provide capability to download an alarm log report via a computer connection to the PLC or to a removable SD (memory) card.

2.11 MANAGED INDUSTRIAL ETHERNET SWITCH

- A. Refer to specification requirements in Section 27 21 29 – Network Switches.

2.12 SOURCE QUALITY CONTROL

- A. Government will inspect and witness testing of new equipment at supplier's manufacturing plant during the factory acceptance testing (FAT). Refer to Section 26 09 18 – Control Boards.

2.13 FACTORY ACCEPTANCE TEST

- A. Coordinate factory acceptance testing of automation systems with factory acceptance testing of the new control boards.

- B. Refer to Section 25 00 01 Automation Control and Monitoring Systems, Section 26 09 16 – Protection Systems, and Section 26 09 18 – Control Boards.

PART 3 EXECUTION

3.01 INSTALLATION

- A. One PLC shall be installed on control board, a distributed I/O system shall be installed on the Turbine level, and one distributed I/O system shall be installed at the Gatehouse.
- B. An HMI shall be installed at the control board, one outside of the front door of the powerplant, one at the Gatehouse, and one at the Turbine level.
- C. Refer to specification requirements in Section 26 09 18 – Control Boards.
- D. Perform installation in accordance with requirements in NFPA 70 and reference drawings.
- E. Install Category 6e UTP and optical fiber cabling in accordance with TIA-568-C.1, TIA/EIA-568-B.2, and TIA-568-C.3.

3.02 FIELD AND COMMISSIONING VERIFICATION

- A. Provide development, integration and installation services required to complete and turn over fully functional systems.
- B. Submit completed, signed, and approved Electrical Commissioning Forms in accordance with the following specification requirements:
1. Section 01 33 00 - Submittal Table 01 33 00A.
 2. Section 26 09 18 – Control Boards.
 3. Section 25 00 01 – Automation Control and Monitoring System.
 4. Section 26 05 90 – Wiring Check out and Tests.
 5. Section 27 20 01 – Fiber Optic Cable and Accessories.
 6. Section 27 21 29 – Network Switches.

END OF SECTION

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SECTION 26 04 10
ELECTRICAL DEMOLITION AND REMOVAL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Demolition and Removal of Electrical Equipment:
1. Payment: Lump sum price offered in the Price Schedule.
 - a. Includes cost of the following:
 - 1) Removal and disposal of equipment and low-voltage cable.
 - 2) Handling, removing, disposing, and salvaging of existing electrical equipment.
 - 3) Providing as-found existing condition drawings.
- B. Removal of Medium Voltage Power Cable Systems:
1. Measurement: Actual quantity of cable removed.
 2. Payment: Linear foot price offered in the Price Schedule.

1.02 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 26 04 10-1, Electrical Demolition and Removal Plan:
1. Include following in demolition plan:
 - a. Area where work will be done (by number or other description).
 - b. Outline procedures for accomplishing work including specific safety precautions to be taken, type and location of barricades, warning signs, and protective grounds and devices to be used.
 - c. Commencement time of work.
 - d. Duration of work.
 - e. Number of personnel and their classification.
 - f. Description of equipment to be used.
 - g. Information indicating that required materials are on site or data indicating shipping dates of materials not on site.
 2. Include following in removal plan:
 - a. Equipment to be removed.

- b. Sequence of removal.
- c. Method of dismantling equipment to remove material from building and site.

1.03 REGULATORY REQUIREMENTS

- A. Refer to Division 02 for proper handling and disposal of hazardous materials.

1.04 PROJECT CONDITIONS

- A. See Specifications Section 51 00 10 – Photographs for photographs of existing equipment.
- B. Contractor shall provide crane or other lifting equipment as needed.
- C. Powerhouse:
 - 1. Existing transformers KSA and KSB:
 - 1) Ratings: 13,800 to 480 volts, 500 kVA.
 - 2) Oil filled:
 - 1) KSA: 460 gallons.
 - 2) KSB: 450 gallons.
 - 3) Manufacturer: Moloney Electric Co.
 - 2. Existing disconnect switches FSA and YSA:
 - 1) Ratings: 15,000 volts, 400 amperes.
 - 2) Oil filled:
 - 1) FSA: 27 gallons.
 - 3) Manufacturer: Moloney Electric Co.
 - 3. Existing switchboard DSA:
 - a. Ratings: 480 volts, 800 amperes.
 - b. (5) section lineup includes:
 - 1) (2) main breakers: 480 volts, 800 ampere.
 - 2) (19) 480 volts breakers in panelboard sections.
 - 3) (24) 240/120V breakers in panelboard section.
 - 4) Control transformers.
 - 5) Relays, contactors, control switches, and meters.
 - 4. Existing transformer KSC:
 - a. Ratings: 75kVA, 480V-240/120V, single phase.

- b. Manufacturer: Matra Electric Inc.
- 5. Existing motor control center M1C:
 - a. Ratings: 480 volts, 300 amperes.
 - b. (1) section with (5) buckets.
 - c. Manufacturer: Trumbull Electric Manufacturing Company.
- 6. Existing Battery Chargers:
 - a. BSA-1 and BSA-2
 - b. Manufacturer: Gould
- 7. Existing medium voltage cables between unit breaker U1A1 and disconnect switch YSA:
 - a. Ratings: 15,000 volts.
 - b. Size: 1/0 AWG.
- 8. Existing low voltage cables between switchboard DSA and loads.
- 9. Existing Control Boards CSA panels and all devices mounted to panels, including side wings on the interior:
 - a. CSA1F
 - b. CSA2F
 - c. CSA3F
 - d. CSA1R
 - e. CSA2R
 - f. CSA5R
 - 1) Do not dispose of Siemens Siprotec Relay. It will be replaced and rewired.
- 10. Annunciator panels
 - a. Including the Annunciator Screens
- 11. Synchronizer Wing Panel and all devices
- 12. Cooling Water High Pressure Switch, Timer, and auxiliary relay
- 13. Unit Brake Air Pressure Switch and Auxiliary Relay
 - a. Location: Turbine Level Shaft
- 14. Creep Detector and Creep Detector Auxiliary relay
 - a. Location: Turbine Level Shaft
- 15. Thrust Bearing Oil Pressure Switch and auxiliary relay
- 16. Sump Shutdown Level Switch

a. Location M1B

D. Gatehouse:

1. Existing panelboard DJA:
 - a. Ratings: 480 volts, 100 amperes.
 - b. (7) 480 volts breakers in panelboard.
2. Existing transformer KJA:
 - a. Ratings: 5kVA, 480V-240/120V, single phase.
 - b. Manufacturer: General Electric.
3. Existing transformer KJB:
 - a. Ratings: 7.5kVA, 480V-240/120V, single phase.
 - b. Manufacturer: General Signal.
4. Existing panelboard LJA:
 - a. Ratings: 240/120 volts, 50 amperes.
 - b. (20) 120 volts breakers in panelboard.
5. Existing panelboard LJB:
 - a. Ratings: 240/120 volts, 100 amperes.
 - b. (14) 120 volts breakers in panelboard.
6. Existing low voltage cables between DJA and LJA and loads.
7. 125VDC Panelboard BJB
 - a. Rating: 125VDC, 40A
8. Existing Breaker Panel BJC
9. Existing control board MJB
 - a. Contactor for Heaters
10. Existing Control Board BJC
11. 48VDC System Panels
 - a. 48VDC Battery Bank in Battery Room
 - b. 48VDC Breaker Panel BJF
 - c. 48VDC Panelboard BJE
 - d. 48VDC Battery Charger BJD
12. Panel BJI
 - a. Including all devices
13. Panel BJJ

- a. Including all devices
- 14. Water Level Switches
 - a. BJE
 - 1) Including Panel
 - b. DJC
 - 1) Including Panel
 - c. HJA
 - 1) Including Panel
- 15. Penstock Relays
 - a. Open and Close
- 16. Penstock Open and Close Pressure Switches and Relays
 - a. Location: MJA
- 17. Penstock Position Limit Switches and Relays
 - a. Location: MJA
- 18. Gatehouse Ambient Temperature Switch and Relay
 - a. Location: MJA
- 19. Contactor for Spillway, Siphon Pipe, and Penstock Gate Heaters
 - a. Location: MJB
- 20. Canal High Water Alarm Unit
 - a. At Canal near Trash Rack
- 21. Gatehouse Door Alarm Intrusion Switch

1.05 SEQUENCING

- A. New equipment shall be onsite with approval drawings and factory test reports approved and complete before removal of existing equipment.

PART 2 PRODUCTS

2.01 MATERIALS AND EQUIPMENT

- A. Provide equipment, materials, slings, cranes, beams, and containers required for disassembly and removal of equipment.

PART 3 EXECUTION

3.01 GENERAL

- A. In accordance with Section 26 05 02 – Basic Electrical Materials and Methods.

3.02 PREPARATION

- A. Request outage in accordance with Section 01 14 30 – Interruption of Service.
- B. Follow instructions related to the Special Work Permit that will be issued by the COR to ensure that the equipment is safely de-energized, such as:
1. All sources of electrical and mechanical energy shall be defined and locked out in accordance with the facility Hazardous Energy Control Program.
 2. Procedures to be implemented by the COR in conjunction with contractor.
 3. After all sources of electrical energy have been locked-out and disconnected, feeder conductors shall be removed from conduits feeding structure.
- C. When all hazardous electrical energy sources have been isolated and removed then removal work may begin.
- D. All new equipment shall be on site before any equipment removal may begin.

3.03 REMOVAL OF EXISTING ELECTRICAL EQUIPMENT

- A. Disconnect all medium voltage circuits from assemblies.
1. Remove from conduits and dispose.
- B. Disconnect all low-voltage cables from terminals in the assemblies.
1. Cables that are to be reused and terminated in the new equipment shall be protected from damage during construction activities.
 2. If there are extra cables that will not be needed for the new installation, disconnect at both ends, remove from conduits, and dispose.
- C. Dismantle and remove existing equipment.
- D. Work includes:
1. Disconnecting all associated cable and bus.
 2. Disconnecting or dismantling electrical equipment cabinets and enclosures.
 3. Cutting anchor bolts flush with plant floor. Spray paint (red) locations for future reference.
 4. Removing equipment.

5. Disposing of equipment.

3.04 DISPOSAL

- A. Any materials deemed to be hazardous to be disposed of in accordance with Section 02 82 20 - Removal and Disposal of Asbestos Containing Materials.
- B. Dispose of removed equipment, cable, and materials in accordance with Section 01 74 00 - Cleaning and Waste Management.
 1. Do not dispose of medium voltage cables until COR agrees to the removal quantities claimed by the Contractor.

END OF SECTION

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SECTION 26 05 02
BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include cost of electrical equipment and materials in applicable prices offered in the Price Schedule.
 - a. Include cost of as-found existing condition drawings in price offered in the Price Schedule for Demolition and Removal of Electrical Equipment.

1.02 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1. ASME BPVC-IX 2023 Boiler and Pressure Vessel Code, Section IX
– Welding and Brazing Qualifications

B. ASTM International (ASTM)

1. ASTM A123/A123M-2017 Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
2. ASTM A153/A153M-2016 Zinc Coatings (Hot-Dip) on Iron and Steel Hardware

C. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C2-2023 National Electrical Safety Code (NESC)

D. InterNational Electrical Testing Association, Inc. (NETA)

1. NETA ATS-2025 Acceptance Testing Specifications

E. National Fire Protection Association (NFPA)

1. NFPA 70-2023 National Electrical Code (NEC)

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals

B. RSN 26 05 02-1, As-Found Existing Condition Drawings:

1. Includes but not limited to the following:

- a. Drawings markups to document as-found conditions at the start of the Work for all equipment inscope. As-found condition documentation to be performed prior to equipment release for construction, or removal of existing equipment.
- b. Provide as-built Government format equipment and interface drawings. This includes updating existing facility drawings to incorporate new equipment information.
- c. Furnish Government format updated drawings for all Government drawings impacted by the Work. Drawings to be updated include the following:
 - a) Facility layout drawings.
 - b) Facility schematic and wiring diagrams.
 - c) Facility single-line drawings.
 - d) Facility three-line drawings.

1.04 QUALIFICATIONS

- A. Welder Qualifications: In accordance with ASME BPVC-IX.
- B. Field testing organization and personnel: As specified in NETA ATS, Section 3, except may be accredited by other than NETA.

1.05 MAINTENANCE

- A. Provide special tools and accessories required for installation or normal operation and maintenance of equipment and installations. Include studs, bolts, and nuts other than United States Standard used in equipment or installation.
- B. Deliver special tools and spare parts for equipment to powerplant as approved by COR. Provide shipments with item list(s).
- C. Special tools and spare parts shall be onsite prior to installation of equipment.

PART 2 PRODUCTS

2.01 MATERIALS AND EQUIPMENT

- A. Conform to NFPA 70 and IEEE C2, and Section 01 60 00 – Product Requirements.
- B. Provide galvanized steel mounting brackets, support structures, bolts, nuts, and washers for major items of electrical equipment such as switchboard, transformers, motor control centers, and battery chargers in accordance with ASTM A 123 and A 153, unless otherwise specified.

- C. Provide silicon bronze nut and bolt hardware for making electrical power connections above 600 V. In addition, use conical Belleville-type washers in combination with a flat washer with a larger diameter and thicker than the conical washer, as described below, and per equipment manufacturer's engineering design.
1. Silicon bronze hardware in accordance with applicable ASTM standards.
 2. Bolt Stack up requirements:
 - a. Top and bottom conical washers per bolt stack-up:
 - 1) Bolt.
 - 2) Conical washer.
 - 3) Large diameter, thick cross-section, flat washer.
 - 4) Conductor pad #1.
 - 5) Conductor pad #2.
 - 6) Repeat of large diameter, thick cross-section, flat washer.
 - 7) Repeat of conical washer.
 - 8) Nut.
 - b. Bottom-conical washer per bolt stack-up:
 - 1) Bolt.
 - 2) Large diameter, thick cross-section, flat washer.
 - 3) Conductor pad #1.
 - 4) Conductor pad #2.
 - 5) Repeat of large diameter, thick cross-section, flat washer.
 - 6) Conical washer.
 - 7) Nut.
- D. Provide galvanized steel or non-corrosive metal for mounting bolts, nuts, and washers for minor items of electrical equipment and lighter weight items. Do not use cadmium-plated mounting hardware.
- E. Change designs as required where Contractor-furnished electrical equipment and materials differ in size, type, ratings, or other physical properties from designs in these specifications. Contracting Officer will approve changes at Contractor's expense, unless Contractor can demonstrate that changes are necessary regardless of manufacturer.
- F. Provide special tools and appliances furnished by manufacturer for maintenance and adjustment of manufacturer's electrical equipment.
- G. Grout materials: In accordance with Section 03 62 20 – Nonshrink Grout and Grouting Mortar for Equipment and Metalwork.

2.02 SOURCE QUALITY CONTROL

- A. In accordance with applicable IEEE standard.

2.03 SOURCE QUALITY ASSURANCE

- A. Government may inspect electrical equipment at supplier's manufacturing plant during fabrication and testing.
- B. Make available copies of equipment standards referenced in Division 26 at the suppliers manufacturing plant, as outlined in Section 01 42 10 – Reference. Standards.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Do not remove existing equipment or install electrical equipment until approval drawings and data and factory test reports for associated equipment have been approved by Government and equipment is onsite, unless otherwise approved by the COR.
- B. Electrical Installations, Assembly Operations, and Adjustments: Comply with NFPA 70 and IEEE C2.
- C. Make electrical installations complete and ready for service.
- D. Install electrical equipment in accordance with directions furnished by manufacturer's instruction books.
- E. Tighten nuts used in electrical equipment assembly with torque wrenches to torque values recommended by equipment manufacturers.
- F. Drill holes in bolted steel structures and provide fastenings required for mounting or installing electrical equipment and materials.
- G. Installation of electrical equipment includes:
 - 1. Leveling and grouting channel bases.
 - 2. Drilling holes, furnishing hardware, and assembling components to each other.
 - 3. Furnishing materials for and making all connections correctly in accordance with final wiring diagrams.
 - 4. Tagging wires and cables at each end.
 - 5. Testing.
- H. Grout electrical equipment installed on concrete foundations in place to provide full and even bearing.

3.02 REPAIR

- A. Repair damaged painted surfaces of equipment to match original finish.

3.03 CONTRACTOR FIELD QUALITY CONTROL AND TESTING

- A. In accordance with requirements of FAR 52.246-12 Inspection of Construction, IEEE C2, and the various equipment Sections.
- B. Field-made bolted power connections:
 - 1. Flat, silver-plated with a resistance value of 10 micro-Ohms or less.
 - 2. Resistance test equipment: Measure down to 0.1 micro-Ohms with an accuracy of +/- 0.3% or better.
 - 3. If connections do not provide acceptable results, re-torque by loosening connections slightly and re-tightening to manufacturer's torque values.

3.04 FIELD QUALITY ASSURANCE

- A. Government will inspect electrical equipment during construction. Acceptance of equipment will be made after equipment is operational.
- B. Government will witness equipment field tests. Provide proof of Government witness on field test reports.

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SECTION 26 05 10
CONDUCTORS AND CABLES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Conductors and Cables:

1. Measurement: Length of installed conductor or cable.
2. Payment: Linear foot price offered in the Price Schedule for applicable conductor or cable size, see Table 26 05 10 A – Conductor and Cable Sizes.

Table 26 05 10 A – Conductor and Cable Sizes

Type and Size of Conductor or Cable
1/C 12 AWG Copper THHN-2
1/C 10 AWG Copper THHN-2
1/C 8 AWG Copper THHN-2
1/C 6 AWG Copper THHN-2
1/C 4 AWG Copper THHN-2
1/C 1/0 AWG Copper THHN-2
1/C 3/0 AWG Copper THHN-2
1/C 350 MCM Copper THHN-2
1/C 500 MCM Copper THHN-2
3/C 12 AWG Copper Fire Rated
12/C 12 AWG Copper THHN-2
5/C 12 AWG Copper THHN-2
5/C 10 AWG Copper THHN-2

- a. Includes cost of work and materials for conductors and cables except as specified.

B. Cost:

1. Include cost of making electrical connections to equipment and devices in prices offered in the Price Schedule for equipment and devices that require connections.

1.02 REFERENCE STANDARDS

- A. ASTM International (ASTM)
 - 1. ASTM B8-2011(2017) Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
- B. National Electrical Manufacturers Association (NEMA)
 - 1. NEMA WC 57-2021 Control, Thermocouple Extension, and Instrumentation Cables
 - 2. NEMA WC 70-2021 Power Cables Rated 2000V or Less for the Distribution of Electrical Energy
- C. National Fire Protection Association (NFPA)
 - 1. NFPA 70-2023 National Electrical Code (NEC)
- D. Underwriters Laboratories, Inc. (UL)
 - 1. UL 486A-486B-2018 Wire Connectors

1.03 DESIGN REQUIREMENTS

- A. Determine conductor sizes, except where shown or specified, in accordance with NFPA 70. In the event of conflict between design requirements and NFPA 70, the more stringent requirement applies.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 10-1, Approval Data:
 - 1. Manufacturer's data.
- C. RSN 26 05 10-2, Field Test Reports and Cable Termination Sheets:
 - 1. Provide test reports for each cable provided by the Contractor.
 - 2. Cable Termination Sheets:
 - a. One termination sheet for each cable provided by the Contractor.
 - b. Termination sheets in computer generated, typed format using Microsoft Word.
 - c. One searchable, consolidated PDF of all termination sheets.
 - d. Termination sheets to include the following information:
 - 1) Name of Project.

- 2) Date terminations were verified.
- 3) Include date and personnel initials for the following:
 - a) Personnel who performed the terminations.
 - b) Personnel who checked the terminations and termination sheets. Must be different from the personnel who performed the terminations.
 - c) Personnel who performed the testing.
 - d) Government personnel who witnessed the testing.
- 4) Cable designation.
- 5) Quantity and size of wires in the cable.
- 6) Distance of cable.
- 7) Cable routing information:
 - a) Equipment designation at origin of cable.
 - b) Equipment designation at destination of cable.
- 8) List of wires in cable including spares.
 - a) Conductor number in the cable in sequential order.
 - b) Color insulation for each wire.
 - c) Termination locations for each wire:
 - i. Equipment designation.
 - ii. Terminal block designation.
 - iii. Terminal number.
 - d) Individual wire designations as indicated on approved Government format wiring diagrams. List spare wires as SPARE.
 - e) Continuity Check: Mark as PASSED for each wire after verification.
 - f) Insulation Resistance Test: Insulation Resistance in Megohms between conductor and ground.

D. RSN 26 05 10-3, As-Built Drawings.

1.05 DEFINITIONS

- A. Cable: Cable, cables, wire, or wires of one or more insulated conductors.
- B. Power Cable: For power loads including receptacle outlets, motors, alternating- and direct-current distribution circuits, HVAC and lighting circuits and equipment.

- C. Multiconductor Control Cable: For control, metering, indication, annunciation, and relaying circuits; and circuits not identified as power circuits.

1.06 PROJECT CONDITIONS

- A. Locations and lengths of existing cables indicated on drawings are approximate.

PART 2 PRODUCTS

2.01 CABLE

- A. Manufactured no more than 24 months prior to Notice to Proceed.
- B. Round, except for 2-conductor cable with parallel conductors.
- C. Conductors:
1. Copper in accordance with ASTM 8, class B or C.
 2. Stranded conductors.
- D. AWG or kcmil designation.
- E. Coverings or insulation: Suitable vertical position without injury to covering or deformation of insulation when supported in accordance with NFPA 70.

2.02 PHASE IDENTIFICATION

- A. Power Cable Phase Conductors:
1. 120/240, single-phase: Black.
 2. 277/480-volt, 3-phase: Black with brown, orange, yellow phase tape.
- B. Neutral (Grounded) Conductors: In accordance with NFPA 70.
- C. Equipment Grounding Conductors:
1. 6 AWG and smaller: Green insulation.
 2. Larger than 6 AWG: Green tape at both ends and visible points including junction boxes.

2.03 COLOR CODING

- A. Control and Instrumentation Cable:
1. Colored insulation or jacket compound. Do not apply color coatings to insulation or jacket surface.
 2. In accordance with NEMA WC 57, Table E-2.

B. Telemetry Cable: In accordance with 7 CFR 1755.390.

2.04 POWER CABLE AND MULTICONDUCTOR CONTROL CABLE

A. 600-volt rated insulation unless specified otherwise.

B. Type:

1. Single-conductor or multi-conductor.
2. Labeled type TC when installed in cable tray.

C. Suitable For:

1. Installation in raceways.
2. Installation in cable trays when required by specifications.

D. Insulation Type: THHN-2/THWN-2 or XHHW-2.

E. Multiconductor Overall Jacket:

1. Type: PVC (Polyvinyl chloride) or better, flame-retardant.
2. Jacket thickness:
 - a. 16 AWG conductors: 45 mils for 12 conductor or less. 60 mils for greater than 12 conductor.
 - b. 14 AWG conductors: 45 mils for 9 conductor or less. 60 mils for greater than 9 conductor.

F. Single-conductor power cable in accordance with NEMA WC 70. Multiconductor cable in accordance with NEMA WC 57. Cable UL listed and labeled.

G. Provide multi-conductor power cable with integral ground conductor. Size ground conductor in accordance with NEMA WC 70.

H. Shielding:

1. Power Cable: None.
2. Control Cable: Conform to NEMA WC 57.

I. Sizes and Number of Conductors Per Cable:

1. Current Transformer Cable: Five 10 AWG conductors per cable.
2. Potential Transformer Cable: Five 12 AWG conductors per cable.
3. Other Control Cables: Twelve 12 AWG conductors per cable or larger.
4. Power Cable: As indicated on drawings.

2.05 FIRE RATED CABLE

- A. UL fire-resistive classified for 2 hours.
- B. 600-volt rated insulation.
- C. Type: Single-conductor or multi-conductor.
- D. Maximum Continuous Temperature: 194 degrees F.
- E. Maximum Exposure Temperature: 1850 degrees F.
- F. Insulation Type: Magnesium Oxide.
- G. Jacket: No smoke, zero halogens.
- H. Copper sheath.

2.06 ACCESSORIES

- A. Single Conductor Markers:
 - 1. Type: Self-laminating-vinyl type, machine printable sleeve.
 - 2. Color: White with black text.
 - 3. Text size: As large as practical but at least 7 point (0.06-inch).
 - 4. 1-inch in length, minimum. Diameter suitable for installation on intended conductor.
- B. Cable Tags:
 - 1. Type: Non-conductive, heat resistant and non-corroding.
 - a. Engraved Valve Tags, as manufactured by R&R Identification, www.rridentification.com, or Valve Tags, as manufactured by Brady, www.bradyid.com, or equal with the following essential characteristics:
 - 1) Minimum 2-ply laminate, engravable.
 - 2) Round shape with attachment hole suitable for tie wraps or similar.
 - 2. Color: White with black text.
 - 3. Text size: As large as practical but at least 16 point (0.16-inch).
- C. Terminal connectors for control cables:
 - 1. Heavy-duty, nylon insulated sleeve, pressure-crimp-type with ring tongues.
 - 2. Tin-plated copper.
 - 3. Serrated inner barrel.

4. UL listed.
5. Compatible with conductor size and type of cable for which it is used.

D. Fire Rated Cable Terminations:

1. Suitable for use with supplied cables and approved for use by cable manufacturer.

PART 3 EXECUTION

3.01 INSTALLATION

- A. When using cable ties, do not over tighten, to ensure tie does not cut conductor insulation or cable outer jacket. Do not use cable ties to support cables.
- B. Block cable opening in sleeves under equipment or passing through blockouts, with silicone-foam, fire-retardant type material.
- C. Use building premises wire for lighting circuits and 120 volt receptacles unless shown otherwise.
- D. Cable Splicing:
 1. Do not splice cables.
- E. Cable Installed In Raceways:
 1. Do not pull cable into conduit until they have been cleaned and are free from obstructions and sharp corners.
 2. Draw a clean, dry, tight-fitting rag through conduit immediately before installing cable.
 3. Install cable using gradual and uniform pulling stresses. Do not exceed allowable pulling tensions and sidewall pressures as recommended by cable manufacturer.
 4. Install cable to prevent cuts or abrasions in insulation or protective covering and kinks in cable.
 5. When a lubricant is needed as an aid to pulling, use only soapstone or other suitable lubricant that is not injurious to cable insulation.
 6. When mechanical cable pulling equipment is used, use a tension meter release mechanism to ensure pulling tension remains below tensile strength of the cable.
 7. Remove and replace any cable damaged during installation.
 8. Where conductors carrying alternating current are installed in ferrous metal enclosures or metal raceways, arrange to avoid heating surrounding ferrous metal by induction. Group phase conductors together.
- F. Cable Installed in Vertical or Inclined Plane:

1. Install in accordance with NFPA 70.
2. Support with cable grips leaving slack spans between supports.
3. Clamp cable entering enclosures, junction boxes, or equipment with commercial cable clamps.

G. Grounding.

1. Provide one nonactive conductor grounded at both ends in each multiconductor control cable.
2. Ground shielding of each shielded control cable on both ends directly to equipment ground bus by method indicated on Standard Drawing 40-D-4335, Figure 8.
3. Ground spare conductors by wire jumpers from equipment terminal block points to equipment ground bus.

H. Wiring methods:

1. Terminate control cable with terminal connectors.
 - a. Make terminations with electric, hydraulic, or ratcheted type pressure crimper.
 - b. Do not use plier crimpers.
2. At termination point of multiconductor cable, form conductors into neat packs and tie conductors with self-locking cable ties.
3. Do not combine alternating- and direct-current circuits, or current and potential transformer circuits in same multiconductor cable.
4. Clamp cable entering equipment with commercial type cable clamps.
5. Leave sufficient length cable ends to make connections conveniently to equipment, fixtures, and devices.
6. Provide 5-conductor cable for each three-phase current and potential transformer circuit.
7. Retain sufficient length of conductors in a current transformer cable length to reach farthest terminal used to select current transformer ratios.
8. Retain spare single conductors at each end of a multiconductor cable in a length equal to longest single conductor of multiconductor cable.
9. Terminate spare conductors of multi-conductor cable at terminal blocks. Ground spare conductors by using wire jumpers between terminations and equipment ground bus.
10. Run both positive and negative leads of a direct-current circuit in same cable.

3.02 CABLE AND CONDUCTOR IDENTIFICATION

A. Power Cable:

1. Identify power conductors that are panelboard and switchboard feeders, motor control center feeders, and motor power conductors with colored insulation or colored phase tape in accordance with this section. An additional conductor marker is not required.
2. Provide a single-line marker on the load end of feeder and branch-circuit conductors that originate at panelboards, switchboards, and motor control centers. The marker to include designation of source equipment and circuit number.

B. Conductor Markers:

1. Text to be machine or computer generated. Handwritten text is not acceptable.
2. Provide individual conductors of multiconductor control cable, instrumentation cable, and telemetry cable with conductor marker. Print conductor designation, as shown on approved schematic diagrams, on first line. Print conductor destination and device terminal number on second line.
3. A single line conductor marker will be acceptable provided the conductor designation is enclosed in parenthesis followed by the conductor destination.
4. Do not heat shrink marker onto conductor. Allow marker to rotate about the conductor.
5. Provide markers for spare conductors of multi-conductor cables. The marker to include cable designation and the word "SPARE". Numerically sequence the spare conductor markers e.g. (1CSA-CSB-SPARE1, 2CSA-CSB-SPARE2, etc.).

C. Cable Tags:

1. Engrave tags with cable designations as shown on approved wiring diagrams. Cable designation to be a single-line of text that includes cable number, originating equipment designation, and destination equipment designation. Develop cable designations when not shown. Examples of cable designations are:
 - a. 1M1C-LOP – First cable from motor control center M1C to lubricating oil pump).
 - b. 3DJA-KJA – Third cable from panelboard DJA to transformer KJA.
 - c. Note, if the second part of the cable designation (destination equipment) would not fit on the tag, it can be eliminated for the tag marking.
2. Securely attach cable tags to multi-conductor cables near ends and single-conductor power cables, as directed. Attach to cable with standard nylon self-locking tie wraps or cable ties. Use two wraps, one firmly around cable, second used to attach tag to tie wrap on cable so tag can dangle freely. Pull tag tie wrap so that loop is approximately ½-inch in diameter.

3.03 CONTRACTOR FIELD QUALITY TESTING

- A. Perform testing after installation, but before connection to equipment.
- B. Test cables in large groups to minimize number of testing sessions.
- C. Notify Contracting Officer's Representative, in writing, of date, time, and cables to be tested at least 5 working days before testing. Testing dates to be mutually agreeable between Contractor and Contracting Officer's Representative.
- D. Perform following tests:
 - 1. Continuity test: Test continuity of each conductor prior to performing insulation resistance test.
 - 2. Insulation resistance tests for power, control, and instrumentation cables:
 - a. Measure insulation resistance between each conductor and station ground with all other conductors in same cable or conduit grounded.
 - b. Test voltage: 1,000 volts direct-current, minimum.
 - c. Minimum acceptable insulation resistance: $R \text{ in } M\Omega = (\text{rated voltage in kV} + 1) \times 1000 / \text{length in ft.}$
 - 3. Insulation resistance test for telemetry cable:
 - a. Test cable before making ground connections,
 - b. Test voltage: 1,000 volts direct-current, minimum.
 - c. Measure insulation resistance between each conductor and remaining conductors, and between conductor and shield.
- E. If any conductor fails continuity or insulation resistance testing, remove and replace cable and conductor.

END OF SECTION

SECTION 26 05 20 GROUNDING AND BONDING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for items requiring grounding and bonding.

1.02 REFERENCE STANDARDS

- A. ASTM International (ASTM)
1. ASTM B 3-13(2018) Soft or Annealed Copper Wire
 2. ASTM B 8-11(2017) Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
 3. ASTM B 228-11a(2016) Concentric-Lay-Stranded Copper-Clad Steel Conductors
- B. Institute of Electrical and Electronic Engineers (IEEE)
1. IEEE 142-2007 Grounding of Industrial and Commercial Power Systems
 2. IEEE 837-2014 Qualifying Permanent Connections Used in Substation Grounding
 3. IEEE C2-2017 National Electrical Safety Code (NESC)
- C. InterNational Electrical Testing Association (NETA)
1. NETA ATS-2025 Acceptance Test Specification for Electrical Power Distribution Equipment and System
- D. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electric Code (NEC)
- E. Underwriters Laboratories (UL)
1. UL 467-2013 Grounding and Bonding Equipment

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.

- B. RSN 26 05 20-1, Approval Data:
 - 1. Manufacturer's technical catalog data
- C. RSN 26 05 20-2, As-Built Drawings:
- D. RSN 26 05 20-3, Field Test Reports.

1.04 PROJECT CONDITIONS

- A. Locations of existing ground conductors indicated on drawings are approximate.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Ground Cable:
 - 1. Annealed bare-copper cable:
 - a. ASTM B 8, class B.
 - b. Solid wires used in forming the copper cable: ASTM B 3.
 - 2. Concentric stranded.
 - 3. Minimum Sizes:
 - a. Ground risers and electrode conductors: 4/0 AWG.
 - b. Equipment grounding conductors:
 - 1) The more stringent of size shown on drawings or required by NFPA 70.

2.02 ACCESSORIES

- A. Cable Connectors:
 - 1. Compression type.
 - 2. Current-carrying capacity: Equal to cable with which they are used.
 - 3. Connectors for ground cables, including fittings, lugs, bolts, nuts, and washers: Copper alloy containing not more than 4 percent zinc.
 - 4. Bolted solderless or compression type connectors: IEEE 837.
 - 5. Ground connectors that will be direct buried or embedded in concrete: Listed for direct burial use in accordance with the requirements of UL 467.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with NFPA 70, IEEE C2, and IEEE 142.
- B. Connect equipment, metal conduit, steel structures, metal pull box covers, exposed metal surfaces, and other miscellaneous metalwork to grounding system as required by NFPA 70.
- C. Ground equipment and miscellaneous metalwork that is required to be grounded but is not indicated on drawings to grounding system.
 - 1. Ground risers: Connect grounding electrode system to ground bus.
- D. Connections:
 - 1. Number of grounding connections for equipment may be one, two, or more in some cases.
 - 2. Remove paint, enamel, scale, oil, grease, or other foreign nonconductive material from point of contact on metal surfaces before making ground connections.
 - 3. Repair damaged or removed paint or galvanizing material on metal finishes.

3.02 CONTRACTOR FIELD QUALITY TESTING

- A. Test integrity of grounding system:
 - 1. In accordance with NETA ATS.
 - 2. Perform as soon as possible after all ground connections are completed.
 - 3. Low-impedance continuity measurement by computer-based grounding multimeter to determine resistance between:
 - a. Main grounding system and each new electrical equipment frame or ground bus.
 - b. Main grounding system and all other connections to the ground mat. This includes but is not limited to ground connections to vault, ground cables, junction boxes, and pull boxes.
 - 4. Investigate and repair locations where acceptable levels of resistance are exceeded.
- B. Test resistance of grounding system:
 - 1. Notify COR at least 24 hours in advance of time of test in order that test may be witnessed by COR.
 - 2. Perform test within 30 days after completion of structural concrete work.
 - 3. Measured resistance: Less than 5 ohms.

END OF SECTION

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SECTION 26 05 13
MEDIUM-VOLTAGE POWER CABLE SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Medium-Voltage Power Cable:
1. Measurement: Length of installed cable.
 2. Payment: Linear foot price offered in the Price Schedule.
 - a. Includes cost of cable accessories.

1.02 REFERENCE STANDARDS

- A. Insulated Cable Engineers Association (ICEA)
1. ICEA S-97-682-2013 Utility Shielded Power Cables Rated 5 Through 46 kV
- B. Association of Edison Illuminating Companies (AEIC)
1. AEIC CS8-2020 Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV
- C. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE 48-2020 Alternating-Current Cable Terminations 2.5 kV Through 765 kV
 2. IEEE 386-2016 Separable Insulated Connectors Systems for Power Distribution Systems Above 600 V.
 3. IEEE 422-2012 Design and Installation of Cable Systems in Power Generating Stations.
 4. IEEE C2-2023 National Electrical Safety Code (NESC)
- D. InterNational Electrical Testing Association (NETA)
1. NETA ATS-2025 Acceptance Test Specification for Electrical Power Distribution Equipment and System
- E. National Electrical Manufacturers Association (NEMA)
1. NEMA WC 74-2017 5-46 KV Shielded Power cable for Use in the Transmission and Distribution of Electric Energy
- F. National Fire Protection Association (NFPA)

- 1. NFPA 70-2023 National Electrical Code (NEC)
- G. Underwriter's Laboratory (UL)
 - 1. UL 486A-486B-2018 Wire Connectors
- H. Bureau of Reclamation (USBR)
 - 1. FIST 4-1B (2020) Maintenance Scheduling for Electrical Equipment available at:
www.usbr.gov/power/data/fist_pub.html

1.03 PERFORMANCE REQUIREMENTS

- A. Furnish, install, and make operable 15-kV cable and cable connections.
- B. Determine actual cable length and cable termination configurations.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 – Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 13-1, Approval Drawings and Data:
 - 1. Manufacturer's catalog data and drawings and bills of material.
 - a. Cable cross-section drawing to identify materials and thicknesses.
 - 2. Field test plan:
 - a. Indicate recommended test methodology and test voltages. Government reserves right to instruct Contractor on methodology and test voltage after submittal.
- C. RSN 26 05 13-2, Factory Production Test Certifications.
- D. RSN 26 05 13-3, Field Test Notification.
- E. RSN 26 05 13-4, Field Test Reports.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Deliver cable on reels which are sufficiently sturdy to withstand normal handling during shipping, hauling, and installation.
- B. Seal ends of cable on reels from moisture with suitable heat-shrinkable caps and mastic compounds.
- C. Do not drag cable across ground, fences, or other sharp projections during shipping or placement of cable reels at the jobsite.

PART 2 PRODUCTS

2.01 POWER CABLE

- A. Conform to NEMA WC-74 and AEIC CS8 and CS9, where applicable.
- B. Ratings:
 - 1. Rated Voltage: 15 kilovolts.
 - 2. Operating Voltage: 13,800 kilovolts.
 - 3. Frequency: 60 hertz.
 - 4. Operating Temp.: 105 degrees C, maximum.
 - 5. Average Insulation Thickness: 115 mils 133% Insulation.
- C. Conductor:
 - 1. Material:
 - a. Class B, stranded, copper conductors; or compact-round, stranded copper conductors.
 - 2. Size: As shown on drawings..
 - 3. Number of Conductors in Cable: One.
- D. Insulation:
 - 1. Solid-dielectric.
 - 2. EPR (ethylene-propylene rubber), rubber-insulated.
- E. Screens:
 - 1. Extruded, semi-conducting EPR conductor strand screen and extruded thermosetting semi-conducting EPR insulation screen.
- F. Shield: 5 mil copper tape, coated, helically applied.
- G. Jacket:
 - 1. Overall PVC compound jacket.
 - 2. Flame-retardant.
- H. Manufacture using triple-extrusion process such that conductor shield and insulation are extruded together and cross-linked simultaneously. Extrude insulation in tandem with and inseparably bond to conductor screen.
- I. Suitable for installation in conduit.
- J. Date of manufacture: No more than 24 months before Notice-to-Proceed.

2.02 ACCESSORIES

A. Cable Terminators (Potheads):

1. Suitable for outdoor use.
2. Equip with skirts.
3. Comply with IEEE 48, Class 1A.
4. Basic impulse insulation level: 95 kilovolts, minimum.
5. Heat shrinkable or cold-shrinkable, molded stress-cone type:
 - a. Continuous maximum rated operating temperature: Not less than 105 degrees C.
 - b. Emergency overload rating: Not less than 130 degrees C.
 - c. Captive termination stress control design with Hi-K stress control compound and EPDM type rubber tubing.
 - d. Installation and termination without requiring silicone grease.
6. Clamp-type line and grounding terminals suitable for conductors furnished:
 - a. One-piece, seamless tin-plated copper.
 - b. Chamfered ends for conductor insertion
 - c. UL 486A-486B Listed.
 - d. Continuous maximum rated operating temperature: Not less than 105 degrees C.
 - e. Emergency overload rating: Not less than 130 degrees C.
7. Insulate portion of base of cable terminator that is connected to cable shield from ground. Bring out insulated grounding lead to readily accessible point. Size lead to carry currents that cable shield is capable of carrying.
8. Use: For terminating cable in equipment.

2.03 CONTRACTOR SOURCE QUALITY TESTING

A. Factory Cable Production Tests:

1. In accordance with AEIC CS8 and CS9.
2. Provide certification that cable has successfully passed tests.
 - a. Provide list of tests conducted.
 - b. Do not submit actual test reports.

B. Factory Cable Terminators (Potheads) Production Tests:

1. In accordance with IEEE 48.

2. Provide certification that equipment has successfully passed tests.
 - a. Provide list of components tested and tests conducted.
- C. Design Tests:
 1. Government reserves the right to request design test certification on any equipment:
 - a. Contractor will be notified by the COR if this option is exercised after submittal of approval drawings and data.
 - b. Design tests in accordance with above referenced standards.

PART 3 EXECUTION

3.01 INSTALLATION

- A. In accordance with:
 1. NFPA 70, IEEE 422, and IEEE C2.
 2. Manufacturer's recommendation on supports and terminators.
- B. Reel procedures:
 1. Remove cable from reel by turning mounted reel.
 2. Do not end feed from reel.
 3. Do not allow cable reel to rest upon reeled cable.
 4. Inspect cable as removed from reel for visible defects.
- C. Do not drag cable across ground, fences, or other sharp projections.
- D. Where conductors carrying alternating current are installed in ferrous metal enclosures or metal raceways, arrange to avoid heating surrounding ferrous metal by induction. Group phase conductors together.
- E. Installation in raceways:
 1. Do not exceed cable pulling tensions and bending radius recommended by manufacturer.
 2. Use lubrication in accordance with manufacturer's recommendations.
 3. Pull cable by hand or use tensiometer or release mechanism on cable pulling equipment to ensure pulling tension remains below tensile strength of cable.
 4. Pull cable in each conduit run at the same time.
- F. Terminate or seal cut ends of cable immediately after cutting operation. Seal ends of cable against moisture with waterproof end caps.

G. Do not splice cables.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Perform testing after cable terminations have been installed, but before connections have been made to bus or apparatus.

B. Notify COR of date, time, and cables to be tested. Test dates shall be mutually agreeable between Contractor and COR.

C. In accordance with NETA ATS.

D. Insulation Resistance Test:

1. Perform test prior to DC high potential test.
2. Measure cable insulation resistance with megohm meter.
3. Test voltage: 1,000 volts direct current.
4. Reject cable if resistance is less than 25 megohms.

E. DC Ramp Test:

1. Request test values from Government after cable has been submitted for approval and prior to testing.
2. To avoid charge injection, test voltage shall not cause internal stress to exceed 10kV/nm (254V/mil).

F. Record the following data on test report for each test performed and for each conductor tested:

1. Date of test.
2. Name or names of Contractor's personnel who performed test.
3. Name of Government inspector who witnessed test.
4. Conductor identification.
5. Provide actual resistance values measured in ohms. Use of "Pass", "Fail", check marks, or infinity symbol are not acceptable.
6. Overall result of tests: Pass or Fail.

END OF SECTION

SECTION 26 05 33 RACEWAYS AND BOXES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Electrical Conduit:

1. Measurement: Length of installed conduit as shown on drawings.
2. Payment: Linear foot price offered in the Price Schedule for applicable conduit size and type, see Table 26 05 33 A – Conduit Sizes.

Table 26 05 33 A – Conduit Sizes

Size of Conduit
1-inch RMC
3-inch RMC
5-inch RMC
4-inch Schedule 80 PVC
1-inch LFMC

- a. Includes cost of work and materials for conduit except as specified.

B. Core Drilling Holes in Concrete:

1. Measurement: Length of holes core drilled.
2. Payment: Linear foot price offered in the Price Schedule for applicable hole size.

C. Cable Vaults:

1. Measurement: Number of vaults installed.
2. Payment: Per each price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI)

1. ANSI C80.1-2020 Electric Rigid Steel Conduit (ERSC)

A. National Electrical Manufacturers Association (NEMA)

1. NEMA FB 1 - 2014 Fittings, Cast Metal Boxes, And Conduit Bodies For Conduit, Electrical Metallic Tubing And Cable

- 2. NEMA TC 2 - 2020 Electrical Polyvinyl Chloride (PVC) Conduit
- 3. NEMA TC 3 - 2016 PVC Fittings for Use With Rigid PVC Conduit and Tubing.
- B. Underwriters Laboratories (UL)
 - 1. UL 360-2013 Liquid-Tight Flexible Metal Conduit
 - 2. UL 651-2011 Schedule 40 and 80 Rigid PVC Conduit

1.03 SYSTEM DESCRIPTION

- A. Design Requirements:
 - 1. Determine routing of exposed conduit when not indicated on drawings.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 33-1 Approval Drawings and Data:
 - 1. Proposed routing of conduit at powerplant and gatehouse.
- C. RSN 26 05 33-2, As-Built Drawings.

PART 2 PRODUCTS

2.01 CONDUIT

- A. General:
 - 1. UL listed.
 - 2. Sizes as shown on drawings.
 - 3. Rate for intended purposes or labeled by nationally recognized testing laboratory (NRTL) accredited by Federal Occupational Safety and Health Administration (OSHA) when applicable.
- B. Rigid steel conduit and attached fittings:
 - 1. Zinc-coated and in accordance with ANSI C80.1.
- C. Plastic Conduit and Fittings:
 - 1. Conform to UL 651 and NEMA TC 2 and TC 3.
 - 2. Schedule 80.

3. Plastic conduit for outdoor use:
 - a. Sunlight resistant.
 - b. Temperature rating: Not less than 90 degrees C.
- D. Liquidtight Flexible Metal Conduit (LFMC):
 1. Interlocked steel construction with flame-retardant PVC jacket.
 2. Sunlight resistant.
 3. Color: Grey.
 4. Conform to NFPA 70 and UL 360.
 5. Fittings conform to NEMA FB 1.

2.02 CONDUIT ACCESSORIES

- A. UL Listed.
- B. Conform to NEMA FB 1 as applicable.
- C. Steel materials and hardware: Galvanized.
- D. Provide fittings required to complete electrical conduit systems. Includes, but not limited to, caps, connectors, couplings (excluding expansion and expansion-deflection couplings), nipples, reducers, elbows, pipe plugs, locknuts, bondnuts, bushings, seals (excluding wall penetration seals), and any other fittings .
- E. Provide devices required to complete electrical conduit systems and to fasten, clamp, attach, and support each conduit in place.
 1. Includes threaded joint compound; protective sealant; materials for sealing ends of conduits terminating at outdoor boxes, panelboards or cabinets; supports and clamps.
 2. Complete with bolts, washers, and nuts.
- F. Expansion Couplings:
 1. Furnish for longitudinal movement only.
 2. Zinc coated, watertight type.
 3. Capable of at least 4-inch expansion/contraction.
 4. Expansion couplings for plastic conduit shall expand at least 4 inches for each 100 feet of run and for each 100 degrees F temperature change.
- G. Embedded Cast or Malleable Metal Outlet Boxes:
 1. Type FD1 (Single-gang with extension):

- a. Rectangular, single-gang, deep-wiring device outlet boxes with threaded hub connections for rigid steel conduit.
 - b. Equip with:
 - 1) Four 1-inch threaded hubs.
 - 2) 1-inch-deep cast or malleable metal extension ring.
 - 3) 1-inch bushed elbows where indicated on drawings.
 - 4) Plugs for any unused openings.
 - 5) Reducers as required.
 - c. Boxes and extension rings: Iron alloy with corrosion-resistant finish.
2. Type FD2 (Two-gang with extension):
- a. Rectangular, two-gang, deep-wiring device outlet boxes with threaded hub connections for rigid steel conduit.
 - b. Equip with:
 - 1) Four 1-inch threaded hubs.
 - 2) 1-inch-deep cast or malleable metal extension ring.
 - 3) 1-inch bushed elbows where indicated on drawings.
 - 4) Plugs for any unused openings.
 - c. Boxes and extension rings: Iron alloy with corrosion-resistant finish.
3. Type GRF:
- a. 4-inch-round outlet boxes approximately 3-1/8 inches deep, with threaded-hub connections for rigid steel conduit.
 - b. Equip with:
 - 1) Four 1-inch threaded hubs.
 - 2) Plugs for any unused openings.
 - 3) Reducers as required.
 - 4) Tap for 3/8-inch fixture stud as required.
 - c. Boxes: Iron alloy with corrosion-resistant finish.

H. Sealing Materials:

- 1. General use:
 - a. Non-toxic.
 - b. Non-shrinking.
 - c. Fire retardant type.
 - d. Seal ends of conduit against liquid, gas, and vapor seepage.

- I. Adhesive: In accordance with conduit manufacturer's recommendations.
- J. Polyethylene warning tape.
 - 1. Width: 6 inches.
 - 2. Color: Red.
 - 3. Copy: Continuously imprinted with "CAUTION BURIED ELECTRIC LINE BELOW".
 - 4. Suitable for direct burial.

PART 3 EXECUTION

3.01 INSTALLATION

- A. In accordance with Section 26 05 02 – Basic Electrical Methods and Materials.
- B. Layout and Procedures:
 - 1. Determine routing of exposed conduit when not indicated on drawings.
 - 2. Ensure that conduit runs and bends are free from kinks, indentations, or flattened surfaces.
- C. Preparation:
 - 1. Remove burrs and sharp corners at ends of metal conduit.
 - 2. Coat male threads of rigid metal conduit joints with suitable graphite or zinc sealing material before making joints.
- D. Install conduit with necessary fittings and supports.
 - 1. Support conduit as indicated on Standard Drawing 40-D-7094.
 - 2. Tighten conduit joints securely to ensure electrical continuity and to prevent entrance of moisture or foreign material.
 - 3. Provide bushings or Chase-type nipples on ends of conduit to protect insulated conductors from abrasion.
 - 4. Use locknuts and bondnuts to provide tight ground connections between conduit and boxes, panelboards, and cabinets.
- E. Bends:
 - 1. Install smooth, gradual bends to permit pulling insulated conductors without undue stress or damage to conductors or conduit.
 - 2. Bend conduit cold to prevent damage to protective coating.
- F. Direct buried conduit:

1. Use plastic (PVC) conduit.
 2. Depth, minimum: 24 inches.
 3. Install 2 inches of sand around each conduit.
 4. Backfill and compact in accordance with Section 31 23 11 – Earthwork for Structures.
- G. Exposed conduit runs:
1. Straight and parallel with each other and with centerlines of room or structure.
 2. Support conduit rigidly from wall or ceiling within 3 feet of each outlet box, junction box, cabinet, or fitting; and at intervals of not more than 5 feet.
 3. Installation includes, where required, drilling holes in bottom, side, or top of enclosures or plates of other electrical equipment.
 4. Tighten conduit securely.
 5. Support conduit rigidly in place.
 6. Make connections to outdoor boxes watertight.
- H. Expansion Couplings: Use where exposed or embedded conduits cross expansion or contraction joints. Install expansion couplings in accordance with manufacturer's instructions.
- I. Liquidtight Flexible Metal Conduit (LFMC):
1. Use for connections to motors and electrical equipment that is subject to movement, vibration, or misalignment; where flexibility is required; or where shown.
 2. Maximum length of conduit section is not to exceed 6 feet. Lengths exceeding 6 feet shall be approved by COR.
 3. Secure and support conduit sections in accordance with NFPA 70.
- J. Seals/Sealing:
1. Seal ends of conduits that terminate at boxes, panelboards, or cabinets with sealing material or sealing bushings to prevent air circulation through conduits into boxes, panelboards, or cabinets. Install seals in accordance with manufacturer's instructions.
 2. Install such that top of handhole is exposed 2" in relation to finished grade level.
 3. Backfill and compact native material around handhole and as approved by COR.

END OF SECTION

SECTION 26 05 36

CABLE TRAYS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cable Trays:
1. Measurement: Length of installed cable tray.
 2. Payment: Linear foot price offered in the Price Schedule:
 - a. Includes cost of work and materials for cable tray.

1.02 REFERENCE STANDARDS

- A. Aluminum Association (AA)
1. AA ADM-2015 Aluminum Design Manual
- B. American Institute of Steel Construction (AISC)
1. AISC 360-2016 Steel Construction Manual 15th Edition
- C. ASTM International (ASTM)
1. ASTM A123/A123M-17 Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
 2. ASTM A153/A153M-16a Zinc Coating (Hot-Dip) on Iron and Steel Hardware
 3. ASTM A385/A385M-20 Providing High-Quality Zinc Coatings (Hot-Dip)
 4. ASTM A563-21a Carbon and Alloy Steel Nuts
 5. ASTM A780/A780M-20 Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
 6. ASTM F844-19 Washers, Steel, Plain (Flat), Unhardened for General Use
- D. American Welding Society, Inc. (AWS)
1. AWS D1.1/D1.1M-20 Structural Welding Code – Steel
 2. AWS D1.2/D1.2M-14 Structural Welding Code - Aluminum
- E. National Electrical Installation Standards (NEIS)
1. NEIS 105-2015 Installing Metal Cable Tray Systems

- F. National Electrical Manufacturers Association (NEMA)
 - 1. NEMA VE 2-2018 Cable Tray Installation Guidelines
- G. National Fire Protection Association (NFPA)
 - 1. NFPA 70-2023 National Electrical Code (NEC)

1.03 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 – Submittals and Section 01 33 26 – Electrical Drawings and data.
- B. RSN 26 05 36-1, Approval Drawings and Data:
 - 1. Approval Drawings:
 - a. Show plan, elevations, and typical cross-sections of cable tray system, with overall dimensions and weights.
 - b. Show locations and types of cable tray supports for both horizontal and vertical run cable trays.
 - c. Show typical connection and splice details.
 - 2. Manufacturer's Product Data:
 - a. Cable trays, fittings, and accessories.
 - b. Supports and hardware.
 - c. Certification that the cable tray system meets the specified requirements.
- C. RSN 26 05 36-2, As-Built Drawings.

1.04 QUALIFICATION OF WELDERS

- A. Qualify welders in accordance with AWS D1.1 and D1.2 using procedures, materials, and equipment of type required for work.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Protect from corrosion, deformation, and other types of damage.
- B. Store items in an enclosed area free from contact with soil and weather.
- C. Remove and replace damaged items with new items.

PART 2 PRODUCTS

2.01 MATERIALS

A. Cable Trays, Fittings, and Accessories:

1. Aluminum or Aluminum Alloy cable trays and accessories as manufactured by Cooper B-Line, Highland IL 62249; Cablofil, PW Industries, Mascoutah IL 62258; or equal, having the following essential characteristics:
 - a. Cable trays and fittings:
 - 1) Formed channel sides or channel with binding strip so there are no sharp edges or projections.
 - 2) Type: As shown:
 - a) Ladder with rungs at 9-inch maximum spacing.
 - i. Rungs: 1-inch-wide channels, minimum.
 - b) Channel/Solid Bottom Type.
 - 3) Side height, minimum: 6 inches.
 - 4) Width: 16 inches.
 - 5) Cable Load: 15 to 20 pounds per lineal foot, minimum.
 - 6) Fitting radius: 12-inches, minimum.
 - 7) Allowable tray deflection: 1/180 of span between supports, maximum, when tray is loaded with manufacturer's design load.
 - 8) Spacing between vertical run supports: 5-feet maximum.

B. Cable Tray Supports: Bracket- or trapeze-type, or combination of both.

1. Bracket-Type:
 - a. Brackets: Brackets equal to Globe Strut No. G-5040; Unistrut Corp. No. P2544; Cooper B-Line Systems No. 297; or equal, having the following essential characteristics:
 - 1) Cantilever or braced.
 - 2) Length: As required to support selected cable tray.
 - 3) Single-channel or double-channel as required.
 - 4) Channel: 1 5/8-inch-wide by 1 5/8-inch-deep with 7/8-inch opening.
 - 5) Finish:
 - a) Galvanized after fabrication, or

- b) Fabricate brackets from material that is coated with 1.25 ounce/ft² of zinc in accordance with ASTM A123 and A385.
 - 2. Trapeze-Type:
 - a. Threaded rods, screws, clamps, nuts, and washers:
 - 1) Threaded rods: Material and size to be determined by cable tray manufacturer.
 - 2) Screws: As recommended by cable tray manufacturer.
 - 3) Clamps: As recommended by cable tray manufacturer.
 - 4) Nuts: ASTM A563.
 - 5) Washers: ASTM F844 (unhardened for general use).
 - 6) Finish: Hot-dip galvanized.
 - C. Arc-Welding Electrodes:
 - 1. Filler metal and required shielding gases or fluxes: AWS D1.1.
 - 2. Filler metal for steel: Minimum tensile strength of 70,000 pounds-per-square-inch (psi).
- 2.02 FABRICATION**
- A. Fabricate metalwork in accordance with AISC 360, AA ADM, and these specifications.
 - 1. Perform welding and related work in accordance with AWS D1.1 and AWS D1.2.
 - B. If straightening is necessary, use methods that will not injure metal.
 - C. Galvanizing:
 - 1. Galvanize threaded rods, screws, clamps, nuts, and washers in accordance with ASTM A153. Remove excess spelter by centrifugal spinning.
 - 2. Galvanize items of metalwork as specified or shown on drawings. Use hotdip galvanizing, where required after fabrication, in accordance with ASTM A123 and ASTM A385.
 - 3. After shop work completion and before galvanizing, if required, clean material of rust, loose scale, dirt, oil, grease, slag from welded areas, and other foreign substances.
 - 4. Fabricator's Galvanizing Repair:
 - a. Repair in accordance with Section 09 96 20 – Coatings.

PART 3 EXECUTION

3.01 PREPARATION

- A. Where locations and dimensions of cable trays shown on drawings are tentative or subject to change dependent upon equipment furnished, confirm locations and dimensions prior to fabrication of cable tray system.

3.02 INSTALLATION

- A. Install cable tray system at locations and elevations shown on drawings and in accordance with manufacturer's instructions.
- B. Ensure that supports for cable tray system are properly attached to structural framing members of metal building system.
- C. Install cable tray system in accordance with NEIS 105 and NEMA VE 2.
- D. Ground cable tray system in accordance with NFPA 70.
- E. Cable tray to equipment/enclosure transitions shall be made using conduit or appropriate box connectors:
 - 1. Ensure equipment integrity is maintained to prevent entrance of foreign material.
- F. Repair galvanizing, if damaged.

END OF SECTION

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SECTION 26 05 39
ENGINE GENERATOR TERMINAL BOX

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Engine Generator Terminal Box:
1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

- A. ASTM International (ASTM)
1. ASTM D1535-2014(2018) Specifying Color of the Munsell System
- B. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE C2-2022 National Electrical Safety Code (NESC)
- C. National Electrical Manufacturers Association (NEMA)
1. NEMA 250-2020 Enclosures for Electrical Equipment (1000 Volts Maximum)
- D. InterNational Electrical Testing Association, Inc. (NETA)
1. NETA ATS-2021 Acceptance Testing Specifications
- E. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electrical Code (NEC)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 39-1, Approval Drawings and Data:
1. Manufacturer's catalog data and terminal box layout drawing, including mounting details.
- C. RSN 26 05 39-2, Field Test Reports.

PART 2 PRODUCTS

2.01 TERMINAL BOX

- A. Fully contained engine generator terminal box with connectors.
- B. Steel: 14 gauge, minimum.
- C. Weatherproof, NEMA 250, Type 4.
- D. Hinged doors:
 - 1. Full length hinge.
 - 2. Pad-lockable.
 - 3. Steel: 14 gauge, minimum.
 - 4. Gasketed.
 - 5. Front Door: Provides access to bussing, lug points, and connectors.
 - 6. Bottom Door: Provides access for cables to enter the terminal box from below.
- E. Mounting bolts, nuts, and washers: Hot-dipped galvanized or non-ferrous, non-corrosive metal.
- F. Bus:
 - 1. Material: Copper, tin plated.
 - 2. Continuous Current Rating: 200A.
 - 3. Short-circuit current rating: 10,000A rms, symmetric.
- G. Ground bus:
 - 1. Material: Copper, tin plated.
 - 2. Suitable for up to No. 4/0 AWG stranded bare copper ground cable.
 - 3. Locate for easy connection to grounding system.

2.02 ACCESSORIES

- A. Designation Nameplate:
 - 1. In accordance with Standard Drawing No. 40-D-6234:
 - a. Type: 'A'.
 - b. Size: 3/32-inch thick.
 - 2. Provide the following nameplates:
 - a. Front of enclosure:

- 1) "Engine Generator Terminal Box ECB".
- 2) "480V - 3 Phase - 3 Wire - 200A".
- 3) "Feeds DSA Breaker No. 1".

2.03 FABRICATION

- A. Provide sufficient space in the terminal box and on bus bars for the following connections:
1. Conductors from terminal box to DSA breaker conductors.
 2. Conductors from terminal box to EG set conductors.
- B. Conduit openings:
1. Provide factory conduit knockouts/openings on bottom of box.
 - a. Quantity and size as required to properly wire up installation and future connection of EG set.
- C. Designation Nameplates:
1. Equipment designation nameplate: Mount on outside of door, centered.
 2. Mount with epoxy cement. Do not use screws.
- D. Suitable for surface mounting.

2.04 FINISHES

- A. Interior of enclosure: White.
- B. Exterior of enclosure: Powder coated, light gray No. 61 in accordance with ASTM D1535.

PART 3 EXECUTION

3.01 INSTALLATION

- A. In accordance with NFPA 70, IEEE C2, and this section.
- B. Mount at on rear wall of powerplant as shown on drawings.
1. Locate and mark exact locations of embedded reinforcement, conduit, and metalwork before installing anchors.
 2. Provide necessary mounting hardware for secure and stable mounting.
 3. Do not use conduits for support.
- C. Seal conduit entrances to prevent water entry after cables are installed.

- D. Ground and bond in accordance with NFPA 70 and Section 26 05 26 - Grounding and Bonding.

3.02 CONTRACTOR FIELD QUALITY TESTING

- A. In accordance with Section 26 05 02 - Basic Electrical Materials and Methods, and this Section.
- B. Perform inspections and tests in accordance with NETA ATS.

END OF SECTION

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SECTION 26 05 90

WIRING CHECKOUT AND TESTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for equipment and devices requiring wiring checkout and tests.

1.02 REFERENCE STANDARDS

- A. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electrical Code (NEC)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 90-1, Field Test Reports.

1.04 RELATED REQUIREMENTS

- A. This section applies, but is not limited to, sections of Division 26 unless specified otherwise in the individual sections.

1.05 DEFINITIONS

- A. External Wiring: Conductor, wire, or cable external to enclosure of equipment being tested.
- B. Internal Wiring: Conductor, wire, or cable internal to enclosure of the equipment being tested.
- C. Qualified Person: In accordance with Article 100 of NFPA 70.
- D. Wiring Checkout: A complete wire-by-wire, contact-by-contact, terminal-by-terminal inspection of wiring associated with a specific piece of equipment.

1.06 PERFORMANCE REQUIREMENTS

- A. Test in addition to tests required by Section 26 05 10 - Conductors and Cables.
- B. Furnish materials, including test equipment, required for checkout and testing.

- C. Make wiring changes and drawing revisions to ensure no discrepancies between check prints, and internal equipment wiring and external cabling.
- D. Replace defective equipment and make device setting adjustments as required for operation of electrical systems in accordance with specifications and manufacturer's instructions.
- E. If required during testing, remove, and replace wire connections.
- F. Make wiring changes, wire tag revisions, and other repairs or revisions during the testing session the discrepancy is discovered.
- G. Perform wiring checkout, and operational and functional tests in presence of the COR.

1.07 QUALIFICATIONS

- A. Testing Personnel:
 - 1. Competent and qualified person to perform wiring checkout, and operational and functional testing.
 - 2. Thorough knowledge of the electrical equipment, installation, and drawings.
 - 3. Qualified to operate equipment being tested.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 FIELD QUALITY CONTROL

- A. Notify COR in writing 10 days in advance of each testing session.
- B. Provide following at time wiring checkouts and tests are performed:
 - 1. Two complete sets of full-size (ANSI D) electrical schematic and wiring diagram check prints for equipment being tested.
 - a. One set to be used and maintained by the COR.
 - b. One set to be maintained by Contractor and used to produce final drawings.
 - 2. Operation and maintenance instruction books and test reports for equipment being tested.
- C. Wiring Checkout:

1. Use electrical schematic and wiring diagram check prints to perform:
 - a. Internal wiring check of equipment.
 - b. Wiring check of external cabling.
2. Perform a complete wire-by-wire, contact-by-contact, terminal-by-terminal check of circuit logic.
3. Verify contact configuration and contact numbering.
4. Verify continuity of wiring.
5. Verify proper wire tagging at ends of conductors and cables.
6. Check conductors for insulation damage. Perform additional insulation resistance test of damaged conductors.

D. Operational and Functional Testing:

1. Perform after completion of wiring checkout.
2. If required, adjust and calibrate protective devices in accordance with specifications, and manufacturer's instructions and recommendations.
3. Check electrical systems and controls for proper sequence of operation, correct adjustments and settings, and agreement with schematic diagram.

3.02 FIELD TEST REPORTS

- A. Provide a letter stating wiring has been checked and tested by the contractor and witnessed by the COR. A signature from the COR is required on this letter.

3.03 SCHEDULE

- A. All electrical equipment supplied requires a wiring checkout and operational test.

END OF SECTION

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SECTION 26 09 16 PROTECTION SYSTEMS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in the prices offered in Price Schedule for items of work requiring protection systems.
 - a. Includes:
 - 1) Protective systems which include but not limited to protective relays, lockout relays, auxiliary relays, switches, and accessories.
 - 2) Associated cables and cable connectors.
 - 3) Settings, Calibrations and Programming.
 - 4) Interface with Excitation Systems and Automation Control Systems.
 - 5) Uploading relay settings for all devices requiring settings.

B. Cost:

1. Include cost of spare parts in lump sum price offered in the Price Schedule for Spare Parts.
2. Include cost of training in the lump-sum price offered in the Price Schedule for Training.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (USBR)

- | | | |
|----|------------------|--|
| 1. | FIST 3-8 – 2021 | Facility Instructions and Standard Techniques (FIST) - Operation, Maintenance, and Field Test Procedures for Protective Relays and Associated Circuits |
| 2. | FIST 4-1B – 2024 | Maintenance Scheduling for Electrical Equipment |
| 3. | PEB 51 – 2024 | Power Equipment Bulletin (PEB) - Update to Protection System Testing Intervals |
| 4. | PEB 52 – 2024 | (PEB) - Update to Protection System Testing Procedures |

B. Institute of Electrical and Electronics Engineers (IEEE)

- | | | |
|----|----------------------------|--|
| 1. | IEEE C37.2 – 2022 | Standard for Electrical Power System Device Function Numbers, Acronyms, and Contact Designations |
| 2. | IEEE C37.90 – 2005 (R2011) | Relays and Relay Systems Associated with Electric Power Apparatus |
| 3. | IEEE C57.13 – 2016 | Instrument Transformers |
| 4. | IEEE 802.3 – 2022 | Standard for Ethernet |

C. National Electrical Manufacturers Association (NEMA)

- | | | |
|----|--------------------------|--|
| 1. | NEMA AB-3 – 2013 (R2023) | Molded Case Circuit Breakers and Their Application |
| 2. | NEMA IA 10039 – 2025 | Industrial Control and Systems: Control Circuit and Pilot Devices. |
| 3. | NEMA ICS 1 – 2022 | Industrial Control & Systems General Requirements |
| 4. | NEMA ICS 6 – 1993 (2016) | Industrial Controls and Systems Enclosures |

1.03 ACRONYMS

- | | |
|----|---|
| 1. | A: Amperes. |
| 2. | CT: Current Transformer |
| 3. | PT: Potential Transformer |
| 4. | SCADA: Supervisory Control and Data Acquisition |
| 5. | VAC: Volts Alternating Current |
| 6. | VDC: Volts Direct Current |

1.04 DESIGN REQUIREMENTS

- A. Design new prefabricated control panel assemblies to replace existing protective relay control panels.
1. Locations:
- a. Panels CSA1R and CSA2R.
- B. Disable the following protection functions in the existing Exciter Relay 111E (24, 40, 59, 64R). The Siemens Siprotec 7UM622 relay will remain in service to provide field overcurrent protection for the excitation system (51C).
- C. Coordinate the design and installation of the relays, auto synchronizer and watthour meters with the automation control and monitoring systems to provide troubleshooting, event recording and operational information.

- D. Provide new electrical protection system for Pole Hill Powerplant. Includes:
1. Replace the following lockout relays:
 - a. 186GP – Generator Protective Lockout Relay Switch.
 - b. 186GD – Generator Differential Lockout Relay Switch.
 - c. 86K1A – GSU Transformer Differential Lockout Relay Switch.
 2. Provide the following digital multi-function protection relays:
 - a. 111GP – Generator Protection Relay (Primary).
 - b. 111GS – Generator Protection Relay (Secondary).
 - c. 111SP – Generator Split Phase Differential Relay.
 - d. 111K1A – GSU Transformer Differential Relay.
 - e. 111KSA – Station Service Transformer Differential Relay.
 - f. 149RTD – RTD Input Module.
 - g. 164F – RTD Input Module.
- E. Provide IRIG-B time synchronization for all relays.
- F. Provide test blocks (FT Series) for inputs and outputs on each relay.
- G. Contractor shall install new relays with the latest firmware.
- H. Provide data logging capability inside protection multifunction relays.
- I. Protection system shall be designed with Time Synchronization.

1.05 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 09 16-1, Approval Drawings and Data – Government Format: Refer to and include in Section 26 09 18 – Control Boards.
1. Provide example drawing for the protection system for each of the following:
 - a. Schematic drawing.
 - b. Wiring drawing.
 - c. Three-line drawing.
 - d. Single-line drawing.
 - e. Equipment layouts.
 - f. Nameplate list.

- g. Manufacturer's Technical Data
 - h. Bill of materials (BOM) list.
- C. RSN 26 09 16-2, Software Development and Cyber Security Documentation:
 - 1. Narrative of the software, cyber security and communications between new PLC system and existing SCADA system.
 - a. Preliminary cyber security documentation.
 - b. Preliminary configuration documentation.
 - 2. Include pertinent descriptive figures, interconnection block diagrams, network diagrams, etc.
 - 3. I/O logic diagram for proposed digital outputs.
 - 4. Narrative describing which devices may operate in the event of a fault.
 - 5. Demonstrate correct relaying-system performance for the operation of all equipment associated with the protective-relaying functions.
- D. RSN 26 09 16-3 - Final Software Files:
 - 1. Provide copy of final software files including a record of all settings and logic values.
- E. Installation Plan and Procedures: Refer to and include in specification requirements in Section 26 09 18 – Control Boards.
- F. Factory Acceptance Test Plans and Procedures: Refer to and include in Section 26 09 18 – Control Boards.
- G. Factory Acceptance Test Reports: Refer to and include in Section 26 09 18 – Control Boards.
- H. Commissioning Test Plans and Procedures: Refer to and include in Section 26 05 90 – Wiring Checkout and Tests and Section 26 09 18 – Control Boards.
- I. Commissioning Test Reports: Refer to and include in Sections 26 05 90 – Wiring Checkout and Tests.
- J. Final (As-built) Drawings and Operation and Maintenance (O&M) Instruction Books: Refer to and include in Section 01 78 30 – Project Record Documents and Section 26 09 18 – Control Boards.

1.06 PROJECT CONDITIONS

- A. Altitude: 6,600 feet
- B. Maximum indoor ambient temperature 50 degrees C.

1.07 QUALIFICATIONS AND RESPONSIBILITIES

A. Installation Supervisor:

1. Responsible for installation of new protection system, equipment, and materials.
2. Technically qualified to supervise installation and testing work, meeting with Project Manager approval.
3. Present or represented at work site during installation period.
4. Report, in writing to Project Manager, work not in accordance with manufacturer's recommendations or other conditions possibly resulting in unsatisfactory job.
5. Maintain daily log of accomplished work.

1.08 SPARE PARTS

A. Table 26 09 16A – Spare Parts

Item	Description	Quantity
1	Digital multifunction protective relay SEL 700G	1
2	Digital multifunction protective relay SEL 787	1
3	SEL 2664 field ground module	1
4	SEL 2600 RTD module	1
5	Lockout relay	1 of each type
6	Test blocks	1 of each type
7	Hardware and software required to make program code changes for controller and PLC	One set

PART 2 PRODUCTS

2.01 DIGITAL MULTI-FUNCTION PROTECTIVE RELAYS

- A. Provide digital multi-function relays that communicate by any ASCII terminal without relay programming software specific to the relays.
- B. Manufactured in accordance with IEEE C37.90.
- C. All protection relays are mounted on the rear panels of control boards, as shown on drawings 245-D- 60136 and 245-D- 60138.
- D. Furnish Protection System relays with read-only inputs configuration.

1. Write access shall not be possible without administrator level access control.
- E. Locations: Control Room Rear Panels, CSA1R and CSA2R.
- F. Provide digital multi-function protective relay systems of same type and manufacture. If a more current model number exists, proposed relay must satisfy or exceed salient characteristics specified in this section.
- G. Protection systems shall be designed with Time Synchronization:
1. Provide time synchronization IRIG B-122 signal distribution to Contractor supplied relays.
 2. Interface with existing satellite synchronization clock (IRIG time demodulator - SEL 2407) and antenna (SEL 9524 GNSS Antenna).
 - a. The clock is physically located in the Pole Hill Powerplant Control Room – CSA4F
 3. Provide IRIG B-122 signal distribution models, cable, raceway and associated accessory components required for the Time Synchronization.
- H. Digital Multi-Function Protection Relay: Primary (111GP) and Secondary (111GS).
1. Type SEL, 700G Model 0700G11ACA1A76850220 (or current equivalent number) Relay manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the following essential characteristics:
 - a. Quantity: 2
 - b. Location: Panel CSA1R
 - c. Control input voltage: 125 VDC
 - d. Power Supply: 110-250 VDC
 - e. Volts per Hertz (24)
 - f. Loss of Field (40)
 - g. Auto Synchronizer (25)
 - h. Synch Check (25SC)
 - i. Negative sequence Overcurrent (46)
 - j. Stator RTD Overtemperature (49)
 - k. Mho impedance phase fault backup (21)
 - l. Generator Overvoltage (59)
 - m. Generator Stator Ground (64G)
 - n. Generator Differential (87GD)
 - o. Out of Step (78)

- p. Breaker Failure
- q. Inadvertent Energization
- r. IRIG-B compatible input, and a rear EIA-232 input for slot B.
- s. Communication ports:
 - 1) Single 10/100BASE-T Ethernet.
 - 2) DNP3 Protocol.
 - 3) One Multimode ST® Fiber optic port.
 - 4) IRIG-B input
 - 5) RS-232 input for slot B
 - 6) Slot C is configured to include four digital inputs and four high-speed, high-current outputs.
 - 7) Slot D is configured to include four digital inputs and four electromechanical outputs.
- t. Inputs and Outputs:
 - 1) Voltage: suitable for use with 125VDC.
 - 2) 10 available inputs per relay.
 - 3) 11 available outputs per relay, minimum, 4 of them specified as fast, high-current.
 - a) Breaker close signal shall be high-speed, high-current output.
 - 4) Provide all inputs and outputs shown on supplied drawings and required in the specifications.
 - 5) Inputs and outputs shall be appropriately rated for the circuit and application where they are used.
 - 6) For 111GS, wire one binary output to 111GPLC digital input for indicating a thermal trip (49) from any stator winding RTD.
- u. Provide capability for auto synchronization.
 - 1) Provide status “Ready for Synchronization” to Unit Control HMI (Human Machine Interface) screen.
- v. Field ground module (64F) and RTD module compatible (49).

I. Field Ground Module (164F).

- 1. Type SEL, 2664 Model 2664XX0XXX Field Ground Module manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the following essential characteristics:
 - a. Quantity: 1.

- b. Location: Generator Air Housing.
 - c. Shall connect to the new primary protective relay (111GP) only, via a dedicated fiber optic link.
- J. Stator Temperature RTD module (149RTD).
 - 1. Type SEL, 2600 Model 2600D01X RTD Module manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the following essential characteristics:
 - a. Quantity: 3.
 - b. Locations: GSU Transformer (K1A), Station Service Transformer (KSA), Generator Air Housing (G1A).
 - c. Shall interface with the 111K1A, 111KSA, and 111GS relays respectively, through dedicated fiber connections.
 - d. Inputs: 12 RTD inputs per module.
- K. GSU Transformer (K1A) Multi-Function Protection Relay (111K1A).
 - 1. Type SEL, 787 Model 07871X1A1A0XA5850220 Relay manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the following essential characteristics:
 - a. Quantity: 1.
 - b. Location: Panel CSA2R
 - c. Control input voltage: 125 VDC.
 - d. Power Supply: 110-250 VDC.
 - e. Differential Current Protection (87).
 - f. Backup Ground Overcurrent (51N).
 - g. Backup Phase Overcurrent (51P).
 - h. DNP3 Protocol.
 - i. IRIG-B compatible.
 - j. Communication ports:
 - 1) RS-232.
 - 2) IRIG-B input.
 - 3) Single fiber input.
 - 4) Single 10/100BASE-T ethernet input.
 - k. Slot C is configured to include four digital inputs and four electromechanical outputs.
 - l. Slot E is configured to include single-phase neutral current input.

- m. Inputs and Outputs:
 - 1) Voltage: suitable for use with 125VDC.
 - 2) 6 available inputs.
 - 3) 7 available outputs.
 - 4) Provide all inputs and outputs shown on supplied drawings and required in the specifications.
 - 5) Inputs and outputs shall be appropriately rated for the circuit and application where they are used.
- n. RTD module compatible (49).

L. Station Service Transformer (KSA) Multi-Function Protection Relay (111KSA).

- 1. Type SEL, 787 Model 07870X1A1A0X0X850220 Relay manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the following essential characteristics:
 - a. Quantity: 1.
 - b. Location: Panel CSA2R.
 - c. Control input voltage: 125 VDC.
 - d. Power Supply: 110-250 VDC.
 - e. Differential Current Protection (87).
 - f. Phase Overcurrent (51P).
 - g. DNP3 Protocol.
 - h. IRIG-B compatible.
 - i. Communication ports:
 - 1) RS-232.
 - 2) IRIG-B input.
 - 3) Single fiber input.
 - 4) Single 10/100BASE-T ethernet input.
 - j. Slot C is configured to include four digital inputs and four electromechanical outputs.
 - k. Inputs and Outputs:
 - 1) Voltage: suitable for use with 125VDC.
 - 2) 6 available inputs.
 - 3) 7 available outputs.
 - 4) Provide all outputs shown on supplied drawings and required in the specifications.

- 5) Inputs and outputs shall be appropriately rated for the circuit and application where they are used.

1. RTD module compatible (49).

M. Generator Split Phase Differential Multi-Function Protection Relay (111SP).

1. Type SEL, 787 Model 07870X1A0X0X0X850220 Relay manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the following essential characteristics:
 - a. Quantity: 1.
 - b. Location: Panel CSA1R.
 - c. Control input voltage: 125 VDC.
 - d. Power Supply: 110-250 VDC.
 - e. Differential Current Protection (87).
 - f. DNP3 Protocol.
 - g. IRIG-B compatible.
 - h. Communication ports:
 - 1) RS-232.
 - 2) IRIG-B input.
 - 3) Single fiber input.
 - 4) Single 10/100BASE-T ethernet input.
 - i. Inputs and Outputs:
 - 1) Voltage: suitable for use with 125VDC.
 - 2) 2 available inputs.
 - 3) 3 available outputs.
 - 4) Provide all inputs and outputs shown on supplied drawings and required in the specifications.
 - 5) Inputs and outputs shall be appropriately rated for the circuit and application where they are used.

N. High-Accuracy IRIG-B Fiber-Optic Transceiver: The following devices are in the new control boards unless stated otherwise.

1. Type SEL, 3405TX, RX Model 3405TXXX, 3405RXXX (or current equivalent numbers) Fiber-optic IRIG-B Transceiver, Receiver Devices manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the same essential characteristics.
 - a. Quantity: one pair in each control boards.

- 1) 6 Transmitters.
 - 2) 6 Receivers.
 - b. Run from existing SEL 2407 to each SEL 3400 (#77T).
 - c. Provide accessory SEL-C942: DB9 (for SEL-3405) to BNC (for IRIG-B) to tinned wires (for power).
2. Type SEL, 3400 Model 3400A100DX (or current equivalent number), IRIG B Distributor Module manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the same essential characteristics.
 - a. Quantity: One per unit, 6 total.
 - b. Location: CP#A2F as shown on 245-D-60054.
 - c. I/O:
 - 1) Two BNC IRIG-B000, B002, B004 Compatible Demodulated Inputs.
 - 2) Twelve Demodulated IRIG-B BNC Outputs.
3. Type SEL C808 Optical fiber Multi mode fiber-optic cable and jumpers manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; having the same essential characteristics.
 - a. Cable type: Multi-Mode.
 - b. Major cable components:
 - 1) UL Riser rated.
 - 2) Dielectric outer cable.
 - 3) Meets IEEE 802.3ae standards.
 - 4) Zip cord and heavy-duty PVC-jacketed fiber.
 - 5) Loose-tube.
 - 6) Gel-free.
 - 7) Attenuation: Directly proportional to cable length for an arbitrary cable length section after steady-state conditions are reached.
 - 8) Color coded per TIA/EIA-598.
 - c. Cable core: Symmetrically stranded buffer tubes around dielectric central member.
 - d. End termination: V-System or ST.
 - e. Cable tensile strength:
 - 1) Long term (installed): 135 pounds.
 - 2) Short term (during installation): 380 pounds.

- f. Operating temperature range: -20 to 80°C.
- 4. Provide necessary IRIG B adapters, splice bushings, adapter cables, and accessories for a functional network.

2.02 RELAY TEST SOFTWARE (RTS)

- A. Type ENOSERV RTS test routine by ENOSERV, 7780 E. 106TH Street, Tulsa, OK, 74133; having the following essential characteristics:
 - 1. Test routine will be located with a relay database (*.MDR) file, and relay tests will be FasTest (*.FST) file format compatible with RTS by Enoserv.
 - 2. Test routine will test each enabled element within relay based on settings internal to relay as recommended for commissioning and testing by relay manufacturer.
 - 3. Test routine shall perform meter test for microprocessor-based relays.
 - 4. Provide automatically generated report indicating pass or fail of all elements within relay, and include relay settings, test date, and test set operator.

2.03 INSTRUMENTATION TRANSFORMERS

- A. Manufacture to fit in existing locations and/or as shown on single-line Drawing.
- B. Conform to IEEE C57.13.
- C. Provide adequate clearance between CTs, cables, insulators and between CTs and adjacent conducting surfaces.
 - 1. Provide silicone insulating sleeve or mechanical bracing to prevent contact with motor terminals if adequate clearance is not feasible.
 - 2. Provide new current transformers and potential transformers where applicable.
 - a. Perform tests in accordance with FIST 3-8.
- D. Potential Transformer:
 - 1. Located on the load side of 52N breaker (between 52N and DSA bus) in switchgear.
 - a. Rated Primary Voltage (V): 480 V
 - b. Ratio: 4:1.
 - c. Accuracy Class: Minimum 0.3 (W, X, M, Y) and 1.2 Z.
 - d. Insulation class: 600 V.
 - e. BIL: 10 kV.
 - f. Thermal Rating: 300 VA (minimum) @ 55°C.
 - g. Configuration: Open Delta.

E. Current Transformers:

1. Located on the load side of 52N breaker (between 52N and DSA bus) in switchgear.
 - a. Ratio: 800:5 ampere.
 - b. Maximum Voltage: 600 V.
 - c. BIL: 10 kV.
 - d. Accuracy Class: C100.
 - e. Continuous Thermal Current Rating Factor: 1.0 @ 55°C, minimum.
 - f. Configuration: 3Y.
2. Located on the 13.8 kV side of the new Station Service Transformer (KSA).
 - a. Ratio: 50:5 ampere.
 - b. Maximum Voltage: 600 V.
 - c. BIL: 10 kV.
 - d. Accuracy Class: C10.
 - e. Continuous Thermal Current Rating Factor: 1.0 @ 55°C, minimum.
 - f. Configuration: 3Y.

2.04 LOCKOUT RELAY

- A. Lockout Relays: Series 24, 78PB06E-002 lockout relays manufactured by Electroswitch, 2010 Yonkers Road, Raleigh, NC 27604; having the following essential characteristics:

1. Device No. 86 as defined by IEEE C37.2.
 - a. Manual-reset type.
2. Coil Voltage: 125 VDC.
3. Burden at rated voltage: 2.5 A.
2. Trip coil Threshold Voltage: 23 VDC.
3. Trip coil operating range: 45-140 VDC.
4. Lighted target nameplate:
 - a. Integral LED lights:
 - 1) Trip: Amber (Left).
 - 2) Reset: Red (Right).
5. Not able to trip by manually turning the external handle.
6. Trip Speed: 8 ms, maximum.
6. Location: Front panels CSA1R and CAS2R control boards.

7. Number of Contacts: Contractor to determine.
 8. Provide 6 decks minimum with at least 3 unused, spare Trip and Reset contacts.
 9. Integral trip coil monitor with electrically isolated normally close contact.
 - a. Digitally driven.
- B. Provide new lockout relays and indicating lights as shown on drawings 245-D-60066 and 245-D-60068.

2.05 RELAY TEST SWITCHES

- A. CT and PT circuit test switch devices shall be provided in accordance with specification requirements in Section 26 09 18 – Control Boards.
- B. Location: Near the respective protective relays as shown on the drawings.

2.06 NAMEPLATES

- A. Nameplates for control board and devices:
1. Provide nameplates at the top of each control board section.
 2. Provide nameplates for all terminal blocks, relays, and contractor supplied devices.
 3. For each device visible from the outside of the control boards (front and rear panels of each section), mount nameplates on the exterior of the enclosure. These devices include, at a minimum:
 - a. Lockout Relays.
 - b. Digital multi-function protection relays.
 - c. Pushbuttons
 - d. Meters
 - e. Control Switches
 - f. Nameplates
 - g. Test Switches.
 4. Attach nameplates with pan head self-tapping screws. Do not use adhesives to attach nameplates.
 - a. Attach nameplates to pushbuttons, indicating lights, and control switches in accordance with component manufacturer's instructions.
 - b. Nameplates shall consist of letters and numbers engraved on phenolic-type material. Size of letters and figures shall be approximately 7/16 inch for cabinet internal device nameplates.

5. Provide nameplates for all devices internal Control Board in accordance with Reclamation device location designations as shown on standard drawings.

2.07 CONTRACTOR SOURCE QUALITY CONTROL

- A. Provide SEL Test Lab Certification on each manufactured relay of each type.
- B. Perform tests in accordance with IEEE 37.90, FIST 3-8, and PEB 52.
- C. Perform test on instrument transformers in accordance with IEEE C57.13:
 - a. Insulation resistance.
 - b. Polarity.
 - c. Excitation current curve (CTs).
 - d. Ratio.
- D. Include test results for each protection system, in the factory and field test reports based on the location of the new protection devices. Refer to specification Section 26 09 18 – Control Boards.
- E. All factory and field test reports shall be included in the Operation and Maintenance (O&M) Instruction Books. Refer to Section 01 78 30 – Project Record Documents and Section 01 33 26 – Electrical Drawings and Data.

PART 3 EXECUTION

3.01 GENERAL

- A. The new protection devices shall be installed on the new control boards. Refer to specification Section 26 09 18 – Control Boards.
- B. Contractor shall reuse raceway routing into and out of the existing control boards. The existing auxiliary cables/conductors shall be replaced up to each auxiliary device. The existing auxiliary devices/equipment shall be reused (butterfly valves, vents, CO2, etc.)

3.02 REMOVAL

- A. Refer to Section 26 25 13 – Removal, Disposal, and Modifying of Existing Electrical Equipment.
- B. The following existing devices shall be removed (for pumps 1 – 6) and provided to the Water Authority.
 1. All Relays,
 2. All Meters,
 3. RTACs,

4. SEL 2600 devices.
5. HMIs

3.03 INSTALLATION

- A. The new protection system shall be installed in the new control boards panels CSA1R and CSA2R as shown on drawings.
- B. Demonstrate correct relaying-system performance: the satisfactory operation of all equipment associated with the protective-relaying functions.
- C. Develop and format detailed instructions and checklists in accordance with 01 33 26 – Electrical Drawings and Data.
- D. Hardwire protective relay tripping outputs into existing SCADA sequence of events recorder. Termination of these signals will be made by Contractor.
- E. Perform coordination of all breaker and relay settings based on the relay settings provided by the Government.
- F. Provide single-point ground for all new PTs and new CTs. Document exact locations of each single-point ground; both new and existing.
- G. Furnish and install patch cords to connect dielectric fiber cable to fiber optic equipment.
- H. Prior to installation of the field ground module, the existing 7XT7100 controller and 7XR6004 resistor unit shall be disconnected from the Exciter Relay 111E analog inputs TD1 and TD2.
- I. Determine the highest temperature RTDs from representative locations within each pump motor and wire up these RTDs into the SEL 2600 modules for thermal protection. The remaining RTDs not wired into a SEL 2600 will be wired into the Flex 5000 or directly into PLC analog input card for instrumentation and alarming.

3.04 CONTRACTOR QUALITY TESTING

- A. Factory Acceptance Tests relating to the Protection Systems:
 1. Protection Systems:
 - a. Serial number identified instrumentation transformer test reports (CTs & PTs).
 - 1) Insulation resistance.
 - 2) Polarity.
 - 3) Excitation current curve (CTs).
 - 4) Ratio.

- b. Digital multifunction protection relay tests:
 - 1) Insulation test voltage: inputs and outputs in accordance with IEC 60255-27.
 - 2) High frequency interference test: in accordance with IEC 60255-27.
 - 3) Electrical discharge and fast transient burst tests.
 - 4) Protection function element tests.
 - 5) Power frequency magnetic field immunity test.
 - 6) Pickup, timing, and logic tests.
- B. Perform fiber-optic network testing per manufacturer's instructions for a fully functional protection system network.
- C. Test new protection system in accordance with Section 26 05 90 - Wiring Checkout and Tests.
- D. Perform visual and mechanical inspection for Contractor installed equipment.
- E. Time Synchronization:
 - 1. Verify that each applicable relay is receiving a time synchronization signal and that appropriate time is displayed in the relay.
- F. Verification of associated annunciation during testing.
- G. Test existing resistive temperature devices (RTDs) for use in protective relaying.
 - 1. If an existing RTD is not working properly, the Contractor shall utilize a nearby existing functional RTD from the stator windings.
 - 2. Document which RTDs are not working properly and which RTDs are interfaced to the new protective relays.
 - 3. Reuse the existing conduit for existing RTD circuits.
- H. Perform tests in accordance with IEEE 37.90, IEEE C57.13, FIST 3-8, FIST 4-1b, and PEB 52.

3.05 FINAL AND AS-BUILTS DRAWINGS

- A. Final As-built Drawings Government Format: refer to Section 01 78 30 - Project Record Documents, Section 01 33 26 – Electrical Drawings and Data and Section 26 09 18 – Control Boards.

3.06 TRAINING

- A. Refer to Section 01 79 20 – Training.

END OF SECTION

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SECTION 26 09 18

CONTROL BOARDS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Control Boards CSA for Unit 1:

1. Payment: Lump sum price offered in the Price Schedule.
 - a. Includes cost of designing, fabricating, furnishing, installing, and testing/commissioning the following:
 - 1) Control boards CSA; 6 sections.
 - 2) Network Communication Cabinet (CP1N) in Control Room: one cabinet shall be provided with all network components.
 - 3) Network Communications at Gatehouse: back panel with all network components mounted onto existing communications rack.
 - 4) Distributed I/O (111TDIO, 111GHIO):
 - a) One cabinet for turbine level controls, instrumentation, and other devices.
 - b) Gatehouse controls, instrumentation, and other devices to be installed on back panel onto existing Communications Rack at Gatehouse Communications Room.
 - b. Includes cost of calibrating, programming, testing the above stated control boards, cabinets, devices, and corresponding systems.

B. Cost:

1. Include spare parts in the lump-sum price offered in the Price Schedule for Spare Parts.
2. Include training in the lump-sum price offered in the Price Schedule for Training.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI)

1. ANSI C37.20-2015 Standard for Metal-Clad Switchgear

B. Facilities Instructions, Standards and Techniques (FIST)

1. FIST 3-3-2000 Electrical Connections for Power Circuits
2. FIST 3-10 Watt-Hour Meter Maintenance and Testing

3. FIST 4-1B-2023 Maintenance Scheduling for Electrical Equipment
- C. International Electrotechnical Commission (IEC)
1. IEC 60255-27 – 2023 Measuring Relays and Protection Equipment
- D. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE C37.118-2011 Standard for Synchrophasor Data Transfer for Power Systems
2. IEEE C37.2-2022 Electrical Power System Device Function Numbers, Acronyms, and Contact Designation
3. IEEE C37.21-2017 Standard for Control Switchboards
4. IEEE C37.81-2017 Guide for Seismic Qualification of Class 1E Metal-Enclosed Power Switchgear Assemblies
5. IEEE 693-2018 Recommended Practice for Seismic Design of Substations
- E. National Electrical Manufacturers Association (NEMA)
1. NEMA 250-(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
2. NEMA ICS 1 – 2000 (2022) Industrial Control & Systems General Requirements
3. NEMA ICS 5 2000 (2017) Industrial Control and Systems: Control Circuit and Pilot Devices.
4. NEMA ICS 6-1993(2016) Industrial Controls and Systems Enclosures
5. NEMA PB 2.1 – 2013 Proper handling, Installation, Operation and Maintenance of Deadfront Switchboards Rated 600 Volts or Less.
6. NEMA WD 1-1999(R 2020) General Requirements for Wiring Devices
- F. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electrical Code (NEC)

1.03 DEFINITIONS

- A. Control Board: A type of switchboard including control, instrumentation, metering, protective (relays), or regulating equipment for remotely controlling other equipment. Control boards do not include the primary power-switching devices or their connections.
- B. Dead-front Switchboard: A switchboard that has no exposed live parts on the front.

- C. Duplex Switchboard: A control switchboard consisting of panels back-to-back and enclosed with a top and ends (not grille). Access space (aisle) with entry doors (at both ends of the aisle) is provided between the rows of panels.
- D. HMI: Human Machine Interface.
- E. PLC: Programmable Logic Controller.
- F. SWC Test: Surge Withstand Capability Test.

1.04 RELATED WORK

- A. Section 25 00 01 – Automation Control and Monitoring Systems
- B. Section 25 12 10 – Programmable Logic Controllers
- C. Section 26 09 16 – Protection Systems
- D. Section 26 33 48 – DC Panelboards
- E. Section 27 20 01 – Fiber Optic Cable and Accessories
- F. Section 27 21 29 – Network Switches
- G. Section 28 05 11 – Cyber Security Requirements for Electronic Safety and Security
- H. Section 51 25 00 – Input/Output List

1.05 SYSTEM DESCRIPTION

- A. Provide new CSA control board panels as indicated on drawings and as specified herein. The design shall consist of a series of stand-alone walk-in duplex dead front boards containing such equipment as digital programmable logic controllers (PLCs), intelligent low-voltage devices, control switches, digital multi-function protection relays, test switch blocks, relays, human machine interface (HMI) touchscreens, annunciator assemblies, and analog meters to name a few. Provide distributed I/O control panels for the turbine level, and another for the gatehouse, including instrumentation and control devices. The system shall retain the ability to send/receive commands from/to the RTU and to operate manually. Manual functions shall be employed through PLC I/O.

1.06 DESIGN REQUIREMENTS

- A. Design of walk-in, duplex (subpanel) control board face panels for CSA (both front and rear) made up of 6 sections in accordance with NEMA ICS and NFPA 70.
- B. Provide new Distribution Input/Output (I/O) Cabinets:
 - 1. (111TDIO) for turbine level including corresponding electrical devices and equipment.

- a. All instrumentation and control devices on Turbine level to connect to this remote I/O cabinet.
- 2. (111GHIO) for gatehouse including corresponding electrical devices and equipment.
 - a. All instrumentation and control devices on Gatehouse level to connect to this distributed I/O panel. Shall be installed onto a metal panel and affixed to the existing communications rack in the communications closet.
- C. Provide one cabinet with HMI only
 - 1. Location: Near front entrance of Powerplant.
- D. Provide one new Network Communication Cabinet (CP1N) to interface to new PLC, distributed I/O, SCADA RTU PLC, and corresponding electrical devices and equipment.
- E. Each PLC and distributed I/O shall be of the same brand.
- F. All local devices shall be connected to an automation control and monitoring system network (Local Auto and Supervisory), enabling digital communication between permissible devices. Local-Manual shall also work through the PLC controls.
- G. Excitation System:
 - 1. Components that will be installed at CSA Control Board that will interface with existing Excitation System and will keep existing functionality.
 - a. Analog meters:
 - 1) Field voltage.
 - 2) Field current.
 - b. Exciter control switches:
 - 1) VAR level adjuster (190ECS).
 - 2) Excitation control mode selector (143EM).
 - 3) Exciter Enable Switch (169ECS).
- H. Refer to Section 26 04 10 – Electrical Demolition and Removal.
- I. Refer to Section 28 05 11 – Cybersecurity Requirements for Electronic Safety and Security. This section includes items that will either be physically located on the new control boards and/or interface to the new control boards. This section applies to all electrical equipment, devices, and electrical systems.
- J. The existing electrical devices and circuits in the existing control board operate at 48 VDC nominal, 125 VDC nominal, and 120 VAC. The new equipment, new circuits and new control boards shall also operate at 125 VDC nominal to the greatest extent possible. Some circuits shall operate at 24 VDC.

- K. Minimize new control board hard wiring, where possible, by integrating signals into the distributed Input/Output (I/O) assemblies.
 - 1. Use distributed I/O data and new/existing field contacts to enhance monitoring and achieve greater flexibility in controls.

1.07 SUBMITTALS

- A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.
- B. RSN 26 09 18-1, Resumes and Qualifications:
 - 1. Resumes and Qualifications:
 - a. Electrical Engineer(s) (EEs) associated with all the electrical systems furnished, installed, and tested control boards.
 - b. EEs associated with the new networking cabinets, panels, and all internal devices.
 - c. EEs associated with the new Distribution I/O (111TDIO, 111GHDIO) cabinets and all internal devices.
 - d. Past experience of EEs and control board facility equal or more than the complexity of this project.
- C. RSN 26 09 18-2, Approval Drawings and Data:
 - 1. Manufacturer's Technical (Catalog) Data: with Bill of Material (BOM) Index/List.
 - 2. Schematic diagrams, equipment layout and nameplate list.
 - 3. Includes the new control boards, new distribution I/O cabinets, the Network Communication Network, and all the devices installed in each control board, cabinet, and panel.
 - 4. Color sample of Control Panel exterior paint.
- D. RSN 26 09 18-3, Proposed Location for Turbine Level Distributed I/O Cabinet and Network Communication Cabinet:
 - 1. Plan and Section drawing for the proposed locations.
- E. RSN 26 09 18-4, Factory Acceptance Test (FAT) Plans and Procedures:
 - 1. For control boards, include FAT plans and procedures for the networking cabinets, panels, and all associated devices and electrical systems.
 - 2. Describe how each electrical device, circuit and electrical system shall be functional tested during the FAT.

3. Include new control board drawings, cabinet drawings and network rack drawings that show all electrical equipment, devices, electrical systems, and corresponding connections.
 4. Include Notification of FAT for unit under clearance, dates, timeframes, location, and facility requirements prior to the FAT.
 5. Describe mock functional test to demonstrate how automation control meet these specifications.
 6. Describe the methods used to simulate I/O feeds to PLCs where required for full testing and demonstration purposes.
 7. Describe the methods, software, and testing metrics used to demonstrate network connectivity, network packet captures, throughput bandwidth measurements, and network locations where these measurements will take place.
 8. Include new network diagrams showing all network connections and device connections for all equipment.
 9. Includes cybersecurity requirements per Section 28 05 11 – Cybersecurity Requirements for Electronic Safety and Security.
- F. RSN 26 09 18-5, Factory Acceptance Test Reports:
1. The FAT test reports shall be included under this submittal.
 1. Check Prints: include any changes made during the FAT tests. Prints (Hard copy – D-size drawings) are shipped with control boards as well as included in this submittal (electronically).
 - a. Includes CSA cabinet and networking cabinets and panels.
 - b. Includes schematic diagrams, wiring diagrams, equipment layout diagrams, nameplate list and Bill of Material list.
 2. Provide a written record of the tests performed and the results of those tests.
 3. Report shall confirm that all electrical devices and circuits were functioned and work properly (up to the outgoing terminal blocks).
 4. Certification of performance and results of system control and monitoring testing, including the results for all network communication paths and network failure scenario testing.
 5. Include network diagram showing all network connections and device connections for all equipment.
 6. Provide the network packet capture files for all locations for which those captures took place.
 - a. Uniquely name the files so that the file contents correspond to, and can be identified, using the network diagram provided in this submittal.
 7. Includes cybersecurity requirements per Section 28 05 11.

G. RSN 26 09 18-6, Installation Plans and Procedures:

1. Narrative of how the as-built drawings showing existing conditions is incorporated into the new control boards as well as the new automation control and monitoring systems.
2. Narrative on installing the new control boards and interfacing them to existing conduit, new and existing cables/auxiliary device wiring, cable trays and existing auxiliary circuits.
3. Narrative of how the new control boards shall interface to the new automation control and monitoring systems.

H. RSN 26 09 18-7, Field and Commissioning Test Plans and Procedures:

1. Detailed written step-by-step procedures of field tests for all electrical equipment, devices and electrical systems physically located in the new control boards, new distributed I/O cabinets, RTU, and new networking cabinet and panel.
2. List compliance tests in accordance with reference standards.
3. Include Contractor's checklists and data logs for equipment, devices, and electrical systems.
4. Pre-Commissioning punch list and pre-start-up procedures.
5. Functional Test Procedures.
6. Commissioning Plan and Procedures with corresponding schedule.

I. RSN 26 09 18-8, Field and Commissioning Test Reports:

1. Refer to Section 26 09 90 – Wiring Checkout.
2. Includes wiring checkout, operational and functional test reports of all electrical equipment, devices, and corresponding electrical systems that interface to control boards, distributed I/O cabinets, and networking cabinet and panel.
3. Provide written description of results for each test performed.
4. Include indication of pass or fail results.
5. Updated layout drawings, schematic diagrams, wiring diagrams utilized during field testing and commissioning.
6. Include all testing equipment used including connections prior to the removal from the network.
7. Include checklists and data logs.
8. Spare Parts: Provide a list of spare parts delivered to site and list of any spare parts used during field testing and commissioning.

J. RSN 26 09 18-9, Final Documents:

1. Final (As-built) Drawings and Operation and Maintenance (O&M) Instruction Books – Control Boards, Distribution I/O Cabinets, and Network Communication Cabinet and Panel: Refer to Section 01 78 30 – Project Record documents.
2. Refer to Section 01 33 26 – Electrical Drawings and Data
3. Refer to requirements in this section as well as other sections identified above.

1.08 DELIVERY, STORAGE, AND HANDLING

- A. Store indoors in a clean, dry space. Maintain factory wrapping or provide an additional heavy canvas or heavy plastic cover to protect from dirt, water, and construction debris.
- B. Handle and store hardware in accordance with manufacturer's instructions and NEMA PB2.1. Include copy of these instructions with equipment at time of shipment.
- C. Deliver furnished equipment to Flatiron Power plant.
 1. Coordinate delivery with COR.

1.09 EXTRA MATERIALS

- A. Spare Parts:
 1. Deliver spare parts to the site prior to the start of the field testing and commissioning.
 2. The Contractor may use any spare part required during field testing/commissioning.
 3. The spare part(s) used during testing/commissioning, shall be replaced with the same spare part(s) by the Contractor, at no additional cost to the project. These replacement spare parts shall be onsite prior to the field testing and commissioning.
 4. After testing/commissioning and in unit is in operation, the contractor shall bench test each spare part, confirm proper operation and document each test performed.
 5. Furnish spare parts listed in Table 26 09 18A – Spare Parts
- B. For spare parts associated with the protection system refer to Section 26 09 16 – Protection Systems.

Table 26 09 18A – Spare Parts

Item	Description	Quantity
1	Control switches	2 of each type
2	Indicating lights	15 of each type
3	Pushbuttons	4 of each type

4	Auxiliary relays	12
5	Test switch blocks – PT, CT	2 of each type
6	Power supply	2
7	Power revenue meter	1
8	Meters	1 of each type
9	Speed Detection Device	1 of each type
10	Spare electrical devices, parts, tools, and software recommended by manufacturers	1 of each type
11	Spare control board parts and tools recommended by the control board manufacturer	1 of each type

1.10 PROJECT CONDITIONS

- A. Maximum indoor ambient temperature 50 degrees C.

1.11 QUALIFICATIONS AND RESPONSIBILITIES

- A. Prime Contractor:

1. Responsible for installation of new unit control board system, equipment, and materials.
 - a. Also includes new distribution I/O cabinets, Network Communication systems, equipment, and material.
2. Technically qualified to supervise installation and testing.
3. Onsite during installation and testing.
4. Report in accordance with manufacturer's recommendations.
5. Maintain daily log of accomplished work.

- B. Control Board Manufacturer:

- a. Minimum of 5 years' experience designing and fabricating control boards utilizing components.
- b. Provide contact information and details of at least two prior projects including design of control boards.
- c. Registered Electrical Engineer(s) resume(s).

- C. Manufacturers, Vendors, Subcontractors associated with the automation control and monitoring systems, PLCs, protection systems, and communication systems:

1. Minimum of 5 years' experience designing and fabricating systems similar in complexity to this project.

2. Provide contact information and details of at least two prior projects including details of similar designs for each electrical system.
3. Registered Electrical Engineer(s) resume(s).

PART 2 PRODUCTS

2.01 CONTROL BOARDS

A. General:

1. Design, fabricate, furnish, install, and test all electrical devices and circuits associated with control boards CSA, distributed I/O assemblies, and new networking system equipment.
 - a. All devices shall be furnished under this procurement as described in the specification sections and drawings, shall be mounted as shown and wired by contractor.
 - b. The contractor shall provide schematic and wiring diagrams in Government format per Section 01 33 26 – Electrical Drawings and Data.
2. Furnish new walk-In duplex control board panel sections (CSA1F, CSA2F, CSA3F, CSA1R, CSA2R, CSA5R, synchroscope panel, and annunciator display panels).
 - a. Coordinate any changes with the COR.
1. Front Panels Exterior:
 - a. Meters, HMI touch screens, annunciation display, control switches, protective relays, test switch blocks, and indicating lights to be sub-panel mounted.
 - b. The control HMI and annunciator screens share the same programmable logic controller (PLC).
 - c. Power Revenue Meter
2. Front Panels Interior wiring:
 - a. Back side of front panels include but are not limited to the low-voltage devices, analog meters, and control and monitoring HMI with corresponding PLC, control switches, lockout relays, indicating lights and test switch blocks.
 - b. All the devices on the front exterior will have their wiring completed from the interior of the front panels.
 - c. Managed switches. Refer to Network Diagram 245-D-60134.
3. Rear Panels (CSA1R, CSA2R, CSA5R): Exterior
 - a. Refer to drawings 245-D-60135 through 245-D-60140.
 - b. Digital multi-function protection relays, and test switch blocks.

4. Rear Panels (CSA1R, CSA2R, CSA5R) – Interior wiring
 - a. Low-voltage devices, PLCs, power supplies, DIN-rails, wireways, I/O cards, and control relays.
 - b. All the devices on the rear exterior panels shall have its wiring completed from the interior of the rear panels.
 5. Interior Wing Panels: Terminal Blocks
 - a. All cables, auxiliary device wiring and conductors leaving or entering front or rear panels, shall first land on a terminal block (located on the left or right wing of the respective panel) before wiring to the electrical devices.
 - b. On CSA3F and CSA2F there shall be 2 wing panels for contractor use to place terminal blocks as necessary.
 - c. On CSA1R and CSA2R there shall be 1 wing panels for contractor use to place terminal blocks as necessary.
- B. Type:
1. Walk-in Duplex Dead-front type of board, metal enclosed.
 2. Fixed front and rear panels.
 3. Aisle between the two rows of panels front and rear, comprised of 5 panel sections.
 4. Exterior Finish: Match color of CSA4F control boards. Provide a sample of the proposed pump control board color to the Project Manager prior to fabricating the first set of control boards.
 - a. All metal surfaces shall be thoroughly cleaned, phosphatized and factory primed prior applying baked enamel or lacquer finish.
 5. Interior Finish: White.
- C. Enclosures:
1. The enclosure shall be constructed from sheet steel no lighter than No. 11 United States Standard Gauge (Manufacturer's Standard Gauge).
 - a. Leveled by Contractor.
 - b. The board assembly shall be braced with reinforcing gussets using black colored bolted connections to assure rectangular rigidity.
 - c. NEMA 250 Type: NEMA-1 Rated.
 2. Overall Dimensions: (H) 9' x (L) 15' x (W) 6'.
 - a. Fit within existing footprint 176" Internal and 179" External.
 - b. Front Panels: subpanels sized to accommodate devices shown.
 - c. Rear Panels: subpanels sized to accommodate devices shown.

3. Conform to NEMA ICS-1 and NEMA ICS-6.
 4. Conduit Entry: Open Bottom and closed top entry.
 - a. For top entry, drill holes as needed for cables/auxiliary device wiring from the existing cable tray.
- D. Receptacles:
1. 2-pole, 3-wire, 20 amperes, 120 volts, 60 hertz (NEMA type 5-20R)
 2. Power supplied from AC source independent of control power.
 3. Quantity: 2 receptacles per pump.
 4. Provide interior, convenience duplex power receptacles; one at each end of the interior aisle way; near each door.
 5. Provide Ground Fault Circuit Interrupter (GFCI) type receptacles.
- E. Include internal cable tray/wire gutter support system. This includes cable tray system in front and rear panels as well as on the ceiling of the new control boards, both front and rear sides of aisle way.
1. Plastic raceway/wire gutter with removable covers or evenly spaced anchors with zip ties will be acceptable.
 - a. Anchors must be mounted by screws. Adhesive type anchors will not be acceptable.
- ## 2.02 NETWORK COMMUNICATION SYSTEM
- A. The control board manufacturer shall fabricate, install required devices, test, and include in the first FAT for the control boards.
- B. Provide a Network Communication Cabinet (CP1N) assembly to mount communications devices and associated components. Includes:
1. Location: Control Room.
 2. Firewall (FW1): Refer to Section 27 21 29 – Network Switches.
Quantity: One
 3. Managed Industrial Ethernet Network Switch. Refer to Section 27 21 29 – Network Switches.
 4. Patch panels: Refer to Section 27 20 01 – Fiber Optic Cable and Accessories.
 5. Interface between Network and the new Data Historian Server (located in the Control Room).
 6. The new cabinet shall interface with the new and existing electrical systems.
 - a. This cabinet shall include but not be limited to communication devices and associated components.

7. Cabinet:
 - a. Provide a smooth and finished look, with no gaps or holes.
 - b. Cabinet shall be shipped to site with components pre-installed on/in the cabinet. The cabinet shall be shipped with the control boards.
 - c. Provide wireway for cable routing.
 - d. Overall Dimensions: Contractor to determine based on adequately housing all required components.
 - e. Fabricate as a stand-alone enclosure.
 - f. Structural Requirements:
 - 1) Hinged door for access to internal equipment.
 - 2) Adequate louvers for ventilation.
 - 3) Floor mounting frames if required.
 - 4) Provide terminal blocks with 20% spare terminal points.
 - 5) Provide pre-drilled copper ground bar.
8. Conform to NEMA ICS-1 and NEMA ICS-6.
9. Power: 120 VAC power from existing Uninterruptible Power Supply (UPS) via existing 125 VDC distribution board in CSA5F.

2.03 DISTRIBUTION INPUT/OUTPUT (I/O) CABINETS (111TDIO, 111GHDIO)

- A. Provide Distribution I/O cabinets. Includes but is not limited to the following:
 1. Location:
 - a. Turbine level
 - b. Gatehouse level – Communication closet
 - 1) Install onto a metal panel and affix to the existing communications rack.
 2. Each cabinet and panel shall connect to generator unit's internal network via network switch. Reference drawing 245-D-60134.
 3. Power: 125 VDC from DC panelboard in CSA5F.
 4. Devices located in each cabinet:
 - a. Refer to Section 25 12 10 – Programmable Logic Controllers.
 - 1) Interface with existing stator winding RTDs and other new non-critical RTDs, including thrust bearing, upper guide bearing, lower guide bearing, turbine guide bearing, turbine stuffing box, generator cooling air (1-6), transformer K1A hot RTDs, KSA hot RTD, and others as needed.
 - b. Power Supply: 125 VDC/24 VDC

5. Each cabinet shall be fabricated, populated, and tested (Factory Acceptance Test (FAT)) at the same time as the control boards.
6. Cabinet:
 - a. Provide a smooth and finished look, with no gaps or holes.
 - b. Cabinet shall be shipped to site with components pre-installed on/in the cabinet or panel.
 - c. Provide wireway for cable routing.
 - d. Exterior Finish: Match existing control board (CSA4F) with final approval from the Project Manager.
 - e. Interior Finish: White
 - f. Overall Dimensions: Contractor to determine based on adequate housing all required components.
 - g. Fabricate distributed I/O cabinets or panels
 - 1) Turbine level distributed I/O
 - a) Stand-alone enclosure on Turbine level.
 - i. Hinged door for access to internal equipment with top and bottom latches.
 - ii. Adequate louvers for ventilation.
 - iii. Wall mounting frames if required.
 - 2) Gatehouse distributed I/O
 - a) Install onto panel and then affix to existing communications rack in communications room of Gatehouse.
 - h. Requirements:
 - 1) Provide terminal blocks with 20% spare terminal points.
 - 2) Provide pre-drilled copper ground bar.

2.04 HMI CABINET

1. Location:
 - a. Next to the door at the entrance of the Pole Hill powerplant building on the southeast corner of the south wall.
 - b. Refer to drawing 245-D-60144.

2.05 DC DISTRIBUTION PANELBOARD

- A. Location: Gatehouse
- B. Refer to Specifications Section 26 24 24 – Distribution Panelboards.

2.06 125V DC BATTERY AND CHARGER

- A. Refer to Section 26 33 46 – 125V DC Battery and Charger.

2.07 AUTOMATION CONTROL AND MONITORING SYSTEMS

- A. Refer to Section 25 00 01 – Automation Control and Monitoring Systems.
1. Data Historian Server – new controls shall interface to this server.

2.08 PROGRAMMABLE LOGIC CONTROLLERS

- A. Refer to Section 25 12 10 – Programmable Logic Controllers.

2.09 PROTECTION SYSTEMS

- A. Refer to Section 26 09 16 – Protection Systems.

2.10 COMMUNICATIONS

- A. Refer to Section 27 20 01 – Fiber Optic Cable and Accessories.
1. Includes but is not limited to the following devices and functions:
 - b. Interface new Network Communication to control boards as well as the respective automation control and monitoring system.
 - c. PLC interfaces to network devices.
- B. Refer to Section 27 21 29 – Network Switches.
1. Includes but is not limited to the following devices and functions:
 - a. Ethernet Switches:
 - 1) Interface excitation system, protection system, and automation control and monitoring system.
 - 2) NTP Time synchronization.
 - b. Network Communications Cabinet (CP1N):
 - 1) Network switch
 - c. Gatehouse Communications Rack
 - 1) Network Switch
 - d. Interface PLCs, HMIs, and other network devices to existing SCADA system.
 - e. Time synchronization of automation control and monitoring systems, Refer to Specifications Section 25 00 01 – Automation Control System.

2.11 UNDERVOLTAGE (27) RELAYS

1. Undervoltage relays shall include the following features:
 - a. Location:
 - 1) CSA Control Board
 - 2) Gatehouse
 - 3) Turbine Level
 - b. Purpose: Provide control circuit undervoltage protection
 - c. Rating: 125 VDC.
 - d. Metal Oxide Varistors (MOV).
 - e. Relay undervoltage element to be provided with adjustable voltage range.
 - f. Selectable Reset Function Options: Self-resetting or manual.
 - g. Relay provided with target or light that indicates relay operation.
 - h. Output Contacts: Single-pole, double-throw (SPDT).
 - i. Four output contacts per relay, minimum.

2.12 AUXILIARY RELAYS

- A. Auxiliary Relays, (or interposing relays): Series CS7 Control Relays manufactured by Sprecher Schuh; Bulletin 700, type CF or P as manufactured by Allen-Bradley; type D15 or M as manufactured by Eaton Cutler-Hammer; Class 8501, type X as manufactured by Square-D; or equal, with the following essential characteristics:
 1. Coils suitable for operating continuously at the voltage required for circuit application.
 - a. 125 VDC or 120 VAC as shown on the drawings.
 2. MOVs (Metal Oxide Varistors), across the coil, sized accordingly.
 3. Contact Rating minimum: P300 for contacts used in direct current circuits or A600 for contacts used in alternating current circuits in accordance with NEMA ICS 5.
 4. Provide normally open or normally closed contacts as shown on the drawings.
 - a. For relays that have only normally closed or normally open contacts the spare contacts should match.
 - b. For relays with a mixture of normally closed or normally open, the spare contacts should also include both types.
 5. Accessories as required.
 6. General purpose “ice cube”, “pluggable socket”, or “machine tool” type relays will not be accepted.
 7. Provide latching relays for the unit auxiliaries start/stop control circuits.

8. Provide interposing relays for PLC discrete outputs as necessary.

2.13 CREEP DETECTOR

- A. Type SpeedSys T20 as manufactured by Istec <https://www.istec.com/en/speed-monitoring/> or equal having the following characteristics:
 1. Redundant sensor inputs.
 2. 2x digital input.
 3. 4 high speed relays:
 - a. 2x DPST
 - b. 2x SPST
 4. Analog output:
 - a. 2x analog output.
 - b. 4 to 20 mA current loop.
 5. 2x frequency output.
 6. Modbus TCP.
 7. DIN rail mounted.
 8. An independent speed sensing device for detection of generator creep.
 - a. Using the voltage regulator control to detect creep is not permissible.
 9. Install two active zero velocity speed pickups.
 - a. Creep shall be detected on both pickup sensors. Cannot operate based on detection from only one sensor.
 - b. Creep detection shall activate with a shaft angular displacement of 3 degrees of arc or less without reference to rate of movement and be immune to unit vibration as defined in IEEE 125 Part 5.3.1.
 - c. Creep sensor failure shall also be detectable when the unit is operating at rated speed.
 - d. Power supplies for speed sensors to be fed from creep detector 125 VDC circuit.
 10. Upon detection of unit creep the same actions shall be performed as shown on drawing 245-D-60127.
 11. Include logic to enable creep detection when unit comes to a dead stop after a time delay, and to disable creep when a generator start is performed.
 12. Creep alarm to be latched.
 13. Provide a soft reset push button on HMI screen.

2.14 MULTIFUNCTION METERS

- A. Multifunction Three Display Meter Bitronics M653: manufactured by Novatech, LLC, 261 Brodhead Road, Bethlehem, PA 18017; or equal, having the following essential characteristics:
1. Subpanel mounted.
 2. Size: 14" (w) x 4.5" (h) x 6.5" (d)
 3. Measurements:
 - a. Generator Voltage
 - b. Generator Amps
 - c. Real Power (W)
 - d. Reactive Power (VAR)
 - e. Power Factor
 4. Power: 120VAC
 5. Accuracy:
 6. UL Listed
 7. Revenue Accuracy: 0.2%
 8. Refresh time: 100 milliseconds
 9. Operating Temperature range: -40 C to 70 C
 10. Output: TCP/IP
 11. Inputs Voltage: 120 VAC, 45-65 Hz

2.15 TRICOLOR BAR GRAPH METERS

- A. Gate Limit Indicator and Gate Position Percent Indicator
1. Tricolor Bar graph BG241: manufactured by Weschler Instruments, 16900 Foltz Parkway, Cleveland, OH 44149; or equal, having the following essential characteristics:
 - a. Bar Display
 - 1) 50 segment LED, 2% resolution
 - b. Digital Display
 - 1) 5-digit display
 - 2) Resolution: 0.01 % of full scale
 2. Input
 - a. Accuracy: 0.3% full scale
 - b. Response Time: <600ms, zero to full scale.

3. Setpoints
 - a. 2 SPDT form C relays
 - b. Hysteresis 0-10% full scale or latching
 - c. Time Delay: 0-10 seconds

2.16 BAR GRAPH METERS

A. Turbine Speed Meter (RPM)

1. The resolution shall encompass the speed of the turbine from 0 to its maximum speed of 450 rpm.
 - a. Bar Graph BG241: manufactured by Weschler Instruments, 16900 Foltz Parkway, Cleveland, OH 44149; or equal, having the following essential characteristics:
 - 1) Bar Display
 - a) 101 segment, 1% full scale resolution.
 - 2) Digital Display
 - a) 3.5-digit display
 - b) Resolution: 0.01% full scale
 - 3) Input
 - a) 4-20mA
 - 4) Response time:
 - a) DC: Less than 600 milliseconds full scale
 - b) AC: Less than 800 milliseconds full scale

B. Turbine Speed Adjustment Percent Meter

1. Bar Graph BG241: manufactured by Weschler Instruments, 16900 Foltz Parkway, Cleveland, OH 44149; or equal, having the following essential characteristics:
 - a. Bar Display
 - 1) 101 segment, 1% full scale resolution.
 - b. Digital Display
 - 1) 3.5-digit display
 - 2) Resolution: 0.01% full scale
 - c. Input
 - 1) 4-20mA
 - d. Response time:
 - 1) DC: Less than 600 milliseconds full scale

- 2) AC: Less than 800 milliseconds full scale

C. Synchronizing Voltage Meters

1. Bar Graph BG241: manufactured by Weschler Instruments, 16900 Foltz Parkway, Cleveland, OH 44149; or equal, having the following essential characteristics:
 - a. Location: (CSA2F)
 - 1) Generator Voltage - Incoming Volts
 - 2) Bus Voltage - Running Volts
 - b. Bar Display
 - 1) 101 segment, 1% full scale resolution.
 - c. Digital Display
 - 1) 3.5-digit display
 - 2) Resolution: 0.01% full scale
 - d. Input
 - 1) 4-20mA

2.17 DIGITAL PROCESS METERS

A. Field Volts and Field Amps Meters

1. Digital Panel meter PD765: manufactured by Precision Digital Corp., 233 South Street, Hopkinton, MA 01748; or equal, having the following essential characteristics:
 - a. Location: CSA3F
 - b. EIA-485 serial port with Modbus RTU communications.
 - c. Power Supply: 90-265VDC
 - d. Function:
 - 1) Generator Field Amps
 - 2) Generator Field Volts
2. Measuring transducers to be provided by the Contractor.
3. Accuracy: .01 percent full scale or better
4. Configurable input range: +/-20mADC, +/- 10VDC
5. Include cabling, terminating resistors, and network adapters as required for EIA-485 Modbus RTU network.

2.18 POWER REVENUE METER

- A. Type SEL, 735 Model 735#04 (or current equivalent number) Relay manufactured by Schweitzer Engineering Laboratories, 2350 NE Hopkins Court, Pullman, WA 99163; or equal, having the following essential characteristics:
1. Class: 0.1 Revenue Metering Accuracy
 2. CL2/10/20 Current Class
 3. 125 VDC.
 4. Communication Ports:
 - a. 10/100BASE-T Ethernet Port
 - b. EIA-232 Rear Serial Port
 - c. USB-C Front Port
 - d. TCP/IP Modbus, DNP3, IEEE C37.118 Synchrophasers
 5. 512 Samples-per-cycle
 6. Voltage and Current Harmonics
 7. Wire Potential Transformer and Current Transformer signals through test switch blocks to the power revenue meter.
 8. Design with power revenue metering for megawatts, megaVARS, current, and voltage that interface to the PLC and existing SCADA system.

2.19 SYNCHROSCOPE

1. Location: CSA2F
2. Synchroscope: Model 007 Synchroscope Switchboard Meter as manufactured by Crompton, 16900 Foltz Parkway, Cleveland, OH 44149; or equal, having the following essential characteristics:
 - a. Voltage: 120VAC
 - b. Size: 4.5 inch by 4.5 inch.
 - c. Meter Face: Slow-Fast

2.20 SYNCHRONIZING LAMPS

1. Location: CSA2F
 - a. The lamps shall switch off when phases and voltages are synchronization-ready.
 - b. Lamps shall be continuously dimmable to reflect when the phases and voltages are synchronization-ready.

2.21 INDICATING LIGHTS

- A. Indicating Lights: 800T-PB-QH2G as manufactured by Rockwell Automation, 1201 South Second Street, Milwaukee, WI 53204; or equal, having the following essential characteristics:
 - 1. Push-to-test, 30.5 mm, LED type, 130 volts AC/ DC, full voltage.
 - 2. Lens color as shown on the drawings.
 - 3. Light assembly lamps shall be replaceable from the front of the panels.
- B. Provide indicating lights as shown on drawings 245-D-60135, 245-D-60136, 245-D-60137, 245-D-60138, and 245-D-60139.
 - 1. Flush mounting, push-to-test lights.

2.22 CONTROL SWITCHES

- A. Control Switches: Electros witch, series 24 manufactured by the Electros witch Unit of Electros witch Corporation, 180 King Avenue, Weymouth, MA 02188; or equal, having the following essential characteristics:
 - 1. Switch Escutcheons & Indicating Lamps: As indicated on drawing 245-D-60148.
 - 2. Type: Heavy-duty rotary switches with double-sided, double-wiping, knife-type contacts.
 - 3. Contacts: 3 Amp interrupting at 125 Volts DC minimum.
 - 4. Provide a minimum of one spare contact in each position.
- B. Handle positions, switch escutcheons and descriptions, type, and rotating action as shown on the drawings.

2.23 TERMINAL BLOCKS

- A. Interface terminal blocks: Type EB-25 manufactured by General Electric Company, 41 Woodford Avenue, Plainville, Connecticut 06062; Buchanan catalog Nos. B104 through B112 manufactured by Tyco Electronics, P.O. Box 3608, Harrisburg, PA 17105-3608; Marathon catalog Nos. 1604 through 1612 manufactured by Marathon Electric Manufacturing Corporation, 13300 Van Camp Road, Bowling Green, OH 43402; or equal, having the following essential characteristics:
 - 1. Type: One piece, molded-block with molded-insulating barrier between terminals. Sectional type or DIN rail mounted terminal blocks are not acceptable.
 - 2. Rating: 600 Volts, 30 Amperes, minimum.
 - 3. Terminals: Binding-head or washer-head screws with serrated or grooved-contact surfaces suitable for use with ring-tongue type terminal connectors.
- B. Terminal blocks for wiring completely contained within the cubicle: Manufacturer's standard type.

1. High-density, sectional or DIN rail mounted terminal blocks will be acceptable.
2. Rating: Suitable voltage and current ratings for the connected circuits.
3. May not be used for external cabling.
4. Arranged and located such that incoming and outgoing cables/auxiliary device wiring can be supported.
5. Separate adjacent rows by at least 6-inches edge to edge.
6. Provide 6-inches from the sides, top, or bottom of control cabinet for wiring that will be made in field.
7. Machine letter circuit designations, as shown on drawings, on white marking strips. Mount marking strip on terminal block.
8. Provide 20 percent spare terminals.
9. Suitable for use in high-vibration areas.

C. Furnish with removable covers and full-size marking strips.

2.24 RELAY TEST SWITCH BLOCKS

- A. Type ABB type FT-1, manufactured by the ABB Company 12040 Regency Parkway Cary, NC 27518: having the following essential characteristics:
1. Provide test block(s) with each protective relay and each lockout relay to facilitate the testing of currents, voltages, inputs, outputs, and trip circuits.
 - a. Provide black handles for PT circuits and red handles for current transformer (CT) circuits.
 - b. Provide 20 percent spares.
 2. Location: control boards as shown on drawings.
 3. Include clear lockable cover with thumbnuts and bracket.
 4. Provide shorting type terminal blocks for CT circuits where wires first enter the new control board.
 - a. Arrange the shorting type of terminal blocks to completely isolate the internal circuits from the external instrument transformer. The current transformer secondary circuits cannot be open circuited in any position while the test plugs or cover plugs are being inserted or removed.
 5. Provide test switches for all connected inputs and outputs for each new multi-function protection relays. Refer to Section 26 09 16 – Protection Systems.
 - a. Utilize potential type test switches for inputs and normally open output contacts and shorting type test switches for normally closed output contacts.
 6. Provide potential type test switches for all connected normally open lockout relay output contacts.

- a. Provide shorting type test switches for all connected normally closed lockout relay output contacts where opening the contact will cause an action.

2.25 FUSES AND FUSE HOLDERS

A. Fuses:

1. Current-limiting type.
2. Cartridge style: glass fuses will not be acceptable.
3. Current ratings determined by contractor.
4. Suitable for 'finger-safe' type fuse holders if not using pullout fuse block.

B. Fuse Holders:

1. Modular 'finger-safe' type or pullout type.
2. Current Rating: Not less than fuse size.
3. Furnish with nameplate identifying protected circuit.

2.26 DIN MOUNTING RAIL

A. Type: Steel, slotted.

B. Dimensions: 35 mm width, 7.5 mm height, length as required.

2.27 NAMEPLATES AND LABELS

A. Contractor to furnish and install nameplates for control switches, indicating lights, meters, pushbuttons, relay test blocks, terminal blocks, and other subpanel mounted devices and in accordance with Standard Drawings 40-D-5324 and 40-D-6234. This includes the nameplates for the cabinets and the cabinet devices.

1. Refer to specification/design drawings: 245-D-60147.
2. The CSA and CP1N cabinets shall each have one nameplate on the exterior (top, center) of each cabinet.

B. Engraved Nameplates:

1. Material: Laminated plastic, stainless steel screws for fastening (with pan head self-tapping screws). Do not use adhesives to attach nameplates.
2. Color: Black, engraved to white core.
3. Each exterior (device visible from the exterior of each panel (both front and rear) device nameplate: material and color as stated above.
4. Each internal device:

- a. Each device mounted within equipment enclosures (interior of each panel both front and rear) may be identified with the device name or standard NEMA or IEEE designation on or adjacent to the device.
 - b. Equipment manufacturer's standard or custom engraved nameplate suitable for mounting directly over device. No adhesives.
 - c. Provide nameplates for all devices internal to the Control Boards in accordance with Reclamation device location designations as shown on standard drawings.
- C. Provide nameplates for all Control Board sections (exterior); CSA3F, CSA2F, CSA1F, and CSA1R, CSA2R, CSA5R. Locate top center, on the exterior side of each respective panel.
- D. Labels for equipment mounted on wiring side of equipment panels.
 - 1. Labels shall be white with black lettering, adhesive type, and machine lettered.
 - 2. Provide double letter device location labels attached to the panel for every device as shown on the drawings.
 - 3. Include additional labels for devices that use the IEEE C37.2 device designation naming convention for example, 186GP, 143UCS, and 111GPLC.
 - a. Attach the device designation label directly on the device itself if possible or in an adjacent location that clearly indicates the equipment that the label is identifying.
- E. Provide labels for terminal blocks. Labels to be white with black lettering, adhesive type, and machine lettered with the single letter terminal block designation as shown on wiring diagrams.
- F. Devices include nameplates on the exterior (outside face) of the panel and device designation labels on the interior (inside face) of the panel as indicated on the drawings.

2.28 FABRICATION

- A. Creep Detector:
 - 1. Remove the existing mechanical shaft creep detector including wiring, shaft brush and solenoid, and support bracket.
- B. Mount control devices on front and rear panels of each set of control boards.
- C. Mount ground bus a bottom of enclosures. Electrically bond to each enclosure.
- D. Where copper bus or other bare metal is used for 125 volt direct-current (DC), control bus circuit connection, it shall be wrapped with insulating tape or otherwise suitably insulated according to the Contractor's standard.

- E. All subpanel mounted control boards, cabinets, and panels shall be constructed at the control boards facility and shall be delivered to the site ready for installation.
- F. All incoming and outgoing cables/auxiliary device wiring for the new control boards shall enter the equipment from below or above.
 - 1. Refer to Section 26 04 10 – Electrical Demolition and Removal.
- G. Wiring Devices: applies to control boards, cabinets and panels.
 - 1. Enclosure wire:
 - a. A wireway duct system (internal cable tray), consisting of overhead horizontal wire-ways (near the ceiling), horizontal wire-ways throughout front and rear sections and vertical wire-ways in the side (wing) sheets, with removable covers shall be provided for the wiring within each panel, both front and rear.
 - b. All wiring within the wire-ways shall be completely accessible for operation, maintenance, removal, and replacement.
 - c. The removable covers shall be completely accessible for removal and replacement.
 - d. Exposed wiring shall be kept to a minimum. Where used, exposed wiring shall be formed into compact groups bound together and firmly supported. Exposed wiring shall be run straight, horizontally, or vertically with short radius right-angle bends.
 - e. Hinge wiring shall be twisted around the longitudinal axis of the wire, wherever possible, instead of being bent laterally.
 - f. Each wire shall be protected from abrasion where it leaves a channel or wire-way. Grommets shall be provided at all holes and all elliptical slots furnished for wires, conductors and cables.
 - g. Splices shall not be permitted in the wiring.
 - h. All connections shall be made at device studs or terminal blocks. A maximum of two connections can be made to any given terminal. All wiring between various devices shall be performed by the most direct method rather than looping back and forth across the five sections of the new control boards.
 - i. Wiring from the subpanel mounted control boards to exterior equipment shall be made using multi-conductor shielded control cables (auxiliary device wiring).
 - j. Wiring between panels of the control board sections shall be routed from device to device via the overhead cable wire-way.
 - k. Circuit/conductor designations as shown on the schematic and wiring diagrams shall be machine printed, stamped, or engraved on a white heat shrink making sleeve at every wire termination other than those at a

terminal block. Terminal block marking shall be as specified in subparagraph below.

2. Ground Bus

- a. Bond mechanically and electrically to control board and plant grounding system.
- b. Drill holes in each bus to allow for attaching equipment ground conductors.
- c. Refer to Section 26 05 20 – Grounding and Bonding.
- d. The ground system bus shall electrically connect the structures in the control board assemblies and shall run the full length of the board.
- e. Circuits connected to the ground bus shall be arranged so that each may be disconnected without disturbing the continuity of the ground bus or any other grounded circuit.

3. Terminal Blocks

- a. Incoming and outgoing cables (auxiliary device wiring) shall be supported and all wiring to the blocks shall be completely accessible without having to move other devices for accessibility.
- b. Separate adjacent rows and columns by at least 6 inches edge to edge.
- c. Provide 6 inches from the sides, top, or bottom of control cabinet for wiring that will be made in field.
- d. Terminal blocks shall be installed in each wing panel of each panel (both front and rear of the 5 sections).
- e. Separate AC and DC terminal blocks.
- f. Provide 20 percent spare terminals in each panel. For example, CSA3F shall have 20 percent spare terminals in the two wing panels combined. This applies to all front panels and all rear panels.
- g. Provide current shorting blocks for current transformer circuits where the wires first enter the control boards. This will facilitate interconnections of wiring between adjacent control board panels, other control board sections and external (current) equipment. These blocks shall be mounted on the side sheets (wing panels; left or right).
- h. The terminals shall be arranged so that the conductors in each cable are fully grouped together possible.

H. Wiring and Wiring Connections: applies to control boards, Distributed I/O cabinets, and Network cabinets and panels.

1. Shields to be terminated at control board receiver end at the field cable termination terminal block.
2. Wire Terminations:

- a. Install ring tongue pressure-crimp-type connectors for terminations at terminal blocks and electrical devices.
 - b. Termination of wire without terminal connector is not acceptable.
 - c. The use of ferrule-type connectors or bare wire connections is not acceptable.
 - d. Ethernet Cat6A terminals shall utilize punch-down keystone jacks.
- I. Wire Markers:
 - 1. Install wire markers on conductor end.
 - 2. Type conductor designation and terminal number on first line.
- J. Nameplates shall be furnished and installed as shown on, but not limited to, arrangement drawings, nameplate lists and standard nameplates drawings. Refer to specification drawing 245-D-60147. The control board panel designation shall also be placed on the inside of each control board panel, using a nameplate identical to that located at top (center) of the exterior side of the panel.
- K. Construction of the subpanel mounted control board panels, distributed I/O cabinets (where applicable) and Network cabinets and panels (where applicable).
 - 1. Structural and Mechanical Features: The equipment shall have the following structural and mechanical features:
 - a. Panels shall be of especially leveled sheet steel no lighter than No. 11 United States Standard Gauge.
 - b. Easily removable subpanels
 - c. The interior structural framework shall be constructed according to the contractor's standard practice and is intended to support subpanel mounted electrical devices and blank sub-panels. The contractor shall provide detailed drawings of the framework for approval prior to construction of the panels.
 - d. Individual sub-panels shall be fabricated according to the drawings and mounted on the framework as shown on the panel arrangement drawings. The contractor shall provide final detailed control board shop fabrication drawings of each sub-panel for approval prior to fabrication of the sub-panels. Blank sub-panels to fill unused space shall be provided by the contractor. Blank sub-panels shall be constructed such that their vertical dimension is an integer multiple of 1-3/4 inches to maintain consistency with the standard subpanel dimensions. Mounting holes for the blank subpanels shall be spaced like the standard subpanels with corresponding dimensions.
 - e. Subpanel mounted control board sections shall be designed so that the control boards can be easily extended/modified in the future.

Additionally, each subpanel mounted control board section shall have the following features:

- 1) Welds, fastenings, and cut edges shall not be visible from the front or the rear of the control boards.
 2. Electrical Features: The following electrical features shall be furnished for new control boards, new distributed I/O cabinets (where applicable) and Network Cabinets and Panels (where applicable):
 - a. Separate 120 volt, single-phase, AC circuits shall be provided for the convenience outlets (receptacles) located within the control boards. One receptacle at each end of the interior aisle near the doors; total of two receptacles. The power supplied from AC source is independent of control power. These circuits shall be wired to outgoing terminal blocks.
- L. Paint:
1. Painting Exterior Surfaces of Control Boards, Distributed I/O Cabinets and Network Cabinet: The exterior shall be painted to match the existing control board color of CSA4F.
 2. Painting Interior Surfaces of Control Boards, distributed I/O Cabinets and Network Cabinet: The interior shall be contractors standard white.

2.29 DESIGN AND MANUFACTURING PROGRESS REVIEW MEETINGS

- A. Refer to Section 01 31 19 – Project Management and Coordination.
- B. Purpose: Pre-Design and Progress Review Meetings are required to ensure that all parties understand all requirements of this contract and that construction of the control boards, Distributed I/O cabinets and Network cabinets and panels are of the highest quality and are accomplished in a timely manner.

2.30 CONTRACTOR SOURCE QUALITY TESTING

- A. Test at factory for correct operation and mechanical adjustment.
- B. Tests: The control board's electrical devices and equipment, CSA cabinet's electrical devices and equipment, and network cabinets and panels' electrical devices and equipment, including wiring shall be subjected to the following tests:
1. The Contractor is responsible for isolating electrical equipment with sensitive components from all tests to avoid damage to or improper operation.
 2. The Contractor shall:
 - a. Prepare a schedule for the tests shown below.
 - b. Be responsible for all testing connections.
 - c. Have performed all testing prior to the arrival of the Government. As stated in Section 01 31 19 – Project Management and Coordination (One

Hundred-Percent Design and Manufacturing Progress Review Meeting), Government shall witness repeats of selected operation and control wiring tests.

3. The Contractor shall furnish two copies of pre-inspection test reports which detail (as a minimum): the type of test performed, date, and tests performed by, equipment tested, failures or unusual results, missing equipment, and corrective measures taken. The test reports shall be provided to the Project Manager at the beginning of the One Hundred-Percent Design and Manufacturing Progress Review Meeting.
4. The Contractor shall furnish two copies of final test reports which include the pre-inspection test report data, and the results of the final inspection tests. These reports shall detail (as a minimum): the type of test performed, date, and tests performed by, equipment tested, failures or unusual results, missing equipment, and corrective measures taken. The final test reports shall be included in the respective submittal per Section 01 33 00 – Submittals, Table 01 33 00A.
5. Examples of tests to be performed are as follows:
 - a. Continuity and Operation: The correctness of the control wiring and logic of the control boards, distributed I/O cabinets and Network cabinet and panel shall be verified by actual electrical operation of the component control and monitoring devices, as follows:
 - 1) All AC current and potential circuits shall be checked by applying three-phase currents or potentials at the external cable connection terminal blocks and verifying the phase angle and magnitude of each quantity at the terminals of each device in the circuit being tested.
 - 2) All DC control circuits shall be checked by operation of each device.
 - 3) Remote devices such as power circuit breakers shall be simulated at the external cable termination blocks in the control boards, distributed I/O cabinets and Network cabinets and panels as applicable. For example, the simulation equipment shall accurately simulate power circuit breaker trip and close coil currents, as well as auxiliary contact operation.
 - 4) Inject test signals into digital and analog inputs of distributed I/O module and PLC to demonstrate that the signal is received at the PLC and RTU system. Simulate data and commands from excitation system for testing.
 - a) Vary the inputs so that the alarms and trips are detected by the protection relays and PLC. Verify that the alarms are displayed correctly in the unit control's HMI window under the annunciation screen tab, as well as on the large window annunciation display. Ensure that both displays respond appropriately to each input.

- b) Verify that the RTU system receives the correct data and alarms/annunciations to each input.
- 5) Demonstrate time synchronization for protection relays and PLC.

2.31 FACTORY ACCEPTANCE TESTS

- A. Government reserves the right to inspect equipment at supplier's manufacturing plant prior to final testing.
- B. Government shall witness FAT for new control boards, new distribution I/O cabinets, and Network cabinets and panels.
- C. The purpose of the Factory Acceptance Test (FAT) for the Control Boards, distribution I/O cabinets, and Network cabinets and panels with the Automation Control and Monitoring System, and Protection System is to demonstrate the functionality, performance and stability of the hardware and software. All Automation Control and Monitoring System components shall be tested during this time.
- D. Coordinate factory acceptance testing of Control Boards, distribution I/O cabinets, and Network cabinets and panels (Section 26 09 18) with the following sections:
 - 1. Section 25 00 01 – Automation Control and Monitoring Systems
 - 2. Section 26 09 16 – Protection Systems
 - 3. Section 26 33 48 – DC Panelboards
 - 4. Section 25 12 10 - Programmable Logic Controllers
 - 5. Section 27 20 01 – Fiber Optic Cable and Accessories
 - 6. Section 27 21 29 – Network Switches
 - 7. Section 28 05 11 – Cyber Security Requirements for Electronic Safety and Security
- E. Notify the Government before acceptance tests are to be conducted and as agreed by the Project Manager in accordance with Section 01 33 00 – Submittals, Table 01 33 00A.
- F. Prior to Factory Acceptance Test of the Control Boards, distribution I/O cabinets, and Network cabinets and panels, the Contractor shall
 - 1. During FAT, relays are tested with all protection elements enabled. Test each relay via the Ennoserve (Doble).
 - 2. Bench test all equipment, devices.
 - 3. Fabricate in accordance with approved drawings.
 - 4. Compare equipment nameplate data with specifications and approved drawings.
 - 5. Specific tests prior shall include but not limited to:
 - a. Simulation of each discrete data point to the Automation System.

- b. Simulation of each analog data signal to the Automation System using current or voltage source.
 - c. Operation simulation of communication between PLC, Distributed I/O devices, meters, network switches (managed) and Data Historian Server.
 - d. HMI display screens and data display.
 - e. Simulation of HMI status functions in Local MANUAL, Local AUTO and Supervisory-control modes.
 - f. Simulation of Annunciation HMI alarms.
 - g. Verify all pre-FAT checks match approved drawings and data.
 6. Perform burn-in test of the equipment for a minimum of 72 hours.
 7. The above items are examples of what will be witnessed during the FAT by the Government.
- G. Other FAT tests shall include but not be limited to:
1. Manually exercise and function test all devices.
 2. Verify that hardware and software meet the requirements of the specifications and approved drawings.
 3. Include point to point checks of all connections and application simulation tests that demonstrate automation controls will operate as described in these specifications.
 4. Current transformers, voltage potential transformers and RTD feeds to digital multi-function protection relays shall be simulated where required for full testing and demonstration purposes.
 - a. Demonstrate CT and PT circuits are wired and terminated to appropriate test switches in accordance with approved drawings.
 5. Equipment shall be subjected to the production tests of IEEE C37.21.
 6. Annunciation:
 - a. Certification of performance and result of annunciation testing, including all network communication paths and network failure scenario testing.
 - b. Provide the network packet capture files for all locations for which those captures took place.
 - 1) Uniquely name the files so that the file contents correspond to, and can be identified, using the network diagram provided in this submittal.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Control Boards, distribution I/O cabinets, and Network cabinets and panels shall be completely assembled, wired, and tested at the factory prior to shipment.
- B. Install new speed sensors for creep detection.
 - a. Contractor to provide location and mounting brackets for sensors and approved by the COR.
 - b. Existing toothed wheel at the governor Permanent Magnet Generator may be used for speed sensing.
- C. Panels shall be pre-wired with all devices connected to field termination blocks except for network communication interfaces.
- D. Devices shall be mounted in parallel even spaced rows and shall present a neat and orderly appearance.
- E. All equipment, including terminal blocks, mounting hardware, and wiring shall be accessible for operation, maintenance, removal, and replacement.
- F. Develop and format detailed instructions and checklists in accordance with 01 33 26 – Electrical Drawings and Data.
- G. Contractor shall reuse existing raceway and cable trays that exist for the new control boards.
- H. Contractor shall furnish new cables/auxiliary device wiring and conductors and raceways in accordance with Section 26 05 10 – Conductors and Cables, and Section 26 05 33 – Raceways and Boxes.
- I. Install new Control Boards, distribution I/O cabinets, and Network cabinets and panels at locations shown on drawings in accordance with NFPA 70 and as recommended by manufacturer.
 - 1. Anchor boards and cabinets with rustproof bolts, nuts, and washers not less than 0.5-inch diameter.
 - 2. Provide conduit and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings.
- J. Control Board Wiring:
 - 1. Install wiring in a neat and workman like manner with wireways provided inside enclosures and cable ties used to secure all conductors installed outside wireways.
 - 2. Support incoming and outgoing cables/auxiliary device wiring.

K. Terminations and Connections:

1. Support and secure wire bundles with cable tie mounting bases. Secure cable tie mounting base with two No. 8 screws. Mounting Cable tie bases with only adhesives is not allowed.
2. Exposed wiring:
 - a. Keep to minimum.
 - b. Support firmly.
 - c. Run straight, horizontally, or vertically with short-radius, right-angle bends.
 - d. Twist hinge wiring around longitudinal axis of wire, wherever possible, instead of being bent laterally.
 - e. Protect each wire from abrasion where it leaves a channel or duct.
3. Maintain daily log of accomplished work, incorporate check-prints into both sets of on-site working drawings for the Control Boards, Protection System

3.02 PRE-COMMISSIONING

- A. Government will witness all tests. Notify Project Manager in writing prior to test, Refer to Section 01 33 00 – Submittal, Table 01 33 00A.
- B. Refer to Specification Section 26 05 90 – Wiring Checkout and Tests.
- C. Communication Links:
 1. Verify communication links between devices.
- D. Visual Inspection and Tests:
 1. Inspect physical, electrical, and mechanical condition.
 2. Verify appropriate anchorage, required area clearances and correct alignment.
 3. Verify electrical device sizes and types corresponding to approved drawings.
 4. Confirm correct operation and sequencing of all electrical components and indicating devices.
 5. Vacuum-clean the interior and exterior of new control boards.
 6. Exercise all active components.
 7. Verify all field cables/auxiliary device wiring are terminated on appropriate terminal blocks within the Control Boards, distribution I/O cabinets, and Network cabinets and panels corresponding to approved drawings.
- E. Perform operational tests on the following equipment or devices to demonstrate that specifications requirements have been met:
 1. Meters.

2. Control switches.
3. Indicating lights.
4. Test switch blocks.
5. Automation Control and Monitoring PLC assembly.
6. Distributed I/O
7. Lockout relays.
8. Communication equipment.
9. Annunciation Display, Gatehouse HMI, Automation Control and Monitoring HMI, Entrance HMI (monitoring only).
10. Digital multifunction protection relays.
11. Control relays and undervoltage relays.

3.03 COMMISSIONING TESTING AND VERIFICATION

- A. Test new Control Boards, distribution I/O cabinets, Network cabinets and panels, and corresponding devices in accordance with FIST 4-1B, and Section 26 09 16 –Protection Systems.
- B. Perform revenue meter testing in accordance with FIST 3-10.
- C. Set field-adjustable switches, auxiliary relays, time-delay relays, timers, and overload-relay pickup and trip ranges.
- D. Submit a Commissioning Test Plan and Procedure for the Government to review per Section 01 33 00 – Submittals, Table 01 33 00A.
 2. Verify that commissioning tests are sufficient to test the new panels and automation controls safely and adequately.
- E. Provide all labor and equipment for all adjustments and modifications during commissioning tests.
- F. Coordinate necessary outages in accordance with COR.
- G. Check process timing sequences and adjust where necessary to improve systems operation.
- H. Adjust logic, settings, and HMIs.
- I. Verify logic and operation of the systems.
- J. Protection System:
 1. Through put functional testing of Unit-specific digital multi-function protective relays test results.

2. Functional testing of lockout relays test results.
- K. Annunciation:
1. Functional testing of annunciation HMI and annunciation windows.
- L. Include checklists and data logs approved by Project Manager with signature of Contractor's Lead Test Engineer.
1. Include list of Contractor and personnel performing and witnessing the field testing and commissioning for each pump.
 2. Provide records of clearances, measurements, test results and position or state of all components at conclusion of commissioning activities.
 3. Include results of all tests and corresponding summary of the commissioning for each set of tests shall be separated by equipment and systems.
 4. Include results for all network communication paths.

3.04 TRAINING

- A. Provide training in accordance with Section 01 79 20 – Training.
1. Training coverage: Pertinent points involved in operating, starting, stopping, and servicing equipment, including major elements of service manuals.
 2. Demonstrate routine maintenance and operations such as tuning, general inspection, and calibration.
 3. Demonstrate calibration and programming changes to each device requiring these changes.
- B. Training sessions shall be provided by the control board OEM discussing maintenance, troubleshooting of fault codes, theory of operation and adjustment of protection system.

3.05 FINAL AS-BUILTS

- A. Final As-built Drawings Government Format: refer to specification Section 01 78 30 and Section 01 33 26 – Electrical Drawings and Data.
- B. Marked prints of specifications drawings indicating “Final As-Built” changes made to electrical components during construction.

3.06 O&M INSTRUCTION BOOK

- A. Submit Final Operation and Maintenance (O&M) Instruction Books as outlined in specification Section 01 78 30 – Project Record Documents.

END OF SECTION

SECTION 26 12 19 PAD MOUNTED TRANSFORMER

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Pad Mounted Transformer:
1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

- A. Environmental Protection Agency (EPA)
1. OPPTS 835.3100-1998 Fate, Transport and Transformation Test Guidelines
- B. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE C2-2023 National Electric Safety Code
 2. IEEE C57.12.00-2021 Liquid-Immersed Distribution, Power, and Regulating Transformers
 3. IEEE C57.12.28-2014 Pad-Mounted Equipment Enclosure Integrity
 4. IEEE C57.12.90-2021 Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers
- C. InterNational Electric Testing Association, Inc. (NETA)
1. NETA ATS-2025 Acceptance Testing Specifications
- D. National Electrical Manufacturers Association (NEMA)
1. NEMA 250-2020 Enclosures for Electrical Equipment
 2. NEMA TR1-2013(R2019) Transformers, Regulators, and Reactors
- E. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electrical Code (NEC)

1.03 QUALITY ASSURANCE

- A. UL Listed.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Handle and store equipment in accordance with manufacturer's instructions. Include copy of these instructions with equipment at time of shipment.

1.05 SUBMITTALS

- A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.
- B. RSN 26 12 19-1, Approval Drawings and Data:
 - 1. Manufacturer's technical data.
 - 2. Equipment layout diagram.
 - 3. Nameplate list.
 - 4. Bill of materials.
 - 5. Schematic diagram.
 - 6. Wiring diagrams.
 - 7. Installation instructions.
- C. RSN 26 12 19-2, Factory Test Reports.
- D. RSN 26 12 19-3, Field Test Reports.
- E. RSN 26 12 19-4, As-Built Drawings.
- F. RSN 26 12 19-5, Operation and Maintenance Instruction Book:

PART 2 PRODUCTS

2.01 TRANSFORMER

- A. Conform to IEEE C57.12.00, C57.12.28, and NEMA TR1.
- B. Suitable for installation outdoors and operation under project conditions.
- C. Type: Self-cooled, outdoor, liquid-filled, three-phase, pad mounted.
- D. Dielectric coolant:
 - 1. Ester-based (soy).
 - 2. Listed less-flammable fluid meeting requirements of NFPA 70 and of IEEE C2.
 - 3. EPA OPPTS 835.3100:
 - a. Non-toxic, non-bioaccumulating.

- b. Readily and completely biodegradable.

E. Ratings and Features:

- 1. Capacity: 500 kVA.
- 2. Primary Specifications
 - a. 13,800 volts, 3-phase, 60-hertz.
 - b. 110-kilovolts BIL
 - c. Delta without neutral bushing.
 - d. Taps: Two 2 1/2 percent FCAN and two 2 1/2 percent FCBN.
- 3. Secondary Specifications:
 - a. 480 volts.
 - b. Grounded Wye with neutral fully insulated with removable ground strap.
 - c. 30-kilovolts BIL.
- 4. Impedance in accordance with IEEE C57.12.00.

F. Integral Fused Disconnect Switch:

- 1. Nominal Voltage: 13,800 volts.
- 2. Rated Maximum Voltage: 15,000 volts.
- 3. Rated Insulation Level: 95-kilovolts BIL.
- 4. Rated Continuous Current: 50 amperes.
- 5. Rated Load Switching Current: 50 amperes.
- 6. Rated Short Time Current (2 Seconds): 38,000 amperes.
- 7. Peak Withstand Current: 98,800 amperes.
- 8. Fuse:
 - a. Nominal Voltage: 13,800 volts.
 - b. Rated Maximum Voltage: 15,000 volts.
 - c. Fuse Size: 30E.
 - d. Rated Interrupting Current: 31,500 amperes, rms, symmetrical.

2.02 CONSTRUCTION

- A. Enclosure: Totally enclosed, dead front, with three-point latching security door with padlock.
- 1. In accordance with IEEE C57.12.28
 - 2. Separate low-voltage compartment.

- B. Core: Grain oriented silicon steel.
- C. Windings: Copper.
- D. Four lifting lugs.
- E. Minimum two stainless steel NEMA 2-hole pattern ground pads.
- F. Primary well.
- G. Secondary voltage bushings with NEMA 2-hole pattern spades.
 - 1. Suitable for connecting at least two parallel sets of conductors.
- H. Sealed tank construction of sufficient strength to withstand a pressure of 7 psig without permanent distortion, and 15 psig without rupturing.
- I. Conduit and cable entry from full length bottom of termination chambers.

2.03 ACCESSORIES

- A. Indicating and Protective Devices:
 - 1. Located outside cable connection compartment.
 - 2. Provide NEMA 250 Type 4 enclosure.
 - 3. Instruments to include:
 - a. Liquid level gauge.
 - b. Dial type thermometer.
 - c. Automatic pressure relief device.
 - d. Pressure/Vacuum gauge.
- B. One-inch drain valve with sampling device and 1-inch upper fill plug.
- C. Nameplates:
 - 1. Manufacturer's nameplate:
 - a. Stainless steel laser engraved.
 - b. Nameplate shall meet IEEE C57.12.00 for Nameplate B.
 - 2. Engraved nameplates:
 - a. Material and size as shown on standard drawing 40-D-6234.
 - b. Provide equipment designation nameplate.
 - c. Provide nameplates for devices mounted on outside of doors that are not furnished with a manufacturer's nameplate or escutcheon.
 - d. Provide nameplate for devices mounted within low-voltage compartments.

2.04 FINISHES

- A. Manufacturer's standard finish.

2.05 SOURCE QUALITY CONTROL

- A. Transformer Factory Tests:
1. Manufacturer's standard factory tests including at a minimum:
 - a. Percent impedance (85 degrees C) at rated current and tap extremes.
 - b. Excitation Current (100 percent voltage) test.
 - c. Winding Resistance measurement tests.
 - d. Ratio Tests using all tap settings.
 - e. Polarity and Phase relation tests.
 - f. Applied and Induced potential tests.
 2. Testing conducted in accordance with IEEE C57.12.90.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Place on raised housekeeping pad where existing KSA is located.
- B. Use galvanized steel channels embedded in housekeeping pad.
1. Determine channel spacing and arrangement from manufacturer's layout plan.
 2. Size channels to evenly support transformer.
 3. Level channels with top of housekeeping pad.
 4. Level and in same plane within manufacturers installation instructions for required tolerances.
 5. Bolt transformer in place on steel channels.
 6. Anchor, support, and mount equipment level and flush to prevent ingress of foreign materials into enclosure or resulting space.
- C. Make pad mounted transformer ready for energizing.
- D. Connect grounds to ground system in accordance with 26 05 26 – Grounding and Bonding and NFPA 70.

3.02 FIELD QUALITY CONTROL

- A. Perform inspections and tests in accordance with NETA ATS.

- B. Measure primary and secondary voltages and make appropriate tap adjustments.

END OF SECTION

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SECTION 26 22 12

DRY-TYPE TRANSFORMER

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Dry-Type Transformer:

1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI)

1. ANSI C57.12.55-1987(R1998) Dry-Type Transformers Used in Unit Installations, Including Unit Substations – Conformance Standard

B. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C57.96-2013 Guide for Loading Dry-Type Distribution and Power Transformers

C. InterNational Electric Testing Association, Inc. (NETA)

1. NETA ATS-2025 Acceptance Testing Specifications

D. National Electrical Manufacturers Association (NEMA)

1. NEMA 250-2020 Enclosures for Electrical Equipment (1,000 Volts Maximum)
2. NEMA TR 1-2013(R2019) Transformers, Regulators and Reactors

E. National Fire Protection Association (NFPA)

1. NFPA 70-2023 National Electrical Code (NEC)

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals and Section 01 33 26 – Electrical Drawings and Data.

B. RSN 26 22 12-1, Approval Data:

1. Manufacturer's technical data.
2. Equipment layout drawings.
3. Nameplate drawing.

- C. RSN 26 22 12-2, Field Test Reports.
- D. RSN 26 22 12-3, Operation and Maintenance Instruction Book.

1.04 REGULATORY REQUIREMENTS

- A. Be UL listed and bear the UL label.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Handle and store equipment in accordance with manufacturer's instructions. Include copy of these instructions with equipment at time of shipment.

PART 2 PRODUCTS

2.01 DRY TYPE TRANSFORMER

- A. Type: General purpose, self-cooled, dry-type.
- B. Enclosure: Ventilated, NEMA 250 Type 2 drip-proof. Fabricated of heavy gauge, sheet steel construction.
- C. Completed coil and core bolted to enclosure base isolated by means of rubber vibration-absorbing mounts.
- D. Provide electrostatic shield for attenuation of voltage spikes, line noise, and transients.
- E. Winding Conductors: Copper.
- F. Insulation: 220-degree C.
- G. Winding Temperature Rise: 150 degrees C above 40-degree C ambient.
- H. Primary Voltage: As shown on drawings.
- I. Secondary Voltage: As shown on drawings.
- J. kVA Rating: As shown.
- K. Primary Taps: Four 2.50 percent full-capacity below rated voltage and two 2.50 percent full-capacity above rated voltage.
- L. Mounting: Floor.

2.02 NAMEPLATES

- A. Material: Type A as described on Standard Drawing 40-D-6234.

- B. Engraving: As shown on drawings.

2.03 FINISHES

- A. Manufacturer's standard Grey finish for intended installation location.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Furnish anchoring, supporting, and mounting materials required to install equipment.
- B. Install in accordance with manufacturer's instructions and recommendations.
- C. Minimum separation of at least 12-inches from any combustible materials,
- D. Make conduit and wiring connections.
- E. Transformer is separately derived alternating-current system as defined in NFPA 70. Ground and bond transformers in accordance with NFPA 70.
- F. Provide require safety labels.
- G. Provide engraved nameplate on transformer.

3.02 ADJUSTING

- A. Measure primary and secondary voltages and make appropriate tap adjustments.

3.03 CONTRACTOR FIELD QUALITY TESTING

- A. In accordance with NETA ATS.

END OF SECTION

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SECTION 26 24 13

SWITCHBOARDS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Switchboards:

1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. American National Standards Institute (ANSI)

1. ANSI C12.20-2018 Electricity Meters 0.1, 0.2, and 0.5 Accuracy Classes

B. InterNational Electric Testing Association, Inc. (NETA)

1. NETA ATS-2025 Acceptance Testing Specifications for Electrical Power Equipment & Systems

C. National Electrical Manufacturers Association (NEMA)

1. NEMA 250-2020 Enclosures for Electrical Equipment (1,000 Volts Maximum)
2. NEMA FU 1-2012 Low-Voltage Cartridge Fuses
3. NEMA PB 2-2011 Deadfront Distribution Switchboards
4. NEMA PB 2.1-2013 General Instructions for Proper Handling, Installation, Operation, and Maintenance of Deadfront Distribution Switchboards Rated 600 Volts or Less

D. National Fire Protection Association (NFPA)

1. NFPA 70-2023 National Electrical Code (NEC)

E. Underwriters Laboratories, Inc. (UL)

1. UL 50-2015 Enclosures for Electrical Equipment, Non-Environmental Considerations
2. UL 489-2016 Safety Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures
3. UL 891-2019 Switchboards

1.03 DESIGN REQUIREMENTS

- A. Free-standing, dead-front type, low-voltage distribution switchboard that utilizes group mounted circuit breakers.
- B. Group mounted assemblies to consist of molded-case circuit breakers mounted on a panelboard type chassis.
- C. Suitable for top cable entry.
- D. Include integrated panelboards to provide power to the required loads.

1.04 PROJECT CONDITIONS

- A. For natural conditions refer to Section 01 14 10 – Use of Site.

1.05 SUBMITTALS

- A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.
- B. RSN 26 24 13-1, Approval Drawings and Data in Manufacturer's Format:
 - 1. Manufacturer's technical data.
 - 2. Layout diagram:
 - a. Switchboard arrangement.
 - b. Anchoring arrangement.
 - 3. Nameplate list.
 - 4. Bill of materials.
 - 5. Manufacturer's equipment installation instructions, including foundation requirements.
 - 6. Factory JHA.
- C. RSN 26 24 13-2, Approval Drawings in Government Format:
 - 1. Three-line Diagram.
 - 2. Schematic Diagrams.
 - 3. Wiring Diagrams.
- D. RSN 26 24 13-3, Factory Test Report.
- E. RSN 26 24 13-4, Check Prints:
 - 1. Schematic and wiring diagrams.

- F. RSN 26 24 13-5, Field Test Reports.
- G. RSN 26 24 13-6, As-Built Drawings.
- H. RSN 26 24 13-7, Operation and Maintenance Instruction Book.

1.06 REGULATORY REQUIREMENTS

- A. UL listed and labeled.
- B. Conform to NEMA PB 2, UL 50, UL 891, and NFPA 70.

1.07 DELIVERY, STORAGE, AND HANDLING

- A. Handle and store equipment in accordance with manufacturer's instructions. Include copy of these instructions with equipment at time of shipment.

PART 2 PRODUCTS

2.01 RATINGS

- A. Switchboard Short-Circuit Rating: 65,000 symmetrical RMS amperes.
- B. Voltage Rating: 480/277-volts, 3-phase, 4-wire, 60-hertz.
- C. Through Bus: 800-amperes, RMS, minimum.
- D. Integrated Panelboards: As shown on drawings.

2.02 CONSTRUCTION

- A. Vertical sections bolted together to form rigid assembly.
 - 1. Form edges of front covers or hinged front panels
 - 2. Provide adequate ventilation within enclosure.
- B. Enclosure Type: Type 3R in accordance with NEMA 250.
- C. Bus and device connections: Accessible from front.
- D. Provide switchboard sections with adequate lifting means.

2.03 BUS

- A. Bus Material: Manufacturer's standard plated copper.
- B. Full capacity neutral bus.

- C. Copper ground bus (minimum 1/4- by 2-inch).
- D. Bus joints with conical spring-type washers.

2.04 MOLDED CASE CIRCUIT BREAKERS

- A. Electronic Trip Unit.
 - 1. Equip the following switchboard breakers with a solid-state tripping system consisting of 3 current sensors and a microprocessor-based trip unit that provides fully adjustable long, short, and instantaneous settings:
 - a. DSA-MAIN:
 - 1) Ability to disable instantaneous setting.
 - 2) Arc flash reduction/maintenance mode.
 - b. DSA-1 (EG Set).
 - c. DSA-32 (MCC M1A).
 - d. DSA-37 (MCC M1B).
 - e. DSA-38 (MCC M1C).
 - f. DSA-43 (Panelboard MSA).
- B. Thermal-magnetic type with inverse-time trip characteristics on overload and instantaneous trip on short circuit.
- C. Voltage: As shown on drawings.
- D. Frequency: 60-Hertz (Hz).
- E. Interrupting Rating: 25,000 symmetrical RMS amperes, minimum.
- F. Arc quenchers.
- G. Quick-make and quick-break toggle mechanisms.
- H. Trip-free operating handle.
- I. Single operating handle. Do not use handle ties on multiple-pole circuit breakers.
- J. Equip with padlock attachments for locking breakers in "OFF" position.
- K. Shunt Trip: Intermittent rated solenoid with tripping plunger and cutoff switch assembled to plug-in module.
- L. Conform to UL 489.

2.05 POWER AND ENERGY METER

- A. Microprocessor based and designed for multifunction electrical measurement on 3-phase power system.
- B. User programmable for voltage range to any PT ratio. Accept direct voltage input range of up to 416-volts line-to-neutral and 721-volts line-to-line.
- C. User programmable for current to any CT ratio and accept current input up to 11-amperes continuous.
- D. Provide true RMS measurements of voltage, phase-to-phase and phase-to-neutral; current, per phase and neutral.
- E. Accuracy: Plus, or minus 0.25 percent for volts, and 0.5 percent for power and energy functions.
- F. Provide Volts, Amps, VAR, VA, PF, Frequency, Watthours, VARh, and VAh.
- G. Provide meter with three-line LED display. Meter display to include a percent of FULL SCALE having not less than 10 segments.
- H. Traceable revenue meter, which contains a utility grade test pulse allowing power providers to verify and confirm the meter is performing to its rated accuracy.
- I. Provide meter with Modbus TCP communication format.
- J. Allow user to set up utility demand profile:
 - 1. Readings for kW, kVAR, kVA, and PF calculated using utility demand features.
 - 2. Other parameters: MAX and MIN capability over a user selectable averaging period.
 - 3. Provide voltage with instantaneous MAX and MIN reading displaying the highest surge and lowest sag seen by meter.
- K. Meter capable of operating on a power supply of 90- to 265-volts, 60-Hz.
- L. UL listed.

2.06 INSTRUMENT TRANSFORMERS

- A. Current Transformers:
 - 1. Three single-phase; 60-hertz; single-ratio; single-secondary; indoor.
 - 2. Voltage rating: 600-volts.
 - 3. Basic impulse insulation level (BIL): 10-kilovolts.
 - 4. Quantity and Ratio: As shown on drawings.

5. Class: As shown on drawings.
6. Thermal Rating: 1.0 at 55-degrees C.
7. Necessary mounting devices, primary and secondary terminal studs, and secondary short-circuiting devices.
8. Shorting blocks for CT wiring that are UL recognized to 105 degrees C and are made up of thermoplastic body material.

B. Potential Transformers:

1. Rated primary voltage: As shown on drawings.
2. Basic impulse insulation level: 10-kilovolts.
3. Quantity/Ratio: As shown on drawings.
4. Rating: 300 VA.
5. ANSI metering accuracy classification and thermal burden ratings: Suitable for this application.
6. Necessary mounting devices and primary and secondary terminal studs.
7. Fuses:
 - a. Current-limiting fuses in primary.
 - b. As indicated on drawings in potential transformer ungrounded secondary leads. Withstand maximum energizing current.
 - c. Locate in accessible areas.
 - d. Potential transformer primary circuit fuses and fuse holder ratings: 600-volts, current limiting with 3-pole block.
 - e. Potential transformer secondary circuit fuses and fuse holder ratings: 6-amperes, 250-volts, with 3-pole block.
 - f. UL listed.
 - g. Fuse-holders: Block type with side barriers.
 - h. Conform to NEMA FU 1.

2.07 WIRING/TERMINATIONS

- A. Control wiring, necessary fuse blocks, pushbuttons, switches, auxiliary relays, potential transformers, and terminal blocks within switchboard as required. Provide 15 percent spare terminals on each terminal block.
- B. Mechanical-type terminals for line and load terminations suitable for copper cable rated for 75 degrees C of size shown.
- C. Lugs in incoming line section for connection of main grounding conductor. Provide additional lugs for connection of other grounding conductors as required.

- D. Control wire: Type SIS, bundled and secured with nylon ties.

2.08 NAMEPLATES

- A. As indicated on standard drawing 40-D-6234.
- B. Provide nameplates for:
1. Overall switchboard assembly. Manufacturer's nameplate:
 - a. Place manufacturer's name, trademark, or other descriptive marking on electrical equipment.
 - b. Other markings: Voltage, current, wattage, or other ratings as specified.
 2. Each section of switchboard assembly. Designations as indicated on drawings or as follows:
 - a. Equipment designation nameplate for each section of switchboard.
 - b. Mount nameplates for each device on a door.
 - c. Each device mounted within switchboard enclosure.
 - d. Each circuit breaker within panelboard sections. Information to identify load served as indicated on associated as-built switchboard schedules.

2.09 FINISH

- A. Manufacturer's standard grey finish.

2.10 FABRICATION

- A. Form 480-volt station service switchboard into an integral assembly complete with channel base.
- B. Where welding is required, weld enclosures and sections along entire length of joint. Do not use skip-welding.
- C. Extend copper ground bus entire length of switchboard:
 1. Mechanically and electrically bond each vertical section.
 2. Equip with No. 4/0 AWG terminal for connection to building ground system.
 3. Drill and tap at 1-inch intervals for 8/32-inch screws.
- D. Provide spare circuit breakers as shown.
- E. Cover blank spaces with removable covers.
- F. All covers secured with captive screws.
- G. Internal Wiring:

1. Wire control circuits in accordance with approved drawings and as necessary for equipment to function as required.
2. Terminate circuits required to be wired to external equipment on terminal blocks.
3. Make wire terminations with ring tongues terminal connectors.
 - a. Use pin type connectors where use of ring tongues is not practical such as terminations at selector switches, pushbuttons, indicating lamps, and auxiliary relays.
 - b. Do not use bare wire connection.
4. Connect current transformer secondary leads to conveniently accessible short-circuit terminal blocks before connecting to other devices.
5. Form wiring into compact groups bound together and firmly supported.
 - a. Run wiring straight, horizontally, or vertically with short-radius, right-angle bends.
 - b. Twist hinge wiring around longitudinal axis of hinge, wherever practicable, instead of bending laterally.
6. Attach nameplates with pan head self-tapping screws. Do not use adhesives to attach nameplates.
7. Install grommets at holes in enclosure for wires and cables.
8. Do not splice wiring within switchboard.

2.11 CONTRACTOR SOURCE QUALITY CONTROL

A. Factory Testing:

1. Production (routine) tests on actual switchboard sections provided in accordance with NEMA PB 2.
2. Test circuits before shipment to demonstrate correctness of completed wiring connections and proper functioning of devices.

2.12 SOURCE QUALITY ASSURANCE

- A. Government may inspect equipment at supplier's manufacturing plant during fabrication and testing.

PART 3 EXECUTION

3.01 INSTALLATION

- A. In accordance with approved manufacturer's instructions and recommendations.
- B. Install on concrete floor as shown:

1. Install switchboard assemblies on embedded galvanized steel channel. Determine channel spacing and arrangement from manufacturer's layout plan.
 2. Level steel channel with top of floor or housekeeping pad. Mounting surface of the steel channels must be:
 - a. Level and in same plane within manufacturer's required tolerances. See manufacturers installation instructions for required tolerances.
 - b. Raised slightly above surface of floor or housekeeping pad. In no case shall any non-supporting surfaces of foundation be higher than tops of steel channels.
 3. Bolt switchboard assemblies in place, on the steel channels.
 4. Furnish anchoring, supporting, and mounting materials required to install equipment level and flush to prevent ingress of foreign materials into enclosure or resulting interstitial space.
- C. Ground and bond distribution board to existing grounding system in accordance with Section 26 05 20 – Grounding and Bonding and NFPA 70.
- D. Make conduit and wiring connections.

3.02 CONTRACTOR FIELD QUALITY CONTROL

- A. Switchboard Assembly:
1. In accordance with NETA ATS.
- B. Circuit Breakers:
1. Visual and mechanical inspection.
 - a. Compare nameplate data with specifications and approved shop drawings.
 - b. Inspect circuit breaker for correct mounting.
 - c. Operate circuit breaker to verify smooth operation.
 - d. Inspect case for cracks or other defects.
 - e. Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted connections or cable connections by using the calibrated torque-wrench method or performing a thermographic survey.
 - f. Inspect mechanism contacts and arc chutes in unsealed units.
 2. Electrical tests.
 - a. Perform contact-resistance tests
 - b. Perform insulation-resistance tests.
 - c. Perform breaker adjustments for final settings.

- d. Verify selective tripping coordination.

C. Current Transformers:

1. Visual and mechanical inspection.
 - a. Compare equipment nameplate data with specifications and approved shop drawings.
 - b. Inspect physical and mechanical condition.
 - c. Verify correct connection.
 - d. Verify that adequate clearances exist between primary and secondary circuit.
 - e. Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by using the calibrated torque-wrench method or performing a thermographic survey.
 - f. Verify that all required grounding and shorting connections provide good contact.
2. Electrical tests.
 - a. Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.
 - b. Perform insulation-resistance tests.
 - c. Perform polarity tests.
 - d. Perform ratio-verification tests.

D. Metering and Instrumentation:

1. Visual and mechanical inspection
 - a. Compare equipment nameplate data with specifications and approved shop drawings.
 - b. Inspect physical and mechanical condition.
 - c. Verify tightness of electrical connections.
2. Electrical tests.
 - a. Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.
 - b. Calibrate watt-hour meters according to the manufacturer's published data.
 - c. Verify all instrument multipliers.
 - d. Electrically confirm that current transformer and voltage transformer secondary circuits are intact.

END OF SECTION

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SECTION 26 24 24

DISTRIBUTION PANELBOARDS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Distribution Panelboards:

1. Payment: Lump-sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. Federal Specifications (FS)

1. FS W-C-375D Circuit Breakers, Molded Case; Branch Circuit and Service

B. Electrical Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C37.13-2015 Low-Voltage AC Power Circuit Breakers Used in Enclosures
2. IEEE C37.16-2009 Low-Voltage Power Circuit Breakers and AC Power Circuit Protectors - Preferred Ratings, Related Requirements, and Application Recommendations

C. InterNational Electric Testing Association, Inc. (NETA)

1. NETA ATS-2025 Acceptance Testing Specifications

D. National Electrical Manufacturers Association (NEMA)

1. NEMA 250-2020 Enclosures for Electrical Equipment (1000 Volts Maximum)
2. NEMA AB 3-2013 Molded Case Circuit Breakers and Their Application
3. NEMA PB 1-2011 Panelboards

E. National Fire Protection Association (NFPA)

1. NFPA 70-2023 National Electrical Code (NEC)

F. Underwriters Laboratories, Inc. (UL)

1. UL 50-2015 Enclosures for Electrical Equipment
2. UL 67-2018 Panelboards

3. UL 489-2016 Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 24 24-1, Approval Drawings and Data:
1. Equipment layout drawings.
 2. Wiring diagrams.
 3. Manufacturer's data.
 4. Bills of material for each device or component of equipment.
 5. Circuit directory for each board.
- C. RSN 26 24 24-2, Factory Test Reports.
- D. RSN 26 24 24-3, Field Test Reports.
- E. RSN 26 24 24-4, Operation and Maintenance Instruction Books.

PART 2 PRODUCTS

2.01 DISTRIBUTION PANELBOARDS

- A. Type: Dead-front, rigid, Type 3R in accordance with NEMA 250], with hinged door.
- B. Voltage and System: As shown on drawings.
- C. Main Breaker Thermal Rating: As shown on drawings.
- D. Branch Circuits: As shown on drawings.
- E. Solid Neutral Bus: Copper, if shown on drawings.
- F. Bus Material: Copper.
- G. Mounting: As shown on drawings.
- H. Conform to IEEE C37.13 and C37.16; NEMA AB 3 and PB 1; UL 50, 67, and 489; NFPA 70; and FS W-C-375.

2.02 NAMEPLATES

- A. As indicated on Standard Drawing No. 40-D-6234.

- B. Engraving: As shown on drawings.

2.03 MAIN BUS

- A. Rating: As shown on drawings, minimum.
- B. Copper, bolted type bars.
- C. Main and Branch Circuit Connections: Unless silver plated, use copper based on 1,000-ampere-per-square-inch current density and contact surfaces with not more than 200 amperes per square inch.
- D. Equip screws and bolts used for making copper connections with lockwashers to prevent loosening. Do not use riveted bus connections.

2.04 CIRCUIT BREAKER

- A. Molded-case type.
- B. Thermal-magnetic type with inverse-time trip characteristics on overload and instantaneous trip on short circuit.
- C. Voltage: As shown on drawings.
- D. Number of Poles: As indicated on drawings.
- E. Interrupting Rating at Specified Voltage: 10,000 amperes, minimum.
- F. Spare breakers: As indicated on drawings.
- G. Arc quenchers.
- H. Quick-make and quick-break toggle mechanisms.
- I. Trip-free operating handle.
- J. Single operating handle. Do not use handle ties on multiple-pole circuit breakers.
- K. Phase arrangement; Conform to NFPA 70.
- L. Mount circuit breaker handles to be operable from front of boards.
- M. Assemble boards so that individual breakers can be removed or added without disturbing adjacent units, without loosening or removing supplemental insulation supplied to meet requirements of UL, and without additional machining, drilling, or tapping.
- N. Assemble boards with molded-case circuit breakers individually mounted on fixed framework behind easily removable covers.

- O. Provide blank spaces for future circuit breakers as indicated on drawings. Cover blank spaces with removable plates.
- P. Padlock attachments for locking breakers in OFF position.
- Q. In accordance with FS W-C-375 and NEMA AB-3 for molded-case circuit breakers.
- R. Equip multiple-pole breakers with common trips so an overload on one pole will automatically cause all poles of breaker to open.
- S. Equip with solderless-type connectors on load side of breakers for conductors No. 8 AWG and smaller. Furnish solder-type or pressure-wire connectors on load side of breakers for conductors larger than No. 8 AWG.
- T. Where indicated on drawings, furnish breakers with bell alarm contacts that are wired to outgoing terminal blocks.

2.05 SOLDERLESS, BOLTED-TYPE LUGS OR TERMINALS

- A. Provide lugs or terminals suitable for copper conductors on breakers for feeder and branch circuit cables, and on bus for supply cables. Size lugs or terminals in accordance with cable sizes indicated on drawings.
- B. Provide neutral buses with sufficient solderless, bolted-type lugs or terminals to accommodate each single-pole branch circuit breaker, each future breaker, and neutral leads indicated on drawings.
- C. Provide solderless, bolted-type copper lugs or terminals at rear base of boards for connecting boards to ground system. Connect lugs or terminals to ground system by No. 4 AWG copper conductor, minimum.

2.06 CONNECTORS BETWEEN BREAKER TERMINALS AND BUSES

- A. Copper bars designed to prevent excessive stress on bus or breaker during expansion due to temperature changes.

2.07 TERMINAL BLOCKS

- A. 600 volts and 25 amperes.
- B. Suitable for use with No. 8 AWG wire.
- C. Molded-block type to accommodate ring connectors 1/2 inch wide (outer diameter) at terminal screws.
- D. Binding-head or washer-head screws having serrated or grooved contact surfaces or having lockwashers.

- E. Molded insulating barriers between terminals.
- F. Removable marking strips. Provide marking strips for terminal blocks with conductor designations engraved or printed with a permanent marking fluid. Furnish one spare blank marking strip with each terminal block.
- G. Removable covers to prevent accidental contact with terminal by operating personnel.
- H. Furnish and install heavy-duty terminal blocks with barriers for external supply cables.
- I. Arrange and locate terminal blocks to support incoming and outgoing cables.
- J. Separate adjacent rows of terminal blocks at least 6 inches edge to edge and at least 6 inches from sides, top, or bottom of the cabinet.
- K. Install grommets at holes for wires and cables.
- L. Furnish sufficient terminal blocks to wire and make ready for operation all equipment on boards, and to terminate incoming control cables.
- M. Provide 25 percent spare terminals on each terminal block.
- N. Mounting: Near the bottom of boards for bottom entry, and near top for top entry.

2.08 GROUND BUS

- A. Copper.
- B. Length: Width of enclosure.
- C. Install at bottom of enclosure.
- D. Bond mechanically and electrically to enclosure.
- E. Drill and tap every inch for 10-32 screws. Install a 10-32 screw in each tapped hole.

2.09 CONDUCTOR IDENTIFYING MARKERS

- A. Tag conductors near each terminal to identify them with their respective studs or terminals.
- B. Vinyl-cloth with silicone plastic overcoating.

2.10 CONTROL WIRING FOR ALARM CIRCUITS

- A. Minimum No. 16 AWG, class K stranded copper conductor, with NEC switchboard type SIS insulation rated for 600 volt service.

- B. Make all terminations of wire with heavy-duty, preinsulated pressure-crimp-type terminal connectors with ring tongues.
- C. Connectors:
 - 1. Tin-plated copper.
 - 2. Serrated inner barrel.
 - 3. 600-volt-rated nylon insulation.
 - 4. Insulation support sleeve for vibration resistance.
 - 5. UL listed.
 - 6. Compatible with conductor size and type on which it is used.
- D. Keep wiring to a minimum.
- E. Form wiring into compact groups, bound together and firmly supported.
- F. Run wiring straight, horizontally or vertically, with short-radius right-angle bends.
- G. Protect wiring from abrasion where it leaves channels or ducts.
- H. Do not splice wiring.
- I. Make connections at device studs or terminal blocks.
- J. Terminate wiring which connects to outgoing control cables at terminal blocks.

2.11 ENCLOSURE

- A. Sheet steel not lighter than No. 12 United States Standard gauge, weatherproof type, for surface mounting on steel structure, furnished without mounting holes and without knockouts or hubs, shall be furnished with a grounding lug for use with 4 AWG stranded copper cable.
- B. Equip enclosure door with flush lock and two keys.
- C. Welded joints.
- D. Galvanize after fabrication.
- E. Finish: Manufacturer's standard finish.

2.12 FABRICATION

- A. Mount ground bus at bottom of panelboard enclosure. Bond bus to enclosure.
- B. Mount circuit directory on inside of door.

- C. Make equipment, removable covers, terminal blocks, wiring, and wiring gutters accessible from fronts of the boards for operation, maintenance, removal, and replacement so boards are capable of being permanently mounted with rear against wall.
- D. Equip ventilation openings with grilles or screens having dust filters.
- E. Provide gutter space on all sides of all boards in accordance with NFPA 70.
- F. Provide boards with easily removable or hinged front panels so incoming cables may be easily connected or disconnected and inspected.
- G. Equip bottom and top of wall-mounted boards with removable plates for field drilling.
- H. Provide access doors for each board.
- I. Equip each door with locks, latches, and concealed hinges to allow doors to open 105 degrees, minimum, from fully closed position.
 - 1. Key all locks alike with identifying number stamped on each key.
 - 2. Provide five keys.
- J. Circuit Directory:
 - 1. Prepare circuit directory for each board, indicating "as-built" conditions.
 - 2. Include board designation, breaker numbers and descriptions of connected loads indicated in their relative physical location.
 - 3. Show breaker number and connected load.
 - 4. Use machine lettering for directory entries. Do not provide pencil or handwritten directory entries.
 - 5. Mount directory in transparent plastic front on inside surface of access doors for each board.
 - 6. Fabricate circuit directory frame such that directory need not be folded when placed in frame, and such that all information on directory is clearly visible and legible.
- K. Fabricate frames and enclosures of sheet steel with sufficient weight and strength to ensure permanent rigidity and alignment.
- L. Hide butt joints on outside surfaces.
- M. Hide holes or fasteners be from front of boards for rear-of- board-mounted equipment.
- N. Provide boards with manufacturer's standard base provided other requirements of this Section are fulfilled.
- O. Support bus bars by structural frames. Do not depend on breakers for support.

- P. Back pan:
1. Provide as foundation for mounting bus and breakers.
 2. Construct rigidly to ensure against damage during shipment and installation.
 3. Provide reinforcement on back pan consisting either of flanging back pan entire length of pan, or securely fastening angle iron to both sides of a flat back pan.
 4. Do not use nonreinforced back pans of ordinary flat sheet steel.
- Q. Construct board frames and enclosures to provide adequate clearance for incoming cables entering from conduits below and above. Make provisions near bottom of boards for supporting incoming and outgoing cables.

2.13 CONTRACTOR SOURCE QUALITY TESTING

- A. Dielectric test in accordance with NEMA PB1.
- B. Test circuits before shipment to demonstrate correctness of completed wiring connections and proper functioning of devices.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Cables entry into enclosure: From beneath or above by means of conduit.
- B. Provide anchoring, supporting, and mounting materials as required.
- C. Install ground cable between panelboard enclosure and grounding system.

3.02 CONTRACTOR FIELD QUALITY CONTROL

- A. Circuit Breakers: In accordance with NETA ATS.

END OF SECTION

SECTION 26 24 43
600 VOLT MOTOR CONTROL CENTER

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. 600 Volt Motor Control Center:
1. Payment: Lump sum price offered in the Price Schedule.
- B. Cost:
1. Include cost of spare parts in lump sum price offered in the Price Schedule for Spare Parts.

1.02 REFERENCE STANDARDS

- A. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE C57.13-2008 Instrument Transformers
- B. InterNational Electric Testing Association, Inc. (NETA)
1. NETA ATS-2025 Acceptance Testing Specifications
- C. National Electrical Manufacturers Association (NEMA)
1. NEMA 250-2014 Enclosures for Electrical Equipment (1000 Volts Maximum)
 2. NEMA FU 1-2012 Low Voltage Cartridge Fuses
 3. NEMA ICS 1-2000(2015) General Requirements
 4. NEMA ICS 2-2000(2020) Controllers, Contactors and Overload Relays Rated 600 V
 5. NEMA ICS 5-2017 Control Circuit and Pilot Devices
 6. NEMA ICS 18-2001(2007) Motor Control Centers
 7. NEMA KS 1-2001(2013) Enclosed and Miscellaneous Distribution Equipment Switches (600 Volts Maximum)
 8. NEMA PB 1-2011 Panelboards
 9. NEMA WC 70-2009 Nonshielded Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy
- D. National Fire Protection Association, Inc. (NFPA)

- 1. NFPA 70-2023 National Electrical Code (NEC)
- E. Underwriters Laboratories, Inc. (UL)
 - 1. UL 845-2005 Motor Control Centers
 - 2. UL 489-2016 Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures

1.03 DESIGN REQUIREMENTS

- A. Design layout of motor control center sections.
- B. Determine size and ratings of the following:
 - 1. Circuit breakers.
 - 2. Fuses.
 - 3. Motor contactors and overload relays. Size overload relays in accordance with requirements of NFPA 70.
 - 4. Control circuit transformers.
 - 5. Instrument transformers.
- C. Determine current transformer ratio if not shown.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 24 43-1, Approval Data:
- C. RSN 26 24 43-2, Approval Drawings:
 - 1. Equipment layout diagram including the method of attaching motor control center to floor.
 - 2. Single line diagram.
 - 3. Nameplate list.
 - 4. Bill of materials.
- D. RSN 26 24 43-3, Schematic Diagrams:
 - 1. Include a separate schematic for each motor control center compartment.
 - 2. Schematic diagram to show all portions of the actual circuitry, regardless of whether the circuitry is located within the motor control center or at another location.

- E. RSN 26 24 43-4, Time-Current Characteristic Curves:
 - 1. Provide characteristic curves for circuit breakers and motor overload relays.
 - 2. Include motor inrush, full-load, and locked-rotor currents.
 - 3. Curves to be plotted on same log-log graph.
- F. RSN 26 24 43-5, Schematic and Wiring Diagram Check Prints.
- G. RSN 26 24 43-6, Factory Test Reports.
- H. RSN 26 24 43-7, Field Test Reports.
- I. RSN 26 24 43-8, Operation and Maintenance Instruction Book:
 - 1. Include the following:
 - a. Title page/index.
 - b. Bill of materials.
 - c. Copies of informational data.
 - d. Half-size prints of As-Built drawings.
 - e. Time-current characteristic curves.
 - f. Motor circuit breaker ratings and settings.
 - g. Motor overload relay settings.
 - h. Undervoltage relay setting.
 - i. Factory test reports.
 - j. Field test reports.

1.05 REGULATORY REQUIREMENTS

- A. UL listed and labeled.

1.06 DELIVERY, STORAGE, AND HANDLING

- A. Handle and store equipment in accordance with manufacturer's instructions. Include copy of these instructions with equipment at time of shipment.

1.07 PROJECT CONDITIONS

A. Ambient Temperature Range: ____ to ____ degrees C.

B. Relative Humidity Range: ____ to ____ percent.

C. Altitude: ____ feet.

1.08 EXTRA MATERIALS

A. Furnish following spare parts:

1. One main operating coil for each different NEMA size contactor furnished.
2. One magnetic motor starter power contact assembly for each different NEMA size contactor furnished.
3. Six fuses of each type and size furnished.
4. Five LEDs of each type furnished.
5. One indicating light lens of each type and color furnished.
6. One control power transformer for each different type furnished.
7. One auxiliary relay for each different size furnished.
8. One timing relay for each different size and type furnished.
9. One overload relay of each type and size furnished.

PART 2 PRODUCTS

2.01 RATINGS AND GENERAL REQUIREMENTS

- A. Conform to referenced IEEE, NEMA, NFPA, and UL standards.
- B. Line-To-Line Voltage: 480 volts.
- C. Horizontal Bus: 600 amperes, silver-plated copper. Bus rating based on 65 degree C maximum temperature rise in a 40 degree ambient.
- D. Vertical Bus Extensions: 300 amperes, silver-plated copper.
- E. Phase and Frequency: 3-phase, 60 Hz.
- F. Power Configuration: 3 wire.
- G. Short Circuit Rating: 22,000 amperes RMS symmetrical, minimum.
- H. NEMA Wiring Class and Type: II-S, B-T in accordance with NEMA ICS 18.

2.02 CONSTRUCTION

A. Enclosure: NEMA 250, Type 3R.

1. Features:
 - a. Provide oil-resistant neoprene gaskets for motor control center unit doors, bottom plates, and cover plates.

- b. Conduit entry: Bottom and top entry.
- B. Incoming Feeder Termination and Device: Terminate incoming cable within motor control center on main lugs in a separate incoming line lug compartment.
- C. Bus and Bus Connectors: Solid-bar type.
- D. Ground Bus:
 - 1. Material: Copper bar at least 1/4 -inch by 1-inch.
 - 2. Continuous along bottom on motor control center sections.
 - 3. Mechanically and electrically bonded to motor control center enclosure.
 - 4. Provide compression lug at each end of ground bus. Size lug to connect ground bus to plant ground system with stranded bare-copper cable as shown.
 - 5. Provide vertical ground bus in each section. Connect vertical ground bus to the horizontal ground bus.
- E. Barriers:
 - 1. Isolate power busses and splice connections from unit compartments and wireways.
 - 2. Mount horizontal bus on a glass filled polyester support assembly that braces the bus against forces generated during short circuit.
 - 3. House vertical bus on a molded glass-filled polyester support that provided bus insulation and braces the bus against forces generated during short circuit. Provide access openings in the support for unit stab-on connections. Provide each access opening with manual shutter to close off the opening.
- F. Wireways:
 - 1. Horizontal Wireway:
 - a. Top and bottom of each section. Extend full depth of section.
 - b. Removable covers held shut with captive screws.
 - c. Isolated from power bus by steel barriers.
 - 2. Vertical Wireway:
 - a. Extend full length of section between horizontal wireways.
 - b. Removable hinged cover.
 - c. Isolated from power bus by steel barriers.

2.03 PLUG-IN UNITS (BUCKETS)

- A. General:

1. Modular plug-in type for starters through NEMA size 5.
 2. Make connection between plug-in unit and vertical bus with plug-in power stabs. Back stabs with spring steel clips designed to maintain high pressure, two-point connection.
 3. Designed to prevent insertion or removal from motor control center unless unit door and disconnect device are open.
 4. Designed to house components of individual motor control circuit.
- B. Mounting:
1. Supported and guided by removable unit support pan. Support pan capable of being moved without tools.
 2. Held in place with two, front accessible, multi-turn latches, minimum.
- C. Disconnect Device Handle:
1. Type: Industrial, heavy-duty flange mounted mechanism that remains engaged with disconnect device regardless of unit door position.
 2. Have up-down motion with down as OFF. ON-OFF position prominently marked on operator.
 3. Provide with means for padlocking in both ON and OFF position.
 4. Interlocked with unit door such that disconnect means cannot be switched to ON unless door is closed and door cannot be opened unless disconnect means is switched to OFF.
 5. Provide mechanical means to purposely defeat interlock that is externally operated with screw driver.
- D. Door:
1. Removable and mounted on pin type hinges.
 2. Latch: 1/4 turn pawl type designed to resist forces during fault conditions.
 3. Door swing: 110 degrees, minimum.
 4. Fastened to stationary structure of motor control center such that door will cover unit space with plug-in unit removed.
 5. Provided with external, low profile motor overload reset pushbutton. Include mechanical extension device to allow overload relay to be reset without opening unit door.

2.04 MOTOR CONTACTOR

- A. Type: Magnetic, full-voltage, non-reversing.
- B. Voltage: 600 volts, 60 Hz.

- C. Coil: 120 volts, 60 Hz.
- D. Provide mechanical interlock between forward and reverse contactors.
- E. Power Contacts: Weld-resistant, double-break, and silver alloy type.
- F. NEMA rated and sized for motor horsepower loads. Contactor rated and sized to IEC standards is not acceptable.
- G. Power contacts, contactor coil, and auxiliary contacts designed to be inspected and replaced.
- H. Provide sufficient auxiliary contacts to perform existing functions.

2.05 MOLDED-CASE CIRCUIT BREAKERS

- A. Voltage: 480 volt, 60 Hz.
- B. Interrupting Rating: 22,000 amperes RMS symmetrical, minimum.
- C. Trip: Thermal-magnetic type with inverse-time trip characteristics on overload and instantaneous trip on short circuit.
- D. Construction: High-strength, molded, glass-reinforced polyester case and cover.
- E. Provide with overcenter, toggle hand-operated, trip free mechanism with quick make and quick break action independent of speed of toggle handle operation.
- F. Provide with common tripping of all poles.
- G. Sized in accordance with NFPA 70.
- H. UL listed and conform to UL 489.

2.06 CONTROL CIRCUIT TRANSFORMER (CCT)

- A. Dry type, copper windings, and sized to have adequate capacity for control power requirements of the motor control circuit.
- B. Primary Voltage: 480 volts, 60 Hz..
- C. Secondary Voltage: 120 volts, 60 Hz.
- D. Integral fuse block mounted on transformer.
 - 1. Primary Fuses: 600 volt, current-limiting type.
 - 2. Secondary Fuses: 250 volt, current-limiting type.

- E. Transformer UL listed. Fuses conform to FU 1 and UL listed.

2.07 PUSHBUTTONS

- A. Bulletin 800T as manufactured by Allen-Bradley; Class 9001 as manufactured by Square-D; type CR104P as manufactured by General Electric; or equal, having the following essential characteristics:
1. Type: Heavy-duty, watertight/oiltight with 30.5 mm mounting hole. Momentary or maintained contacts as required.
 2. Contact rating: NEMA A600,
 3. START pushbutton: Momentary contact with black, flush button.
 4. STOP pushbutton: Momentary contact with red, extended head button.
- B. Nameplate: Equipment manufacturer's standard or custom engraved nameplate suitable for mounting directly over device. Engraving to match existing.
- C. Conform to NEMA ICS 5.

2.08 INDICATING LIGHTS

- A. Type: Heavy-duty, watertight/oiltight, push-to-test with 30.5 mm mounting hole. Full voltage for operation on 120 volt, 60 Hz.
- B. Color
1. On: Red.
 2. Off: Green.
- C. Lamp:
1. High visibility, LED type, 120 volts, 60 Hz.
 2. Minimum of 5 LEDs in a cluster. Lamps with one LED are not acceptable.
- D. Nameplate: Equipment manufacturer's standard or custom engraved nameplate suitable for mounting directly over device.

2.09 PROTECTIVE RELAYS

- A. General Requirements:
1. Suitable for use on 120 volt, 60 Hz. control circuits.
 2. Relay output contacts rated for intended application.
 3. Relays to be mounted in cubicle that contains the associated control circuit.
- B. Motor Overload Relays:

1. Type: Eutectic alloy or bi-metallic, manual reset.
2. Provide overload heater element in each phase of motor power circuit.
3. Trip class designation: NEMA Class 20.
4. Output contacts: One NC and one NO.
5. Conform to NEMA ICS 2.

C. Undervoltage Relay:

1. Provide low voltage protection.
2. Relay undervoltage element to be provided with an adjustable voltage range.

2.10 AUXILIARY RELAYS

- A. Bulletin 700, type P as manufactured by Allen-Bradley; type M as manufactured by Eaton Cutler-Hammer; Class 8501, type X as manufactured by Square-D; or equal, having the following essential characteristics:
1. Type: Heavy-duty, self-resetting, machine tool relay with convertible contacts.
 2. Latching type auxiliary relay to be electrically reset.
 3. Contact rating: A600 in accordance with NEMA ICS 5.
 4. Coil voltage: 120 volts, 60 Hz.

1.01 MOTOR START COUNTER

- A. Display: Six digit LCD. Digits 0.2-inches (5 mm) in height, minimum. Continuously lit.
- B. Non resettable counter.
- C. Provide with nonvolatile memory (EEPROM), which retains data for at least 25 years.
- D. Suitable for operation at 120 volts, 60 Hz.
- E. Case: Rectangular suitable for mounting on motor starter compartment door.

1.02 TERMINAL BLOCKS (PULL-APART TYPE)

- A. Pull apart assembly with male and female component held together with captive screws.
- B. Recess terminals to prevent accidental contact when withdrawn.
- C. Stationary portion of terminal block to be used for factory connections and to remain attached to unit.
- D. Rating: 600 volts, 10 amperes, minimum.

2.11 ENCLOSURE WIRE

- A. 600 volt, stranded copper, type SIS.
- B. Class K stranded in accordance with NEMA WC 70.
- C. Size: 16 AWG, minimum.

2.12 TERMINAL CONNECTORS

- A. Type: Heavy-duty, preinsulated, pressure-crimp type with serrated inner barrel.
- B. Style: Ring-tongue or pin type.
- C. Compatible with conductor and termination point for which it is used.
- D. UL listed.

2.13 WIRE AND CABLE MARKERS

- A. Type: Machine-printable, tubular heat-shrink sleeve label.
- B. Color: White.

2.14 NAMEPLATES, LABELS, WARNING SIGNS

- A. Provide manufacturer's nameplates, engraved nameplates, labels, and warning signs.
- B. Manufacturer's Nameplates. As a minimum provide:
 - 1. Manufacturer's name.
 - 2. Required power supply (voltage, phase, frequency).
 - 3. Load (amperes).
 - 4. Short-circuit rating expressed in RMS symmetrical amperes.
- C. Engraved Nameplates:
 - 1. Provide nameplates of appropriate size and that conform to drawing 40-D-6234 using type A material.
 - 2. Provide nameplate for each motor control center compartment and plug-in unit, which describes function of compartment or plug-in unit.
- D. Provide labels for terminal blocks. Labels to be white, adhesive type, and machine lettered with the single letter terminal block designation as shown on wiring diagrams.

2.15 FABRICATION

A. Structural and Mechanical Features:

1. Assemble sections using corrosion resistant hardware.
2. Connect vertical sections together to form rigid lineup.
3. Attach neoprene gaskets with oil-resistant adhesive.
4. Provide barrier in incoming line compartment to prevent accidental contact with incoming lugs.
5. Mount pushbuttons and indicating lights to be readily accessible and no higher than 78-inches above finished floor.
6. Mount motor overload reset pushbutton on door of contactor plug-in unit.

B. Internal Wiring:

1. Wire power and control circuits in accordance with approved drawings and as necessary for equipment to function as required.
2. Terminate circuits wired to external equipment on terminal blocks.
3. Make wire terminations with terminal connectors with ring tongues.
 - a. Use pin type connectors where the use of ring tongues is not practical such as terminations at selector switches, pushbuttons, indicating lamps, and auxiliary relays.
 - b. Do not use bare wire connections, except for motor power circuit conductors within the motor control center.
4. Keep exposed wiring to a minimum.
 - a. Where used, form wiring into compact groups bound together and firmly supported. Run straight, horizontally, or vertically with short-radius, right-angle bends.
 - b. Twist hinge wiring around longitudinal axis of wire, wherever practicable, instead of bending laterally.
5. Where wire leaves channel or duct, protect from abrasion. Provide grommets for holes used for wires and cables.
6. Do not splice wiring.
7. Make connections at device studs or terminal blocks.
8. Do not terminate more than two wires at each terminal point.
9. Make wiring between devices by most direct method. Do not loop back and forth across panels and between switchboard sections.
10. Use cable tie mounting bases to firmly support and secure exposed wire bundles.
 - a. Mount cable tie base with two No. 8 screws.

- b. Do not mount cable tie bases with only an adhesive.
- 11. Arrange wires or grouped on terminal blocks so that number of individual conductors associated with external cables can be minimized.
- C. Pill-Apart Terminal Blocks:
 - 1. Use in plug-in units with NEMA motor contactors.
 - 2. Terminals to be readily accessible from inside unit when unit door is open.
 - 3. Terminations for field connections to be accessible such that connections can be made without removing the unit or its components.
 - 4. Provide machine lettered conductor designations on the terminal block marking strips.
- D. Nameplates, Legend Plates, and Labels:
 - 1. Attach nameplates with cadmium plated, slotted, pan head self-tapping screws.
 - 2. Attach legend plates for pushbuttons and indicating lights directly to the device.
 - 3. Adhesive type labels may be used on wiring side of hinged equipment doors and back panels of motor control center compartments.
- E. Cable and Wire Identification:
 - 1. Provide computer-generated or labeling machine lettering.
 - 2. Provide wire markers on both ends of each conductor. Wire marker to include conductor designation on the first line and conductor destination on second line. A single line wire marker is acceptable provided the conductor designation is enclosed in parenthesis followed by the conductor destination.
 - 3. Conductor designation and conductor destination to be the same as shown on approved schematic and wiring diagrams.
 - 4. Do not heat shrink wire marker onto conductor. Leave marker loose in order to easily rotate about the conductor.

2.16 FINISHES

- A. Provide manufacturer's standard interior and exterior coating.

2.17 SOURCE QUALITY CONTROL

- A. Perform production tests in accordance with manufacturer's standard shop and quality control procedures.
- B. Test for correct operation and mechanical adjustment.
- C. Perform AC dielectric test on power circuit in accordance with NEMA ICS 1.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with manufacturer's instructions and recommendations.
- B. Motor control center to be installed in location of existing motor control center.
- C. Select and install motor overload heater elements. Size heater in accordance with NFPA 70.
- D. Make cable and conduit connections.
- E. Make grounding and bonding connections in accordance with Section 26 05 20 - Grounding and Bonding and NFPA 70.
- F. Control Cable and Wire Identification: In accordance with Section 26 05 33 – Raceways and Boxes.

3.02 FIELD QUALITY CONTROL

- A. Perform field wiring checkout and test in accordance with Section 26 05 90 - Wiring Checkout and Operational Tests.
- B. Field tests in accordance with NETA ATS.

END OF SECTION

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SECTION 26 27 40

WIRING DEVICES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Wiring Devices:
 - 1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

- A. National Electrical Manufacturers Association (NEMA)
 - 1. NEMA WD 1-1999(R2015) General Requirements for Wiring Devices
 - 2. NEMA WD 6 -2016 Wiring Devices - Dimensional Specifications
- B. National Fire Protection Association, Inc. (NFPA)
 - 1. NFPA 70-2023 National Electrical Code (NEC)

1.03 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 27 40-1, Approval Data.

PART 2 PRODUCTS

2.01 GROUND FAULT CIRCUIT INTERRUPTING (GFCI) RECEPTACLES

- A. Type: Specification grade, heavy duty, duplex plug with integral ground fault circuit interrupter.
- B. NEMA Configuration: 5-20R.
- C. Voltage: 120 volts, 60 hertz.
- D. Current: 20 amperes
- E. Body: White, impact resistant plastic.
- F. Terminals: Screw type for line wiring and grounding.
- G. Pushbuttons: Test and reset.

H. Conform to NEMA WD 1 and NEMA WD 6.

2.02 COVER PLATES

A. Weatherproof Receptacle Coverplates:

1. Type: Rain-tight, "in-use" cover per NEC.
2. Material: Metal.
3. Latching plastic door.
4. Suitable for mounting on type FS cast box.
5. Lockable.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with NFPA 70.
- B. Install weatherproof gang boxes and cover plates.
- C. Install receptacles with grounding pole on top.
- D. Connect wiring devices by wrapping solid conductor around screw terminal. Do not use push-in terminals for wiring connections if screw terminals are available.
- E. Bond wiring devices to equipment grounding conductor.

END OF SECTION

SECTION 26 29 13

ENCLOSED CONTACTORS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Enclosed Contactors:
1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

- A. InterNational Electric Testing Association, Inc. (NETA)
1. NETA ATS-2025 Acceptance Testing Specifications for Electrical Power Equipment & Systems
- B. National Electrical Manufacturers Association (NEMA)
1. NEMA EN 10250-2024 Enclosures for Electrical Equipment (1000 Volts Maximum)
 2. NEMA ICS 2-2000(2020) Controllers, Contactors and Overload Relays Rated 600 V
 3. NEMA WC 70-2021 Power Cables Rated 2000 Volts or Less for Distribution of Electrical Energy
- C. Underwriter's Laboratories, Inc., (UL)
1. UL 508-2024 Industrial Control Equipment

1.03 DESIGN REQUIREMENTS

- A. Replace heater contactors associated with Powerplant and Gatehouse.
- B. Design to match functionality of existing heater contactors.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 29 13-1, Approval Drawings and Data:
1. Manufacturer's technical data.
 2. Layout drawing.
 3. Wiring diagram.

- 4. Installation and maintenance instructions.
- C. RSN 26 29 13-2, Factory Test Reports.
- D. RSN 26 29 13-3, Field Test Reports.
- E. RSN 26 29 13-4, Operation and Maintenance Instruction Books.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Handle in accordance with manufacturer's recommendations.
- B. Store in clean, dry, secure space. Protect from spills or other moisture by using appropriate coverings.

PART 2 PRODUCTS

2.01 ENCLOSURE

- A. Completely factory assembled and wired.
- B. Type 3R in accordance with NEMA 10250.
- C. Door: Hinged, pad-lockable.
- D. UL listed.
- E. Mounting: Surface-mounted.
- F. Nameplate:
 - 1. As indicated on standard drawing 40-D-6234.
 - 2. Copy:
 - a. Line 1: "Contactor".
 - b. Line 2: Designation indicated in Table 26 29 13 A – Enclosed Contactor Ratings.
- G. Finish: Manufacturer's standard.

2.02 CONTACTORS

- A. Type: Magnetic, full-voltage, non-reversing.
- B. Ratings: As shown in Table 26 19 13 A – Enclosed Contactor Ratings.
- C. Power Contacts: Weld-resistant, double-break, and silver alloy type.

- D. NEMA rated and sized for applicable heater or fan loads. Contactor rated and sized to IEC standards is not acceptable.
- E. Power contacts, contactor coil, and auxiliary contacts designed to be inspected and replaced.
- F. Control Power:
1. Powerhouse: From control circuit transformers.
 2. Gatehouse: From panelboard LJA.

Table 26 29 13 A – Enclosed Contactor Ratings

Designation	Associated Equipment	NEMA Size	Poles	Voltage	System	Coil Rating	Continuous Current
NCA	N1E	0	3	480V	3 Phase, 60 Hz	120 V	20 A
	N1H	0	3	480V	3 Phase, 60 Hz	120 V	20 A
NCB	NSA	0	3	480V	3 Phase, 60 Hz	120 V	20 A
		0	3	480V	3 Phase, 60 Hz	120 V	20 A
		0	3	480V	3 Phase, 60 Hz	120 V	20 A
	NSB	0	3	480V	3 Phase, 60 Hz	120 V	20 A
		0	3	480V	3 Phase, 60 Hz	120 V	20 A
		0	3	480V	3 Phase, 60 Hz	120 V	20 A
MJB	NJG/NJD	0	2	240V	1 Phase, 60 Hz	120 V	20 A
	NJK/NJF	0	2	240V	1 Phase, 60 Hz	120 V	20 A
	NJGJ/NJC	0	2	240V	1 Phase, 60 Hz	120 V	20 A

Table 26 29 13 A – Enclosed Contactor Ratings

Designation	Associated Equipment	NEMA Size	Poles	Voltage	System	Coil Rating	Continuous Current
	NJE/NJH	0	2	240V	1 Phase, 60 Hz	120 V	20 A

2.03 CONTROL CIRCUIT TRANSFORMERS (CCT)

- A. Dry type, copper windings, and sized to have adequate capacity for control power requirements of the motor control circuit.
- B. Primary Voltage: 480 volts, 60 Hz..
- C. Secondary Voltage: 120 volts, 60 Hz.
- D. Integral fuse block mounted on transformer.
 - 1. Primary Fuses: 600 volt, current-limiting type.
 - 2. Secondary Fuses: 250 volt, current-limiting type.
- E. Transformer UL listed. Fuses conform to FU 1 and UL listed.

2.04 PROTECTIVE RELAYS

- A. General Requirements:
 - 1. Suitable for use on 120 volt, 60 Hz. control circuits.
 - 2. Relay output contacts rated for intended application.
 - 3. Relays to be mounted in cabinet that contains the associated control circuit.
- B. Thermal Overload Relays:
 - 1. Type: Eutectic alloy or bi-metallic, manual reset, or solid state, self powered.
 - 2. Relay trip on heater or fan overload.
 - 3. Provide overload element in each phase of load power circuit.
 - 4. Trip class designation: Field selectable NEMA Class 10, 20, and 30.
 - 5. Output contacts: One NC and one NO.
 - 6. Conform to NEMA ICS 2.

2.05 AUXILIARY RELAYS

- A. Bulletin 700, type P as manufactured by Allen-Bradley; type M as manufactured by Eaton Cutler-Hammer; Class 8501, type X as manufactured by Square-D; or equal, having the following essential characteristics:
 - 1. Purpose: To energize all four gatehouse contactors. Existing thermostat will energize auxiliary relay.
 - 2. Type: Heavy-duty, self-resetting, machine tool relay with convertible contacts.
 - 3. Latching type auxiliary relay to be electrically reset.
 - 4. Contact rating: A600 in accordance with NEMA ICS 5.
 - 5. Coil voltage: 120 volts, 60 Hz.

2.06 MOLDED-CASE CIRCUIT BREAKERS

- A. Voltage: 480 volt, 60 Hz.
- B. Interrupting Rating: 22,000 amperes RMS symmetrical, minimum.
- C. Trip: Thermal-magnetic type with inverse-time trip characteristics on overload and instantaneous trip on short circuit.
- D. Construction: High-strength, molded, glass-reinforced polyester case and cover.
- E. Provide with overcenter, toggle hand-operated, trip free mechanism with quick make and quick break action independent of speed of toggle handle operation.
- F. Provide with common tripping of all poles.
- G. Sized in accordance with NFPA 70.
- H. UL listed and conform to UL 489.

2.07 INDICATING LIGHTS

- A. Type: Heavy-duty, watertight/oiltight, push-to-test with 30.5 mm mounting hole. Full voltage for operation on 120 volt, 60 Hz.
- B. Color
 - 1. On: Red.
 - 2. Off: Green.
- C. Lamp:
 - 1. High visibility, LED type, 120 volts, 60 Hz.
 - 2. Minimum of 5 LEDs in a cluster. Lamps with one LED are not acceptable.

- D. Nameplate: Equipment manufacturer's standard or custom engraved nameplate suitable for mounting directly over device.

2.08 TERMINAL BLOCKS

- A. 600 volts and 25 amperes.
- B. Suitable for use with size of cables associated with the contactor circuits.
- C. Molded-block type to accommodate ring connectors 1/2-inch wide (outer diameter) at terminal screws.
- D. Binding-head or washer-head screws having serrated or grooved contact surfaces or having lockwashers.
- E. Molded insulating barriers between terminals.
- F. Removable marking strips. Provide marking strips for terminal blocks with conductor designations engraved or printed with a permanent marking fluid. Furnish one spare blank marking strip with each terminal block.
- G. Removable covers to prevent accidental contact with terminal by operating personnel.
- H. Arrange and locate terminal blocks to facilitate connection to incoming and outgoing cables.
- I. Install grommets at holes for wires and cables.
- J. Furnish sufficient terminal blocks to wire and make ready for operation all equipment on boards, and to terminate incoming control cables.
- K. Provide 25 percent spare terminals on each terminal block.

2.09 WIRING

- A. 600-volt, stranded copper, type SIS.
- B. Class K stranded in accordance with NEMA WC 70.
- C. Size: 12 AWG, minimum.
- D. UL listed.

2.10 FABRICATION

- A. Pre-wire contactors to terminal blocks before shipment to jobsite.

2.11 CONTRACTOR SOURCE QUALITY TESTING

- A. Test circuits before shipment to demonstrate correctness of completed wiring connections and proper functioning of devices.

PART 3 EXECUTION

3.01 INSTALLATION

- A. In accordance with approved manufacturer's instructions and recommendations.
- B. Connect to existing thermostat contacts to match existing functionality.
- C. Mount in locations shown on drawings.
- D. Ground and bond to existing grounding system in accordance with Section 26 05 20 – Grounding and Bonding and NFPA 70.
- E. Make conduit and wiring connections.

3.02 CONTRACTOR FIELD QUALITY CONTROL

- A. Perform field wiring checkout and test in accordance with Section 26 05 90 - Wiring Checkout and Operational Tests.
- B. In accordance with NETA ATS.

END OF SECTION

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SECTION 26 33 46
125v DC BATTERY AND CHARGER

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. 125v DC Station Battery Charger:

1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

A. Federal Specifications (FS)

1. FS W-C-375E Circuit Breakers, Molded Case; Branch Circuit and Service

B. Electrical Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE 484-2019 Recommended Practice for Installation Design and Installation of Vented Lead-Acid Batteries for Stationary Applications
2. IEEE 485-2020 Sizing Lead-Acid Batteries for Stationary Applications
3. IEEE 450-2020 Recommended Practice for Maintenance, Testing, And Replacement of Vented Lead-Acid Batteries for Stationary Applications
4. IEEE 1375-1998(2003) Guide for Protection of Stationary Battery Systems

C. National Electrical Manufacturers Association (NEMA)

1. NEMA PE5-97(2003) Utility Type Battery Chargers
2. NEMA AB 3-13 Molded Case Circuit Breakers and Their Application
3. NEMA PB 2-11 Dead Front Distribution Switchboards

D. National Fire Protection Association (NFPA)

1. NFPA 70-23 National Electrical Code (NEC)

E. Underwriters Laboratories, Inc. (UL)

1. UL 50-20 Enclosures for Electrical Equipment

2. UL 489-19 Molded-Case Switches, and Circuit-Breaker Enclosures

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 – Submittals and Section 01 33 26 – Electrical Drawings and Data.
- B. RSN 26 33 46-1, Approval Data:
1. Equipment layout drawings, manufacturer's data.
 2. Bill of material for each device or component of equipment, and circuit directory for each board.
- C. RSN 26 33 46-2, Proposed location of New Battery Chargers
1. Plan and Section drawing showing proposed location on wall outside of battery room.
- D. RSN 26 33 46-3, Factory Test Report.
- E. RSN 26 33 46-4, Operation and Maintenance Instruction Books.

1.04 DESIGN REQUIREMENTS

- A. Design layout of station battery charger in accordance with IEEE, NEMA and NFPA 70.
- B. Station Battery Information:
1. Location: Governor Gallery level
 2. Type: Flooded-cell, heavy-duty, lead-acid stationary type with calcium-lead alloy grids.
 3. Voltage: 125 VDC
 4. Cell: 58 cells.
 5. Ampacity: 585 Amperes Hours.
 6. Nominal cell specific gravity: 1.205 – 1.220 at 77 degrees F.
 7. Float charge voltage range: 2.17 - 2.25 volts per cell.

PART 2 PRODUCTS

2.01 CHARGERS

- A. Voltage: 480 volts AC 3 phase ± 10 percent variation in input voltage.
- B. Frequency: 60 Hertz.

- C. Ampacity: 35 Amperes.
- D. Efficiency: 80 percent and 70 percent or better power factor at full load.
- E. Chargers connect to battery rated: 585 AH per 8 hours, 125-volt lead acid lead-calcium flooded cell battery system.
- F. Voltage regulation: ± 0.5 (one-half) percent overload range of 1 to 100 percent.
- G. Float voltage control range: 125.86 to 131.08 volts, provide equalizing voltage control of 145 volts.
- H. Current Limiting: Adjustable from 90 to 110 percent. Full output (i.e., 100 percent) current rating of charger at 6,581 feet.
- I. Filtering: Limit charger output ripple to 30 millivolts rms or less.
- J. Parallel and Load Sharing Operation:
1. Without interconnecting control wiring: Capable of parallel operation where two compatible battery chargers are connected in parallel.
 2. With interconnecting control wiring (forced load sharing) 0-full rated current:
 - a. No more than 20 percent of sum of combined rated charger output currents.
 3. Be free from output current instability (hunting) from 20 percent to rated output current.
 4. Any single charger:
 - a. Continue to operate without interruption or failure in event its counterpart fails.
- K. Stand-Alone/No Load Sharing:
1. Capable of operating in a stand-alone" mode (no load sharing - one charger "OFF," one charger "ON"), even with load sharing wiring connected.
- L. Effect of Altitude: Provide full rated output at 6,581 feet above sea level (and lower), with an ambient air temperature range of 0 to 50 degrees C.
- M. Battery Eliminator Capability:
1. Capable operate effectively with battery disconnected from system.
 2. In addition, provide a "soft start" automatic startup with resistive load only (i.e., battery disconnected) or when connected to battery.
- N. Discharge Protection:

1. AC power failure, being turned OFF (whether via local or remote), or other charger shut down situations.
 2. Extremely low current requirements of load-side powered controls need not be considered a cause of battery discharge.
- O. Protection:
1. AC Circuit breaker: Properly sized AC circuit breaker, UL listed.
 2. DC Circuit breaker or fuse:
 - a. Properly sized two pole AC circuit breaker or fuse, UL listed.
 3. Protect both AC input and DC output circuits from surge and transient voltage and current occurrence, including that related to powering-up charger from a "cold" start (no previous charge on surge and filter capacitors).
- P. Automatic Charger Restart: Restart (in float mode) automatically upon loss and restoration of AC input power.
- Q. Accessories:
1. AC Pilot Lamp/LED: Provide continuous-duty type blue LED indication of energized AC input.
 2. Local manual ON/OFF control: Provided control switch to be accessible from front (exterior) panel of charger.
 3. Float Level Control: microprocessor system equivalent.
 4. Equalize Level Control: microprocessor system equivalent.
 5. Equalize "Timer Switch" LEDs/LCD display:
 - a. 0- to 24-hour timer (wider range is acceptable) with automatic return to float mechanism and operational mode indicator lamps/LEDs.
 - b. Equalize mode is manually operator selected and is NOT to occur automatically after loss and restoration of AC power source.
 6. Battery Temperature Compensation: Provide temperature compensation that varies charger float voltage (measured at battery terminals) to compensate for battery temperature variations from 77 degrees F (25 degrees C):
 - a. $V_{\text{corrected}} = V_{77^{\circ}\text{F}} - ((T_{\text{actual}} - 77^{\circ}\text{F}) \times (0.003\text{V}/^{\circ}\text{F}))$ or $V_{\text{corrected}} = V_{25^{\circ}\text{C}} - ((T_{\text{actual}} - 25^{\circ}\text{C}) \times (0.0055\text{V}/^{\circ}\text{C}))$.
 - b. Provide proper voltage for 60 cell 125-volt battery, with rated float voltage of 2.17 - 2.26 volts per cell at 77 degrees F.
 - c. Provided remote (for mounting on battery) sensor with 50 feet of insulated jacket shielded cable per sensor:
 - 1) Sensor will be connected by customer to negative post of a middle cell, using intercell terminal connection bolt, washer, and straps provided by battery manufacturer.

- 2) Sensor type using an adhesive for attachment/mounting to battery is NOT acceptable.
 7. DC output ammeter and voltmeters: Provide 3.5-inch digital display types with one percent accurate of full scale.
 - R. Alarms with Red indication LED light:
 1. Charger Failure Alarm: Isolated form "C" contacts to indicate battery charger failure, such as tripped circuit breaker, overvoltage shutdown, and no output voltage.
 2. Low DC Voltage Alarm: Isolated form "C" contacts for alarm indication when DC output voltage falls below (customer adjustable) preset level:
 - a. Minimum voltage adjustment range for pickup and dropout: 98.6 to 127.6 volts.
 3. High DC Output Voltage Alarm Relay: Isolated form "C" contacts for alarm indication when DC output voltage rises above (customer adjustable) preset level:
 - a. Minimum voltage adjustment range for pickup and dropout shall be 127.6 to 145 volts.
 4. AC Power Failure Alarm: Isolated form "C" contacts for alarm indication in case of low AC input voltage or interrupted AC input.
 - S. Audible Noise:
 1. Noise level of a single charger: 65 dB maximum at any point 5 feet away.
- 2.02 CHARGER ENCLOSURE**
- A. Enclosure: NEMA-1 with full-length hinged door.
 - B. Ventilation Class 1 (i.e., natural convection cooling, without fans).
 - C. Floor-mounted enclosures:
 1. Height: 4 foot 8-inch minimum.
 2. Entry: Top or sides via conduit knockouts or removable cover plates.
 - D. Easy access to component parts in interior with cabinet back against the wall.
 - E. Circuit breaker:
 1. Molded-case type.
 2. Thermal-magnetic type with inverse-time trip characteristics on overload and instantaneous trip on short circuit.
 3. DC Voltage: 125V minimum.
 4. Ratings: Manufacture sizing.

5. Arc quenchers.
6. Quick-make and quick-break toggle mechanisms.
7. Trip-free operating handle.
8. Single operating handle. Do not use handle ties on multiple-pole circuit breakers.
9. Assemble so that individual breakers can be removed or added without disturbing adjacent units, loosening or removing supplemental insulation supplied to meet requirements of UL, and without additional machining, drilling, or tapping.

F. Terminal blocks:

1. 600 volts and 25 amperes.
2. Suitable for use with No. 8 AWG wire.
3. Molded-block type to accommodate ring connectors 1/2-inch wide (outer diameter) at terminal screws.
4. Binding-head or washer-head screws having serrated or grooved contact surfaces or having lock washers.
5. Molded insulating barriers between terminals.
6. Removable marking strips.
 - a. Provide marking strips for terminal blocks with conductor designations engraved or printed with a permanent marking fluid.
 - b. Furnish one spare blank marking strip with each terminal block.
7. Removable covers to prevent accidental contact with terminal by operating personnel.
8. Arrange and locate terminal blocks to support incoming and outgoing cables.

G. Solderless, Bolted-Type Lugs or Terminals:

1. Provide lugs or terminals suitable for copper conductors on breakers. Size lugs or terminals in accordance with cable sizes indicated on drawings.
2. Provide solderless, bolted-type copper lugs or terminals for connecting enclosure to ground system. Connect lugs or terminals to ground system by No. 4 AWG copper conductor.

2.03 CHARGER FABRICATION

A. Mount:

1. Ground bus at bottom of enclosure. Bond bus to enclosure.
2. Equalize voltage adjustments, control, and selection on front of enclosure.
3. Circuit breaker handles to be operable from front of enclosure.
4. Meters at approximately normal eye level on front of enclosure.

- B. Make equipment, removable covers, terminal blocks, wiring, and wiring gutters accessible from fronts of the panel for operation, maintenance, removal, and replacement.
- C. Make provisions for anchoring backs of floor-mounted boards to walls for additional support.
- D. Equip ventilation openings with grilles or screens having dust filters.
- E. Equip each door with locks, latches, and concealed hinges to allow doors to open 105 degrees, minimum, from fully closed position:
 - 1. Key all locks alike:
 - a. With identifying number and “Do Not Duplicate” stamped on each key.
 - 2. Provide five (5) keys.

2.04 SOURCE QUALITY TESTING

- A. Design and production tests in accordance with NEMA PE-5.
 - 1. Dielectric test.
 - 2. Voltage adjustment test.
 - 3. No load test.
 - 4. Current limit test.
 - 5. Ripple voltage measurement.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Charger:
 - 1. Cable entry into enclosure: From bottom of enclosure.
 - 2. Provide anchoring, supporting, and mounting materials as required.
 - 3. Install ground cable between enclosure and grounding plate.

END OF SECTION

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SECTION 26 36 23

AUTOMATIC TRANSFER SWITCH

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Automatic Transfer Switch:
1. Payment: Lump sum price offered in the Price Schedule.

1.02 REFERENCE STANDARDS

- A. InterNational Electrical Testing Association (NETA)
1. NETA ATS-2025 Acceptance Test Specification for Electrical Power Distribution Equipment and System
- B. National Electrical Manufacturer's Association (NEMA)
1. NEMA 250-2017 Enclosures for Electrical Equipment (1000 Volts Maximum)
- C. National Fire Protection Association (NFPA)
1. NFPA 70-2017 National Electric Code (NEC)

1.03 SYSTEM DESCRIPTION

- A. Transfer switch shall be automatic so that when normal service power has been interrupted, a future engine-generator will start, and when the generator has attained the rated voltage and frequency of the distribution source the load will be transferred.
- B. Switch shall have mechanical and electrical interlocks.
- C. Switch shall have time delays.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 – Submittals.
- B. RSN 26 36 23-1, Approval Drawings and Data:
1. Commercial product data.
 2. Electrical interconnection wiring diagrams, and schematics showing wire connections and conduit layout for all electrical products. Provide electrical control schematic diagrams in ladder-logic format.
- C. RSN 26 36 23-2, As-Built Drawings.

- D. RSN 26 36 23-3, Field Test Reports.
- E. RSN 26 36 23-4, Operation and Maintenance Instruction Books:
 - 1. Electronic copies in Adobe pdf format of Service manuals:
 - a. Operating and maintenance instructions.
 - b. Installation and detailed assembly drawings.
 - c. Approved commercial product data.
 - d. As-built drawings.
 - e. Field Test Reports.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Protect equipment from corrosion, deformation, and other types of damage.

PART 2 PRODUCTS

2.01 AUTOMATIC TRANSFER SWITCH

- A. Ratings:
 - 1. 800 amperes.
 - 2. 480 Volts, 3-Pole with switched neutral.
 - 3. Withstand Current Capability: 22,000 amperes.
- B. Wall mounted.
- C. Solid-state control circuitry.
- D. Exercise Clock: Solid-state, fully programmable.
- E. Attach schematic wiring diagrams to inside of door panel.
- F. Enclosure: NEMA 250, Type 3R.
- G. Provide switch with mechanical and electrical safety interlocks.
- H. Provide auxiliary form "C" contacts, one normally open and one normally closed.
 - 1. Provide output contact to annunciator to indicate status and position of the automatic transfer switch.
- I. Adjustable Time Delays:
 - 1. Start Delay: Prevents generator set from start-up during momentary utility disturbance.

2. Transfer Delay: Allows the generator set to reach a stable running speed before transfer.
3. Re-Transfer Delay: Allows the utility to stabilize before load re-transfer, thereby preventing multiple attempts to transfer as a result of an unstable utility source.
4. Stop Delay: Maintains the generator set for immediate reconnection if the normal source fails shortly after re-transfer, and allows gradual generator set cool-down by running with no load before shutdown,

PART 3 EXECUTION

3.01 INSTALLATION

- A. Mount the transfer switch at rear of powerplant as shown on drawings.
- B. Install transfer switch in accordance with manufacturer's recommendations and NFPA 70.
- C. Ground the switch in accordance with Section 26 05 26 - Grounding and Bonding and NFPA 70.
- D. Install ground cable between the switch and the grounding system.

3.02 CONTRACTOR FIELD QUALITY TESTING

- A. In accordance with NETA ATS.
- B. On-Site Test:
 1. Conduct on-site test to test the ability of the switch to operate properly:
 - a. Sense failure of the normal power source.
 - b. Send a start command to temporary engine-generator set and have the engine-generator set respond properly.
 - c. After a time delay, switch should sense proper voltage and frequency from the generator then transfer the load to the generator.
 - d. Sense the return of proper voltage of the normal power source.
 - e. After a time delay, transfer the load back to the normal power source.
 - f. After the stop time delay (cool down cycle), shut down the engine-generator.
 - g. Disconnect and return temporary engine generator set to Government.
 2. Correct defects disclosed by the tests to the satisfaction of COR.

END OF SECTION

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SECTION 27 05 43
OUTSIDE PLANT RACEWAYS FOR COMMUNICATIONS SYSTEMS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Outdoor Conduit and Accessories.
1. Payment: Lump sum prices offered in the Price Schedule.
 2. Includes:
 - a. Conduit, conduit supports and pull boxes.

1.02 DEFINITIONS AND ACRONYMS

- A. ID: Inside diameter.
- B. OD: Outside diameter.

1.03 REFERENCE STANDARDS

- A. American National Standards Institute (ANSI)
1. ANSI C80.1-2020 Electrical Rigid Steel Conduit (ERSC)
- B. American Society for Testing and Materials (ASTM)
1. ASTM A563/A563M-2024 Standard Specification for Carbon and Alloy Steel Nuts (Inch and Metric)
 2. ASTM F463/463M-2024 Standard Specification for Hardened Steel Washers Inch and Metric Dimensions
 3. ASTM F1554-2020 Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
 4. ASTM F2160-22a Solid Wall High Density Polyethylene (HPDE) Conduit Based on Controlled Outside Diameter (OD)
- C. American Society of Mechanical Engineers (ASME)
1. ASME B1.1-2024 Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)
- D. Manufacturers Standardization Society (MSS)
1. MSS SP-58-2025 Pipe Hangers and Supports

- E. National Electrical Manufacturers Association (NEMA)
 - 1. NEMA FB 1-2014 Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable
- F. National Fire Protection Association (NFPA)
 - 1. NFPA 70-2023 National Electrical Code (NEC)
- G. Underwriters Laboratories (UL)
 - 1. UL 2024-14 Safety Cable Routing Assemblies and Communications Raceways

1.04 SUBMITTALS

- A. In accordance with Sections 01 33 00 – Submittals, 01 33 26 – Electrical Drawings and Data, and 01 78 30 - Project Record Documents.
- B. RSN 27 05 43-1, Approval Drawings and Data:
 - 1. Items in Part 2 Products.
 - 2. Bill of materials.
 - 3. Manufacturer's technical product data, including for:
 - a. Conduit.
 - b. Conduit fittings.
 - c. Conduit support.
 - d. Concrete anchors.
 - e. Pull boxes.
 - 4. Conduit and pull box layout drawings.
- C. RSN 27 05 43-2, Installation Plan and Procedures:
 - 1. Provide installation plan and procedures for fiber optic conduit installation from the Powerplant to the Gatehouse.

1.05 DESIGN REQUIREMENTS

- A. Determine routing of exposed and embedded conduit.
 - 1. Refer to drawings 245-D-6180 through 245-D-245-D-6182.
- B. Determine quantity and placement of pull boxes required.

PART 2 PRODUCTS

2.01 CONDUIT

A. General:

1. UL listed.
2. Sizes as shown on drawings.
3. Rate for intended purposes or labeled by nationally recognized testing laboratory (NRTL) accredited by Federal Occupational Safety and Health Administration (OSHA) when applicable.

B. Rigid steel conduit and attached fittings:

1. Conduit:
 - a. Zinc-coated and in accordance with ANSI C80.1.
 - b. Size: 2 inch.
2. Fittings:
 - a. Zinc-coated to match conduit.

2.02 CONDUIT ACCESSORIES

A. General:

1. Conform to NEMA FB 1 as applicable.
2. Steel materials and hardware: Galvanized

B. Provide fittings required to complete conduit systems. Includes caps, connectors, couplings (excluding expansion and expansion-deflection couplings), nipples, reducers, elbows, pipe plugs, locknuts, bondnuts, bushings, seals (excluding wall penetration seals), and other fittings required for a complete system.

C. Conduit support:

1. Galvanized steel pipe clamps per MSS SP-58

D. Adhesive Anchors: Drilled type, installed with a chemical adhesive system:

1. Seismic qualified per IBC and AC308 for seismic design category A through F and cracked concrete conditions.
2. Adhesive: HIT-HY 200 epoxy adhesive manufactured by HILTI, P.O. Box 21148, Tulsa, OK 74121; or equal, having the following essential characteristics:
 - a. Injectable two-component epoxy adhesive.
 - b. Dual packaging system designed for automatic mixing during injection.
 - c. Weather resistant.

- d. Resistant against high temperatures.
 - e. Suitable for use in diamond cored or pneumatic drilled holes.
 - f. Extended temperature range from -40 to 104-degrees F.
- 3. Threaded carbon steel anchor rod: HAS Anchor rod manufactured by HILTI, P.O. Box 21148, Tulsa, OK 74121; or equal, having the following essential characteristics:
 - a. Rod material: ASTM F1554, Grade 36.
 - b. Thread: ASME B1.1, continuously threaded (all-thread).
 - c. Fabricate with 45-degree chisel point on one end to facilitate insertion into adhesive-filled hole.
 - d. Hot-dip galvanized.
- 4. Standard Steel Nuts: ASTM A563, Grade A. Hot-dip galvanized.
- 5. Standard Steel Washers: ASTM F436, Type 1. Hot-dip galvanized.

2.03 PULL BOXES

- A. Location: on Penstock support ring concrete structures.
- B. Stainless Steel Type 304
- C. Screw-Cover
- D. Type 4X

2.04 PULL TAPE

- A. Polyester pull tape
- B. Strength: minimum 1000 lbs.

PART 3 EXECUTION

3.01 INSTALLATION

- A. In accordance with NFPA 70 and manufacturer's requirements.
- B. Cable and wiring in accordance with Section 26 05 10 – Conductors and Cables.
- C. Repair coatings of galvanized conduit damaged during installation as necessary.
- D. Fiber Optic cable installation in accordance with Section 27 20 01 – Fiber Optic Cable and Accessories.

- E. Completed in a neat and workmanlike manner in accordance with NFPA 70. Damage to enclosures or installation area shall be repaired and left in a finished condition.
- F. Installation should provide a continuous run between the Gatehouse and the Powerplant.
- G. Layout and Procedures:
 - 1. Determine routing of exposed or buried conduit.
 - 2. Ensure that conduit runs, and bends are free from kinks, indentations, or flattened surfaces.
 - 3. Conduit shall adjust angle to align with pull box knockouts 10 feet or further from the pull box to allow unimpeded entry into the pull box knockouts.
 - 4. There shall be a minimum of three pull boxes.
- H. Install conduit with necessary fittings and supports: Tighten conduit joints securely to prevent entrance of moisture or foreign material.
- I. Bends: Install smooth, gradual bends to permit pulling insulated conductors without undue stress or damage to conductors or conduit.
- J. Install and leave in place a pull tape in conduit for future use.
- K. Dissimilar metals: Protected from galvanic corrosion by means of pressure tapes, coatings, or isolators.
- L. Protection: Properly cap open ends of pipe at end of each day's work or other stopping point throughout construction. Tightly cover equipment and protect against dirt, water, chemical, and mechanical damage.
- M. Install pipe supports such that pipe lengths between supports rest on grade with minimal bending stresses induced from pipe support.
- N. Adhesive Anchor Bolts:
 - 1. Adhesive capsules of different diameters may be used to obtain proper volume for embedment, but no more than two capsules per anchor may be used.
 - 2. When installing different diameter capsules in same hole, install larger diameter first.
 - 3. Any extension or protrusion of capsule from hole is prohibited.
 - 4. Two-part injection adhesives are to be supplied in individually sealed, equal volume cartridges.
 - 5. Epoxy components: Thoroughly mixed when dispensed through static spiral mixing/dispensing nozzle:
 - a. Bulk hand mixed and measured adhesives: Not be allowed.

6. Follow installation recommendations by anchor system manufacturer carefully, including maximum hole diameter.
7. Holes shall have rough surfaces, such as can be achieved using rotary percussion drill.
8. Holes: Blown clean with compressed air and free of dust or standing water prior to installation.
9. Anchor: Left undisturbed and unloaded for full adhesive curing period.
10. Concrete temperature (not air temperature): Compatible with curing requirements of adhesives per adhesive manufacturer:
 - a. Anchors: Not be placed in concrete below 40 degrees F.

END OF SECTION

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SECTION 27 20 01

FIBER OPTIC CABLE AND ACCESSORIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Fiber Optic Cable and Accessories:
 - 1. Payment: Lump sum price offered in the Price Schedule.
 - a. Includes:
 - 1) Industrial Ethernet Switches.
 - 2) Data Collector.
- B. Cost:
 - 1. Include cost of spare parts in lump sum price offered in the Price Schedule for Spare Parts.

1.01 DEFINITIONS

- A. Attenuation: The reduction in power of the light signal as it is transmitted. Measured in dB/km.
- B. dB/km: Decibel per kilometer.
- C. Installed test: Testing for optical fiber that has been installed in conduit or a pull box. Installed tests shall require testing of the entire length of fiber installed for the project.
- D. Optical Fiber Cable: One or more fiber strands and strand bundles, single-mode or multimode transmission mode.
- E. OS2: 9/125 μm single-mode fiber, low-water peak.
- F. OSP: Outside plant.
- G. OTDR: Optical Time Domain Reflectometer used for testing fiber.
- H. Reel test: Testing for optical fiber still on the wheel or reel it was shipped to site on.
- I. Single-mode optical fiber: Optical fiber designed to carry a single mode of light.
- J. Strand or Fiber: Individual light carrying optical fiber used to make up optical fiber cables.
- K. Telecommunications Cable: For supervisory computer control, voice communication, and logic level signals (Fiber, Ethernet, Coax, etc.).

- L. Wavelength (Fiber): The distance between two successive crests or troughs of a light wave. Denoted by the Greek letter lambda (λ) and measured in nanometers (nm).

1.02 REFERENCE STANDARDS

- A. Building Industry Consulting Service International (BICSI)
1. OSPDRM-2018 Outside Plant Design Reference Manual, 6th Edition
- B. Institute of Electrical and Electronic Engineers (IEEE)
1. IEEE C2-2023 National Electrical Safety Code (NESC)
- C. Insulated Cable Engineer Association (ICEA)
1. ICEA S-87-640-2023 Optical Fiber Outside Plant Communications Cable
 2. ICEA S-104-696-2019 Indoor-Outdoor Optical Fiber Cable
- D. National Fire Protection Association (NFPA)
1. NFPA 70-2023 National Electrical Code (NEC)
- E. Telecommunications Industry Association (TIA)
1. TIA-455-226-2022 Calibration of Optical Time-Domain Reflectometers (OTDR's)
 2. TIA-455-231-2015 Calibration of Fibre-Optic Power Meters
 3. TIA-526-7-A-2015 Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant
 4. TIA-568.0-E-2020 Generic Telecommunications Cabling for Customer Premises
 5. TIA 568.3-E-2022 Optical Fiber Cabling Components Standard
 6. TIA 598-2021 Optical Fiber Cable Color Coding.
- F. Underwriters Laboratories, Inc. (UL)
1. UL 1666-2021 Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts

1.03 DESIGN REQUIREMENTS

- A. Provide single-mode fiber for all outdoor installations.
- B. Design OSP in accordance with BICSI OSPDRM.

- C. Two OSP parallel runs of 12-strand Single-mode fiber optic cable shall be installed in conduit between the powerplant and gatehouse.
- D. Ground rods and conductors shall be provided for grounding of armored fiberoptic cable.
- E. Provide UL listed plenum or tray cable as required.
- F. Optical fiber communications transmission and distribution cabling, also known as outside plant (OSP), consists of optical fiber cable and associated components and accessories required to provide data communication signal paths between buildings and plant facilities.
- G. Optical fiber cable shall be suitable for indoor/outdoor environments and rated for general purpose horizontal, duct, or direct burial installation. The fiber cable shall be armored. At a minimum cable shall be rated for both wet and dry applications.
- H. Work consists of providing, testing, and making operational optical fiber cable and interconnecting components necessary to form a complete communications transmission and distribution (OSP) cabling system for continuous use.
- I. Optical fiber shall be installed and spliced between rack-mounted fiber patch panels in the control room, gatehouse and turbine level.
- J. Reuse existing embedded concrete conduit to the furthest extent possible.
 - 1. Applicable only for conduit inside of powerplant and gatehouse.
- K. Provide connection between new IRIG-B Distribution Module to the existing facility satellite clock/antenna.
 - 1. The IRIG-B signal shall be established in network switches and distributed to the PLC.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 – Submittals, Section 01 33 26 - Electrical Drawings and Data, and 01 78 30 – Project Record Documents.
- B. RSN 27 20 01-1, Approval Data:
 - 1. Items and systems in Part 2 Products.
 - 2. Manufacturer's technical data.
- C. RSN 27 20 01-2, Installation Plan and Procedures:
 - 1. Provide installation plan and procedures for fiber optic cable installation from the Powerplant to the Gatehouse through new conduit.
- D. RSN 27 20 01-3, Field Test Reports:

1. Test Reports and Wire Termination Sheets.
 - a. Provide test reports for each cable provided by Contractor.
 2. Include results of all fiber optic cable and accessory tests performed in the field.
 3. Include completed, signed, and approved electrical commissioning forms.
- E. RSN 27 20 01-4 Final Drawings:
1. Items and systems in Part 2 Products.
 2. Show type and locations of all optical fiber cabling and optical fiber termination points.
 3. Include identifying information as indicated on cable labels.
 4. Fiber schedule table detailing final termination points within splice closure.

1.05 SPARE PARTS, EXTRA MATERIALS AND SPECIAL TOOLS

- A. Include fiber optic cable accessories and jumpers of each type of component or model used.

1.06 DELIVERY STORAGE AND HANDLING

- A. Furnish fiber optic cable on reels labeled with lengths as described herein.
1. Length of cable on a reel shall not be spliced.
 2. Provide protection terminations for shipping.
- B. Deliver cable to the Pole Hill Powerplant.
- C. Deliver fiber optic cables with control cables as listed in Section 26 05 10 – Conductors and Cables.

PART 2 PRODUCTS

2.01 GENERAL

- A. Manufactured no more than 24 months prior to Notice to Proceed.
- B. Coverings or insulation: Suitable for installation in vertical position without injury to covering or deformation of insulation when supported in accordance with NFPA 70.

2.02 OPTICAL FIBER CABLE

- A. Provide cable in accordance with ICEA S-87-640, and TIA-568.3, as appropriate.
- B. Cable performance shall not be impacted by immersion in water.
- C. UL listed, where available.

- D. Single-Mode OS2 Ultra Fiber Optic Cable: Model 096ZUV-T4101D20, manufactured by Corning, Inc., One Riverfront Plaza, Corning, New York, 14831; or equal, having the following essential characteristics:
1. Attenuation of 1310/1383/1550 nanometer wavelength light: Less than 0.4/0.4/0.3 decibels per kilometer.
 2. Attenuation: Directly proportional to cable length for an arbitrary cable length section after steady-state conditions are reached.
 3. At a minimum, includes 2 ripcords.
 4. Designed Service Life: 40 years at specified installation and operating conditions.
 5. Maximum Age at Installation: 24 months.
 6. Flame-retardant inner and outer jacket material.
 7. Designed for indoor/outdoor installations.
- E. Minimum Number of Fiber Strands per Cable: 12 fibers.
- F. Buffer Tubes:
1. Symmetrically stranded buffer tubes around dielectric central member.
 2. Gel free/stranded, loose tube.
 3. Diameter: 2.5-millimeter.
 4. Protect each fiber in a buffer tube.
- G. Strength Members:
1. With expansion and contraction characteristics similar to glass fibers.
 2. Central dielectric strength member.
 3. Tensile strength sufficient for installation and residual rated loads per ICEA S-87-640.
 4. Strength member is included to serve as a cable core foundation to reduce strain on fibers. It shall not serve as a pulling strength member.
- H. Shielding or Other Metallic Covering:
1. Armored Cable: Required throughout the project, corrugated steel tape armor.
 2. Single tape covering or shield.
 3. Per ICEA S-87-640.
- I. Jacket:
1. Type: PE (polyethylene) or better, UV resistant, flame-retardant.
 2. Cable jacket marking shall be legible and contain the following information:
 - a. Manufacturer's name and trademark.

- b. Strand count.
 - c. Cable type.
 - d. Pair count.
 - e. UL listing.
- J. Color Coding: Per TIA-598.
- K. Water blocking. Performance shall not be impacted by immersion in water.
- L. Suitable for aerial, direct buried, and general-purpose horizontal installations.
- M. Bend Radius, minimum: 10 times outer diameter of cable or larger as recommended by cable manufacturer.
- N. Cable Tensile Strength, minimum:
- 1. Long term (installed): 200 pound-force (lbf).
 - 2. Short term (during installation): 600 lbf.
- O. Operating Temperature Range: Minus 40 to plus 158 degrees Fahrenheit.

2.03 ACCESSORIES

- A. Fiber Optic Patch Panels:
- 1. Location:
 - a. Control Board (CSA).
 - 1) Refer to Specification Section 26 09 18 – Control Boards.
 - b. Gatehouse.
 - 1) Refer to Specification Section 25 00 01 - Automation Control System.
 - 2. Type: Rack mount or wall mount.
 - 3. Adaptor Plates:
 - a. Same manufacturer as enclosure.
 - b. Mount at minimum 24 fibers.
 - c. Use LC connectors where possible.
 - d. Use dust covers where no fiber optic termination is made on adaptor plate.
- B. Fiber Optic Cable Connectors:
- 1. Fusion splice OSP fiber optic cable to fiber optic pigtails with factory-terminated LC connector and connect to patch panels. Make connections to components or SFPs as required.

2. Provide patch cords to connect fiber optic cable to fiber optic equipment.
- C. Cable Ties:
1. General purpose black Nylon 6/6.
 2. UV resistant.
 3. Self-locking.
 4. Stainless steel tab.
- D. Single Strand Markers:
1. Type: Heat-shrink or self-laminating-vinyl type, machine printable sleeve.
 2. Color: White with black text.
 3. Text size: As large as practical but at least 7 point (0.06 inch).
 4. 1 inch in length, minimum, and have diameter suitable for installation on intended conductor.
- E. Cable Tags:
1. Type: Cross-laminated polyethylene film or Nomex. Resistant to water, oil, solvent, abrasion, and tearing.
 2. Rectangular shape and have minimum dimensions of 2 by 0.5 inches.
 3. Color: White with black text.
 4. Text Size: As large as practical but at least 16 point (0.16 inch).
 5. Attach to cable with standard nylon self-locking tie wraps or cable ties.

PART 3 EXECUTION

3.01 OPTICAL FIBER INSTALLATION

- A. Install in accordance with BICSI OSPDRM, IEEE C2, NFPA 70, TIA-568.0, TIA-568.3, and manufacturer's requirements and recommendations.
- B. Install all fiber optics cable in circuit raceway and conduit or similar protected installation in accordance with **Section 27 05 43 - Outside Plant Raceways for Communications Systems**.
- C. Grounding and bonding of cable armor in accordance with Section 26 05 20 – Grounding and Bonding.
- D. Test each fiber optic cable once it is delivered to site and before it is installed. Then, test each fiber optic cable after it is installed, but before connecting to various electrical components and devices.

- E. Maintain manufacturer recommended minimum bending radius for cables.
- F. Fiber optic cabling shall provide a designated and easily identified to-and-from location marker at each end of the cable.
- G. Fiber optic cable installed outside structures shall be installed in ducts.
- H. Install fiber optic cable without exceeding allowable pulling tensions and sidewall pressures recommended by cable manufacturer.
- I. Clamp and lube fiber optic cable per manufacturer's recommendations and as necessary.
- J. Seal cable penetrations into equipment or passing through block outs, with silicone-foam, fire-retardant type material.
- K. Provide service loop on each end of the cable.
- L. Termination:
 - 1. Terminate fiber optic cable at fiber patch panels with applicable connector.
 - 2. Terminate both ends of each fiber with an appropriate fusion spliced pigtail.
 - 3. Fibers shall be terminated in strict compliance with the manufacturer's printed instructions.
 - 4. Furnish and install patch cords to connect fiber cables to fiber optic equipment.
- M. Cable Splicing:
 - 1. Do not splice optical fiber cables outside of optical fiber splice closures.
 - 2. Method: Arc-fusion only. No mechanical splices.
 - 3. Splice loss: 0.1 decibels, average bi-directional, or less.
- N. Leave sufficient length of fiber optic cable to make arc fusion splices to adaptor plate pigtails.
- O. Patch Panels
 - 1. Coil 10 feet of fiber cable inside each patch panel.
 - 2. Terminate spare fibers in each patch panel.
- P. Cable Installed in Conduit:
 - 1. Do not pull cable into conduit until cleaned and are free from obstructions and sharp corners.
 - 2. Draw clean, dry, tight-fitting rag through conduit immediately before installing cable.
 - 3. Install cable using gradual and uniform pulling stresses. Do not exceed allowable pulling tensions and sidewall pressures as recommended by cable manufacturer.

4. Install cable to prevent cuts or abrasions in insulation or protective covering and kinks in cable.
5. When a lubricant is needed as an aid to pulling, use only soapstone or other suitable lubricant that is not injurious to cable insulation.
6. When mechanical cable pulling equipment is used, use a tension meter release mechanism to ensure pulling tension remains below tensile strength of the cable.
7. Remove and replace any cable damaged during installation.

Q. Cable Installed in Vertical or Inclined Plane:

1. Support with cable grips leaving slack spans between supports.
2. Clamp cable entering enclosures, junction boxes, or equipment with commercial cable clamps.

R. Cable in Boxes and Enclosures:

1. Provide optical fiber cable coils inside pull boxes, junction boxes, and other electrical and control enclosures, and where recommended by the manufacturer.
2. Provide slack in the cable in all boxes, and enclosures. Cable should not be pulled tight against any edge or component in enclosures and boxes.

- S. Clamp cable entry and support cable with fiber type cable clamp. Do not use cable ties on optical fiber cable.

3.02 CONTRACTOR SOURCE QUALITY TESTING

- A. Cables shall be tested at the supplier in accordance with the requirements of ICEA S-87-640 and ICEA S-104-696, as appropriate.

3.03 FIELD TESTING

A. Perform testing for each of the following situations:

1. Each reel of fiber shall be tested within 24 hours of shipment to the staging site.
2. Each reel of fiber shall be tested if it has not been tested in the previous 7 calendar days prior to installation.
3. Each reel of fiber shall be tested after installation and prior to splicing.

B. Conduct power loss measurements in accordance with TIA-526-7.

C. Notify COR, in writing, of date, time, and cables to be tested at least 7 working days before testing. Testing dates shall be mutually agreeable between Contractor and COR.

D. 100 percent of optical fiber cabling shall be tested and shall meet all requirements and specifications as stated by manufacturer and these specifications.

E. Test each fiber using an Optical Time Domain Reflectometer (OTDR):

1. Purpose and intent of testing:
 - a. Determine if installed cable is free from defects.
 - b. Compare measured attenuation with cable manufacturer's specified maximum attenuation per wavelength.
 - c. Compare measured bi-directional splice loss with specified splice loss of 0.1 decibels per splice.
 - d. Compare measured connector loss with specified connector loss of 0.3 decibels.
 - e. Verify cable length.
 2. OTDR Requirements:
 - a. Calibrated in accordance with TIA-455-226 and TIA-455-231.
 - b. Fiber Type: Single-mode.
 - c. Dynamic Range: 32 dB or better.
 - d. Resolution: 0.01 dB or better.
 - e. Wavelength Range: 1310 to 1650 nm in 1 nm steps or better.
 - f. Calibrated Wavelengths: 1310 and 1550 nm at a minimum.
 - g. Measurement Range: Plus 5 to minus 50 dBm.
 - h. Bare fiber adapter.
 - i. Reflectance Accuracy: ± 2 decibels.
 3. Obtain an OTDR waveform for each fiber to:
 - a. Determine actual cable length.
 - b. Determine attenuation and splice loss.
 - c. Identify and locate splices and step discontinuities and possible fiber breaks.
- F. Prepare test reports:
1. PDF copy and paper copy of each trace made during testing for inclusion in report.
 2. OTDR waveform for each fiber showing "A" and "B" range marks: Do not move or change location of either mark.
 3. Include an Analysis Summary Results Table for each fiber:
 - a. Indicate numerical values associated with each event.
 - b. Do not use question marks, dashes, or blanks.
 4. Include following information for each trace:
 - a. Trace name.

- b. Date of test.
 - c. Name or names of Contractor's personnel who performed test.
 - d. Name of Government inspector who witnessed test.
 - e. Fiber type being tested.
 - f. Bundle color.
 - g. Fiber color.
 - h. Fiber number.
 - i. Launch reel length.
 - j. OTDR resolution.
 - k. Pulse width.
 - l. Wavelength.
 - m. Index of refraction.
 - n. Averaging number.
 - o. Receiver bandwidth.
 - p. Sampling point count.
 - q. End to end loss, decibels.
 - r. Reflectance, decibels.
 - s. Fiber break, decibels.
 - t. Backscatter.
5. Traces shall be viewable and reproducible on a computer running a Microsoft Windows operating system.
6. Indicate the following for each test:
- a. The test is a reel test upon arrival to the staging area.
 - b. The test is a reel test prior to installation if it has not been tested in the previous 7 calendar days.
 - c. The test is a reel test after installation but prior to splicing.
 - d. The test is an end-to-end test after installation and splicing are completed.
 - e. Contractor elected reel test.
 - f. Include a unique reel number when reels are tested individually as well as for the most recent reel spliced when performing an end-to-end test.
- G. If any cable or fiber fails testing, remove, and replace cable.
- H. If after installation a cable is determined to be faulty and it is determined that the Contractor failed to perform any testing requirements outlined in these specifications or

in Section 26 05 90 – Wiring Checkout and Tests, the Contractor shall replace that cable in addition to any faulty equipment at cost to the Contractor.

3.04 COMMISSIONING AND VERIFICATION

A. Submit completed, signed, and approved Electrical Commissioning Forms or similar in accordance with the Submittal Table:

1. ECF-27 20 01 – Fiber Optic Cable and Accessories – Installation.

B. Contractor may utilize the ECF verification checklists supplied under specification Section 26 08 00 in accordance with the procedures defined and/or generate their own checklists that are similar.

3.05 FINAL AS-BUILTS

A. Final As-built Drawings Government Format: refer to Specification Section 01 78 30 and Section 01 33 26 – Electrical Drawings and Data.

3.06 O&M INSTRUCTION BOOK

A. Submit Final Operation and Maintenance Instruction Book Manuals as outlined in Specification Section 01 78 30 – Project Record Documents prior to Training. Refer to Section 01 79 20 – Training.

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END OF SECTION

SECTION 27 21 29

NETWORK SWITCHES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include network switches in the lump sum prices offered in the Price Schedule for Fiber Optic Cables and Accessories.
 - a. Include:
 - 1) Industrial Ethernet Switches.
 - 2) Data Collector.

B. Cost:

1. Include cost of spare parts in lump sum price offered in the Price Schedule for Spare Parts.
2. Include cost of training in the lump-sum price offered in the Price Schedule for Training.

1.02 REFERENCE STANDARDS

A. Federal Register

1. EO 13800 Strengthening the Cybersecurity of Federal Networks and Critical Infrastructure

B. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE 802.3-2022 Ethernet
2. IEEE 1588-2021 Precision Clock Synchronization Protocol for Networked Measurement and Control Systems
3. IEEE 3003.2-2014 IEEE Recommended Practice for Equipment Grounding and Bonding in Industrial and Commercial Power Systems

C. National Fire Protection Association (NFPA)

1. NFPA 70-2023 National Electrical Code (NEC)

1.03 DEFINITIONS AND ACRONYMS

A. ACDM: Automated Continuous Diagnostics and Mitigation

- B. DRAM: Dynamic Random Access Memory.
- C. GB: Giga Byte.
- D. HSR: High-Availability Seamless Redundancy.
- E. IGMP: Internet Group Management Protocol.
- F. I/O: Input/Output.
- G. LFM: Linear Feet per Minute.
- H. mA: milliampere.
- I. MAC: Media Access Control
- J. MB: Mega Byte.
- K. MTBF: Mean Time Between Failures.
- L. PLC: Programmable Logic Controller.
- M. SCADA: Supervisory Control and Data Acquisition.
- N. SFP: Small Form-factor Pluggable.
- O. SNMP: Simple Network Management Protocol.
- P. STP: Spanning Tree Protocol.
- Q. USB: Universal Serial Bus.
- R. VDC: Direct Current Voltage.
- S. VLAN: Virtual Local Area Network.

1.04 SUBMITTALS

- A. In accordance with Sections 01 33 00 - Submittals, 01 33 26 - Electrical Drawings and Data, and 01 78 30 - Project Record Documents.
- B. RSN 27 21 29-1, Approval Drawings and Data:
 - 1. Data: switch configurations:
 - a. List of open ports and services.
 - b. IP addresses of networked devices
 - c. Software and firmware versions of switch.

- C. Final As-Built Drawings and Operation and Maintenance (O&M) Instruction Books:
Refer to Section 01 78 30 – Project Record Documents.

1. Includes final switch configuration.

1.05 DESIGN REQUIREMENTS

- A. Industrial Ethernet Switches:

1. Provides ethernet and/or fiber optic connectivity of automation control, annunciation systems, RTU, Turbine level Remote I/O, powerplant entrance HMI, and Gatehouse Remote I/O, to the facility network and PLC internal control network as shown on drawing 863-D-60048.
2. Duplex connectivity.
3. Network devices and protection relays will be entirely isolated from the Internet and external servers.

- B. Ethernet Communications:

1. Integration of limited Zero-Trust Tenets with Ethernet communications will meet secured protection from cyberattacks.

1.06 EXTRA MATERIALS

- A. Spare Parts:

1. Deliver to Flatiron powerplant.
2. All spare parts on site prior to field testing.
3. Furnish in accordance with Table 27 21 29A – Spare Parts.

Table 27 21 29A - Spare Parts

Item	Description	Quantity
1	Managed Network Switch	2 of each type

PART 2 PRODUCTS

2.01 MANAGED INDUSTRIAL ETHERNET SWITCH

- A. Managed Switch, Model 5800, Allen-Bradley product manufactured by Rockwell Automation, 1201 South Second Street, Milwaukee, WI 53204-2496: having the following essential characteristics:

1. Quantity: 2
2. Location:
 - a. Main Control Room
 - b. Gatehouse Communications Room

3. Layer 3 switch.
4. Total Ports: Modular to accept 24 connections.
5. Contain the following installed minimum hardware:
 - a. Alarm relay.
 - b. I/O
 - c. Memory card
6. Din rail mountable.
7. Software License: Perpetual.
8. Managed ethernet switches shall have capability for multiple user access controls with 17 characters minimum passwords.
9. Communication:
 - a. Ethernet/IP.
 - b. Port Types:
 - 1) RJ-45 console connector, 10/100/1000BASE-T Copper.
 - a) 8 minimum PoE ports.
 - 2) Fiber 10G/1G/100Mb.
 - 3) 100/1000 SFP ports.
10. Annunciate trouble alarm to the new Annunciation Systems; Section 26 09 17.
 - a. Alarm Contacts: Trouble alarm.
11. Full SNMP.

2.02 NETWORK TRAFFIC MONITORING SOFTWARE

- A. Per EO 13800, Strengthening the Cybersecurity of Federal networks and Critical Infrastructure:
 1. Continuous Monitoring (Automated Continuous Diagnostics and Mitigation (ACDM): The Water Authority/Government will monitor network traffic to identify and analyze suspicious activity with appropriate actions taken in response that is not reactionary, but rather continuous.
 2. Continuous Vulnerability Assessment.
 3. Risk management Framework:
 - a. Facility developed Roles and Responsibilities.
 - b. Configuration and Patch management.
 - c. On-site Maintenance:
 - 1) Maintenance and support practices must be performed on-site, not remotely.

- 2) Plan for only Water Authority/Government owned computers connected to the new automation control system.
- d. Maintain the cybersecurity practices and procedures.
- e. Uninstall any programs, applications, and services not necessary or prohibited.
- f. Provide system administrators dedicated to managing network cybersecurity features and funding for contract software support.
4. Managing Identity, Access, and Authentication.
5. Detecting Security Events and Incident Response Plan.
- B. Passive Security Monitoring:
 1. Traffic protocols with network conditions and anomalous conditions monitoring.
 2. Employ network traffic analyzers to detect network assets and possible vulnerabilities.
- C. Network Monitoring:
 1. Non-AI (Artificial Intelligence).
 2. Utilize active scanning tools to connect to approved networked assets.

2.03 FIREWALL (FW1)

- A. NSA 2700 SU #02-SSC-7370 as manufactured by Sonic Wall, Inc, 1033 McCarthy Boulevard, Milpitas, CA 95035 having the following essential characteristics:
 1. Location: in the Server Room next to the Main Control Room.
 2. RJ-45 Ports: 4 (minimum), gigabit speed (or faster)
 3. SFP: 1 (minimum)
 4. Configurable firewall rules based on TCP/IP, UDP, ICMP
 5. FIPS 140-2 compliant
 6. 120 V AC power
 7. Rack Mounted – will mount on existing rack.
- B. New firewall between existing main PLC and new control network to establish security zones:
 1. Utilized to support network isolation and as boundary protection devices.
 2. For any reason a firewall blocks traffic that it should not, potentially impacting operations; the operator will depend on configuration and rulesets:
 - a. Detection of blocked traffic:
 - 1) Continuously monitor incoming and outgoing traffic.

- 2) Predefined rulesets and security policies to determine whether traffic should be allowed or blocked.
 - 3) Identify traffic that violates rulesets.
 - a) Commissioning shall involve joint collaboration between Contractor and Water Authority/Government to configure firewall rulesets.
 - b. Operator notification:
 - 1) Generate firewall logs and alarm indication for certain blocked traffic.
 - c. Investigation:
 - 1) System Administrator will investigate blocked traffic and assess for legitimacy or potentially malicious activities.
 - d. Ruleset modification and exceptions capability:
 - 1) System Administrator has ability to modify rules and temporary exceptions that will not inhibit critical day-to-day operations.
- C. Network segmentation.
 - 1. Zero-Trust Gateways:
 - a. Employ Zero-Trust philosophy with firewalls as access checkpoints that segment the network by allowing only verified access.
 - b. System administrators to develop firewall rulesets configured to monitor traffic in line with the facility's needs.
 - c. Implement network segmentation into tiers or demilitarized zones to manage access to sensitive data while also maintaining operational performance.

2.04 TIME SYNCHRONIZATION

- A. Refer to specifications Section 25 00 01 – Automation Control System

2.05 FACTORY ACCEPTANCE TESTING

- A. FAT will be completed in accordance with requirements outlined in Specification Section 25 00 01 – Automation Control System and 26 09 18 – Control Boards.

2.06 NETWORK ASSET CONFIGURATION MANAGEMENT LIST

- A. Refer to specification Section 25 00 01 – Automation Control System.

2.07 SECURITY

- A. Refer to specification Section 25 00 01 – Automation Control System and Section 28 05 11 – Cybersecurity Requirements for Electronic Safety and Security.

PART 3 EXECUTION**3.01 INSTALLATION**

- A. Install in accordance with NFPA 70 and manufacturer's requirements.
- B. Grounding in accordance with IEEE 3003.2, NFPA 70, and Section 26 05 26 - Grounding and Bonding.
- C. Cable and wiring in accordance with Section 26 05 10 - Conductors and Cables.
- D. Optical fiber cabling in accordance with Section 27 20 01 – Fiber Optic Cable and Accessories.

3.02 COMMISSIONING AND VERIFICATION

- A. Commissioning will be completed in accordance with requirements outlined in Specification Section 25 00 01 – Automation Control System and 26 09 18 – Control Boards.

3.03 FINAL AS-BUILTS

- A. Final As-built Drawings: refer to Specification Section 01 78 30 and Section 01 33 26 – Electrical Drawings and Data, and 01 78 30 - Project Record Documents.

END OF SECTION

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SECTION 28 05 11

CYBERSECURITY REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. ¹[{Cybersecurity Requirements for Electronic Safety and Security:
1. Payment: Lump sum price offered in the Schedule.}
- B. {Cost:
1. Include in prices offered in the schedule for Complete Control System.}]

1.02 REFERENCES

- A. National Institute of Standards and Technology (NIST)
1. NIST SP 800-53 - 2020 Security and Privacy Controls for Information Systems
 2. NIST SP 800-40 - 2022 Guide to Enterprise Patch Management Planning: Preventive Maintenance for Technology
 3. NIST SP 800-41 - 2009 Guidelines on Firewalls and Firewalls Policy
 4. NIST SP 800-115 - 2008 Technical Guide to Information Security Testing and Assessment
 5. NIST SP 800-77 - 2020 Guide to IPsec VPNs
 6. NIST SP 800-82 - 2015 Guide to Industrial Control System (ISC) Security
- B. North American Electric Reliability Corporation (NERC)²
1. CIP-004-6 - 2016 Cyber Security - Personnel & Training
 2. CIP-005-7 - 2022 Cyber Security - Electronic Security Perimeter(s)
 3. CIP-006-6 - 2016 Cyber Security - Physical Security of BES Cyber Systems

¹ Select appropriate method of payment. If cost to be included in item other than Complete Control System, revise accordingly.

² Applicable only to power plants. Delete all references to NERC/CIP if not a power plant project

- | | | |
|----|------------------|--|
| 4. | CIP-007-6 - 2016 | Cyber Security - System Security Management |
| 5. | CIP-010-4 - 2022 | Cyber Security - Configuration Change Management and Vulnerability Assessments |
| 6. | CIP-012-1 - 2022 | Cyber Security – Communications between Control Centers |
| 7. | CIP-013-2 - 2022 | Cyber Security - Supply Chain Risk Management |
| 8. | CIP-014-3 - 2022 | Cyber Security - Physical Security |

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 – Submittals, and 01 33 26 – Electrical Drawings and Data.
- B. RSN 28 05 11-1, Preliminary Cybersecurity Documentation
 - 1. Documents listed in paragraph 3.01 – Preliminary Cybersecurity Documentation.
- C. RSN 28 05 11-2, Preliminary Configuration Documentation
 - 1. Documents listed in paragraph 3.03 – Preliminary Configuration Documentation.
- D. RSN 28 05 11-3, Factory Acceptance Test Procedure³
 - 1. Documents listed in paragraph 3.04 A – Factory Acceptance Test Procedure.
- E. RSN 28 05 11-4, Factory Acceptance Test Report⁴
 - 1. Documents listed in paragraph 3.04 D – Factory Acceptance Test Report.
- F. RSN 28 05 11-5, Site Acceptance Test Procedure
 - 1. Documents listed in paragraph 3.06 A – Site Acceptance Test Procedure.
- G. RSN 28 05 11-6, Site Acceptance Test Report
 - 1. Documents listed in paragraph 3.06 D – Site Acceptance Test Report
- H. RSN 28 05 11-7, As-Installed Cybersecurity Documentation
 - 1. Documents listed in paragraph 3.07 – As-Installed Cybersecurity Documentation.
- I. RSN 28 05 11-8, Final Documentation

³ Delete if FAT not needed.

⁴ Ibid.

1. Documents listed in paragraph 3.08 – Final Documentation.

1.04 DEFINITIONS AND ACRONYMS

A. Definitions:

1. PII: Personally Identifiable Information
2. Services:
3. Wireless Technologies: Synonyms with Wireless Devices. Technology and associated protocols that allow analog and digital communication without wires, such as radio, microwave, Wi-Fi, Bluetooth, infrared, and ZigBee.

B. Acronyms:

1. ACL: Access Control List
2. ACIO: Associate Chief Information Officer
3. AV: Antivirus
4. CAPEC: Common Attack Pattern Enumeration and Classification
5. DMZ: demilitarized zone
6. FAT: Factory Acceptance Test
7. ICS: Industrial Control Systems
8. IRO: Information Resources Office
9. MAC address: Media access control address
10. OS: Operating System
11. PLC Programmable Logic Controller
12. RTU: Remote Telemetry Unit
13. SAT: Site Acceptance Test
14. SCAP: Security Content Automation Protocol
15. SCC: SCAP Compliance Checker
16. SPAN: Switch Port Analyzer
17. STIG: Security Technical Implementation Guide
18. TCP/IP: Transmission Control Protocol/Internet Protocol
19. VLAN: Virtual Local Area Network
20. VPN: Virtual Private Network

1.05 CYBERSECURITY ADMINISTRATIVE REQUIREMENTS

A. Access Agreements:

1. Individuals requiring access to Government information or information systems are required to sign appropriate access agreements prior to being granted access.
 2. Access requests for Government information or information systems will be evaluated on a case-by-case basis, and approval is not guaranteed.
 3. Access agreements must be re-signed to maintain access when access agreements have expired or been update.
 4. Only approved and trained users shall be allowed access to the information systems in accordance with CIP-004-6.
- B. Personally Identifiable Information (PII): Sensitive PII shall not be stored on the information system.

1.06 INFORMATION SYSTEM SECURITY REQUIREMENTS

- A. Configure system to meet NIST and NERC/CIP standards referenced in paragraph 1.02. Where requirements conflict, the more stringent requirement applies.
- B. Access Control: Provide system administration mechanism for changing user role (e.g., group) associations.
- C. Account Management:
1. Provide centralized and local account management capability.
 2. Implement currently approved Department of the Interior (DoI) password policy.
 3. Passwords shall not be stored in clear text or hardcoded into software or scripts.
 4. Multiple concurrent logins using the same authentication credentials, retaining login information between sessions, auto-fill during login, or anonymous logins shall not be allowed unless specifically requested by the government.
 5. Provide account-based and group-based configurable session logout and timeout settings.
- D. Logging and Auditing:
1. Currently approved government audit logging policy shall be implemented.
 2. The logging and analysis tool shall:
 - a. Automatically collect and store log data.
 - b. Ability to search and filter log data using specific criteria.
 - c. Method for alerts and notifications to be displayed.
 - d. Store log data for no less than 90 days.

3. The following events shall be logged:⁵
 - a. Successful and unsuccessful authentication and access attempts
 - b. Account management changes
 - c. Administrative and privileged account use
 - d. Application start-up and shutdown
 - e. Application failures
 - f. Major application configuration changes
 4. Provide log file confidentiality and integrity security protection.
 5. The government shall be allowed to configure the audit logging capability to support future security auditing requirements.
 6. Provide standard time synchronization in the product or an authoritative time source and configure all system components to synchronize to the authoritative time source.
- E. Applications, Software and Services:
1. Software must be vendor supported and not end-of-life for at least five years from delivery to government.
 2. Software not required for operation, maintenance or support of the ⁶[{Industrial Control System (ICS)}, {Information System}, etc.] shall not be installed on the system.
 3. Open-source software requires government approval.
 4. Freeware and shareware software is not authorized.
 5. Operating Systems and applications shall not require Internet access for activation and updates.
 6. System components shall never connect to the Internet or external networks. Including during initial provisioning.
- F. Antivirus, and Malware Detection and Protection:
1. Host-based malware detection capability shall be installed on all systems (including servers, workstations, desktops) where technically feasible.
 2. Capabilities shall include:
 - a. Scan for malicious code
 - b. Malware Detection

⁵ Edit to fit specific project requirements

⁶ Insert appropriate name of project system

- c. Scan in real-time
 - d. Set scheduled scans
 - e. Alerting
 - f. Quarantine, not automatically delete, suspected infected files
3. Provide capability to automatically scan any removable media.

G. Network Communication:

- 1. Network segmentation and boundary protection shall be implemented in accordance with CIP-005-6.
- 2. Connection to the Reclamation enterprise network, DoI network, other government networks, or internet is not allowed unless approved by the Regional Director, Senior Advisor of ICS Cybersecurity and Associate Chief Information Officer (ACIO).
- 3. A network monitoring mechanism is required at security zone ingress and egress points. Capabilities include:
 - a. Customizable filtering and monitoring rules
 - b. Rule-based alarms or alerts
 - c. Traffic restriction
- 4. Detect and alert on known or suspected malicious communications for both inbound and outbound network traffic.
- 5. Allow unique routable network spaces used within the government's network. Address spaces other than 192.168.0.0/16, 172.16.0.0/12, and 10.0.0.0/8 shall be supported.
- 6. Remote access to the system and/or network is not allowed unless approved by the Regional Director, Senior Advisor of ICS Cybersecurity and ACIO. Remote access is defined as access to an organizational system by a user (or a process acting on behalf of a user) communicating through an external network.
- 7. Restrict communication traffic between different network security zones.

H. Network Components⁷:

- 1. Network components shall be managed. Include ability to change hardware and software configurations, such as addressing schemes.
- 2. Network switches shall have the capability to configure, manage, and monitor network traffic.
- 3. Unused ports and protocols must be disabled.

⁷ Edit or delete if network components such as switches and routers are not being provided as part of project.

4. Network components shall never connect to the Internet or external networks. Including during initial provisioning.
 5. Network components must be vendor supported and not end-of-life for at least five years from delivery to government.
- I. Virtual Private Network (VPN):
1. [{Provide}, {Use existing}]⁸ security isolating environment, such as a demilitarized zone (DMZ), outside the control network in which the communications tunneling server resides.
 2. End-to-end encryption shall be FIPS 140-3 compliant.
 3. Credentials for control network access via communication tunneling shall be unique and different from in-network authentication credentials.
 4. All network traffic between control centers and remote sites shall be encrypted.
 5. Communication tunneling components must be vendor supported and not end-of-life for at least five years from delivery to government.
- J. Wireless Technology/Wireless Devices⁹:
1. Wireless device frequencies and frequency bands shall be coordinated through the ACIO, Information Resources Office (IRO).
 2. Wireless devices not required for normal or emergency operation shall be disabled.
- K. Configuration Management:
1. Provide ability to create full and incremental backup files of system component configurations.
 2. Apply government furnished STIGs to system components where technically feasible.
- L. Portable Media: Unauthorized portable storage devices shall not be used on Reclamation systems.
- M. Physical Security: Provide physical security of system components either in locked cabinets or enclosures in accordance with CIP-006-6.

PART 2 PRODUCTS

Not Used

⁸ Select "Use existing" if there is an existing DMZ, etc. to be used, otherwise select "Provide".

⁹ Delete of wireless devices are not being provided as part of project.

PART 3 EXECUTION

3.01 PRELIMINARY CYBERSECURITY DOCUMENTATION

- A. Provide documentation listed below in RSN 28 05 11-1, Preliminary Cybersecurity Documentation.
- B. Asset Management List:
1. Provide as spreadsheet with ability to sort based on column contents.
 2. Hardware device information. List, by row, hardware devices connected to the communication networks. For each device, include the following information in separate columns:
 - a. Device description
 - b. Manufacturer
 - c. Model Number
 - d. Operating system (if applicable)
 - e. Host Name (if applicable)
 - f. Physical Location
 - g. Device description
 - h. ¹⁰Power and frequency requirements
 3. Port information. On a separate spreadsheet tab for each device, list by row electronic access ports required for normal operation, emergency operation, and troubleshooting. List both active and inactive ports. For each port, include the following information in separate columns:
 - a. Port identification
 - b. Enabled or disabled status.
 - c. Communication protocol used on this port.
 - d. If the port is enabled, list the service, node, device, and/or system requiring this communication port to be enabled.
 4. Software and firmware information. On a separate spreadsheet tab for each device, list by row software and firmware assets. For each software and firmware asset, include the following information in separate columns:
 - a. Software/firmware designation
 - b. Software/firmware description
 - c. Version

¹⁰ Applies only to wireless technologies. Delete if no wireless technology/devices provided

- d. Patch version
- 5. System operation services. On a separate spreadsheet tab for each device, list by row services required for normal operation, emergency operation, and troubleshooting. For each service, include the following information in separate columns:
 - a. Service identification
 - b. Service description
 - c. Enabled or disabled status. If enabled, include the following information in separate columns:
 - 1) Network node on which service resides (if applicable)
 - 2) Hardware device running service (if applicable)
 - 3) Service software (if applicable)
 - 4) System component requiring service (if applicable)
- C. Account Management List:
 - 1. Provide as spreadsheet with ability to sort based on column contents.
 - 2. On a separate spreadsheet tab for each asset in the Asset Management List, list by row accounts associated with the asset. Including generic and default accounts. For each account, include the following information in separate columns:
 - a. Account identifier
 - b. Enabled or disabled status. If enabled, include the following information in separate columns:
 - 1) Service requiring account to be enabled (if applicable)
 - 2) Program requiring account to be enabled (if applicable)
 - 3) Tasks requiring account to be enabled (if applicable)
 - 4) Function requiring account to be enabled (if applicable)
- D. Logging and Auditing Documentation:
 - 1. List system log management capabilities
 - 2. Identify the log file format
 - 3. Identify system logs that are enabled by default and all additional logging configured to meet requirements in Section 1.06 D
- E. Network Documentation:
 - 1. Provide System Architecture Diagram. Include, at a minimum:

- a. ¹¹Control system and maintenance components.
- b. Component information, to include device type, (e.g., workstation, server, PLC, etc.), hostname, manufacturer, and model.
2. Provide Network Diagram. Include, at a minimum:
 - a. Devices on the network that communicate via TCP/IP protocol.
 - b. Logical communication paths between devices.
 - c. Communication ports and interfaces for each connection.
 - d. Device host names.
 - e. IP address for each device.
 - f. Virtual Local Area Network (VLAN) segregation of devices.
 - g. Remote access entry pathways (if approved).
3. Provide Data Flow Diagram. Include, at a minimum:
 - a. Normal system communications between devices on the network. Indicate direction of communication.
 - b. Ports and protocols used.
 - c. Device information, to include type, (e.g., workstation, server, PLC, etc.), hostname, manufacturer, and model.

3.02 SYSTEM CONFIGURATION

- A. Access Control: Configure each component to operate using the principle of least privilege.
- B. Account Management and Authentication:
 1. Change default accounts and passwords to Government specified settings. Changed account information shall not be published.
 2. Implement currently approved DoI inactive session logout and unsuccessful logon attempt policy.
- C. Logging and Auditing:
 1. Collect and store security log files. Provide means to transfer or forward log files.
 2. Log data for each event shall include, at a minimum:
 - a. Time stamp
 - b. Source address

¹¹ Edit description for project specific system, e.g., control system, information system, communication system, etc.

- c. Destination address
 - d. User/process identifier
 - e. Event description
 - f. Filenames involved
 3. Configure system components to synchronize with an authoritative time source.
- D. Applications, Software and Services:
1. Disable services and ports not required for normal operation, emergency operation, or troubleshooting.
 2. Update firmware and software to latest versions in accordance with CIP-007-6.
 3. Implement Government provided System Use Notification banner.
- E. Malware Detection and Protection: Verify that cybersecurity services (e.g., virus checking and malware detection) do not interfere with other services running on the system.
- F. Licensing:
1. Provide licenses required to meet the content of this specification.
 2. Provide licenses to cover applicable components of the system.
 3. Provide licenses that are valid: no less than one year from the date of system delivery to the government.
 4. Provide licenses that shall not require Internet access for activation and updates.
 5. Transfer ownership of all licenses to the Government upon system delivery.
- G. Network Communications:
1. Restrict communication traffic between different network security zones. Provide disconnection points between network security zones. Provide method to isolate zones to all limited, continued operation.
 2. Firewalls:¹² Configure initial firewall rule sets to “deny all.” Set firewall rule exceptions as approved by the Government.
 3. Allow Government access, including administrative as required, to network components [and ¹³firewalls].
 4. If remote access is approved, ensure that remote access entry paths can be enabled and disabled by the Government as needed.

¹² Delete if firewalls are not used.

¹³ Ibid.

H. Network Components¹⁴:

1. Configure network switches to manage and monitor network traffic.
2. Secure network configuration management interface.
3. Apply Security Technical Implementations Guides (STIGs) as furnished by the Government.

3.03 PRELIMINARY CONFIGURATION DOCUMENTATION

A. Provide documentation listed below in RSN 28 05 11-2, Preliminary Configuration Documentation.

B. Applications, Software and Services:

1. List and track software and software licenses
2. List all disabled ports, connectors, and interfaces
3. Document software and firmware version updates
4. List known methods of bypassing system authentication, e.g., backdoors.

C. Network Communications:

1. Document that disconnection points are established between network security zones.
2. Document remote access entry paths (if approved).

D. Asset Configuration Documentation¹⁵:

1. For all firewalls listed in the Asset Management List, document, at a minimum:
 - a. Host Name (where applicable).
 - b. All firewall rulesets used for normal and emergency operations.
 - c. All enabled logging capabilities including log format.
 - d. Configured password policy settings.
 - e. Evidence that government-provided STIGs were applied.
2. For all routers and switches listed in the Asset Management List, document, at a minimum:
 - a. Host Name (where applicable).
 - b. All enabled logging capabilities including log format.
 - c. Evidence that all network management interfaces are secured.

¹⁴ Edit or delete if network components such as switches and routers are not being provided as part of project.

¹⁵ Edit as needed to cover components provided as part project scope.

- d. Configured password policy settings.
 - e. Evidence that government-provided STIGs were applied.
 3. For all hosts listed in the Asset Management List, document, at a minimum:
 - a. Host Name (where applicable).
 - b. All enabled logging capabilities including log format.
 - c. Configured password policy settings.
 - d. Evidence that government-provided STIGs were applied.
 4. Uniquely name the documents so that contents correspond to, and can be identified, using the Asset Management List.
- E. STIG Exception Document: For each device, list the following:
 1. STIGs applied.
 2. STIG configuration items that do not match the Government-provided STIG settings.
 3. Detailed technical justification risk mitigation recommendations for STIG settings not applied.

3.04 FACTORY ACCEPTANCE TEST¹⁶

- A. Factory Acceptance Test Procedure:
 1. Provide documentation listed below in RSN 28 05 11-3, Factory Acceptance Test Procedure.
 2. Describe procedures for conducting testing requirements identified in subparagraph 3.04 C – Factory Acceptance Testing.
 3. Describe methods and tools used to gather required information identified in subparagraph 3.04 C – Factory Acceptance Testing.
 4. Include procedure to validate and document software development process and/or code review.
 5. Include copy of network diagram listed in subparagraph 3.01 E – Network Documentation. Show test equipment connection locations, and Switch Port Analyzer (SPAN) port and network tap traffic capture points.
- B. Factory Acceptance Test Notification: Notify COR in writing at least 60 days prior to beginning factory acceptance test. Government reserves right to witness test.
- C. Factory Acceptance Testing:

¹⁶ Delete if FAT not required.

1. Verify that components are configured to operate using the principle of least privilege.
2. Verify that account and configuration management requirements have been met.
3. Verify that default usernames and passwords are changed, disabled, or removed.
4. Verify that audit logging is configured as required.
5. Verify that software, services, and ports not required for normal system operation or maintenance are removed or disabled.
6. Verify that validated software and hardware updates and patches are installed and tested.
7. Verify that software is vendor supported and not end-of-life.
8. Verify that system baseline has been documented, to include, at a minimum, the following:
 - a. Hardware
 - b. Software
 - c. Protocols, ports, and services
9. Use tools such as Nipper Studio, Nmap, Nessus, or any other equivalent security auditing software tool to demonstrate that network meets encryption and security requirements.
10. Verify that vulnerability scans have been conducted and findings have been remediated.
 - a. Cybersecurity components shall be free of vulnerabilities.
 - b. Document false positive vulnerability scan results.
11. Capture network traffic from a configured SPAN port or network tap to demonstrate network communications among cybersecurity components. Captured traffic shall be representative of normal operations and be captured for no less than 4 hours.
 - a. List hosts identified in the traffic capture.
 - b. List conversations identified in the traffic capture. Include source address, destination address, port numbers and protocol names.
12. Verify that STIGs furnished by the Government have been applied. Provide SCC (SCAP Compliance Checker) tool results showing STIG settings.

D. Factory Acceptance Test Report:

1. Provide documentation listed below in RSN 28 05 11-4, Factory Acceptance Test Report.

2. Document verifications and results of factory acceptance tests conducted as required in subparagraph 3.04 C – Factory Acceptance Testing. Include results from any other tests performed during the Factory Acceptance Test.
3. Provide updated Asset Management List. Highlight changes from approved Asset Management List.
4. Provide updated Asset Configuration Files.
 - a. Highlight changes from the approved Asset configuration files.
 - b. Uniquely name files so contents can be identified using the updated Asset Management List.
5. Provide updated network diagram. Identify changes from the approved network diagram provided in RSN 28 05 11-3, Factory Acceptance Test Procedure
6. Provide network traffic capture files for all locations where captures took place. Uniquely name the files so that the file can be identified using the updated network diagram.
7. Provide an updated STIG Exception Document. Highlight changes from the approved STIG Exception Document.

3.05 SITE ACCEPTANCE TEST

- A. Site Acceptance Test Procedure:
 1. Provide documentation listed below in RSN 28 05 11-5, Site Acceptance Test Procedure.
 2. Describe procedures for conducting testing requirements identified in subparagraph 3.06 C – Site Acceptance Testing.
 3. Describe methods and tools used to gather required information identified in subparagraph 3.06 C – Site Acceptance Testing.
 4. Include copy of network diagram listed in subparagraph 3.01 E – Network Documentation. Show test equipment connection locations, and SPAN port and network tap traffic capture points.
- B. Site Acceptance Test Notification: Notify COR in writing at least 60 days prior to beginning site acceptance test. Government reserves right to witness test.
- C. Site Acceptance Testing:
 1. Verify that components are configured to operate using the principle of least privilege.
 2. Verify that account and configuration management requirements have been met.
 3. Verify that default accounts, usernames, passwords, security settings, security codes, and other access methods are changed, disabled, or removed.
 4. Verify that audit logging is configured as required.

5. Verify that software, services, and ports not required for normal system operation or maintenance are removed or disabled.
6. Verify that validated software and firmware updates and patches are installed and tested.
7. Verify software is vendor supported and not end-of-life.
8. Verify that testing procedures include validation and documentation of the software development process and/or code review.
9. Verify that the system baseline has been documented to include the following.
 - a. Hardware
 - b. Software
 - c. Protocols, ports, and services,
10. Use tools such as Nipper Studio, Nmap, Nessus, or any other equivalent security auditing software tool to demonstrate that network meets encryption and security requirements.
11. Verify that vulnerability scans have been conducted and findings have been remediated.
 - a. Cyber components shall be free of vulnerabilities.
 - b. Document false positive vulnerability scan results.
12. Capture network traffic from a configured SPAN port or network tap to demonstrate network communications among cyber components. Captured traffic shall be representative of normal operations and be captured for no less than 4 hours.
 - a. List hosts identified in the traffic capture.
 - b. List conversations identified in the traffic capture. Include source address, destination address, port numbers and protocol names.
13. Verify that STIGs furnished by the Government have been applied. Provide SCC tool results showing STIG settings.
14. Demonstrate that the communication network meets encryption and security requirements.

D. Site Acceptance Test Report:

1. Provide documentation listed below in RSN 28 05 11-6, Site Acceptance Test Report.
2. Document verifications and results of site acceptance tests conducted as required in subparagraph 3.04 C – Site Acceptance Testing. Include results from any other tests performed during the Site Acceptance Test.
3. Provide an updated Asset Management List. Highlight any changes from the approved Asset Management List

4. Provide updated Asset Configuration Files.
 - a. Highlight changes from the approved Asset configuration files.
 - b. Uniquely name files so contents can be identified using the updated Asset Management List.
5. Provide updated Account Management List. Highlight any changes from the approved Account Management List
6. Provide updated network diagram. Identify changes from the approved network diagram provided in RSN 28 05 11-5, Site Acceptance Test Procedure.
7. Provide network traffic capture files for all locations where captures took place. Uniquely name the files so that the file can be identified using the updated network diagram.
8. Provide an updated STIG Exception Document. Highlight any changes from the approved STIG Exception Document

3.06 AS-INSTALLED CYBERSECURITY DOCUMENTATION

- A. Provide documentation listed below in RSN 28 05 11-7, As-Installed Cybersecurity Documentation
- B. Provide the following documentation:
 1. Final Hardware/Software/Firmware inventory list including software and firmware versions and patch versions.
 2. Summary of the security features and instructions on product maintenance, support, and reconfiguration of default settings.
 3. All user and group accounts for each component.
 4. All ports, protocols, services, connectors, and interfaces required for normal operation.
 5. All disabled ports, protocols, services, connectors, and interfaces.
 6. All communications (e.g., protocols) required between network security zones, whether inbound or outbound, and identify each network component initiating communication.
 7. All methods or equipment used to restrict communication traffic.
 8. Disconnection points between the network security zones and the methods to isolate the zones to continue limited operations.
 9. All firewall rule sets and/or access control lists for normal and emergency operations. (The basis of the firewall rule sets shall be “deny all,” with exceptions explicitly identified by the Contractor)
 10. All remote access entry pathways and instructions for enabling and disabling them.

11. A complete Network Diagram that shows all the devices on the control network that communicate via TCP/IP protocol and/or via serial connections. Any VLAN segregation of the devices must be indicated on the diagram.
12. Provide a Data Flow Diagram that indicates the normal system communications between devices on the network, including the ports and protocols utilized for communications.

C. Asset management list

1. Include documentation for all modifications made following the approval of RSN 28 05 11-4, Factory Acceptance Test Report.
2. Provide a spreadsheet with ability to sort based on column contents.
3. Hardware device information. List, by row, hardware devices connected to the communication networks. For each device, include the following information in separate columns:
 - a. Device description
 - b. Serial Number
 - c. Manufacturer
 - d. Model Number
 - e. Firmware revision number (if applicable)
 - f. IP or other communication addresses
 - g. Operating system (if applicable)
 - h. Host Name (if applicable)
 - i. MAC Addresses
 - j. Physical Location
 - k. Device description
 - l. Default usernames and passwords
 - m. Power and frequency requirements (only applicable to wireless technologies)

D. Account management list

- E. Software Patch and Firmware Update Documentation: Document security software patches and firmware updates applied to system during implementation and integration. Document the following:
1. Firmware updates
 2. Software updates
 3. Security patch updates

- F. Network Diagram
 - 1. Final system architecture diagram
 - 2. Final data flow diagram
- G. Asset configuration files
- H. Configuration Backups:
 - 1. Provide configuration backup files for each asset in the Asset Management List contained in RSN 28 05 11-7, As-Installed Cybersecurity Documentation.
 - 2. tested, verify, and demonstrate to the government configuration backups ability to fully restore system to initial configuration.
 - 3. Provide electronic files on encrypted removable media.
 - 4. Provide evidence that backups were tested and verified as functional for restoring the system.
 - 5. Provide technical justification for each asset that does not have an accompanying configuration backup.
- I. STIG Exception Documentation

3.07 FINAL DOCUMENTATION

- A. Provide documentation listed below in RSN 28 05 11-8, Final Documentation.
- B. Licensing: list the following:
 - 1. Licenses required to meet the content of this specification
 - 2. Licenses to cover applicable components of the system
 - 3. Licenses transferred to the government upon system delivery.
- C. Supply Chain Risk Assessment: Complete a Supply Chain Risk Questionnaire in support of a Deep Dive Supply Chain Risk Assessment. Questionnaire to be provided by Government.
- D. Backdoor Attestation
 - 1. Signed PDF
 - 2. Attest that there are no known methods for bypassing computer authentication, often referred to as “backdoors”, in the Contractor-provided system. And that all such backdoors created by the Contractor have been permanently removed.
- E. Wireless Technology Documentation
 - 1. Provide comprehensive documentation for all wireless technologies used in the Contractor-provided system to include, at a minimum:

- a. Document use, capabilities, and limits for the wireless devices.
 - b. Document that the wireless technology and associated devices comply with standard operational and security requirements in accordance with IEEE 802.11.
 - c. Provide summary test data demonstrating that known attacks, as documented in Common Attack Pattern Enumeration and Classification (CAPEC) lists, such as malformed packet injection, man-in-the middle attacks, or denial-of-service attacks do not cause receiving wireless devices to crash, hang, be compromised, or otherwise malfunction.
 - d. Document configuration control options that enable verifying the security level of the devices.
2. Document protocols and other detailed information required for wireless devices to communicate with the control network.
 3. Document the power and frequency requirements of the wireless devices (e.g., VHF, UHF, microwave, and satellite devices meet the frequency requirements of the Radio Frequency Authorization assigned, and Generic Requirements [GR]-63 Network Equipment Building System [NEBS] and GR-1089).

END OF SECTION

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SECTION 51 00 00
INFORMATION AVAILABLE TO OFFERORS

PART 1 GENERAL

1.01 ORDER OF PRECEDENCE

- A. In case of differences between sections in Division 51 – Information Available to Offerors and sections in other divisions of the specifications, the requirements in the other divisions will govern.
- B. Government assumes no responsibility for deductions, interpretations, or conclusions made by Contractor based on information made available by the Government, in accordance with the clause at FAR 52.236-3, Site Investigations and Conditions Affecting the Work.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 51 00 50
ELECTRONIC MEDIA RELEASE

U.S. Bureau of Reclamation
Electronic Media Release

U.S. Bureau of Reclamation, Contract Number/Solicitation Number: _____

This release for electronic media is dated the ____ day of _____ (month), ____ (year), between _____ (RECIPIENT) and U.S. Bureau of Reclamation, for the exchange of electronic media (CD ROMs, DVD, Flash Drives, External Hard Disk Drives, etc.) containing information on the _____ project (hereinafter referred to as the PROJECT) for use by the RECIPIENT.

Therefore, RECIPIENT and U.S. Bureau of Reclamation (Reclamation), agree as follows:

1. All data and intellectual property contained on the electronic files remains the property of U.S. Bureau of Reclamation and may not be copied or distributed to any other party without the written permission of U.S. Bureau of Reclamation.
2. The electronic files provided to RECIPIENT by U.S. Bureau of Reclamation, for the PROJECT may be used or adapted by RECIPIENT for their own use. If RECIPIENT chooses to alter in any way, in whole or in part, the electronic files provided for the PROJECT or any future project(s), RECIPIENT agrees that the use of the altered files shall be without liability or legal exposure to U.S. Bureau of Reclamation.
3. Because information and data provided electronically may be altered, whether inadvertently or otherwise, U.S. Bureau of Reclamation, reserves the right to retain copies of the electronic file(s) and to remove from the electronic files provided to RECIPIENT all identification (such as logo, surveyor's seal, etc.) reflecting the involvement of U.S. Bureau of Reclamation, in their preparation.
4. The electronic files are provided solely as a convenience to RECIPIENT by U.S. Bureau of Reclamation and shall not be considered "Drawings of Record" or "Construction Documents." All documents considered "Drawings of Record" or "Construction Documents" will be hardcopy drawing sheets with signatures or scanned files of the hardcopy drawings with signatures.
5. RECIPIENT is advised to check all electronic media for viruses and malware before loading the files. RECIPIENT is solely responsible for intercepting and disabling viruses, if any, that may be transmitted with the electronic files and hereby agrees to indemnify and hold U.S. Bureau of Reclamation harmless from and against all claims of any type or nature asserted by RECIPIENT or any party as a result of viruses transmitted with the electronic files.
6. Files distributed electronically are subject to data erosion, erasure, and/or alteration, and computer systems and software become obsolete in time. By accepting these electronic files, RECIPIENT acknowledges these risks and agrees to waive all claims against U.S. Bureau of Reclamation, should data erosion, erasure, and/or alteration of these electronic files occur.

7. RECIPIENT agrees to defend, indemnify, and hold U.S. Bureau of Reclamation, harmless from all claims, injuries, damages, losses, expenses, and costs, including attorney's fees, arising out of breach of this agreement, or the modification or reuse of the electronic files.

ACCEPTED FOR RECIPIENT:

RELEASED BY U.S. BUREAU OF RECLAMATION:

By: _____

Title: _____

Date: _____

END OF SECTION

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SECTION 51 02 82

HAZARDOUS MATERIALS SURVEY AND ANALYTICAL RESULTS

PART 1 GENERAL

1.01 SURVEY AND ANALYTICAL RESULTS

- A. Hazardous Materials Survey and Analytical Results are provided.
- B. The following is a summary of the analytical results collected during hazardous materials surveys conducted at Pole Hill Powerplant in 2025.
1. Table 1 and Table 2 are summaries of the analytical results.
 2. Table 3 provides a summary of locations where mercury containing switches and relays were found.
 3. Some materials could not be sampled as the panels were still energized. These are summarized in Table 4.
 4. In Article 1.02 are tables of the complete analytical results. In Article 1.03 are pictures of the features and sample locations. In Article 1.04 are copies of the laboratory analytical reports.
 5. Materials which tested positive for asbestos are non-friable.
- C. Summary of findings:
1. Powerplant, Generator Level, Control Room.
 - a. Control panels CSA 1F, 1R, 2F, 2R, 3F, 5R were to be inspected. During the inspection, suspect wiring insulation was noted on panel 2R and samples were collected. This insulation tested positive for asbestos.
 - b. Panels BSC, BSD, and CSC were inspected; no suspect asbestos materials were noted.
 - c. Control panels DSA3, DSA4, and DSA5 control the station service switchgear (SSS). Samples of electrical equipment were collected from the SSS equipment in panels DSA4 and DSA5 however the wiring and components were the same in the other cabinet. This electrical equipment and wiring were tested for asbestos and some tested positive (see Table 2). The control panels had the original manufacturer coating on them and were therefore not sampled and analyzed.
 - d. The interior wall panels of the control room contain asbestos and have stickers on them denoting this. See Figure 1.
 2. Powerplant, Generator Level, Generator.

- a. Two mercury containing switches were noted mounted on the air housing, under the stairs leading up to the exciter.
3. Powerplant, Turbine Level.
 - a. One mercury containing switch was noted mounted onto the Pelton pressure regulating valve across from the turbine. Two mercury containing cases are still in place; one still contains a mercury ampule and one has been replaced with solid state switches. See Figure 17 and Figure 18.
 - b. Another mercury containing switch was noted in the upper turbine platform. See Figure 19.
4. Powerplant, Pipe Gallery Level: A mercury containing switch was noted on the wall away from the penstock. See Figure 20.
5. Powerplant, SSS Transformers.
 - a. The two SSS transformers, KSA and KSB, had blue "No PCB" stickers on them so the oil they contain was considered to have PCB concentration less than 50 parts per million per 40 CFR Part 761 and therefore not regulated. The electrical wiring and components inside the transformers could not be accessed for observation or sample collection. Given that the transformers were built in the 1950's, it is likely the electrical equipment inside will contain electrical components and wiring containing asbestos.
6. Gate House, Gate House Room.
 - a. 171B3 Forebay Low Water Level Switch: A mercury containing switch was noted attached to the arm. See Figure 29.
 - b. Panel DJA: 480 volt power cabinet. Sample PHSS102 and replicates were collected from breakers and panel blanks. No asbestos was detected.
 - c. Box BJJ, alarm point panel, had five mercury wetted contact relays.
 - d. Box FJA: Sample PHSS101 and replicates were collected from the mounting board and tested positive for asbestos.
 - e. Panel LJA, lighting cabinet, had the same manufacturer and type of breakers as Panel DJA. No asbestos was detected from the samples collected from Panel DJA.
 - f. Panel MJA, penstock gate panel: suspect asbestos materials were noted in this panel: the mounting board and the resistor. No samples could be collected as the equipment was energized.
 - g. No suspect materials were noted in the following panels and cabinets:
 - 1) Level switch 171B1 Forebay low water level switch,
 - 2) Penstock full level switches 171PG1 and 171PG2,
 - 3) Panel BJB, 125 volt direct current (VDC) shunt trip and alarms,
 - 4) Panel BJD, 48 VDC charger,

- 5) Panel BJE, the 48 VDC breaker panel,
 - 6) Panel BJF, 48 VDC battery disconnect,
 - 7) Box BJI, low 48 VDC alarm point,
 - 8) Box DJB, loss of annunciator alarms,
 - 9) Box DJC, no name on the box,
 - 10) Box DTB, no name on the box,
 - 11) Transformer KJA, lighting,
 - 12) Transformer KJB, lighting.
7. Gate House, Battery Room: In this room, there are 23 lead acid batteries for the 48 VDC system that will be removed and recycled.
 8. Gate House, Communications Room: Panel LJB had no suspect materials.

Table 1. October 2025 Summary of Analytical Results.

Sample ID (PHSS)	Powerplant or Gate House	Sample Location	Description	Material	Asbestos
100	Powerplant	CSA2R panel in Control Room	Dark gray	Wire insulation	X
101	Gate House	Gate House room, panel FJA	Black	Mounting board	X
102	Gate House	Gate House room, panel DJA	Black	Circuit breakers and panel blanks	ND

Notes:

X: detected

ND: tested but not detected

Table 2. April 2025 Summary Sampling Results.

Sample ID	Sample location	Description	Material	Asbestos
PHSSS001	Panel DSA4	Gray wire	Wire insulation	X
PHSSS002	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	X
PHSSS003	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	X
PHSSS004	Panel DSA4	Black	Terminal block	ND
PHSSS005	Panel DSA5	Black	Westinghouse, 15 ampere, circuit breaker	ND
PHSSS006	Panel DSA4	Black	Westinghouse, 125 ampere, circuit breaker	X

Notes:

ND: tested but not detected

X: detected

Table 3. Summary of Mercury Containing Switches and Relays.

Paragraph Reference	Figure reference	Powerplant or Gate House	Switch Location	Switch ID/Description
1.01C.2.a	Figure 11, Figure 12	Powerplant	On the side of the generator	163GW3 Start Enable
1.01C.2.a	Figure 11, Figure 12	Powerplant	On the side of the generator	163GW Low Pressure Shutdown
1.01C.3.a	Figure 17, Figure 18	Powerplant	Turbine level	163 P2 Penstock Pressure Low
1.01C.3.b	Figure 19	Powerplant	Upper Turbine Level	AE 163BK
1.01C.4	Figure 20	Powerplant	Pipe Gallery	163GW2
1.01C.6.a	Figure 28, Figure 29	Gate House	Gate House	171FB3 Forebay Starting Level Switch
1.01C.6.c	Figure 27, Figure 35, Figure 36	Gate House	Gate House: BJJ Panel	KA: PGA27DC Penstock gate DC voltage alarm
1.01C.6.c	Figure 27, Figure 35, Figure 36	Gate House	Gate House: BJJ Panel	KC: 133PG4X penstock not open...?
1.01C.6.c	Figure 27, Figure 35, Figure 36	Gate House	Gate House: BJJ Panel	KE: 171FBIX Forebay low water level
1.01C.6.c	Figure 27, Figure 35, Figure 36	Gate House	Gate House: BJJ Panel	KF: 171FB2X Forebay high water level
1.01C.6.c	Figure 27, Figure 35, Figure 36	Gate House	Gate House: BJJ Panel	KG: no additional information.

Table 4. Summary of Panels Not Surveyed for Asbestos.

Paragraph Reference	Figure reference	Powerplant or Gate House	Location	Description
1.01C.5.a		Powerplant	Inside both SS transformers	SSS wiring
1.01C.3.c	Figure 13, Figure 14	Powerplant	Turbine level, Lube oil room	Panel M1C, MCC
1.01C.6.f	Figure 27, Figure 30	Gate House	Gate House, Panel MJA	Mounting board and resistor

1.02 COMPLETE ANALYTICAL RESULTS TABLES

Table 5. October 2025 complete laboratory analytical results for asbestos using EPA Method 600/R-93/116

Sample ID (PHSS)	Powerplant or Gate House	Sample Location	Description	Material	Results	Units
100A	Powerplant	CSA2R panel in Control Room	Dark gray	Wire insulation	40	%
100B	Powerplant	CSA2R panel in Control Room	Dark gray	Wire insulation	40	%
100C	Powerplant	CSA2R panel in Control Room	Dark gray	Wire insulation	40	%
101A	Gate House	Gate House panel FJA	Black	Mounting board	50	%
101B	Gate House	Gate House panel FJA	Black	Mounting board	50	%
101C	Gate House	Gate House panel FJA	Black	Mounting board	50	%
102A	Gate House	Gate House panel DJA	Black	Trumbull circuit breaker No. 2, 100 ampere	ND	%
102B	Gate House	Gate House panel DJA	Black	Trumbull circuit breaker No. 3, 15 ampere	ND	%
102C	Gate House	Gate House panel DJA	Black	Trumbull panel blank for No. 1	ND	%

Notes:

ND: not detected, below method detection limit.

?: percent.

Table 6. April 2025 complete laboratory analytical results for asbestos using EPA Method 600/R-93/116.

Sample ID	Sample location	Description	Material	Results	Units
PHSSS001A	Panel DSA4	Gray	Wire insulation	35	%
PHSSS001B	Panel DSA4	Gray	Wire insulation	35	%
PHSSS001C	Panel DSA4	Gray	Wire insulation	35	%
PHSSS002A	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	25	%
PHSSS002B	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	15	%
PHSSS002C	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	20	%
PHSSS003A	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	20	%
PHSSS003B	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	25	%
PHSSS003C	Panel DSA4	Dark gray	Electrical relay, Allen Bradley	35	%
PHSSS004A	Panel DSA4	Black	Terminal block	ND	%
PHSSS004B	Panel DSA4	Black	Terminal block	ND	%
PHSSS004C	Panel DSA4	Black	Terminal block	ND	%
PHSSS005A	Panel DSA5	Black	Westinghouse, 15 ampere circuit breaker	ND	%
PHSSS005B	Panel DSA5	Black	Westinghouse, 15 ampere circuit breaker	ND	%
PHSSS005C	Panel DSA5	Black	Westinghouse, 15 ampere circuit breaker	ND	%
PHSSS006A	Panel DSA4	Black	Westinghouse, 125 ampere circuit breaker	ND	%
PHSSS006B	Panel DSA4	Black	Westinghouse, 125 ampere circuit breaker	ND	%
PHSSS006C	Panel DSA4	Black	Westinghouse, 125 ampere circuit breaker	2.75	%

Notes:

ND: not detected.

%: percent.

1.03 SAMPLING PHOTOS, OCTOBER 2025

A. Powerplant, Generator Level, Control Room, CSA panel



Figure 1. View of the panel CSA front side. The subpanels are numbered from left to right. CSA1F is on the left.



Figure 2. This is the rear side of the CSA panel. The subpanel to the right is CSA1R and CSA5R is on the left.

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Figure 3. This is a view of inside subpanel CSA1F, top half.

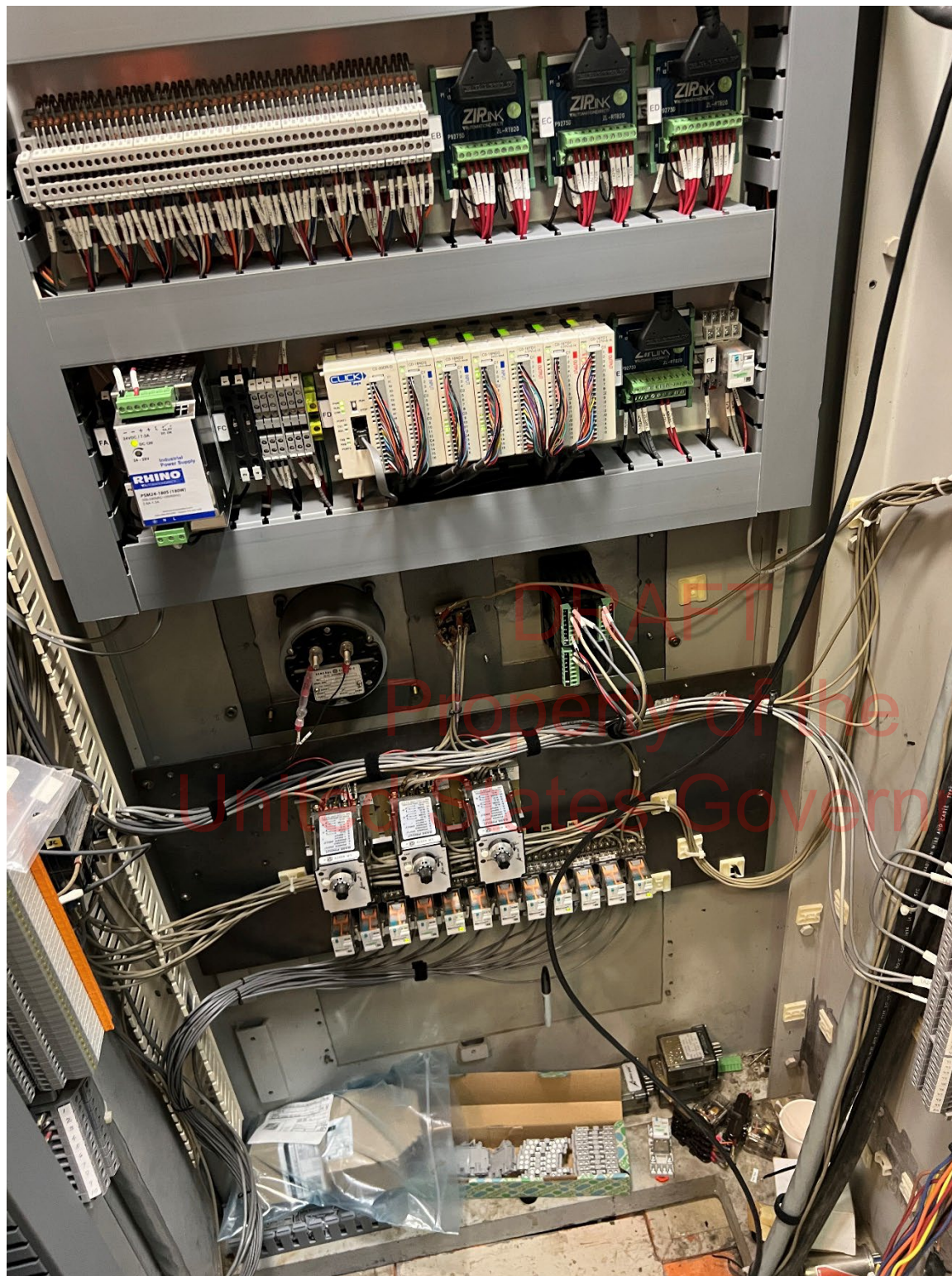


Figure 4. This is a view of subpanel CSA1F, bottom half.



Figure 5. This is a view of inside subpanel CSA1R.

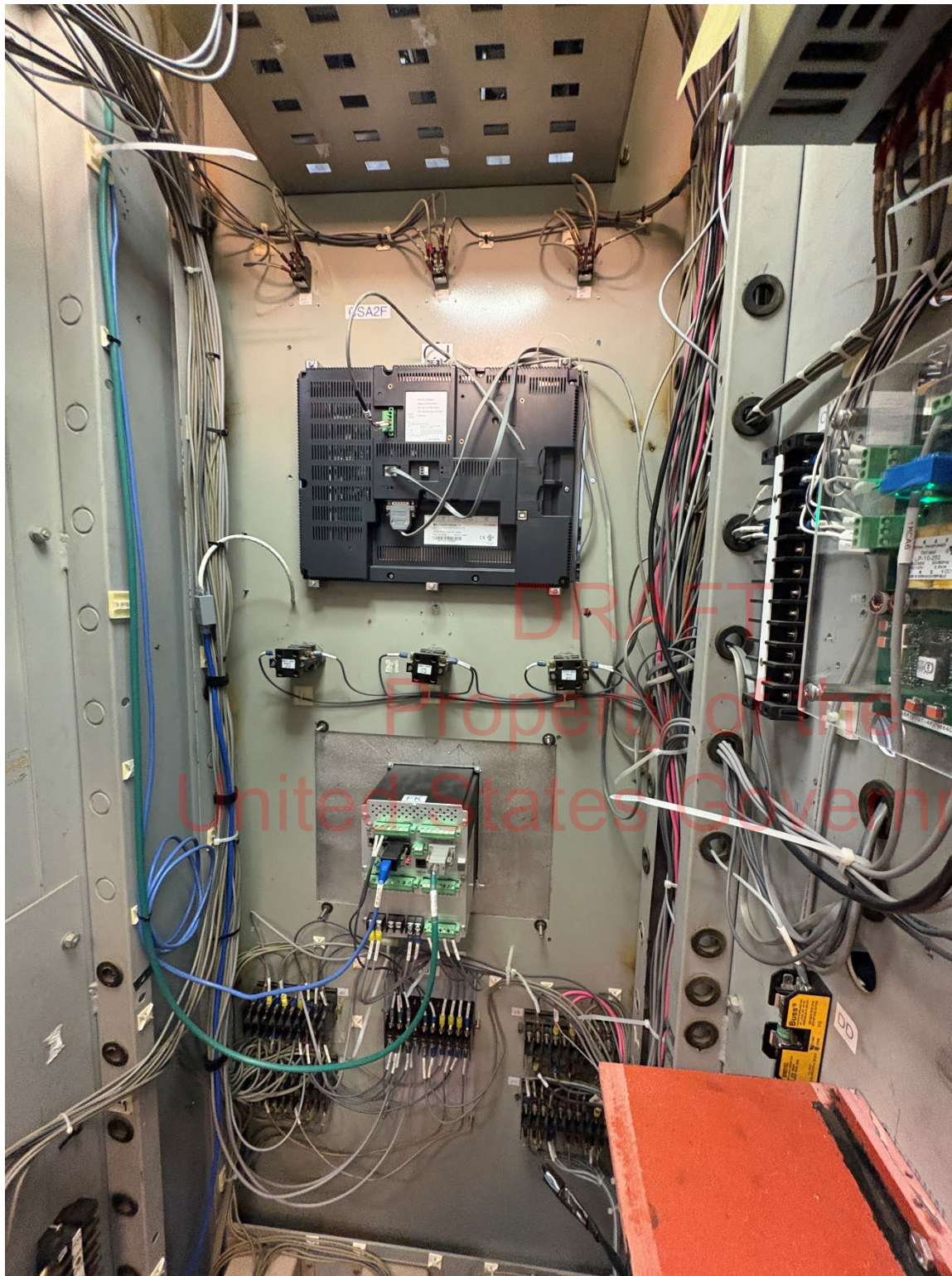


Figure 6. This is a view of inside subpanel CSA2F.

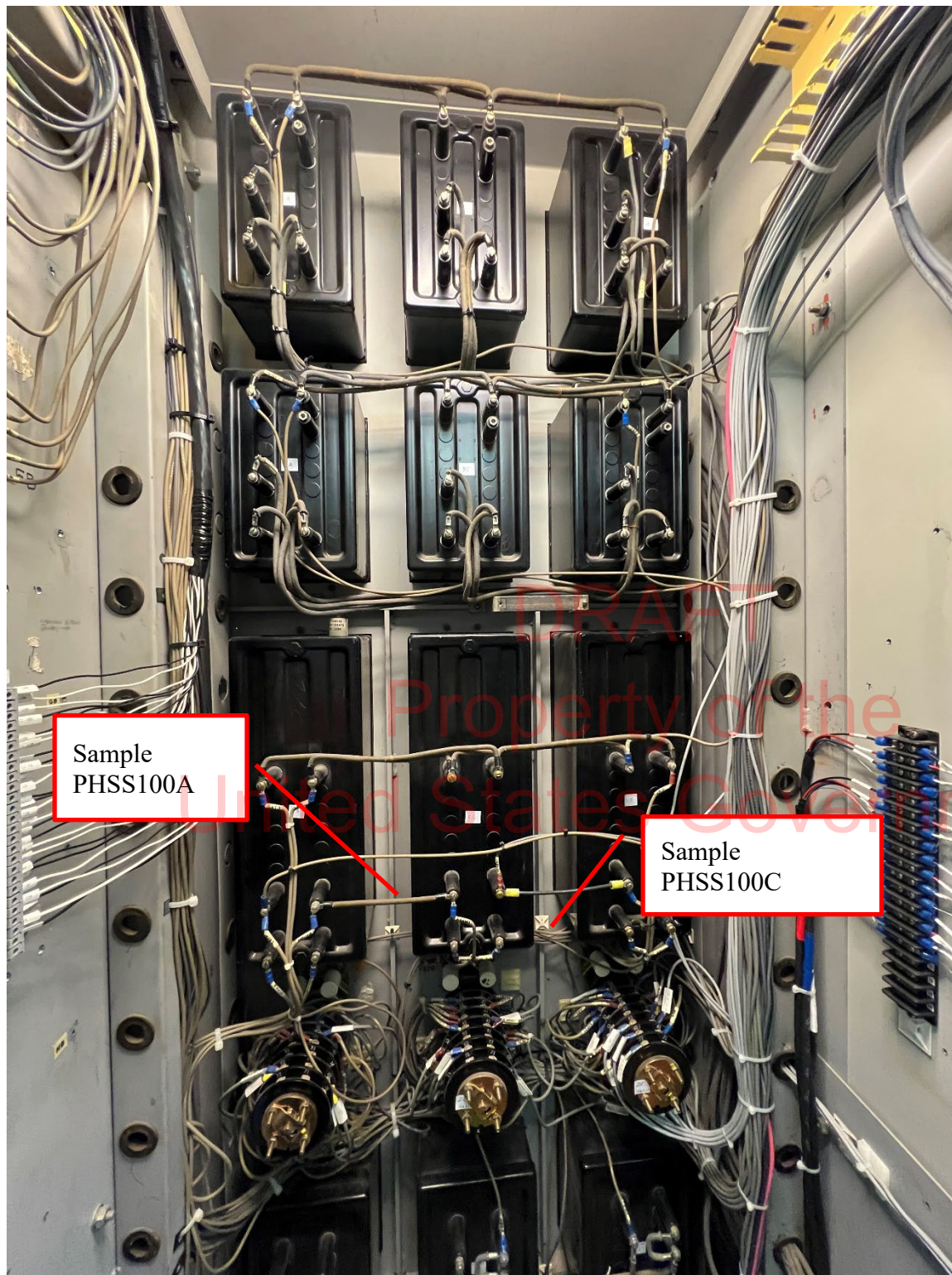


Figure 7. This is a view of panel inside subpanel CSA2R. Note the sample locations of PHSS100A and PHSS100C.



Figure 8. This is where sample PHSS100B was collected from CSA2R.



Figure 9. This is a view inside subpanel CSA3F.

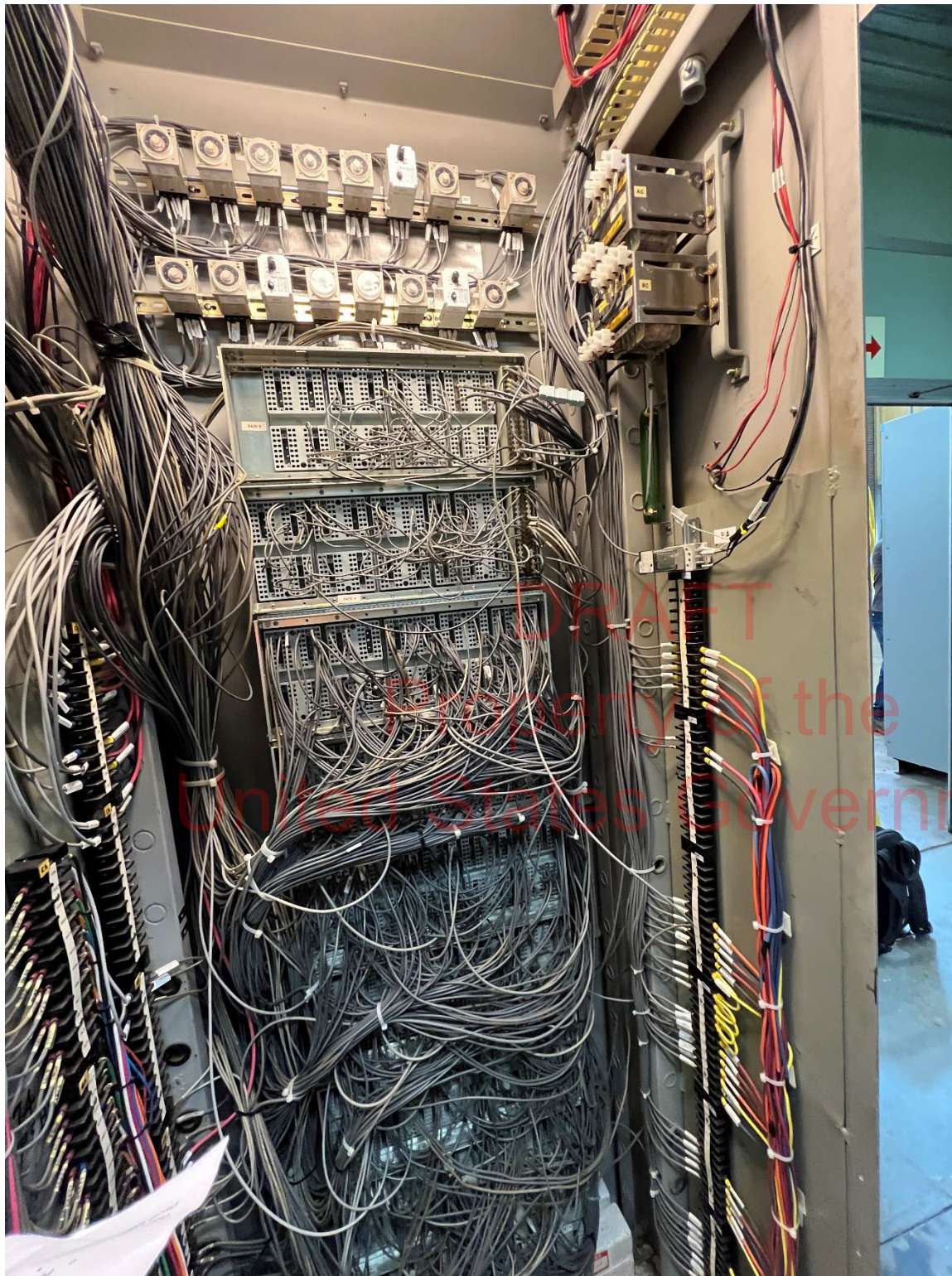


Figure 10. This is a view inside subpanel CSA5R.



Figure 11. This is a view of the generator. The mercury switches are marked with an arrow.

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Figure 12. This is a view of the mercury containing switches on the generator, under the stairs.

B. Turbine Level: Lube Oil Room



Figure 13. This is the M1C panel, the Motor Control Center (MCC).

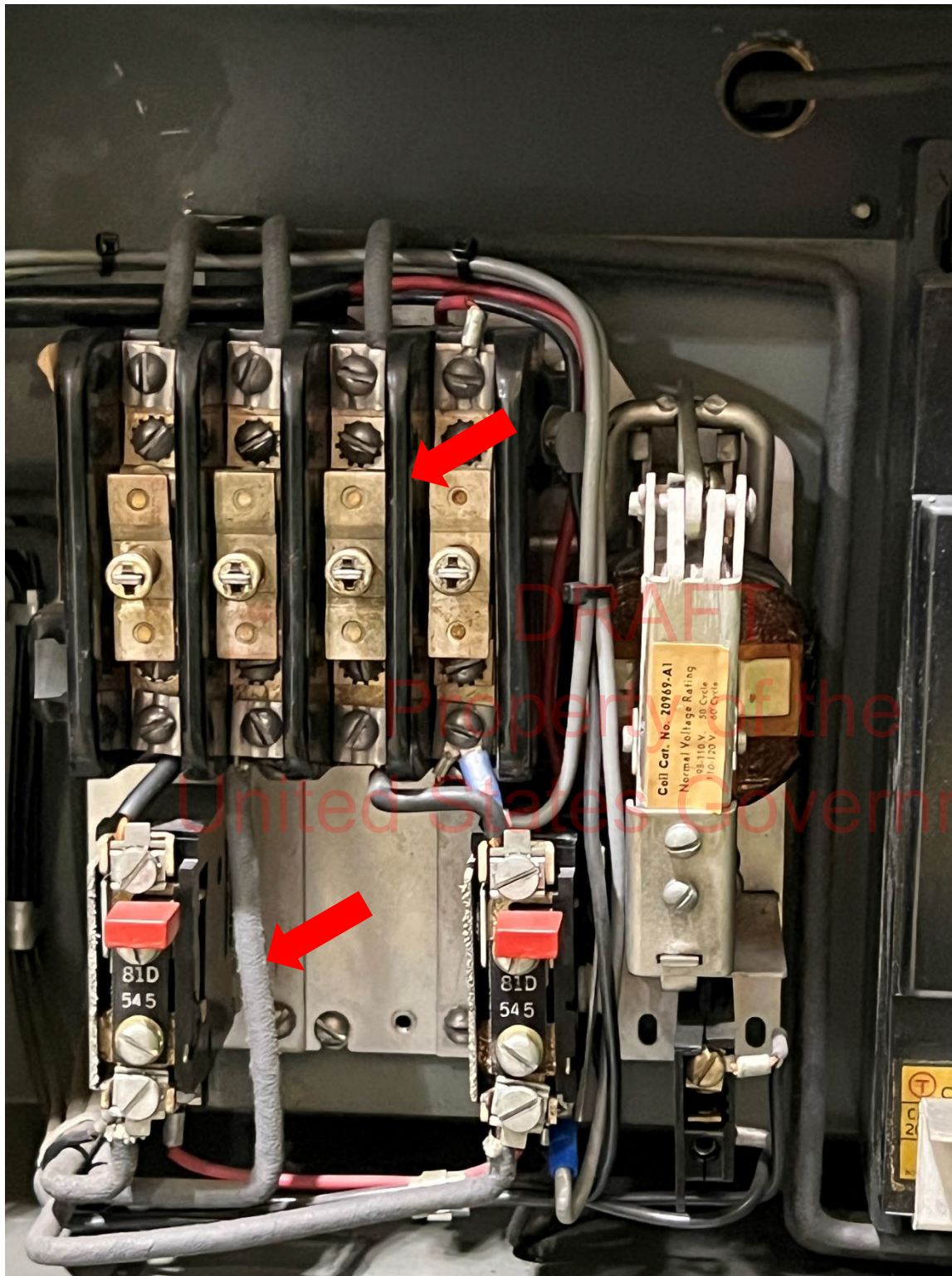


Figure 14. This is a view inside the Lube Oil Transfer Pump panel where there is suspect ACM. Note the arrows. There is more suspect ACM in the other MCC subpanels as well.

C. Turbine Level, Battery Room



Figure 15. This is a view inside the battery room. The four batteries to the right will be removed. Note the arrow.

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D. Turbine Level: Turbine, and Pressure Regulating Valve



Figure 16. This is a view of the turbine and wicket gates. There are pneumatic gauges on the left. Note the arrow pointing to them. They do not contain mercury.

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Figure 17. On the opposite side of the turbine is the pressure regulating valve. There are three mercury switch casings however only one still contains mercury. Two are blocked from view by the stairs. Note the arrows.



Figure 18. This is a view of the two mercury switch casings. The one marked with the arrow still contains mercury.



Figure 19. Upper turbine level with mercury containing switch. Note the arrow.

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E. Powerplant, Pipe Gallery Level.



Figure 20. The penstock is above. The mercury containing switch is marked by an arrow.

F. Powerplant, Station Service Switchgear (SSS) Transformers.



Figure 21. View of the two SSS transformers, located behind the powerplant. One is in front of the other.



Figure 22. Blue "No PCB" sticker on KSA SSS transformer.



Figure 23. SSS bus connection into the powerplant. KSA is to the right and KSB is to the left.



Figure 24. Blue "No PCB" sticker on SSS transformer KSB. Note the arrow.

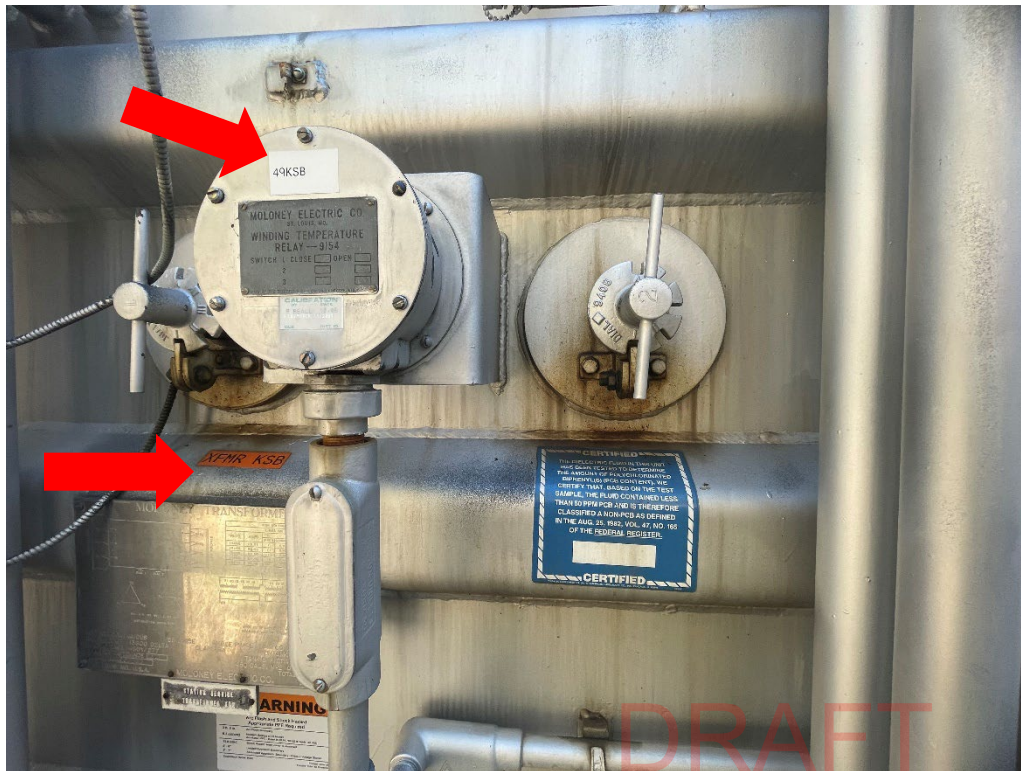


Figure 25. Here is a closer view of the sticker and markings that it is from transformer KSA. Note the arrows.

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G. Gate House.



Figure 26. The Gate House is on the right, and the Communications (Comm) Room is on the left. The Battery Room is behind the Comm Room.

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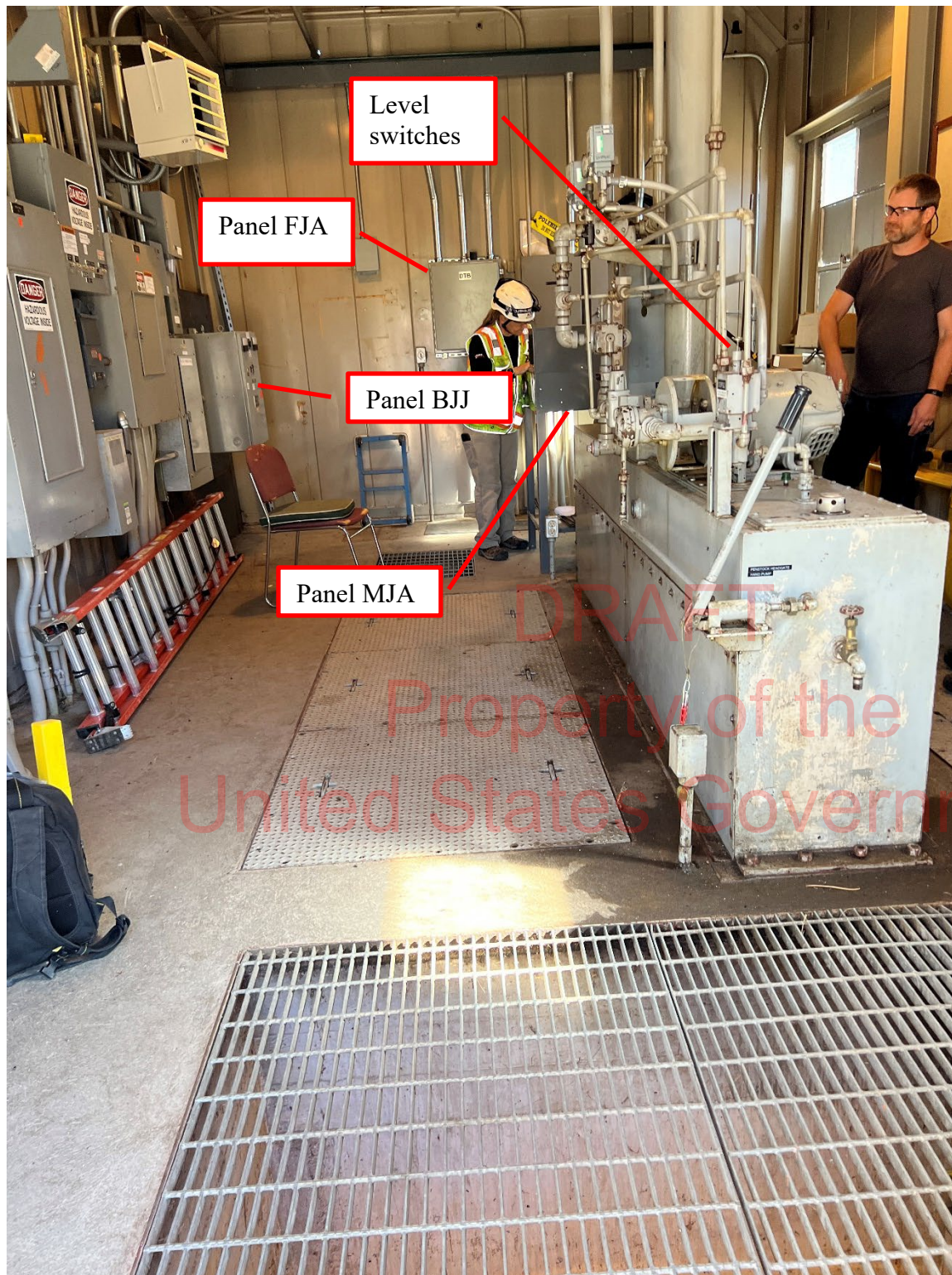


Figure 27. This is a view inside the Gate House. The water level switches are in the far right corner.

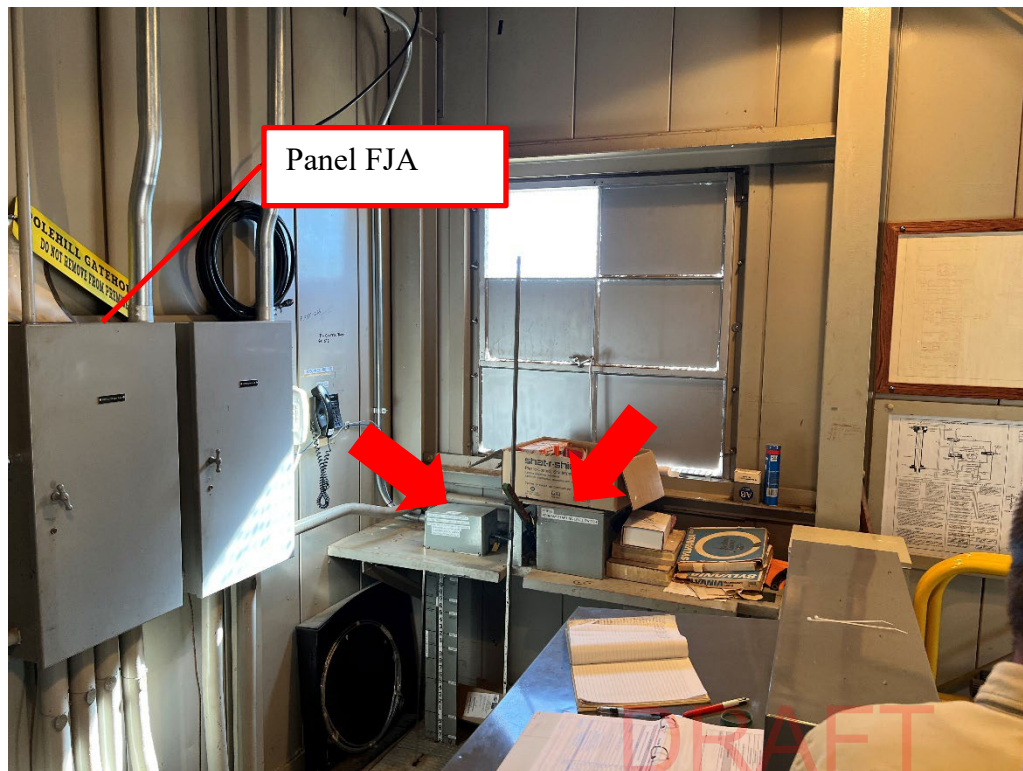


Figure 28. These are forebay water switches. The one on the right has a mercury ampule.

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Figure 29. This is a view of the mercury ampule on the forebay water level switch which was noted in Figure 28. Note the arrow pointing to the ampule containing mercury.

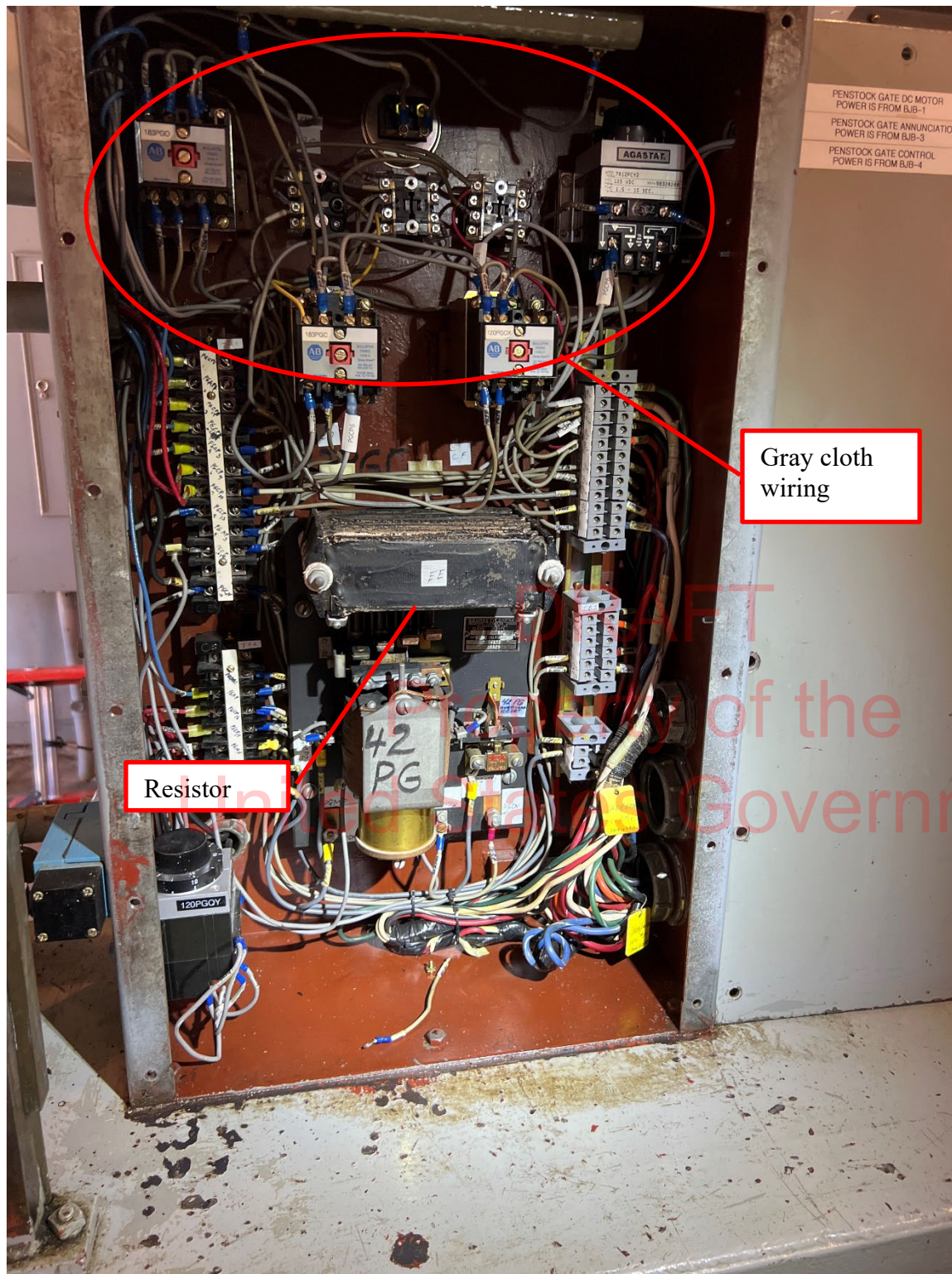


Figure 30. This is a view inside panel MJA, Headgate Control Panel, with suspect ACM (resistor, gray cloth wire insulation).

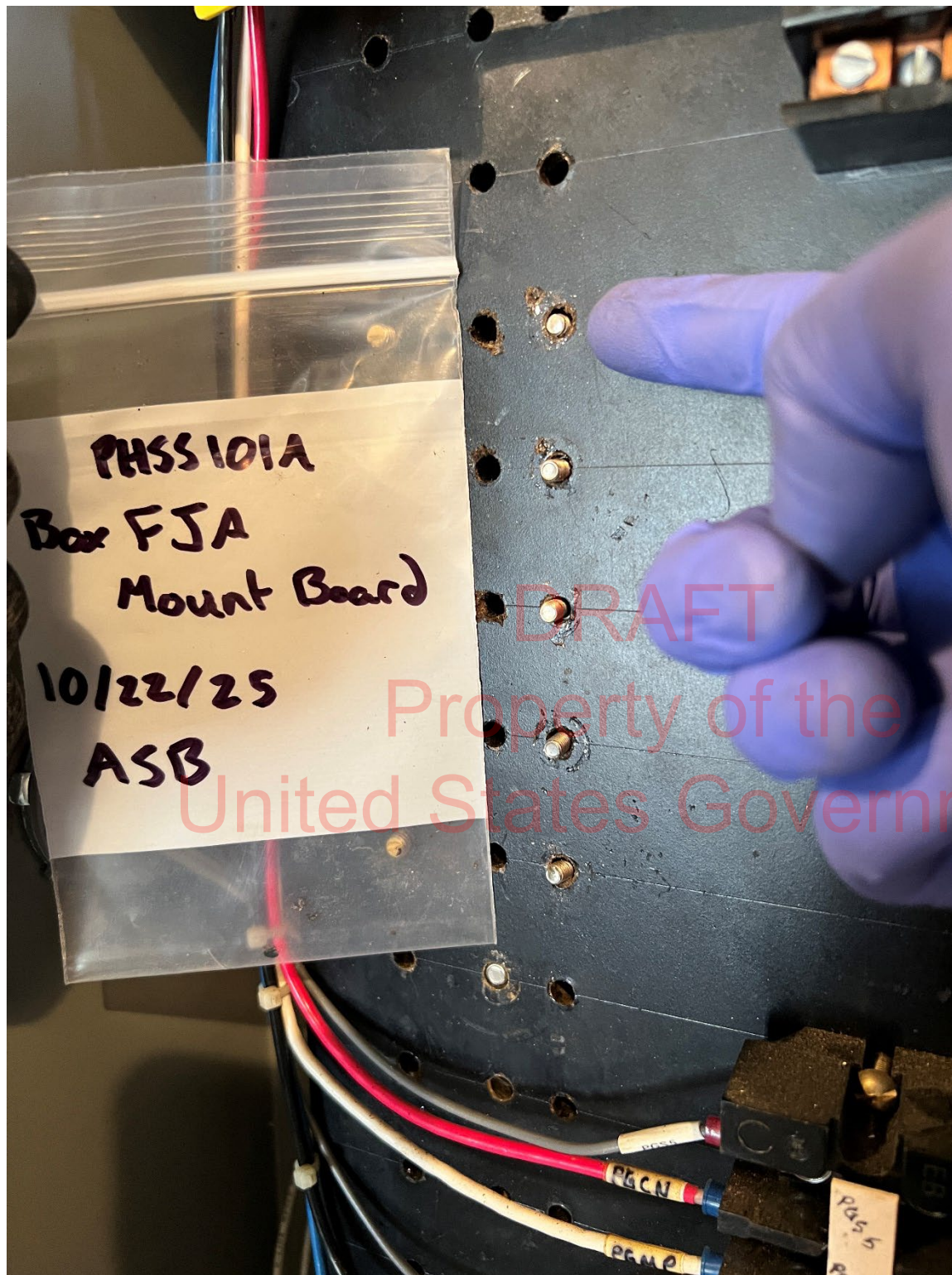


Figure 31. This is a view inside panel FJA and sample PHSS101A.



Figure 32. This is a view of sample PHSS101B from the mounting board inside panel FJA.



Figure 33. This is a view of sample PHSS101C from the mounting board inside panel FJA.

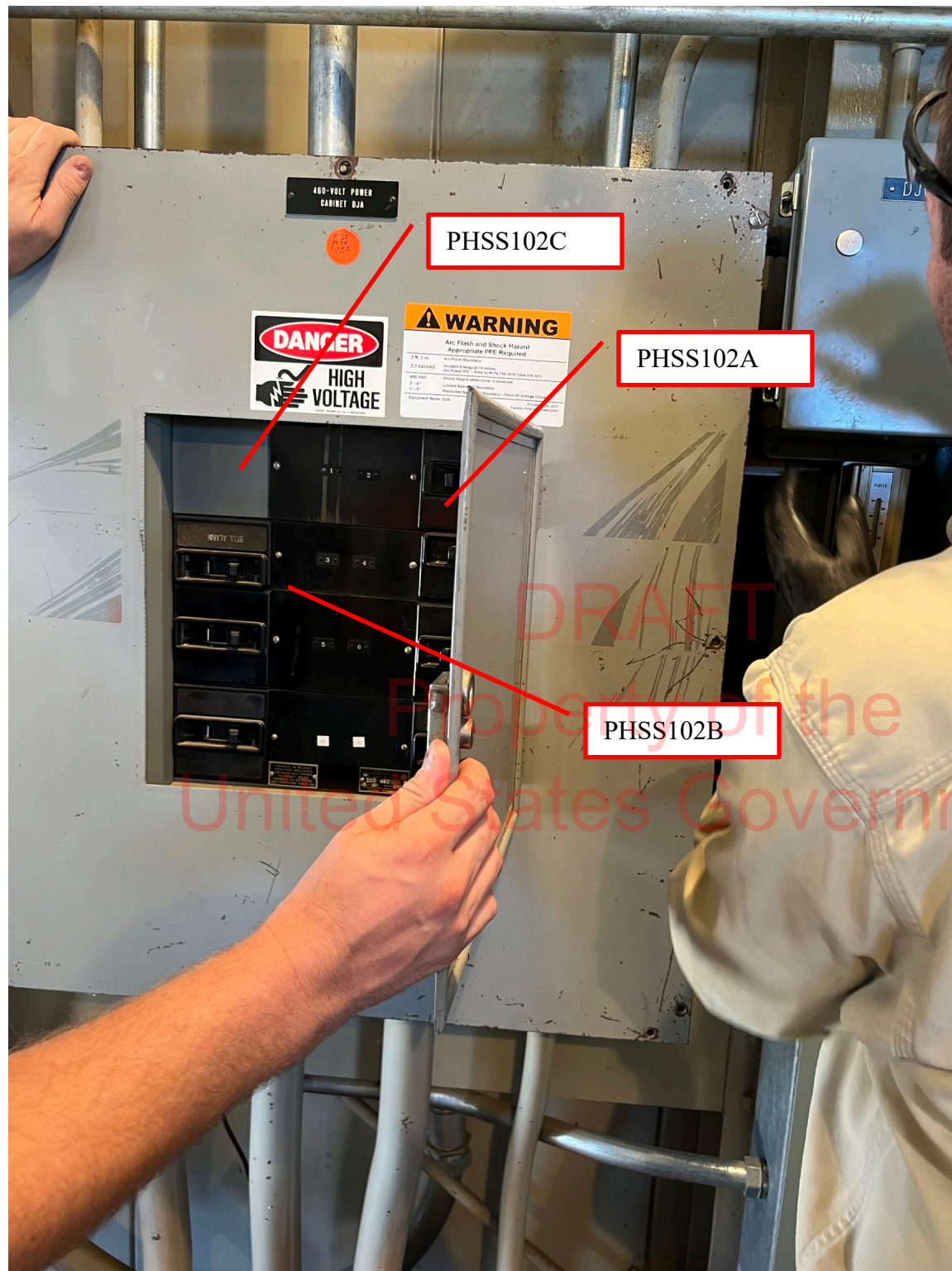


Figure 34. This is panel DJA, 480 volt power cabinet. Samples PHSS102A through C were collected here.

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Figure 35. This is panel BJJ. Power.



Figure 36. This is a view inside panel BJJ. There are five mercury relays inside. Note the red circle.

H. April 2025

1. Control Room showing panels DSA3, DSA4, and DSA5



Figure 37. Sample PHSSS001A was collected from panel DSA4, gray wiring.

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Figure 38. Sample PHSSS001B was collected from panel DSA4, gray wiring.



Figure 39. Sample PHSS001C was collected from panel DSA4, gray wiring.

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Figure 40. Sample PHSSS002A was collected from an Allen Bradley heater relay body.



Figure 41. Sample PHSS002B was collected from an Allen Bradley heater relay body in panel DSA4.



Figure 42. Sample PHSS002C was collected from an Allen Bradley heater relay body in panel DSA4.



Figure 43. Sample PHSS003A was collected from an Allen Bradley heater relay body in panel DSA4.

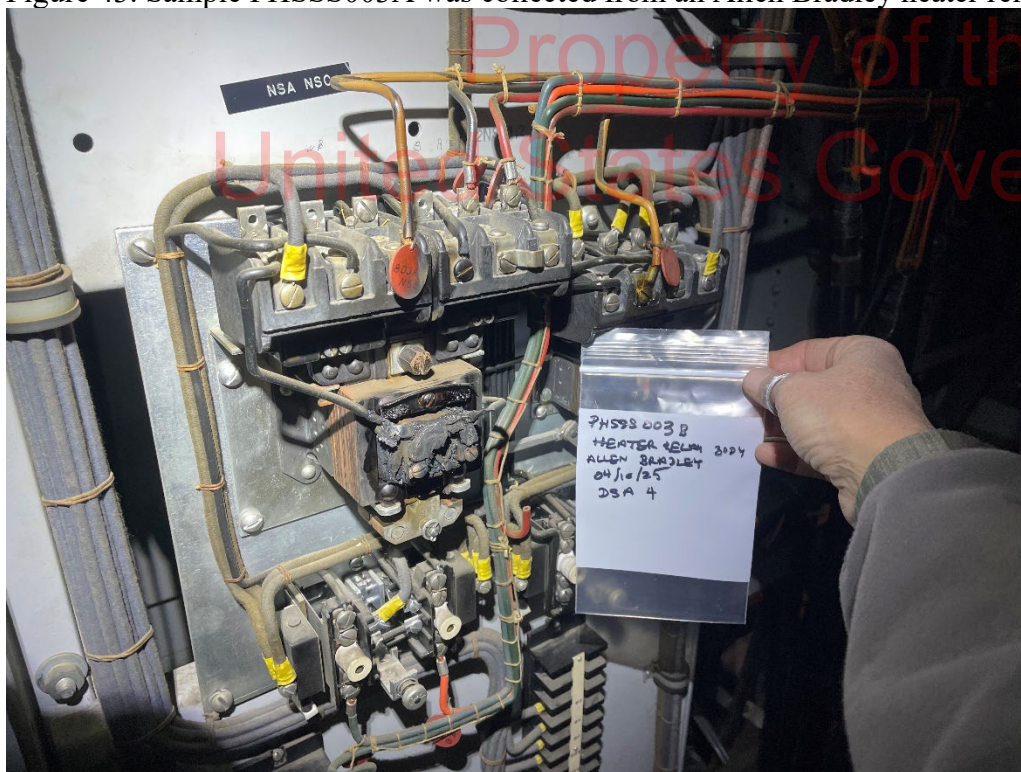


Figure 44. Sample PHSS003B was collected from an Allen Bradley heater relay body in panel DSA4.

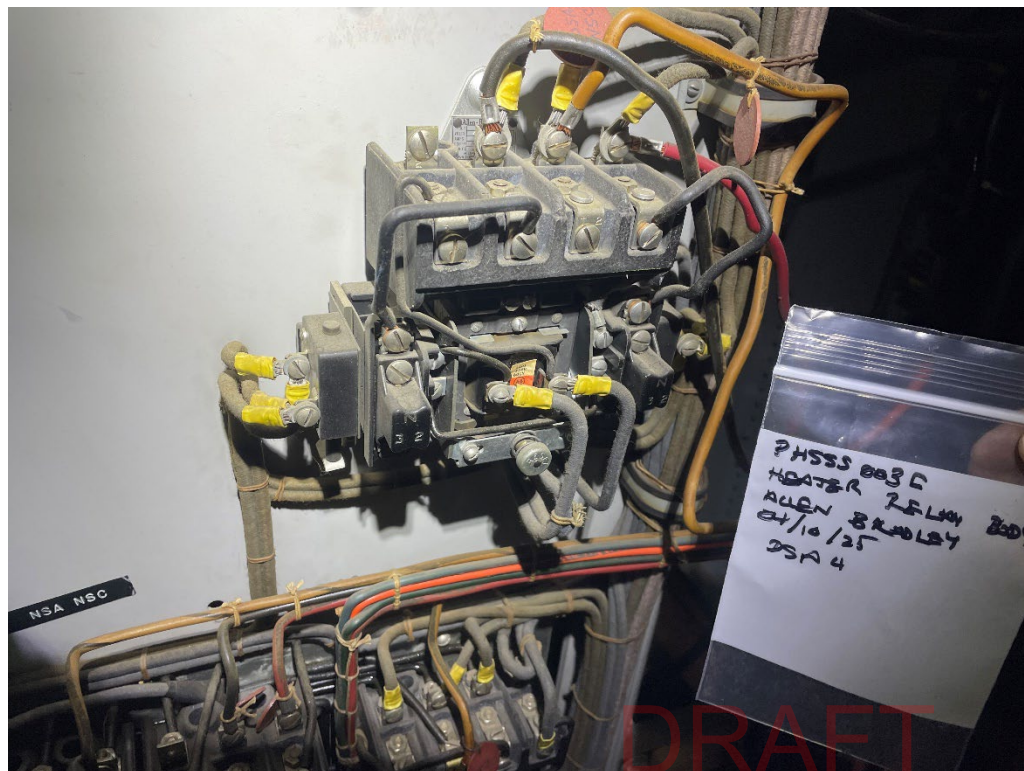


Figure 45. Sample PHSSS003C was collected from an Allen Bradley heater relay body in panel DSA4.

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Figure 46. Sample PHSSS004A was collected from a terminal block in panel DSA4.



Figure 47. Sample PHSSS004B was collected from a terminal block in panel DSA4.

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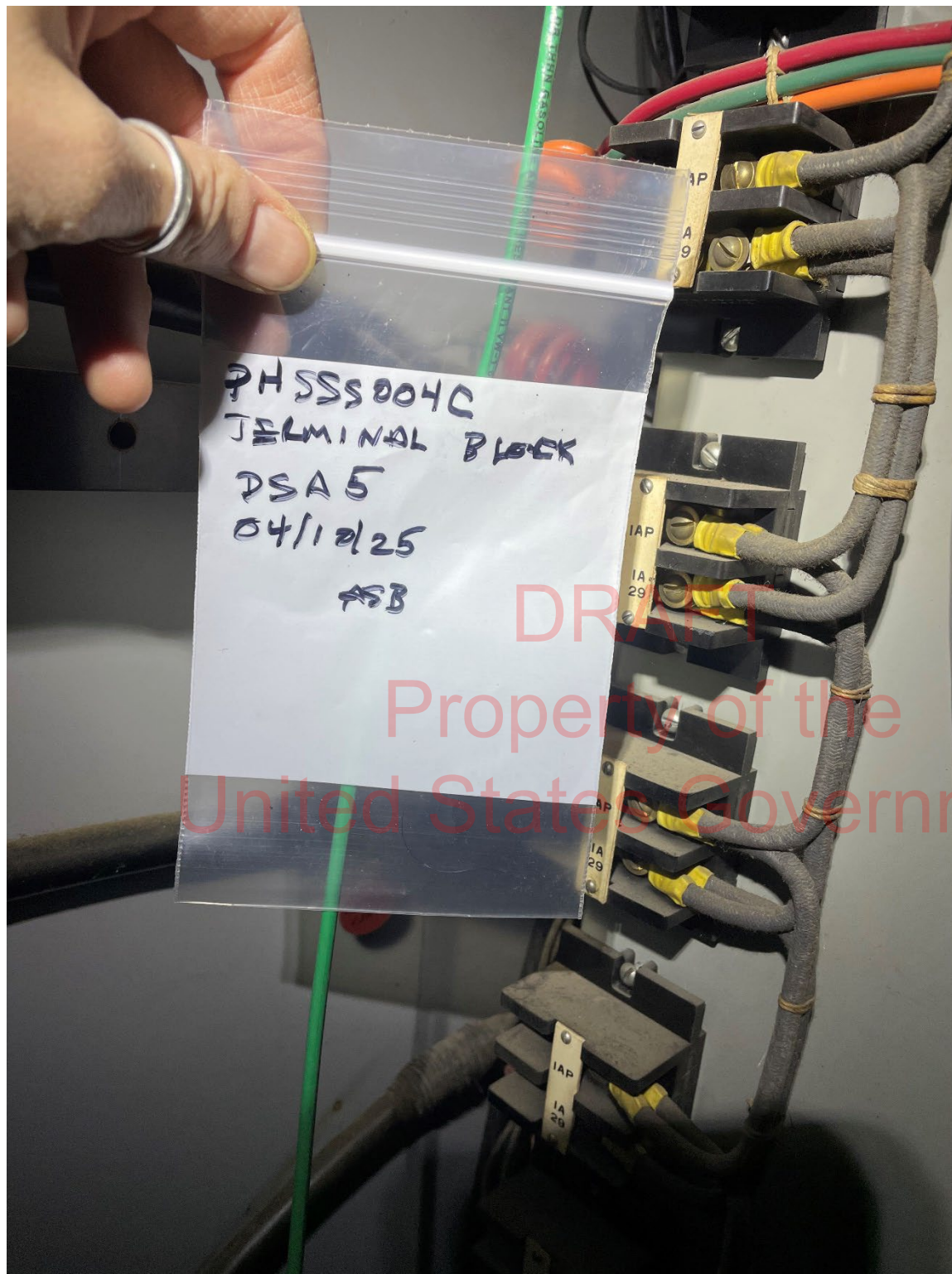


Figure 48. Sample PHSSS004C was collected from a terminal block in panel DSA4.



Figure 49. Sample PHSS005A was collected from a Westinghouse 15 ampere circuit breaker in panel DSA5.

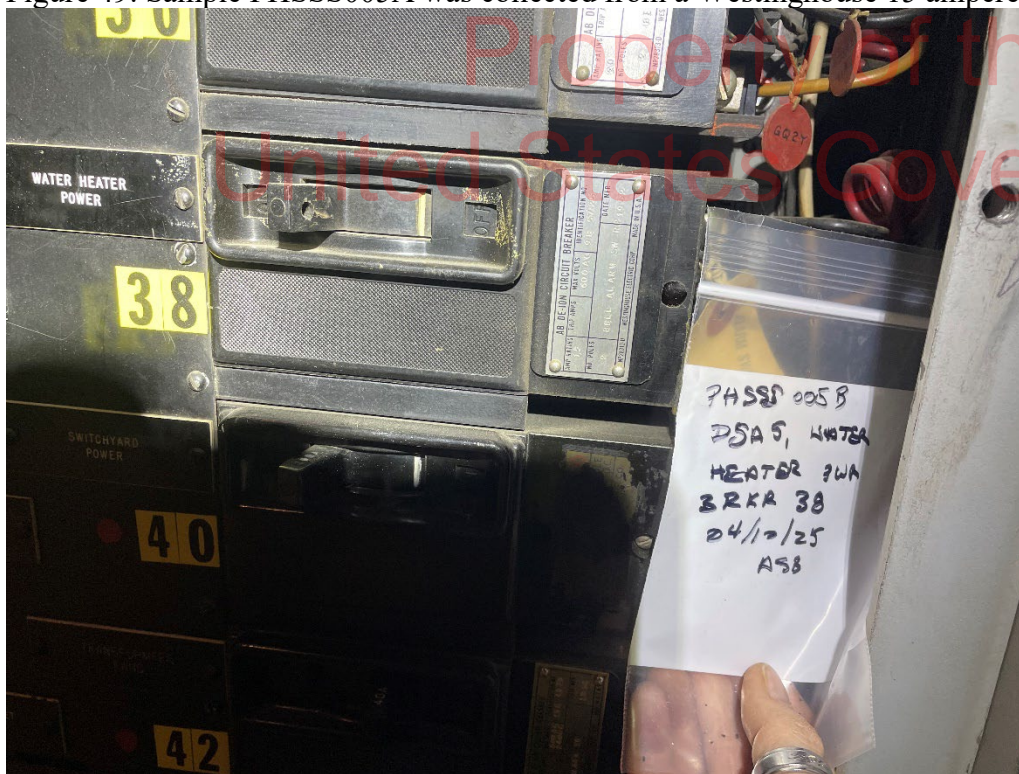


Figure 50. Sample PHSS005B was collected from a Westinghouse 15 ampere circuit breaker in panel DSA5.

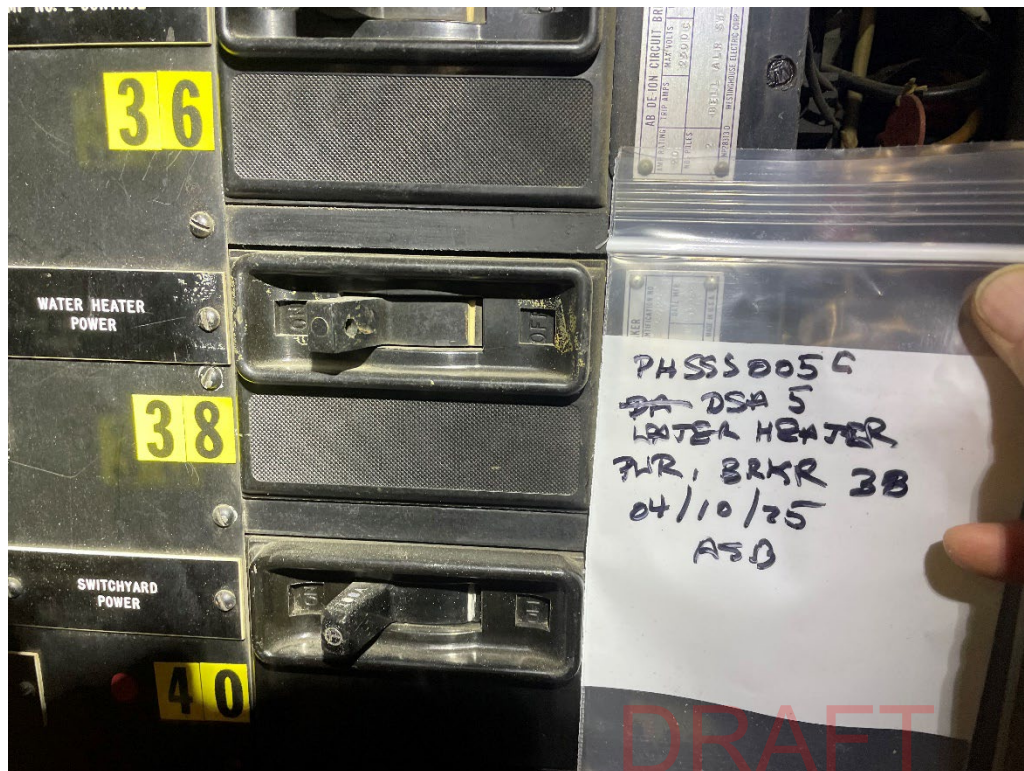


Figure 51. Sample PHSS005C was collected from a Westinghouse 15 ampere circuit breaker in panel DSA5.



Figure 52. Sample PHSS006A was collected from a Westinghouse 125 ampere circuit breaker in panel DSA4.



Figure 53. Sample PHSSS006B was collected from a Westinghouse 125 ampere circuit breaker in panel DSA4.



Figure 54. Sample PHSSS006C was collected from a Westinghouse 125 ampere circuit breaker in panel DSA4.

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1.04 COPIES OF ANALYTICAL RESULTS

A. Asbestos bulk analysis.

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Built Environment Testing
Reservoirs

April 12, 2025

Subcontractor Number:**Laboratory Report:** RES 650648-1**Project #/P.O. #:** G501P**Project Description:** Pole Hill PP SSS

Lise Pederson
US Bureau of Reclamation (Den. Fed. Ctr.)
P.O. Box 25007, Mail Stop 86-68540
Denver CO 80225

Dear Lise,

Eurofins Reservoirs is an analytical laboratory accredited for the analysis of Industrial Hygiene and Environmental matrices by the National Voluntary Laboratory Accreditation Program (NVLAP), Lab Code 101896-0 for Transmission Electron Microscopy (TEM) and Polarized Light Microscopy (PLM) analysis and the American Industrial Hygiene Association (AIHA LAP, LLC), Lab ID 101533 for Phase Contrast Microscopy (PCM) analysis. This laboratory is currently proficient in both Proficiency Testing and PAT programs respectively.

Eurofins Reservoirs has analyzed the following samples for asbestos content as per your request. The analysis has been completed in general accordance with the appropriate methodology as stated in the attached analysis table. The results have been submitted to your office.

RES 650648-1 is the job number assigned to this study. This report is considered highly confidential and the sole property of the customer. Eurofins Reservoirs will not discuss any part of this study with personnel other than those of the client. The results described in this report only apply to the samples analyzed, as received and with the information provided by the customer. This report must not be used to claim endorsement of products or analytical results by NVLAP or any agency of the U.S. Government. This report shall not be reproduced except in full, without written approval from Eurofins Reservoirs. Samples will be disposed of after sixty days unless longer storage is requested. If you have any questions about this report, please feel free to call 303-964-1986.

Sincerely,

by Tyler Hutchinson

Jeanne Spencer
President



Poll Hill Powerplant
Controls, Protection, and Station Service Upgrade
Colorado-Big Thompson Project, Colorado

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EUROFINS RESERVOIRS ENVIRONMENTAL, INC

NVLAP Lab Code 101896-0
AIHA LAP, LLC. LAB ID 101533

TABLE: I ANALYSIS: PLM BULK ANALYSIS, PERCENTAGE COMPOSITION BY VOLUME

RES Job Number: **RES 650648-1**
Client: **US Bureau of Reclamation (Den. Fed. Ctr.)**
Client Project/P.O.: **G501P**
Client Project Description: **Pole Hill PP SSS**
Date Samples Received: **April 11, 2025**
Analysis Type: **EPA 600/R-93/116 - Short Report, Bulk**
Turnaround: **Priority**
Date Samples Analyzed: **April 12, 2025**

NA = Not Analyzed
NR = Not Received
ND = None Detected
TR = Trace; <1 % Visual Estimate
Trem-Act = Tremolite-Actinolite

Laboratory Sample ID Client Sample Number	L A Y E R	Physical Description	Sub Part (%)	Asbestos Content		Non- Asbestos Fibrous Components (%)	Non- Fibrous Components (%)
				Mineral	Visual Estimate (%)		
650648 - PHSSS001A	A	Black resinous tape	15		ND	0	100.0
	B	Black/tan wire wrap	25		ND	40	60.0
	C	Gray wire insulation	30	Chrysotile	35	40	25.0
	D	Gray fibrous woven material	30		ND	85	15.0
650648 - PHSSS001B	A	Black/tan wire wrap	20		ND	40	60.0
	B	Black resinous tape	25		ND	0	100.0
	C	Gray wire insulation	25	Chrysotile	35	40	25.0
	D	Gray fibrous woven material	30		ND	85	15.0
650648 - PHSSS001C	A	Black resinous tape	5		ND	0	100.0
	B	Gray wire insulation	20	Chrysotile	35	40	25.0
	C	Black/tan wire wrap	25		ND	40	60.0
	D	Gray fibrous woven material	25		ND	85	15.0
	E	Gray paper	25		ND	85	15.0
650648 - PHSSS002A	A	Black fibrous resinous material	100	Chrysotile	25	0	75.0
650648 - PHSSS002B	A	Black fibrous resinous material	100	Chrysotile	15	0	85.0
650648 - PHSSS002C	A	Black fibrous resinous material	100	Chrysotile	20	0	80.0

TEM Analysis recommended for organically bound material (i.e. floor tile) if PLM results are <1%.

Poll Hill Powerplant
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Colorado-Big Thompson Project, Colorado

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EUROFINS RESERVOIRS ENVIRONMENTAL, INC

NVLAP Lab Code 101896-0
AIHA LAP, LLC, LAB ID: 101533

TABLE: I ANALYSIS: PLM BULK ANALYSIS, PERCENTAGE COMPOSITION BY VOLUME

RES Job Number: **RES 650648-1**
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NA = Not Analyzed
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ND = None Detected
TR = Trace; <1 % Visual Estimate
Trem-Act = Tremolite-Actinolite

Laboratory Sample ID Client Sample Number	L A Y E R	Physical Description	Sub Part (%)	Asbestos Content		Non- Asbestos Fibrous Components (%)	Non- Fibrous Components (%)
				Mineral	Visual Estimate (%)		
650648 - PHSSS003A	A	Black fibrous resinous material	100	Chrysotile	20	0	80.0
650648 - PHSSS003B	A	Black fibrous resinous material	100	Chrysotile	25	0	75.0
650648 - PHSSS003C	A	Black fibrous resinous material	100	Chrysotile	35	0	65.0
650648 - PHSSS004A	A	Black tar	100		ND	0	100.0
650648 - PHSSS004B	A	Black fibrous resinous material	100		ND	25	75.0
650648 - PHSSS004C	A	Black fibrous resinous material	100		ND	20	80.0
650648 - PHSSS005A	A	Black fibrous resinous material	100		ND	25	75.0
650648 - PHSSS005B	A	Black fibrous resinous material	100		ND	20	80.0
650648 - PHSSS005C	A	Black fibrous resinous material	100		ND	25	75.0
650648 - PHSSS006A	A	Black fibrous resinous material	100		ND	20	80.0
650648 - PHSSS006B	A	Black fibrous resinous material	100		ND	20	80.0
650648 - PHSSS006C	A	Black fibrous resinous material	100	Chrysotile Point Count	4 2.75	10	86.0

TEM Analysis recommended for organically bound material (i.e. floor tile) if PLM results are <1%.

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Tyler Hutchinson
Analyst

Poll Hill Powerplant
Controls, Protection, and Station Service Upgrade
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**Built Environment Testing
Reservoirs**

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RES Job #: 650648

SUBMITTED BY	INVOICE TO	CONTACT INFORMATION	SERIES
Company: US Bureau of Reclamation (Den. Fed. Ctr.) Address: P.O. Box 25007, Mail Stop 86-68540 Denver, CO 80225 Project Number and/or P.O. #: G 501P Project Description/Location: Pole Hill PP SSS	Company: US Bureau of Reclamation (Den. Fed. Ctr.) Address: P.O. Box 25007, Mail Stop 86-68540 Denver, CO 80225 Project Zip Code:	Contact: Lise Pederson Phone: (303) 445-3095 Fax: Cell: (303) 987-8794 Final Data Deliverable Email Address: lpederson@usbr.gov (+ 2 ADDNL. CONTACTS)	- 1 PLM Priority

ASBESTOS LABORATORY	REQUESTED ANALYSIS										VALID MATRIX CODES				LAB NOTES
PLM / PCM / TEM DTL RUSH PRIORITY STANDARD											Air = A	Bulk = B			
CHEMISTRY LABORATORY											Dust = D	Food = F			
Dust RUSH PRIORITY STANDARD											Paint = P	Soil = S			
Metals RUSH PRIORITY STANDARD											Surface = SU	Swab = SW			
Organics* SAME DAY RUSH PRIORITY STANDARD											Tape = T	Wipe = W			
MICROBIOLOGY LABORATORY											Drinking Water = DW Waste Water = WW				
Viable Analysis** PRIORITY STANDARD											**ASTM E 1782 approved wipe media only**				
Medical Device Analysis RUSH STANDARD															
Mold Analysis RUSH PRIORITY STANDARD															
Turnaround times establish a laboratory priority, subject to laboratory volume and are not guaranteed. Additional fees apply for afterhours, weekends and holidays.															
Special Instructions: 400 point count on 10% and less.															
Client Sample ID Number (Sample ID's must be unique)															
1 PHSS001A	X											B	1	04/11/25	10:41
2 PHSS001B	X											B	1	04/11/25	10:41
3 PHSS001C	X											B	1	04/11/25	10:41
4 PHSS002A	X											B	1	04/11/25	10:41
5 PHSS002B	X											B	1	04/11/25	10:41
6 PHSS002C	X											B	1	04/11/25	10:41
7 PHSS003A	X											B	1	04/11/25	10:41
8 PHSS003B	X											B	1	04/11/25	10:41
9 PHSS003C	X											B	1	04/11/25	10:41
10 PHSS004A	X											B	1	04/11/25	10:41
11 PHSS004B	X											B	1	04/11/25	10:41
12 PHSS004C	X											B	1	04/11/25	10:41
13 PHSS005A	X											B	1	04/11/25	10:41

Eurofins Reservoirs Environmental, Inc. establishes a unique Lab Sample ID, for each sample, by preceding each unique Client Sample ID with the laboratory RES Job Number.

Eurofins Reservoirs Environmental, Inc. will analyze incoming samples based on information received and will not be responsible for errors or omissions in calculations resulting from the inaccuracy of original data. By signing, client/company representative agrees that submission of the following samples for requested analysis as indicated on this Chain of Custody shall constitute an analytical services agreement with payment terms of NET 30 days. Failure to comply with payment terms may result in a 1.5% monthly interest surcharge.

Relinquished By:	Lise Pederson	Date/Time: 04/11/2025 12:45:24	Sample Condition: Acceptable
Received By:	Emily Creasey	Date/Time: 04/11/2025 12:58:37	Carrier: Hand

(303) 964-1986
(888) RES-ENV

4655 Yarrow St, Suite 100, Anvada, CO 80002
Page 1 of 2

<https://www.eurofins.us.com/builtenv>
<https://clients.rei.eurofins.us.com>

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Colorado-Big Thompson Project, Colorado

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Built Environment Testing
Reservoirs

RES Job #: 650648

Submitted By: US Bureau of Reclamation (Den. Fed. Ctr.)

 Built Environment Testing Reservoirs	RES Job #: 650648	Submitted By: US Bureau of Reclamation (Den. Fed. Ctr.)	PLM-PLM Short Report (EPA/600/R-03/116)	REQUESTED ANALYSIS							VALID MATRIX CODES				LAB NOTES
				TEM	PCM	DUST	METALS	ORGANICS	VIABLES	MEDICAL	MOLD	Air = A	Bulk = B		
												Dust = D	Food = F		
												Paint = P	Soil = S		
												Surface = SU	Swab = SW		
												Tape = T	Wipe = W		
												Drinking Water = DW			
												Waste Water = WW			
												ASTM E1792 approved wipe media only			
												Sample Volume(L)/Area	Sample Temperature (°C)	Length(or Algebra) x Width(or Area)(sq.ft)	
Laboratory Analysis Instructions															
Client Sample ID Number	(Sample ID's must be unique)			ASBESTOS	CHEMISTRY	MICROBIOLOGY	ICO								
14 PHSS005B		X								B	1	04/11/25	10:41		
15 PHSS005C		X								B	1	04/11/25	10:41		
16 PHSS006A		X								B	1	04/11/25	10:41		
17 PHSS006B		X								B	1	04/11/25	10:41		
18 PHSS006C		X								B	1	04/11/25	10:41		

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PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 51 25 00

INPUT/OUTPUT LISTS

PART 1 GENERAL

1.01 INPUT / OUTPUT AND COMMUNICATION LISTS

- A. Lists are for information of Contractor.

1.02 INPUT / OUTPUT LISTING FOR GENERATOR UNIT 1 AUTOMATION PLC AT CONTROL SWITCHBOARD CSA

- A. Inputs identified as Automation Control Logic, Start Permissive, and Annunciation Circuit are used in the PLC Ladder Logic Diagram drawings 245-D-60125 through 214-D-60129.
- B. All input and output status to be displayed on the HMI screens.
- C. These I/O points have been extracted from Pole Hill CSA Control Boards.
- D. Explanation:
1. "IO Desc" PLC assigned designation
 2. DI – Discrete Input.
 3. DO – Discrete Output.
 4. AI – Analog Input

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Controls, Protection, and Station Service Upgrade

Colorado-Big Thompson Project, Colorado

Solicitation No.

TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	ACCESS NAME	REPLACE DEVICE	FUNCTION		Annunciation Circuit
						AUTOMATION CONTROL LOGIC	START PERMISSIVE	
METERING ANALOG AND DIGITAL POINTS – GENERATOR UNIT 1								
AI/DI		T U1 KWHM	Power Meter Status	10001		✓		
AI/DI		T U1 A 51	UNIT FIELD AMPS,U1 PHASE A	10002		✓		
AI/DI		T U1 A 51	UNIT FIELD AMPS,U1 PHASE B	10003		✓		
AI/DI		T U1 A 51	UNIT FIELD AMPS,U1 PHASE C	10004		✓		
AI/DI		UTV-1	UNIT PHASE A VOLTS	10005		✓		
AI/DI		UTV-2	UNIT PHASE B VOLTS	10006		✓		
AI/DI		UTV-3	UNIT PHASE C VOLTS	10007		✓		
AI/DI		T U1 A MW	UNIT MEGAWATTS,U1	10008		✓		
AI/DI		T U1 A MVAR	UNIT MEGAVAR,U1	10009		✓		
AI/DI		T U1 A 59 E	EXITER FIELD VOLTS,U1	10010		✓		
AI/DI		T U1 A 51 E	EXITER FIELD AMPS,U1	10011		✓		
AI/DI		URPM	UNIT SPEED RPM	10012		✓		
AI/DI			BUTTERFLY VALVE POSISTION INDICATER ;U1	10013		✓		
AI/DI		T U1 KWHM	UNIT KWHMETER,U1	10014				
AI/DI			LIFTING & JACKING AIR PSI;U1	10016		✓		
SEL 735 POWER QUALITY & REVENUE METER								
DI								
AUTOMATION CONTROL LOGIC I/O (111MPLC)								
DI		143UCS-MANUAL	Control Selector Switch in Local Manual	111MPLC DI:A.001	✓			
DI		143UCS-AUTO	Control Selector Switch in Local Auto	111MPLC DI:A.002	✓			
DI		143UCS-SUPERVISORY	Control Selector Switch in Supervisory	111MPLC DI:A.003	✓			
DI		127GCDC	G1 DC Control Power Undervoltage Relay	111MPLC DI:A.004	✓			

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Controls, Protection, and Station Service Upgrade

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	ACCESS NAME	REPLACE DEVICE	FUNCTION		Annunciation Circuit
						AUTOMATION CONTROL LOGIC	START PERMISSIVE	
DI		127GPDC	Generator DC Protection Power Low Voltage Relay	111MPLC DI:A.005				
DI		122CS-TRIP	Breaker 122 Control U1A1 – Trip	111MPLC DI:A.006	✓			
DI		122CS-CLOSE	Breaker 122 Control U1A1 – Close	111MPLC DI:A.007	✓			
DI		JY1CL	Breaker JY1A (164) Auxiliary Close Position Relay	111MPLC DI:A.008				
DI		JY2CL	Breaker JY2A (264) Auxiliary Close Position Relay	111MPLC DI:A.009				
DI		165SCCS-RAISE	Governor Speed Change – Raise	111MPLC DI:A.010	✓			
DI		165SCCS-LOWER	Governor Speed Change – Lower	111MPLC DI:A.011	✓			
DI		165GLCS – RAISE	Governor Gate Limit – Raise	111MPLC DI:A.012	✓			
DI		165GLCS – LOWER	Governor Gate Limit – Lower	111MPLC DI:A.013	✓			
DI		125SS-AUTO	Generator Synchronizer Selector – Auto	111MPLC DI:A.014	✓			
DI		190ECS – RAISE	Exciter Field Control Switch – Raise	111MPLC DI:A.015	✓			
DI		190ECS – LOWER	Exciter Field Control Switch – Lower	111MPLC DI:A.016	✓			
DI		169ECS – ENABLE	Exciter Enable Switch – Enabled	111MPLC DI:B.001	✓			
DI		143EM-AUTO	Voltage Regulator Control Mode Selector Switch – AUTO	111MPLC DI:B.002	✓			
DI		152ECS – TRIP	Exciter Breaker Control Switch – Trip	111MPLC DI:B.003	✓			
DI		152ECS – CLOSE	Exciter Breaker Control Switch – Close	111MPLC DI:B.004	✓			
DI		105CS	Emergency Shutdown Control Switch	111MPLC DI:B.005	✓			
DI		186GP	Generator Protective Lockout Switch	111MPLC DI:B.006	✓			
DI		B27DC	125VDC Bus Low Voltage Trip	111MPLC DI:B.007	✓			
DI		186K1A	Transformer K1A Differential Lockout Switch	111MPLC DI:B.008	✓			

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Colorado-Big Thompson Project, Colorado

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	ACCESS NAME	REPLACE DEVICE	FUNCTION		Annunciation Circuit
						AUTOMATION CONTROL LOGIC	START PERMISSIVE	
DI		186GD	Generator Differential Lockout Switch	111MPLC DI:B.009	✓			
DI		K734	Exciter Field Breaker Open Auxiliary Relay	111MPLC DI:B.010				
DI		K2	Excitation is ON	111MPLC DI:B.011				
DI		K5	Exciter Group Alarm	111MPLC DI:B.012				
DI		K6	V/Hz Limiter Active	111MPLC DI:B.013				
DI		K7	Under Excitation Limiter Active	111MPLC DI:B.014				
DI		K8	Over Excitation Limiter Active	111MPLC DI:B.015				
DI		K14	7UM Siprotec Malfunction	111MPLC DI:B.016				
DI		143E L/R	Exciter Local/Remote Switch (E1A1)	111MPLC DI:C.001				
DI		SPARE	SPARE	111MPLC DI:C.002				
DI		SPARE	SPARE	111MPLC DI:C.003				
DI		SPARE	SPARE	111MPLC DI:C.004				
DI		SPARE	SPARE	111MPLC DI:C.005				
DI		SPARE	SPARE	111MPLC DI:C.006				
DI		SPARE	SPARE	111MPLC DI:C.007				
DI		SPARE	SPARE	111MPLC DI:C.008				
DI		SPARE	SPARE	111MPLC DI:C.009				
DI		SPARE	SPARE	111MPLC DI:C.010				
DI		SPARE	SPARE	111MPLC DI:C.011				
DI		SPARE	SPARE	111MPLC DI:C.012				

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	ACCESS NAME	REPLACE DEVICE	FUNCTION		Annunciation Circuit
						AUTOMATION CONTROL LOGIC	START PERMISSIVE	
DI		SPARE	SPARE	111MPLC DI:C.013				
DI		SPARE	SPARE	111MPLC DI:C.014				
DI		SPARE	SPARE	111MPLC DI:C.015				
DI		SPARE	SPARE	111MPLC DI:C.016				
AI	4- 20 mA	I RPM	RPM Indicator at Control Room – Fed from PMG	111MPLC AI:A.001				
AI		SPARE	SPARE	111MPLC AI:A.002				
AI		SPARE	SPARE	111MPLC AI:A.003				
AI		SPARE	SPARE	111MPLC AI:A.004				
AI		SPARE	SPARE	111MPLC AI:A.005				
AI		SPARE	SPARE	111MPLC AI:A.006				
AI		SPARE	SPARE	111MPLC AI:A.007				
AI		SPARE	SPARE	111MPLC AI:A.008				
AI		SPARE	SPARE	111MPLC AI:A.009				
AI		SPARE	SPARE	111MPLC AI:A.010				
AI		SPARE	SPARE	111MPLC AI:A.011				
AI		SPARE	SPARE	111MPLC AI:A.012				
AI		SPARE	SPARE	111MPLC AI:A.013				
AI		SPARE	SPARE	111MPLC AI:A.014				
AI		SPARE	SPARE	111MPLC AI:A.015				
AI		SPARE	SPARE	111MPLC AI:A.016				

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	ACCESS NAME	REPLACE DEVICE	FUNCTION		Annunciation Circuit
						AUTOMATION CONTROL LOGIC	START PERMISSIVE	
D O		186GPT	186GP trip command from PLC	111MPLC DO:A.001				
D O		186K1AT	186K1A trip command from PLC	111MPLC DO:A.002				
D O		186GDT	186GD trip command from PLC	111MPLC DO:A.003				
D O		SPARE	SPARE	111MPLC DO:A.004				
D O		SPARE	SPARE	111MPLC DO:A.005				
D O		SPARE	SPARE	111MPLC DO:A.006				
D O		SPARE	SPARE	111MPLC DO:A.007				
D O		SPARE	SPARE	111MPLC DO:A.008				

1.03 INPUT/OUTPUT LISTING FOR DISTRIBUTED I/O AT GENERATOR UNIT 1 TURBINE LEVEL – 111TDIO

- A. Inputs identified as Automation Control Logic, Start Permissive, and Annunciation Circuit are used in the PLC Ladder Logic Diagram drawings 245-D-60125 through 214-D-60129.
- B. All input and output status to be displayed on the HMI screens.
- C. These I/O points have been extracted from Pole Hill CSA Control Boards.
- D. Explanation:
 - 1. “IO Desc” PLC assigned designation
 - 2. DI – Discrete Input.
 - 3. DO – Discrete Output.
 - 4. AI – Analog Input

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ANNUNCIATION CIRCUIT	FUNCTION	ACCESS NAME	REPLACE DEVICE	DESCRIPTION	DESIGNATION	ANALOG SIGNAL TYPE	TYPE	START PERMISSIVE	AUTOMATION CONTROL
DIGITAL/ANALOG INPUTS TO DISTRIBUTED I/O 111TDIO									
					120GCWVLS – OPEN	Generator Cooling Water Valve Open Limit Switch		111TDIO DI:A.001	
					120GCWVLS – CLOSE	Generator Cooling Water Valve Close Limit Switch		111TDIO DI:A.002	
					163TBQ	G1 Thrust Bearing Oil Pressure Switch		111TDIO DI:A.003	
					171Q5	Governor Oil Level Switch		111TDIO DI:A.004	
					143BKCS	G1 Brakes Control Switch		111TDIO DI:A.005	
			✓		TBQ127ACX	Thrust Bearing Oil Pump AC Control Voltage Low Relay		111TDIO DI:A.006	
			✓		143TBQ1 – AUTO	Thrust Bearing Oil Pump Selector Switch at Contactor – AUTO		111TDIO DI:A.007	
			✓		143TBQ1 – MANUAL	Thrust Bearing Oil Pump Selector Switch at Contactor – Manual		111TDIO DI:A.008	
					126TSBX	Turbine Stuffing Box Temperature Auxiliary Relay		111TDIO DI:A.009	
					138TGX	Generator Turbine Guide Bearing Temperature Aux Relay		111TDIO DI:A.010	
					138TGQX	Generator Turbine Guide Bearing Oil Temperature Aux. Relay		111TDIO DI:A.011	
					171TPX	Turbine Pit Water Level Auxiliary Relay		111TDIO DI:A.012	
			✓		71S	Sump Shutdown Level Switch		111TDIO DI:A.013	
					GFCS	Gen. Fire CO2 Control Switch		111TDIO DI:A.014	
					126FX1	Gen. CO2 Discharge Auxiliary Relay		111TDIO DI:A.015	
			✓		127GDDC	Gen. Diff. D.C. Control Voltage Relay		111TDIO DI:A.016	
					163CDX	Gen. CO2 Discharge Pressure Auxiliary Relay		111TDIO DI:B.001	

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
DI		SPARE	SPARE		111TDIO DI:B.002			
DI		SPARE	SPARE		111TDIO DI:B.003			
DI		SPARE	SPARE		111TDIO DI:B.004			
DI		SPARE	SPARE		111TDIO DI:B.005			
DI		SPARE	SPARE		111TDIO DI:B.006			
DI		SPARE	SPARE		111TDIO DI:B.007			
DI		SPARE	SPARE		111TDIO DI:B.008			
DI		SPARE	SPARE		111TDIO DI:B.009			
DI		SPARE	SPARE		111TDIO DI:B.010			
DI		SPARE	SPARE		111TDIO DI:B.011			
DI		SPARE	SPARE		111TDIO DI:B.012			
DI		SPARE	SPARE		111TDIO DI:B.013			
DI		SPARE	SPARE		111TDIO DI:B.014			
DI		SPARE	SPARE		111TDIO DI:B.015			
DI		SPARE	SPARE		111TDIO DI:B.016			
DI		SPARE	SPARE		111TDIO DI:B.016			
AI	4-20 mA	163GW2	Generator Cooling Water Pressure	✓	111TDIO AI:A.001			
AI	4-20 mA	163BK	G1 Brake Air Pressure	✓	111TDIO AI:A.002			

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
AI	4-20 mA	198C	G1 Creep Detector	✓	111TDIO AI:A.003			
AI	4-20 mA	165AL	Gate Limit	✓	111TDIO AI:A.004			
AI	4-20 mA	165AP	Gate Position	✓	111TDIO AI:A.005			
AI	4-20 mA	165AS	Speed Changer	✓	111TDIO AI:A.006			
AI	4-20 mA	171TBQ	Thrust & Upper Guide Bearing Oil Level Auxiliary Relay		111TDIO AI:A.007			
AI	4-20 mA	171TGQ	Turbine Guide Bearing Oil Level Auxiliary Relay		111TDIO AI:A.008			
AI	4-20 mA	180TSBW	Turbine Stuffing Box Cooling Water Flow Switch		111TDIO AI:A.009			
AI		SPARE	SPARE		111TDIO AI:A.010			
AI		SPARE	SPARE		111TDIO AI:A.011			
AI		SPARE	SPARE		111TDIO AI:A.012			
AI		SPARE	SPARE		111TDIO AI:A.013			
AI		SPARE	SPARE		111TDIO AI:A.014			
AI		SPARE	SPARE		111TDIO AI:A.015			
AI		SPARE	SPARE		111TDIO AI:A.016			
A O	4-20 mA	I RPM	RPM Indicator at Governor A1A		111TDIO AO:A.001			
A O		SPARE	SPARE		111TDIO AO:A.002			

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
A O		SPARE	SPARE		111TDIO AO:A.003			
A O		SPARE	SPARE		111TDIO AO:A.004			
A O		SPARE	SPARE		111TDIO AO:A.005			
A O		SPARE	SPARE		111TDIO AO:A.006			
A O		SPARE	SPARE		111TDIO AO:A.007			
A O		SPARE	SPARE		111TDIO AO:A.008			
D O		165S	Governor Shutdown Solenoid		111TDIO DO:A.001			
D O		165SNL	Governor Speed-No-Load Solenoid		111TDIO DO:A.002			
D O		142TBQ	Thrust Bearing High Pressure Oil Pump Contactor		111TDIO DO:A.003			
D O		SPARE	SPARE		111TDIO DO:A.004			
D O		SPARE	SPARE		111TDIO DO:A.005			
D O		SPARE	SPARE		111TDIO DO:A.006			
D O		SPARE	SPARE		111TDIO DO:A.007			
D O		SPARE	SPARE		111TDIO DO:A.008			

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
D O		SPARE	SPARE		111TDIO DO:A.009			
D O		SPARE	SPARE		111TDIO DO:A.010			
D O		SPARE	SPARE		111TDIO DO:A.011			
D O		SPARE	SPARE		111TDIO DO:A.012			
D O		SPARE	SPARE		111TDIO DO:A.013			
D O		SPARE	SPARE		111TDIO DO:A.014			
D O		SPARE	SPARE		111TDIO DO:A.015			
D O		SPARE	SPARE		111TDIO DO:A.016			

1.04 INPUT / OUTPUT LISTING FOR DISTRIBUTED I/O AT GATEHOUSE

- A. Inputs identified as Automation Control Logic, Start Permissive, and Annunciation Circuit are used in the PLC Ladder Logic Diagram drawings 245-D-60125 through 214-D-60129.
- B. All input and output status to be displayed on the HMI screens.
- C. These I/O points have been extracted from Pole Hill CSA Control Boards.
- D. Explanation:
 - 1. “IO Desc” PLC assigned designation
 - 2. DI – Discrete Input.
 - 3. DO – Discrete Output.
 - 4. AI – Analog Input

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ANNUNCIATION CIRCUIT	FUNCTION	ACCESS NAME	REPLACE DEVICE	DESCRIPTION	DESIGNATION	ANALOG SIGNAL TYPE	TYPE	START PERMISSIVE AUTOMATION CONTROL
DIGITAL/ANALOG INPUTS TO DISTRIBUTED I/O 111TDIO								
				✓	Penstock Gate Structure Door Alarm Relay at Polehill Plant	DOOR ALARM	DI	111TDIO DI:A.001
			✓	Penstock Gate Structure DC Alarm Voltage Monitoring Relay	PGA27DC		DI	111TDIO DI:A.002
				Battery Charger A.C. Voltage Check Relay	PGBC27AC		DI	111TDIO DI:A.003
				Battery Charger D.C. Voltage Check Relay	PGBC27DC		DI	111TDIO DI:A.004
				Battery Ground Detector Relay	PG65B		DI	111TDIO DI:A.005
			✓	125 Volt DC Bus Voltage Check Relay	PGB27DC		DI	111TDIO DI:A.006
			✓	Penstock Gate Control D.C. Voltage Check Relay	PGC27DC		DI	111TDIO DI:A.007
				Penstock Gate Control Pushbuttons	PGPB OPEN		DI	111TDIO DI:A.008
				Penstock Gate Control Pushbuttons	PGPB CLOSE		DI	111TDIO DI:A.009
				Penstock Gate Control Pushbuttons	PGPB STOP		DI	111TDIO DI:A.010
			✓	Penstock Gate Open Position Limit Switch	133PG2		DI	111TDIO DI:A.011
			✓	Penstock Gate Oil Pump Motor D.C. Voltage Check Relay	PGM27DC		DI	111TDIO DI:A.012
			✓	Alarm Control Voltage Check Relay	PGA27DC		DI	111TDIO DI:A.013
			✓	Penstock Gate Close Position Limit Switch	133PG1		DI	111TDIO DI:A.014
				Battery D.C. Breaker	52 PGB DC BKR		DI	111TDIO DI:A.015
				Battery Charger D.C. Breaker	52 PGBC DC BRKR		DI	111TDIO DI:A.016
				Communications Battery D.C. Breaker	52 PGBCB DC BKR		DI	111TDIO DI:B.001

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
DI		52 PGBCB DC BKR	Communications Battery Charger D.C. Breaker		111TDIO DI:B.001			
DI		PGSS27AC-1	Penstock Gatehouse 480 Volt Bus Relays	✓	111TDIO DI:B.002			
DI		PGSS27AC-2	Penstock Gatehouse 480 Volt Bus Relays	✓	111TDIO DI:B.003			
DI		PGSS27AC-3	Penstock Gatehouse 480 Volt Bus Relays	✓	111TDIO DI:B.004			
DI		26NJA	Thermal cutout for Heater NJA		111TDIO DI:B.005			
DI		26NJB	Thermal cutout for Heater NJB		111TDIO DI:B.006			
DI		43MJB	Selector Switch for Contactor MJB - Manual		111TDIO DI:B.007			
DI		SPARE	SPARE		111TDIO DI:B.008			
DI		SPARE	SPARE		111TDIO DI:B.009			
DI		SPARE	SPARE		111TDIO DI:B.010			
DI		SPARE	SPARE		111TDIO DI:B.011			
DI		SPARE	SPARE		111TDIO DI:B.012			
DI		SPARE	SPARE		111TDIO DI:B.013			
DI		SPARE	SPARE		111TDIO DI:B.014			
DI		SPARE	SPARE		111TDIO DI:B.015			
DI		SPARE	SPARE		111TDIO DI:B.016			
AI	4-20 mA	171PG1	Penstock Water Level		111TDIO AI:A.001			
AI	4-20 mA	FBEL71W	Forebay Elevation		111TDIO AI:A.002			

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
AI	4-20 mA	163PGQ1	Penstock Gate Pressure		111TDIO AI:A.003			
AI	4-20 mA	PG23	Penstock Gatehouse Ambient Temperature		111TDIO AI:A.004			
AI	4-20 mA	23MJC	Thermostat for Heater NJA		111TDIO AI:A.005			
AI	4-20 mA	23MJD	Thermostat for Heater NJD		111TDIO AI:A.006			
AI	4-20 mA	71CHL	Canal Water Level		111TDIO AI:A.007			
AI		SPARE	SPARE		111TDIO AI:A.008			
AI		SPARE	SPARE		111TDIO AI:A.009			
AI		SPARE	SPARE		111TDIO AI:A.010			
AI		SPARE	SPARE		111TDIO AI:A.011			
AI		SPARE	SPARE		111TDIO AI:A.012			
AI		SPARE	SPARE		111TDIO AI:A.013			
AI		SPARE	SPARE		111TDIO AI:A.014			
AI		SPARE	SPARE		111TDIO AI:A.015			
AI		SPARE	SPARE		111TDIO AI:A.016			
DO		171PG1	Penstock Low Level		111TDIO DO:A.001			
DO		171PGF	Penstock Full		111TDIO DO:A.002			

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TYPE	ANALOG SIGNAL TYPE	DESIGNATION	DESCRIPTION	REPLACE DEVICE	ACCESS NAME	FUNCTION		ANNUNCIATION CIRCUIT
						START PERMISSIVE	AUTOMATION CONTROL	
D O		PGCY	Penstock Gate Close		111TDIO DO:A.003			
D O		183PGO	Penstock Gate Selective Control Contactor - Open		111TDIO DO:A.004			
D O		183PGC	Penstock Gate Selective Control Contactor – Close		111TDIO DO:A.005			
D O		120PGQ	Penstock Gate Solenoid Valve		111TDIO DO:A.006			
D O		142PG	Penstock Gate Oil Pump Motor Contactor		111TDIO DO:A.007			
D O		42MJC	Contactor for Heater NJA		111TDIO DO:A.008			
D O		42MJD	Contactor for Heater NJB		111TDIO DO:A.009			
D O		42MJB	Contactor for Spillway, Siphon Pipe, and Penstock Gate Heaters		111TDIO DO:A.010			
D O		SPARE	SPARE		111TDIO DO:A.011			
D O		SPARE	SPARE		111TDIO DO:A.012			
D O		SPARE	SPARE		111TDIO DO:A.013			
D O		SPARE	SPARE		111TDIO DO:A.014			
D O		SPARE	SPARE		111TDIO DO:A.015			
D O		SPARE	SPARE		111TDIO DO:A.016			

PART 2 EXECUTION

Not Used

END OF SECTION

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SECTION 52 00 00

DRAWINGS

PART 1 GENERAL

1.01 DISCREPANCIES, ERRORS, OR OMISSIONS

- A. Inform the CO of discrepancies discovered on drawings in accordance with clause at FAR 52.236-21, Specifications and Drawings for Construction.
- B. In accordance with clause at FAR 52.236-21, Specifications and Drawings for Construction, in case of discrepancies, written specifications take precedence over drawings unless otherwise specified.

- 1. Highlighting provided on drawings is solely for convenience of the Contractor to help identify areas of work defined in the written specifications.

1.02 PROJECT CONDITIONS

- A. Where there are differences as determined by the CO between details and dimensions shown on drawings and details and dimensions of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.03 INFORMATION DRAWINGS

- A. Drawings marked "For Information Only" in the drawing list are included to show existing features about which knowledge is required to perform work under this contract. These drawings do not show work to be performed under this contract.
- B. If there are differences as determined by the CO between details and dimensions shown on these drawings and those of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.04 STANDARD DRAWINGS

- A. Standard drawings may show details which are not a part of work under this contract. Disregard details shown on these drawings which are not applicable to work under this contract.

1.05 COPIES OF DRAWINGS

- A. One set of full-size (22 inches by 34 inches) drawings, except standard drawings, will be furnished to the Contractor.
- B. Upon request, additional half-size (11 inches by 17 inches) copies of standard drawings will be furnished to the Contractor.

1.06 LIST OF DRAWINGS

- A. Drawings listed in Table 52 00 00A – List of Drawings are made a part of Section C – Description/Specifications.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

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Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
Map:		
1	245-D-5679	Vicinity Map
2	245-713-2275	Location Details
3	245-D-5631	General Plan Dam Profile
Demolition:		
4	245-D-60095	Powerplant – Electrical Demolition - General Plan
5	245-D-60096	Powerplant – Electrical Demolition – Single Line Diagram
6	245-D-60109	Gatehouse – Electrical Demolition - General Plan
7	245-D-60110	Gatehouse – Electrical Demolition – Single Line Diagram
8	245-D-7027_D	Switchyard Oil Circuit Breaker (JY1A) 164
9	245-D-9000	Station Annunciator CSA1FA & CSA3T Green - Schematic Wiring Diagram
10	245-D-9001	Generator Auxiliaries Schematic Green - Schematic Wiring Diagram
11	245-D-9002_D	Plant Miscellaneous Circuits - Schematic Wiring Diagram
12	245-D-9003_D	Service Water and Generator Cooling Water Control - Schematic Wiring Diagram
13	245-D-9007	Generator and Transformer K1A Differential Relaying - Schematic Wiring Diagram
14	245-D-9142_D	Penstock Gate, Alarm and Heater Control - Schematic Wiring Diagram
15	245-D-9212	Metering and Relaying – C.T. Circuits Schematic Wiring Diagram
16	245-D-9213_D	Potential and Synchronizing Circuits - Schematic Diagram
17	245-D-9214	Generator Annunciator Device CSA1F/B Schematic Wiring Diagram
18	245-D-9215	Unit 1 Control Schematic Wiring Diagram
19	245-D-9217	Generator Protection – Schematic Wiring Diagram

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
20	245-D-9218_D	PCB U1A1 Control and Plant Relaying – Schematic Diagram
21	245-D-10673_D	Unit 1 Excitation System – Schematic Diagram Place Holder
22	245-D-10674_D	Unit 1 Excitation System – Schematic Diagram (Sht. 4 of 18) Place Holder
23	245-D-10678_D	Unit 1 Excitation System – Schematic Diagram (Sht. 8 of 18)
24	245-D-5630_D	AC and DC Distribution Systems – Single Line Place Holder
Installation:		
25	245-D-5630_I	AC and DC Distribution Systems – Single Line
26	245-D-7027_I	Switchyard Oil Circuit Breaker (JY1A) 164 Place Holder
27	245-D-9002_I	Plant Miscellaneous Circuits - Schematic Wiring Diagram Place Holder
28	245-D-9003_I	Service Water and Generator Cooling Water Control - Schematic Wiring Diagram Place Holder
29	245-D-9142_I	Penstock Gate, Alarm and Heater Control - Schematic Wiring Diagram Place Holder
30	245-D-9213_I	Potential and Synchronizing Circuits - Schematic Diagram Place Holder
31	245-D-9218_I	PCB U1A1 Control and Plant Relaying – Schematic Diagram Place Holder
32	245-D-10673_I	Unit 1 Excitation System – Schematic Diagram Place Holder
33	245-D-10674_I	Unit 1 Excitation System – Schematic Diagram (Sht. 4 of 18) Place Holder
34	245-D-10678_I	Unit 1 Excitation System – Schematic Diagram (Sht. 8 of 18)
35	245-D-60097	Switchboard DSA – Single Line Diagram
36	245-D-60098	Switchboard DSA – Panel Schedule

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
37	245-D-60099	Panelboards DSA4 and MSA and Motor Control Centers M1A, M1B, and M1C – Single Line Diagram
38	245-D-60100	Panelboard DSA4 – Panel Schedule
39	245-D-60101	Switchboard DSA – Power Cable Schedule
40	245-D-60102	Switchboard DSA – Power Conduit Schedule
41	245-D-60103	Motor Control Center M1C – Cable and Conduit Schedule
42	245-D-60104	Panelboard DSA4 – Power Cable Schedule
43	245-D-60105	Panelboard DSA4 – Power Conduit Schedule
44	245-D-60106	Switchboard DSA – Conduit Plan View
45	245-D-60107	Switchboard DSA – Conduit Layout Plan
46	245-D-60108	Control Room – Conduit Layout Plan
47	245-D-60111	AC Distribution – Single Line Diagram
48	245-D-60112	Panelboards DJA and LJA – Panel Schedules
49	245-D-60113	Gatehouse – Power Cable Schedule
50	245-D-60114	Gatehouse – Power Conduit Schedule
51	245-D-60115	Gatehouse –Conduit Layout Plan
52	245-D-60123	Place Holder
53	245-D-60124	Place Holder
54	245-D-60125	Unit 1 Control – Schematic Diagram (1/3)
55	245-D-60126	Unit 1 Control – Schematic Diagram (2/3)
56	245-D-60127	Unit 1 Control – Schematic Diagram (3/3)
57	245-D-60128	Unit 1 Control – Schematic Diagram (3/3)
58	245-D-60129	Generator Local Auto Supervisory Start Permissive – Ladder Logic Diagram Place Holder
59	245-D-60130	Generator HMI Unit Start Sequence Display Information – Ladder Logic Diagram Place Holder
60	245-D-60131	Generator HMI Unit Stop Sequence Display Information – Ladder Logic Diagram Place Holder

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
61	245-D-60132	Penstock Gate, Alarm and Heater Control – Schematic Diagram Place Holder
62	245-D-60133	Plant Miscellaneous Circuits – Schematic Diagram Place Holder
63	245-D-60134	Unit Control Communication – Network Diagram Place Holder
64	245-D-60135	Main Control Board CSA Panel 1F – Layout Diagram
65	245-D-60136	Main Control Board CSA Panel 1R – Layout Diagram
66	245-D-60137	Main Control Board CSA Panel 2F – Layout Diagram
67	245-D-60138	Main Control Board CSA Panel 2R – Layout Diagram
68	245-D-60139	Main Control Board CSA Panel 3F – Layout Diagram
69	245-D-60140	Main Control Board CSA Panel 3F/4F Top Section – Layout Diagram
70	245-D-60141	Turbine Distributed I/O Cabinet (G1T) – Layout Diagram
71	245-D-60142	Generator Air Housing Cabinet (G1A) – Layout Diagram
72	245-D-60143	Gatehouse HMI Cabinet (MJB) – Layout Diagram
73	245-D-60144	Main Entrance HMI Cabinet – Layout Diagram
74	245-D-60145	K1A Cabinet (149K1A) – Layout Diagram
75	245-D-60146	KSA Cabinet (149KSA) – Layout Diagram
76	245-D-60147	Main Control Board Panel 3F – Layout Diagram
77	245-D-60148	CSA1R & CSA3F Ctrl. & Instrument SW. Escutcheons & Indicating Lamps – Designation Drawing
78	245-D-60149	Generator Differential Lockout 186GD – Schematic Diagram
79	245-D-60150	Transformer K1A Differential Lockout 186K1A – Schematic Diagram
80	245-D-60151	SEL-700G & 787 Relays Power Supply – Schematic Diagram
81	245-D-60152	External RTU Connections – Schematic Diagram
82	245-D-60153	Generator Protective Lockout 186GP – Schematic Diagram

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
83	245-D-60154	PCB U1A1 Control and Plant Relaying – Schematic Diagram Place Holder
84	245-D-60155	Compressor Relocation – Plans and Sections
85	245-D-60156	Compressor Relocation - Sections
86	245-D-60157	Fiber Conduit Support - Sections
87	245-D-60158	Place Holder
88	245-D-60159	Place Holder
89	245-D-60160	Place Holder
90	245-D-60161	Place Holder
91	245-D-60162	Transformer K1A and KSA Protection/Diff. Relaying CT's and PT's – 3-Line Diagram
92	245-D-60163	Generator 111GP/111GS Protection and Diff. Relaying CT's and PT's – 3-Line Diagram
93	245-D-60164	Gatehouse 125 VDC Distribution – Single Line Diagram
94	245-713-1852	Generator Electronic Speed Switches – Schematic Diagram Place Holder
Informational Drawings:		
95	245-D-5354	Power Plant and Switchyard – Switching Diagram
96	245-D-5589	General Arrangement Transverse Sections
97	245-D-5590	General Arrangement Plans and Sections
98	245-D-5591	General Arrangement Plans – Floor El. 6597.25 and El. 6581.00
99	245-D-6179	Steel Penstock – General Plan and Profile
100	245-D-6180	Welded and Riveted Steel Penstock Sta. 498+57.70 to Sta. 504+77.29
101	245-D-6181	Welded and Riveted Steel Penstock Sta. 504+77.29 to Sta. 511+12.77
102	245-D-6182	Welded and Riveted Steel Penstock Sta. 511+12.77 to Sta. 517+03.25

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
103	245-D-9005	Plant Gatehouse Control & Telemetry – Schematic Diagram
104	245-D-9009	125V Station Battery Chargers BSA and BSB and Cabinet B1A – Schematic and Wiring Diagram
105	245-D-9094	Main Control Board CSA Panels CSA1FY & Z Side Panels – Wiring Diagram
106	245-D-9096	Main Control Board CSA Panel CSA1R Rear Panel – Wiring Diagram
107	245-D-9097	Main Control Board CSA Panels CSA1RY & Z Side Panels – Wiring Diagram
108	245-D-9098	Main Control Board CSA Panel CSA2F Front Panel – Wiring Diagram
109	245-D-9099	Main Control Board CSA Panels CSA2RY & Z – CSA2RY & Z Side Panels – Wiring Diagram
110	245-D-9100	Main Control Board CSA Panel CSA2R Rear Panel – Wiring Diagram
111	245-D-9101	Main Control Board CSA Panel CSA3F Upper Half Front Panel – Wiring Diagram
112	245-D-9102	Main Control Board CSA Panel CSA3F Lower Half Front Panel – Wiring Diagram
113	245-D-9103	Main Control Board CSA Panels CSA3FY & Z Side Panels – Wiring Diagram
114	245-D-9105	Main Control Board Panel CSA4F – Wiring Diagram
115	245-D-9106	Main Control Board CSA Panels CSA4FY & Z Side Panels – Wiring Diagram
116	245-D-9107	Main Control Board CSA – Panel 4R Rear & Side Panels – Wiring Diagram
117	245-D-9108	Main Control Board Panels CSA5F & Panels CSA5FY & Z Front & Side Panels – Wiring Diagram
118	245-D-9109	Main Control Board CSA5R Auxiliary Relay Rack 1 – Wiring Diagram

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
119	245-D-9110	Place Holder
120	245-D-9111	Place Holder
121	245-D-9112	Place Holder
122	245-D-9113	Place Holder
123	245-D-9114	Place Holder
124	245-D-9115	Place Holder
125	245-D-9116	Place Holder
126	245-D-9117	Place Holder
127	245-D-9118	Place Holder
128	245-D-9119	Main Control Board CSA5R Auxiliary Relay Rack 11 – Wiring Diagram
129	245-D-9120	Main Control Board Panels CSA5R Terminal Block Rear Panel CSA5RY & CSA5RZ – Wiring Diagram
130	245-D-9121	Main Control Board CSA – Front Panels Equipment Arrangement
131	245-D-9122	Main Control Board CSA – Rear Panels Equipment Arrangement
132	245-D-9129	Governor A1A – Wiring Diagram
133	245-D-9136	Control Center M1A – Schematic and Wiring Diagram
134	245-D-9141	Penstock Gate AC and DC Distribution Systems – Single Line Diagram
135	245-D-10697	Unit 1 Excitation Cubicle E1A – Panel 1R – Wiring Diagram Sht. 7

Table 52 00 00A – List of Drawings

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136	245-D-10698	Unit 1 Excitation Cubicle E1A – Panel 1R – Wiring Diagram Sht. 8
137	245-D-10707	Unit 1 Excitation Cubicle E1A – Panel 1D – Wiring Diagram Sht. 17
138	245-704-3778	Lighting System El. 6564.75 to El. 6581.00 – Conduit Installation
139	245-704-3779	Lighting System El. 6581.00 to El. 6597.25 – Conduit Installation
140	245-704-3780	Lighting System El. 6581.00 to El. 6659.25 – Conduit Installation
141	245-704-3781	Lighting System El. 6597.25 to El. 6610.71 – Conduit Installation
142	245-704-3808	Cable and Duct System Plan and Sections – EL. 6577.75 to EL. 6581.00 – Conduit Installation
143	245-704-3810	Cable and Duct System Sections – EL. 6581.00 to EL. 6597.25 – Conduit Installation
144	245-704-3858	Grounding System Network Plans – EL. 6597.25 to EL. 6581.00
145	245-704-3956	Penstock Gate Structure Electrical System – Plans and Sections – Conduit Installation
146	245-704-4052	Cable and Duct System -Plan and Sections – EL. 6597.25 – Lines D to E – Conduit Installation
147	245-704-4282	Gate Motor Board MJA – Schematic and Detail Wiring Diagrams
148	245-704-4414	Penstock Gate Structure – Electrical System – Sections – Conduit Installation
149	245-704-5085	Power Plant to Terminal Pole and Gate Structure to Terminal Pole – Conduit Installation
150	245-704-5721	Governor A1A – Wiring Diagram
151	245-704-5722	13.8/115 KV. GRD Y Transformer K1A – Wiring Diagram
152	245-704-5899	Cabinets M1D and M1J – Schematic and Detail Wiring Diagram

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
153	245-704-5900	Cabinet M1E – Schematic and Detail Wiring Diagram
154	245-704-5909	Pressure Regulator and Turbine Auxiliaries – Wiring Diagram
155	245-704-5916	Annunciator Relay Cabinet CSC Terminal Blocks – Wiring Diagram
156	245-704-6723	Main Control Board CSA – Panel 5R – Wiring Diagram
157	245-704-7899	Main Control Board CSA Relays – Wiring Diagram
158	245-704-7900	Main Control Board CSA Relays – Wiring Diagram
159	245-713-0959	Turbine Greasing System Automatic Control – Schematic Diagram
160	245-713-1773	Excitation Control Cabinet C1B – Wiring Diagram
161	245-713-60049	RTU Cabinet C1CA – Layout Diagram
162	245-713-60052	RTU Cabinet C1CA Right Side – Wiring Diagram
163	245-713-60053	RTU Cabinet C1CA Back Panel (Left Side) – Wiring Diagram
164	245-713-60054	RTU Cabinet C1CA Back Panel (Right Side) – Wiring Diagram
165	245-713-60055	RTU Cabinet C1CA Left Side – Wiring Diagram
166	245-713-60057	RTU Cabinet C1CC – Layout Diagram
167	245-713-60058	RTU Cabinet C1CC Right Side – Wiring Diagram
168	245-713-60059	RTU Cabinet C1CC Back Side – Wiring Diagram
169	245-713-60060	RTU Cabinet C1CC Left Side – Wiring Diagram
Standard Drawings:		
170	40-D-4334	Typical Grounding Details – Sht. 1 of 2
171	40-D-4335	Typical Grounding Details – Sht. 2 of 2
172	40-D-5324	Control and Instrument Switch Escutcheons and Indicating Lamps
173	40-D-6187	Unit Control Board – Controls, Instruments and Relaying Standard Nameplates
174	40-D-6234	Standard Nameplates

Table 52 00 00A – List of Drawings

Sheet No.	Drawing No.	Title
175	40-D-7102	Government Drawing Format - Drawing Border, Sample Title Block, and Signature Lines
176	104-D-1150	Device Designations and Symbols to be Used on Single-Line and Schematic Diagrams
177	104-D-1165	Typical Powerplant Main Control Board CSA - Panel 7R Wiring Diagram
178	104-D-60001	Design Data – Electrical – Government Format – Breaker Control - Schematic Diagram
179	104-D-60002	Design Data – Electrical – Microprocessor Relay – Schematic Diagram
180	104-D-60003	Design Data – Electrical – Microprocessor Relay – Wiring Diagram
181	104-D-60005	Design Data – Electrical - Electrical Government Format – Standards
182	104-D-60006	Switchyard – Three Line Diagram – Standards

END OF SECTION